



Ministry of Works and Transport

ROAD DESIGN AND CONSTRUCTION MANUAL

Volume V: Low Volume Sealed Roads



July 2018



CrossRoads
Supporting for national road construction industry



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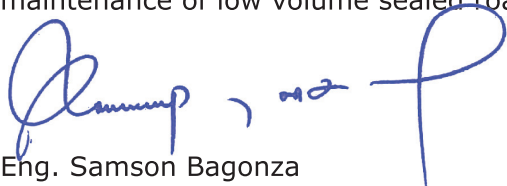
Foreward

Uganda's road network comprising about 145,000km of national, urban, district and community access roads, provides the dominant mode of freight and passenger transport. Unsealed low volume roads constitute over 90% of the Country's road network and largely impact negatively on the lives of majority of the population who live and work in the rural areas. Therefore, a significant portion of the road network, especially of the low volume roads, needs to be sealed in pursuit of Vision 2040 and the National Development Plan.

The main purpose of the Specifications Series 3000LVR, 4000LVR and the Road Design and Construction Manual is to provide stakeholders with local, regional and international best practices for the design, construction and maintenance of low volume sealed roads. Specifications Series 3000LVR and 4000LVR have been prepared as an addition to the General Specifications for Road and Bridge Works and contain cross references to other sections therein. Series 3000LVR and 4000LVR cater for pavement and surfacing requirements of low volume sealed roads.

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Preface

This manual has been prepared for the Ministry of Works and Transport and the Uganda Road Sector in general for the design and construction of low volume roads paved using low cost seals. The manual covers standards and guidelines for the design of road geometry, drainage, materials, pavements and surfacing. The basic principle is to deliver more for less which entails greater value for money for the clients.

The manual was prepared as part of the national capacity building in the training of practitioners and managers including provision of facilities and equipment. The manual shall be the main referral document for design, tendering and execution of related road works.

The manual conforms to both local and international standards for the key activities covered in the document including innovations and simplification of the processes relating to procurement of works and quality control.

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Abbreviations

AADT	Annual Average Daily Traffic
AASHTO	American Association of State Highway Transportation Officials
ADT	Average Daily Traffic
AHD	Average Haul Distance
AIV	Aggregate Impact Value
ASTM	American Standard and Testing Materials
BoQ	Bill of Quantities
BS	British Standards
CBR	California Bearing Ratio
DCP	Dynamic Cone Penetrometer
DSD	Double Surface Dressing
ESA	Equivalent Standard Axles
ICL	Initial Consumption of Lime
LCS	Low Cost Seals
LVR	Low Volume Roads
LVSR	Low Volume Sealed Roads
MAE	Measurement After Execution
MDD	Maximum Dry Density
MELTC	Mount Elgon Labour Based Training Centre
MESA	Million Equivalent Standard Axles
MOD	Modified
MoWT	Ministry of Works and Transport
OMC	Optimum Moisture Content
OMC	Optimum Moisture Content
PCU	Passenger Car Units
Pen	Penetration
SN	Structural Number
SS	Sand Seal
SSD	Single Surface Dressing
TLC	Traffic Loading Class

PART A: DESIGN OF LOW VOLUME ROADS

1.0 Introduction

Low volume roads are defined as roads or sections of road carrying low traffic and traffic is defined and used in a number of ways:

- i. The traffic volume in terms of the number of vehicles passing a particular point or section per day. There is no international standard on the limit of traffic. However, there is general consensus that the maximum limit should be 300 vehicles per day in the middle year of the design life of the roads which is usually 15 to 20 years;
- ii. The traffic volume in terms the loading in equivalent standard axles (ESAs). The consensus is that the maximum limit for low volume roads is one million equivalent standard axles (1MESA);

The traffic volume in number of vehicles per day is important for geometric design influencing decisions on road width and other elements of geometric design; and

The traffic volume in terms of traffic loading is important for pavement design and also the design of surfacings.

This manual provides detailed standards and guidelines for design covering:

- i. Geometric design: The basic principle is to follow alignments of existing roads as much as possible in order to minimise earthworks and take advantage of the in-situ strength built up through prolonged periods of consolidation under traffic and repetitive wetting and drying;
- ii. Drainage design: The service life of a low volume road is dependent largely on the efficiency of the drainage system. The manual covers drainage in great detail and this is also in recognition of the impact of climate change in sustainability of low volume roads;
- iii. Materials selection:– Material selection is also very important because the materials that are often used are non-conventional and at times marginal. The approach to the selection of materials for the pavement design is different from the conventional approach in that designs are tailor-made for the locally available materials;
- iv. Pavement design: Low cost seals are generally thin and contribute little to the strength of the pavement. The structural considerations for the different scenarios obtained as a result of weak subgrade and thin surfacings have resulted in the development of appropriate pavement structures or catalogues which are detailed in the manual;
- v. Design of low cost surfacing: The cost of surfacings is usually the predominant one and minimising the cost of surfacing is important in the design of low volume roads. The cost of surfacing is minimised through the use of locally available materials, simpler construction techniques and innovation; and
- vi. Work standards and guidelines: This section provides the descriptions, the list of tasks, the standards and guidelines which instruct the execution of the activities.

– This manual also aims at reducing the life cycle costs of road surfacing. Life-cycle costs are summation of the construction and maintenance costs incurred over the whole life of the roads. The cost of construction is minimised through the use of locally available materials and the employment of appropriate methods of construction. Also, long life surfacings lead to lower maintenance and in turn reduce life-cycle costs. Minimising the costs of construction and the overall cost of maintenance leads to affordability and sustainability of the low volume sealed road networks which are fit for purpose.

The manual is developed to be used by all practitioners including the contractors, consultants, and the public sector.

2.0 Road Investigations

2.1 Introduction

This section covers road investigations which are appropriate for low volume roads. The approach depends largely on whether the works involve:

- i. New construction: This is a situation where a new road and a new alignment is required;
- ii. Upgrading an existing earth/gravel road to a low volume sealed road standard: This is where the road must follow the existing alignment as much as possible in order to take advantage of the inherent strength and existing drainage; and
- iii. Upgrading a lower class road to a higher class road with low cost sealing: This may include realignment and widening to accommodate increased traffic.

This section is focussed more on investigations which should be carried out when upgrading unpaved roads to paved road standards. When a decision has been made to upgrade a road from earth/gravel to a low volume sealed road, information about the existing road needs to be gathered.

The information gathered at this stage assists the designer to get an overall representation of the road so as to be able to produce a cost effective design. This process therefore has a huge bearing on the final cost of the road and how well it will serve its purpose.

2.2 Preliminary Assessment

Driving or walking along the roads should provide good preliminary information to the designer on nature of each of the road and help in planning for a more detailed assessment.

In such an assessment the extent of swamps, the general topography and geometry, the state of drains and the type and level of traffic may be obtained.

Existing records from various sources may also be obtained at this stage of the assessment. From the maintenance data, information about the materials borrow pits, the date for the last re-gravelling and the properties of the gravel used can be obtained.

Information from the preliminary assessment helps in guiding the detailed assessment.

2.3 Visual Condition Assessment

Prior to conducting the survey, chainages should be marked along the roadside using wooden pegs or preferably iron pegs. A measuring wheel may be used to determine the distances and every 500m the chainage should be painted on an immovable object. The pegs should be located in such a way that they do not pose a danger to pedestrians and cannot be easily vandalised.

This assessment is important in making sure that any necessary improvements are carried out before the execution of surfacing works. It is not uncommon to find rutting and potholing on gravel roads. A visual condition assessment helps to identify any defects and to quantify the extent of such defects. Extensive presence of rutting may be an indication that the layers of gravel present are not strong enough to withstand the existing traffic levels. Potholes may require filling up before any scarification; re-shaping and re-compaction can take place.

The following defects should be noted:

- i. Areas of excessive subsidence;
- ii. Lengths and locations of low-lying sections;
- iii. Extent of areas with oversize materials;

- iv. Rut depths, lengths and locations;
- v. Extent, size and number of potholes; and
- vi. The lengths and extents of gullies on the carriageway.

These defects and quantities are later used for determining work activities and in computing work quantities..

2.4 Geometry Assessment

Road geometry greatly affects the quantity of earthworks and hence the general cost of the works. If a high standard of geometry is desired, then a full topographic survey should be carried out to compare the existing vertical and horizontal alignment to the geometric standards appropriate for low volume sealed roads.

However if the focus is on providing a basic standard, then a less rigorous survey may be carried out to correct adverse areas such as extremely steep grades, sharp bends, poor sight distance and low-lying areas.

Geometric assessments involve the measurement of cross-sectional profile in order to determine whether carriageway widths, shoulders and drain dimensions are adequate and meet the minimum design standards of the particular class of road.

2.5 Structural Assessment

It is vital to ascertain the strength of any existing layers of materials on the road so that decisions can be made on whether to incorporate them in the new pavement and at what level in the structure.

A Dynamic Cone Penetrometer (DCP) is ideal for the purpose of ascertaining the strength and thicknesses of the existing layers at in-situ conditions. More details are available in TRL publication "Dynamic Cone Penetrometer Tests and Analysis, by Colin Jones. PR/INT/277/04" which may be obtained from www.transport-links.org for instructions on how to use the DCP and <http://www.transport-links.org/ukdcp/index.htm> for a software for analysis of the data. Forms for recording field data are also available in the former. The software includes tutorials on the analysis. Analysis may also be done using Microsoft Excel.

An alternative method for determining the strength of the materials is the DCP-DN method which is described in (Chapter 6)

2.6 Climatic Assessment

The performance of low volume roads is affected more by environmental factors than traffic as seen in the Figure 21. Because of this, it is important to understand the climatic factors particularly rainfall and evaporation in the area that the road will be constructed. Knowing the rainfall and the evaporation enables the engineer to be able to compute the Weinert value that is later used to select the design charts and tables to obtain the pavement layer thicknesses.

The Weinert Number can be obtained from:

$$N = 12 \times \frac{E_y}{P_n}$$

Where E_y is the evaporation in the hottest month of the year (mm)

P_a is the annual precipitation (mm)

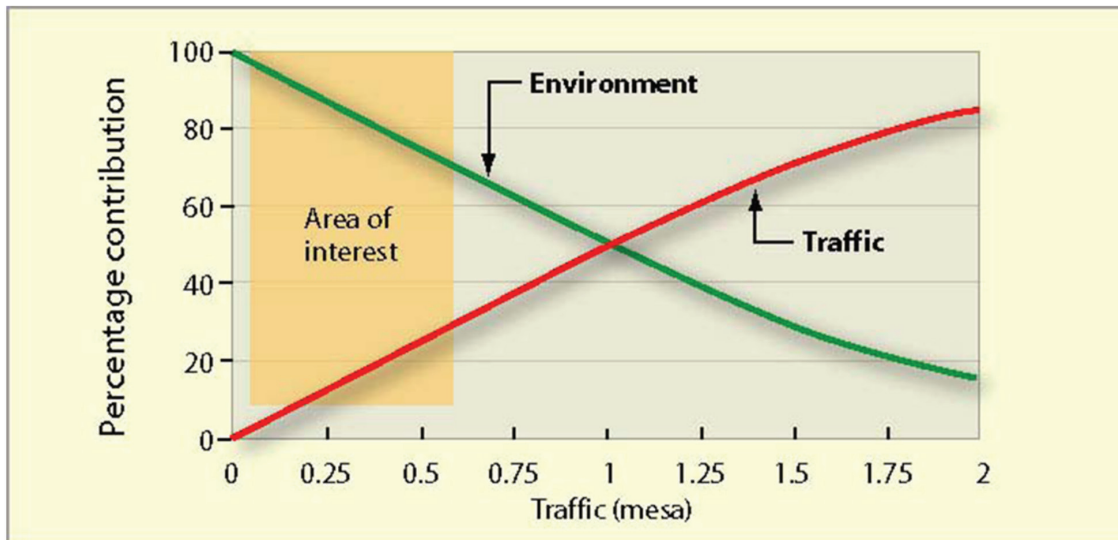


Figure 2-1 Comparison of the Effect of Environment and Traffic on LVR

Source: Pinard & Overby, 2006.

A guide of the Weinert values for various climates is shown in Table 21.

Table 2-1 Indicative Weinert N Values

Description	Weinert N Value	Thornthwaite Moisture Index, I_m	Typical Mean Annual Rainfall (mm)
Arid	5+	< -40	< 250
Semi-arid	4 to 5	-20 to -40	250-500
Semi-arid to Sub-tropical (Savanna)	2 to 4	-20 to +20	500-1000
Humid Tropical (Equatorial)	< 2	+20 to +100	> 1000

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2.7 Drainage Assessment

The drainage conditions should be good with scour checks on steep sections. This is because when the road is surfaced, there will be increased runoff that can lead to scouring. During the assessment, personnel should walk along the length of road to be surfaced and check the condition of the drainage system including scour checks and the condition of the existing structures.

Local residents usually have useful information about locations of flood prone areas and the flood levels. This can greatly simplify the amount of hydraulics calculations needed to size culverts and other drainage structures.

If side drains exist, then they should be checked for scouring, collapsed sides, collected debris, and missing rip-rap. When no side drain exists and the section is not on a substantially raised embankment, consideration should be made to provide side and mitre drains. Existing culverts should be checked for silting or damage. Where water cuts across the carriageway and in valleys, provision should be made for culverts. For other structures such as box culverts and small bridges, the following aspects should be assessed:

- i. General condition of the structures:

- ii. Original design life:
- iii. As-built records if they exist:
- iv. Indications of maximum flood levels on structures:
- v. Signs of post construction settlement:
- vi. Main problems with the existing structures:
- vii. Failures in any of the structural elements:
- viii. Current level of scour around structures:
- ix. Indications of excessive loading or abuse: and
- x. Dimensions and any possibility of refurbishment or adaptation.

2.8 Traffic Assessment

Before selecting the appropriate geometric standard or the required pavement structure, the traffic composition, numbers and axle loads need to be determined and aggregated. This can be achieved by conducting a minimum of 7 days traffic count with at least 2 days count conducted for 24 hours of which one is during the week and the other during the weekend.

It may not be possible to measure axle loads, but these can be obtained from records and equivalent factors recommended by the roads authority. This also applies to traffic growth rates.

The following aspects should be considered when determining design traffic.

- i. Existing vehicle and axle load data;
- ii. Final road width;
- iii. Traffic density: Whether it varies seasonally or on market days in the local area;
- iv. Review of standards used elsewhere in order to adopt appropriate ones;
- v. Changes in traffic including vehicle size or loading when the road is improved (generated and/or diverted traffic);
- vi. Exceptional loads which may be transported—including logging, quarries, mining or other industries in the area; and
- vii. Other traffic, economic and safety implications.

2.9 Other Assessment Notes

A good assessment form for collecting field data during visual condition is given in Figure 22. A similar form can be produced and customised for particular needs, for example instead of recording data in 10m increments, data may be recorded in 100m increments. For critical sections such as where realignment needs to be done or where levels need to be readjusted or where drainage needs to be re-aligned, a strip map may need to be produced.

Condition Survey Form												
ROAD			ROAD TYPE					DATE				
SECTION			ROAD WIDTH									
KILOMETRE			TERRAIN					SURVEYOR				
			1	2	3	4	5	6	7	8	9	10
SECTION	CHAINAGE	From	0	20	40	60	80	100	120	140	160	180
		To	20	40	60	80	100	120	140	160	180	200
	GRADIENT (%)											
	CURVATURE (DEG)											
LEFT	VEGETATION											
	DRAINAGE	Existence										
		Scouring										
		Blockage										
	SPECIAL DRAINAGE											
	SLOPES	Type										
		Damaged										
	SHOULDER	Type										
Level												
CARRIAGEWAY	SHAPE											
	EFFECTIVE WIDTH											
	CROWN HEIGHT											
	SURFACE	Ruts										
		Corrugations										
		Potholes										
		Loose Material										
Oversize Material												
RIGHT	VEGETATION											
	DRAINAGE	Existence										
		Scouring										
		Blockage										
	SPECIAL DRAINAGE											
	SLOPES	Type										
		Damaged										
	SHOULDER	Type										
Level												
URGENT												
SPOT IMPROVEMENT												
GRAVEL THICKNESS												
CULVERTS	Repair											
	New											
BRIDGES	Repair											
	New											
RIVER / STREAM												

Figure 2-2 Visual Condition Survey Form

3.0 Geometric Design

3.1 Introduction

Geometric design is the process whereby the layout of the road through the terrain is designed to meet the needs of all the road users. The geometric standards are intended to meet two important objectives namely: i) to provide minimum levels of safety and comfort for drivers by provision of adequate sight distances, coefficients of friction and road space for manoeuvres and, ii) to minimise earthworks to reduce construction costs.

Geometric design covers road width; cross-fall; horizontal and vertical alignments and sight lines; and the transverse profile or cross-section. The cross-sectional profile includes the design of the side drainage ditches, embankment heights and side slopes, and is a vital part of geometric design for low volume roads. The cross-section essentially adapts the pavement to the road environment and is part of the drainage design.

3.2 Geometric Design process.

Three basic geometric elements of cross section, horizontal alignment and vertical alignment are defined for LVRs based on the AADT. These basic elements are largely determined by the design speed but may be modified, sometimes substantially, depending upon:

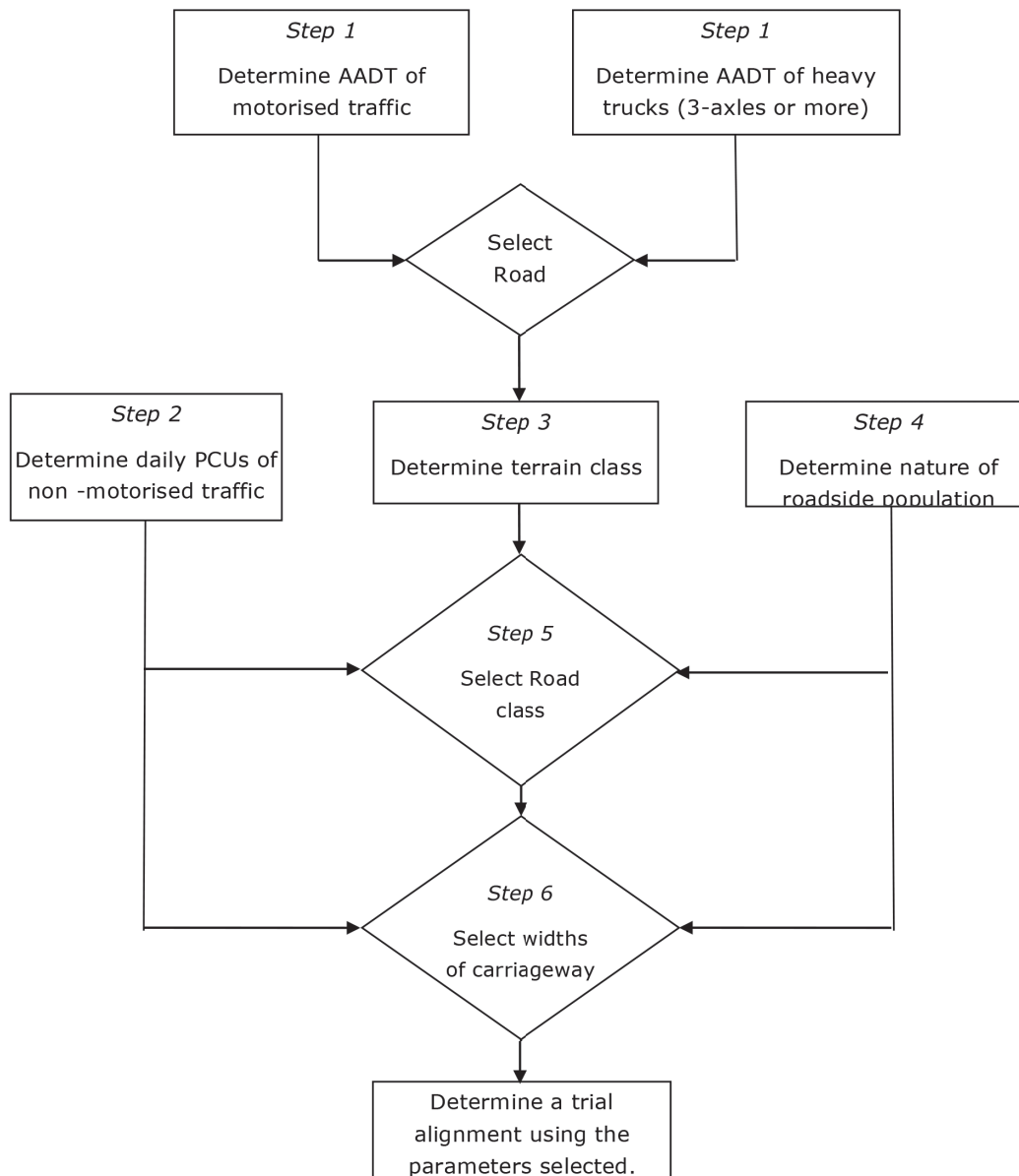
- i. Traffic composition;
- ii. Topography or terrain class;
- iii. Nature of road side population;
- iv. Road surface type (or structure);
- v. Land use and physical features;
- vi. Construction technology;
- vii. Safety;
- viii. Economic and financial considerations; and
- ix. Environmental considerations.

Figure 3-1 shows the process for geometric design

3.2.1 Traffic Volume and Composition

Assessment of traffic volume and axle loading is discussed in Section 2.8. The AADT of motorised vehicles with two or more axles provides the basic method of defining the different geometric road standards but these are modified based on the proportion of heavy vehicles in the traffic stream and also on the combined number of non-motorised vehicles, two-wheeled vehicles and pedestrians. These are assessed in terms of the effective road space that they occupy (dependent on size and speed) measured in terms of PCUs.

The impact of non-motorised and two wheeled traffic is significant and should be taken account of in the geometric design. In order to incorporate this traffic, each type is converted to equivalent passenger car units. The conversion is given in Table 31. Once converted, they should be added to the normal AADT of passenger cars for geometric design purposes.

**Figure 3-1 Geometric Design Process****Table 3-1 Equivalent Passenger Car Units (PCUs) Values**

Vehicle	PCU value
Pedestrian	0.15
Bicycle	0.20
Motor cycle	0.25
Bicycle with trailer	0.35
Motor cycle taxi	0.40
Motor cycle with trailer	0.45
Small animal-drawn cart	0.70
Bullock cart	2.0
All based on a passenger car = 1.0	

3.2.2 Terrain/Topography

Terrain has a major influence on geometric standards, both because of the very much higher costs that would occur if the same standards were used in hilly and mountainous terrain as in flat terrain and because of the additional safety aspects required for roads in such terrain.

Terrain class is determined by the number of 5-metre contours crossed by a straight line connecting the two ends of the road section in question. Four terrain classes are defined in Table 3-2.

Table 3-2 Terrain Classes

Classification	Definition
Flat	0 to 10 five-metre contours per km. The natural ground slopes perpendicular to the ground contours are generally below 3%.
Rolling	11 to 25 five-metre contours per km. The natural ground slopes perpendicular to the ground contours are generally between 3 and 25%.
Mountainous	26 to 50 five-metre contours per km. The natural ground slopes perpendicular to the ground contours are generally above 25%.
Escarpment	Escarpments are geological features that require special geometric standards because of the engineering risks involved. Typical gradients are greater than those encountered in mountainous terrain.

It should be noted that the terrain class is not dependent on the alignment chosen for the road.

3.2.3 Roadside population

When a road passes through a village, town or other populated or market areas an extra carriageway is provided in each direction for parking and for passenger pick-up and a pedestrian footpath is also specified. The latter is essentially the road shoulder. In addition, the main carriageway is paved and it should be 4.5-6.0 metres wide unless specified otherwise.

These standards for the lower traffic levels of LVSRs which are single carriageway when the road is passing through a particularly well-populated area should include widened shoulder (up to 2.5 metres) for the extent of the populated area.

3.2.4 Road surface type

The friction factors for paved and unpaved roads are significantly different and this affects the distance required to stop safely and the safe speed for negotiating curves. Thus the geometric standards differ according to the type of surface.

3.2.5 Land use and physical features

Land use influences the design of the drainage features of a road and access to the road. Dealing with water run-off and potential erosion for example, forms an important aspect of design, much of which is geometric in nature. These aspects are also dealt with under construction standards and guidelines in Part B.

3.2.6 Construction technology

In a labour-abundant economy it is usually beneficial to maximise the use of labour rather than rely predominantly on equipment-based methods of road construction. In such a situation the choice of technology affects the standards that can be achieved, especially in hilly and mountainous areas. This is because:

- i. Cut and fill need to be minimised;
- ii. Economic haul distances are limited to those achievable using wheel-barrows;
- iii. Mass balancing is achieved by transverse rather than longitudinal earth movements;
- iv. Maximum gradients follow the natural terrain gradients; and

- v. Horizontal alignments may be less direct.

The standards in hilly and mountainous terrain are always lower than in flat terrain but this reduction in standards need not necessarily be greater where labour-based methods are used. Following the contour lines more closely will make the road longer but the gradients can be less severe. Every effort should be made to preserve the same standards in the particular terrain encountered irrespective of construction method.

3.2.7 Safety

The mix of traffic, including relatively old, slow-moving and usually overloaded vehicles, a large number of pedestrians, animal drawn carts and possibly motorcycle-based forms of transport, plus poor driver behaviour and poor enforcement of regulations means that methods to improve safety through engineering design assume paramount importance. Figure 32 shows some designs that improve safety.

The following factors affect safety:

- i. Traffic level and composition;
- ii. Vehicle speed;
- iii. Horizontal curvature;
- iv. Vertical curvature;
- v. Width of shoulders;
- vi. Embankment slopes;
- vii. Design of drainage ditches; and
- viii. Inappropriate public transport pick-up/set-down areas,

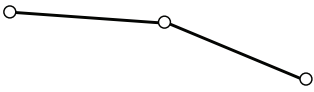
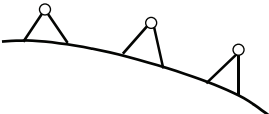
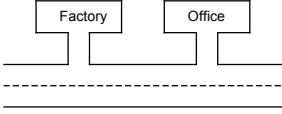
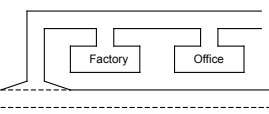
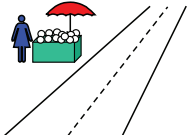
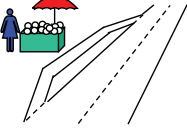
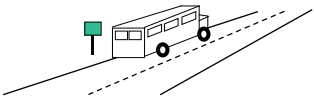
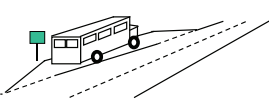
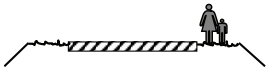
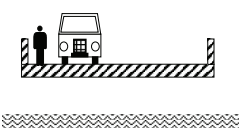
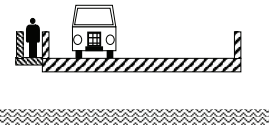
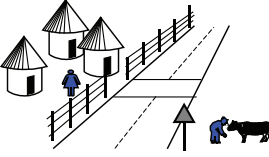
	Undesirable	Desirable	Principle applied
Route location			Major routes should by-pass towns and villages
Road geometry	(i)  (ii) 		Gently-curving roads have lowest accident rates
			Prohibit direct frontal access to major routes and use service roads
Roadside access			Use lay-bys or widened shoulders to allow villagers to sell local produce
			Use lay-bys for buses and taxis to avoid restriction and improve visibility
			Seal shoulder and provide rumble divider when pedestrian or animal traffic is significant
Segregate motorised and non-motorised traffic			Construct projected footway for pedestrians and animals on bridges
			Fence through villages and provide pedestrian crossings

Figure 3-2 Safety Features for LVSR

These factors are all inter-related and part of geometric design. In addition, safety is also affected by:

- Road surface type and surface condition (e.g. potholes);
- Dust (poor visibility); and
- Slippery unsealed road surfaces.

3.3 Design-by-Eye

The design-by-eye method is best suited to rehabilitation or upgrading projects where a road alignment already exists and is the preferred method for developing a design for a track or undesignated road under a community roads programme where a walking track is being improved to enable it to carry occasional vehicles. Nevertheless, considerable experience and skill is needed to carry out the design-by-eye method and the approach should only be used under the guidance and supervision of an experienced engineer.

3.4 Design Situations

There are three design situations namely:

- i. Upgrading from a lower class of road to a higher class;
- ii. Designing a road to replace an existing track; and
- iii. Designing a completely new road where no road existed before.

3.4.1 *Upgrading an existing gravel/earth road*

The basic alignments will already exist but the standards will probably be those applicable to a road of lower class, i.e. earth or gravel, Table 33. The existing road would therefore have an unfavourable camber of 6-8% which is too steep for sealed roads. The design speed for unpaved roads is lower and sharper curves are permissible. However, the camber and curvature should be assessed properly and corrected where necessary. The new sealed road may require higher standards which may involve a wider cross section, higher design speeds and therefore larger horizontal and vertical radii of curvature. In flat and rolling terrain, larger horizontal radii of curvature are usually achieved by means of minor realignments at the curves themselves. Larger vertical radii of curvature are usually more difficult but, depending on the terrain, can often be achieved by additional fill rather than deeper cutting. In more severe mountainous terrain it may be necessary to make substantial realignments to avoid deep cuts, for example, following a contour more closely to avoid a steep hill with inadequate sight distances over a crest.

The most difficult aspect is likely to occur in mountainous terrain when substantial widening is required. Under these circumstances it may not always be possible to meet the standards of the new road class and therefore adequate warning signs will need to be employed to alert drivers to the lower standards.

In general, the main improvements, apart from overall widening, are essentially spot improvements and do not require sophisticated design methods.

Table 3-3 Geometric Standards of Unpaved District Roads

	Flat Terrain			Rolling Terrain			Hilly Terrain		
	Design Speed			Design Speed			Design Speed		
	50	60	70	40	50	60	30	40	50
Recommended minimum vertical grade (%)	2	2	2	2	2	2	2	2	2
Recommended maximum vertical grade (%)	8	8	7	10	10	8	12	11	10
Maximum grade length at maximum vertical grade (m)	650	800	100	500	550	660	250	300	500
Absolute maximum vertical grade (%)	10	10	8	12	12	10	15	12	10
Maximum grade length at absolute maximum vertical grade (m)	325	400	500	250	275	300	125	150	250
Design camber (as-built)	8	8	8	8	8	8	8	8	8
Minimum curve radius below which super elevation is applied (m)	80	125	190	40	80	125	20	40	80
Minimum curve radii (m)	65	95	145	35	65	95	20	35	65
Safe stopping sight distance (m)	65	85	110	45	65	85	30	45	65
Crest curve K-value (for stopping sight distance)	12	20	35	8	12	20	5	8	12
Sag K-value (for stopping sight distance)	12	18	25	8	12	18	5	8	12
Sag K-value for comfort	2.5	4	5	1.5	2.5	4	1	1.5	2.5

3.4.2 Designing a new road

Designing a geometric alignment for an entirely new road is a considerably more complex process because of the many different route alignments that are possible and the relative lack of information available at the beginning of the process. In some cases there is need for a feasibility study to identify possible corridors for the road and to decide whether the project is likely to be viable. This will be followed by a detailed design study based on the route selected. The level of detail in this process depends critically on the class of road being designed and the terrain through which it will pass. Errors at this stage can be costly and, once the road is built, can also impose serious burdens in the future if the road requires excessive maintenance. Route selection is described in Section 3.2. It is based on surveys of various kinds that provide information about all the likely technical engineering issues related to the new road but also surveys concerned with environmental and social issues.

The final design will inevitably be a compromise between many competing factors such as terrain and there is no formal way of resolving all of them. Engineering judgement and consensus will be required to arrive at a satisfactory alignment.

3.5 Horizontal Alignment Design Standards

The horizontal alignment is composed of tangents (segments of a straight line), circular curves and in some instances spiral transition curves. The manner in which these components are assembled into a horizontal alignment, will significantly affect the safety, operational efficiency and aesthetics of the road. They therefore need to be determined based on specific standards. The standards are defined based on sight distances, cross falls, turning radius and super-elevation among other factors.

3.5.1 Sight distance

A good sight distance enables drivers to have a clear view of not only the oncoming vehicles, but also any physical features or obstacles around the corner or over the crest that may compromise

the safety of the road. The minimum sight distance requirements for districts roads for particular design speeds and in the different terrain are as provided in Table 33.

This table was adopted from the Uganda MoWT design standards for district roads. Therefore, for the safety of the low volume sealed roads, the horizontal curves designed have got to satisfy these requirements.

3.5.2 Crossfall

On unpaved district roads the cross fall is designed to remove rainwater quickly and effectively. An as-built camber of 8% is chosen to allow for consolidation of the road body over time with a minimum residual camber of 5% which is sufficient to achieve rapid shedding of rain water into the side drains. The 8% as-built camber also ensures balancing of cut and fill starting with a level plateau.

For low volume sealed roads, water ingress into the pavement layer is limited. Therefore cross-falls ranging between 2.5% and 4% are provided for.

3.5.3 Horizontal curves

For small changes of direction, it is desirable to use large turning radii for horizontal curves. This improves the appearance of the road by removing rapid changes in edge profile. Additionally, with large curve radii, sight distances will be greatly improved and may allow for safe overtaking.

Long sections of small radius curves should be avoided where possible since drivers travelling at speeds higher than the design speeds may find it difficult to remain in the carriageway. Curve widening could solve this problem although this would mean added costs.

Abrupt changes in direction from successive curves should be avoided where possible. Successive curves in the same direction should also be separated by an appropriate straight, as drivers are unlikely to anticipate what may be an abrupt change in radial acceleration. Curves should start and end with transition curves.

3.5.4 Super-elevation

The normal cross fall on a road will result in vehicles on the outside lane of a horizontal curve needing to develop high levels of frictional force to resist sliding, the amount of increase being dependent on speed, curve radius and cross fall. The road therefore has to be super-elevated. Super-elevation is aimed at calculating and providing banking needed to be applied to curves in the horizontal alignment to help them offset centrifugal force while the vehicles are turning. Provision of super-elevation helps the road to make the transition from normal crown to a fully banked curve and back again.

Transition sections are required both before and after the curve over which the super elevation is introduced and withdrawn. These sections are normally 40 metres long and the super elevation is phased in and out as shown in Figure 3-3.

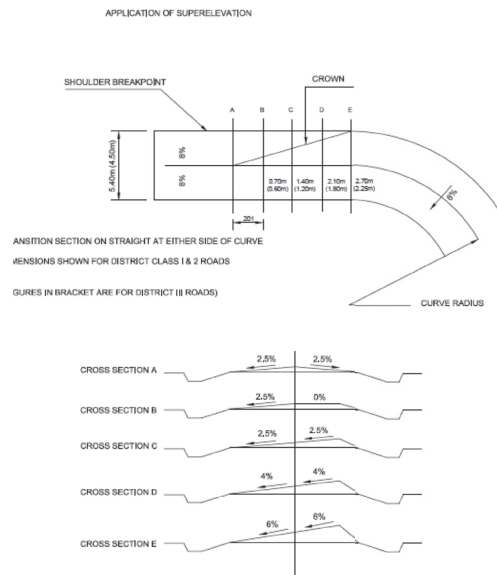


Figure 3-3 Illustration of Transition Curves and Super-elevation

3.5.5 Curve widening

Widening of the carriageway will be essential on horizontal curves to allow for the swept paths of trucks and to allow drivers to manoeuvre when approaching other vehicles. The required amount of widening is dependent on the characteristics of the vehicles using the road, the radius and length of the curve, and lateral clearances. The following levels of widening are recommended as listed in Table 3-4.

Table 3-4 Recommended Curve Widening for Different Radii of Curvature

Curve radius (m)	20	30	40	60
Increase in width (m)	1.5	1.0	0.75	0.5

The above values are guidelines only and there will be many situations where widening will be neither necessary nor cost-effective.

Widening should be applied on the inside of a curve and be gradually introduced over the length of the transition.

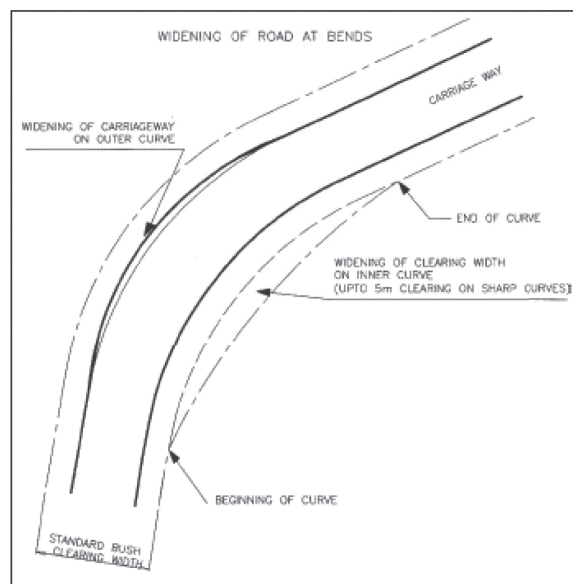


Figure 3-4 Curve Widening Standards

3.5.6 Radius of curvature

In road alignment, the use of the sharpest curve permitted for the particular design speed being used should be avoided wherever possible. Generally, the flattest curve practical should be used. Sharp curves should not be used at the end of long straights (i.e. straights more than 500 metres long).

Abrupt changes from flat curvature to sharp curvature should be avoided. Change from flat curvature to sharp curvature should be approached by compatible curves. Compatible curves are achieved when the radius of curvature of one curve is not more than 50% greater and/or not more than 33.3% less than a curve immediately adjacent to it. The minimum curve radii requirements for the different speeds in different terrain are provided in Table 3-8 and Table 3-9

3.6 Vertical Alignment Design Standards

3.6.1 Vertical curvature

A vertical alignment profile is a series of tangents connected by parabolic arcs (vertical curves). Vertical curves are applied to effect the transition between straight gradients. The design of the vertical alignment involves defining vertical alignment standards and designing the crests and sag vertical curves using appropriate formulae. The alignment standards defined include gradients, sight distance and K-values among others.

3.6.2 Gradients

A desirable maximum gradient is required for adequate drainage. The gradients requirements for unpaved district roads are provided in Table 33. The maximum gradient specifications for LVSRs are given in Table 37 and Table 38. If these requirements cannot be achieved, the Engineer should consider options such as hairpin bends, alternative surfacing, etc. Warning signs should be erected in advance of any potential hazards. Gradients can be checked using an ordinary level instrument or line level and string.

3.6.3 Sight distance requirements

Sight distances are provided to vertical curves to enable drivers observe all relevant features on the road and traffic conditions ahead in a vertical dimension. This is to enable them take necessary actions and precautions to avoid hazards and proceed in a safe and orderly way. The sight distance can either be stopping sight distance or passing sight distance. The required sight distance requirements are provided in Table 3-8 and Table 3-9.

3.6.4 K-values

The rate of change of grade along a vertical curve is constant and is measured by dividing the algebraic difference between the grades (G) by the length of the curve in meters (L). This value (G/L) gives the percentage change in grade per horizontal meter of curve. The reciprocal of this value (G/L), which is (L/G), represents the horizontal distance required to effect a 1% change in the gradient along the curve. The expression L/G is termed as K-value and is very useful in determining the minimum length of the curve and is also related to the minimum sight distance required.

The **K** factor for **crest** vertical curves is given by the formula:

$$K = \frac{L}{A} = \frac{S^2}{(372.665)}$$

This applies where $S < L$

Where,

L = length of vertical curve

S = sight distance

A = algebraic difference in grade (%)

The K factor for sag vertical curves for each design speed is defined by the formula:

$$K = \frac{L}{A} = \frac{S^2}{(120 / 3.491S)}$$

This applies where $S < L$

Where,

L = length of vertical curve

S = sight distance

A = algebraic difference in grade (%)

The minimum K-values for the different terrain and speeds are provided in Table 38 and Table 39.

3.7 Cross-Sections

The MoWT district manuals provides different classification for cross sections based on existing terrain, types of soil and embankments among other factors. Broadly, there are three classes I, II and III. For LVSRs the cross-sections are modified to allow of shoulders of minimum 0.5m on either side to allow for drainage, compaction and lateral stability as shown in Table 3-5

Table 3-5 Typical Cross-section dimensions

Design Standard	Design Traffic Flow (AADT) (Mid-life)	Surface Type	Carriageway Width (m)	Shoulder Width (m)
LVSR 1 ⁽²⁾	25 – 150	Paved	4.5 - 5.5	0.5 – 1.0
LVSR 2 ⁽²⁾	150 – 300	Paved	5.5 - 6.0 ⁽¹⁾	0.5 – 1.25
LVSR 3	300 – 1,000	Paved	6.0 - 7.0	0.5 – 1.5

3.8 Junctions

Junctions or intersections are necessary to allow vehicles to access and exit from the road. Access points should not be located at following positions:

- i. Where earthworks are heavy;
- ii. Near bridges;
- iii. On small radius curves; and
- iv. On steep gradients.

The provision of safety standards is either expensive or impossible to obtain at such positions. Adequate visibility under night driving shall be considered when planning junctions. Entry and exit radii shall be between 10 and 15 metres. However, entry radii from Community Access Roads can be reduced to as little as 6metres.

The alignment of the district road shall run smoothly into the major road cross-fall. Cross-falls of the bell-mouth formed by the feeder road junction are to be provided to ensure that surface water is shed effectively towards the side drains.

3.9 Provisions for Road Safety – Ancillaries

The safety and efficiency of operation of a road depends to a considerable degree upon the geometric design. However, the physical layout must also be supplemented by effective road markings, road signs and other road accessories which would greatly improve the driver's perception and comprehension of the continually changing appearance of the road.

3.9.1 Road signs

Road signs are grouped into three categories depending on the purpose they serve:

- i. **Regulatory or mandatory signs**

These inform road users of the special regulations applying on a section of a road. These signs **MUST** be observed by all motor vehicle traffic using the road section. These signs are usually surmounted by a red circle and a triangle, the latter usually having its apex downwards.

- ii. **Warning signs**

These warn drivers of potentially hazardous road conditions and allow drivers sufficient time to take the necessary precautionary measures before arriving at the hazardous location. They are recognized by a triangle with its apex upwards.

- iii. **Informative signs**

These allow drivers to know where they are and which route to take as they drive along.

3.9.2 The full range of road signs in Uganda is defined in the specifications of the Ministry of Works and Transport. Road marking

The road marking would primarily comprise of centreline marking, edge line marking and directional markings. Road signs would include warning signs, prohibition signs and information signs.

3.10 Geometric Design Standards for Low Volume Roads

The geometric design standards for low volume sealed roads are given in Table 3-6, Table 3-7, Table 3-8 and Table 3-9.

Table 3-6 Road Classification, AADT, Carriageway Widths and Design Speeds

Design Standard	Design Traffic Flow (AADT) (Mid-life)	Surface Type	Width (m)	Design Speed (km/h)				
			Carriageway	Flat	Rolling	Mountainous	Escarpment	Urban/ Peri-Urban
LVS R 1 ⁽²⁾	25 – 150	Paved	4.5 – 5.5	70	60	50 45 ⁽³⁾	30	50
LVS R 2 ⁽²⁾	150 – 300	Paved	5.5 – 6.0 ⁽¹⁾	70	60	50 45 ⁽³⁾	30	50
LVS R 3	300 – 1,000	Paved	6.0 – 7.0	85	70	60	50	50

Notes:

- 1) Choice of *carriageway and shoulder widths* also depends on numbers of Large Heavy Vehicles defined as vehicles with 3 or more axles and with GVW > 10 tonnes.
- 2) The choice of *design standard* depends on the numbers of Large Heavy Vehicles as well as AADT.
- 3) Design speed is adjusted slightly to provide the same minimum radius of horizontal curvature as for the paved road option in mountainous terrain.

Table 3-7 Shoulder Widths on Each Side of the Carriageway

Design Standard for Paved Roads	Rural Terrain/Shoulder Width (m)				High numbers of PCUs ⁽²⁾	Town Section Widths ⁽¹⁾ (m)		
	Flat	Rolling	Mountainous	Escarpment		Parking Lane	Footway	Median ¹
LVS R1	1.0	1.0	0.5	0.5	2.5	3.5	2.5	n/a
LVS R2	1.25	1.25	0.5	0.5	2.75	3.5	2.5	n/a
LVS R3	1.5	1.5	0.5	0.5	2.75	3.5	2.5	n/a

Notes:

- 1) Additional widths not applicable on escarpments and may not be possible in mountainous terrain.
- 2) Where finances are constrained the shoulder width may be reduced to 0.5m paved or unpaved for traffic less than 150 vehicles per day in flat and rolling terrain.

Table 3-8 7 Geometric Parameters for Design Class Paved LVSR 2⁽¹⁾ (AADT 150-300) and LVSR 3 (AADT 300-1000)

Design Element		Unit	Flat	Rolling	Mountain	Escarp't	Urban Peri-Urban
Design Speed		km/h	70	60	50	25	50
Width of Carriageway		m	6.0 ⁽²⁾	6.0 ⁽²⁾	5.5	5.5	6.5 ⁽²⁾
Width of Shoulders		m	Table 35 and Table 36				
Minimum. Stopping Sight Distance	g = 0%	m	110	85	65	25	65
	g = 5%	m	120	90	70	25	70
	g = 10%	m	140	105	75	25	75
Min Passing Sight Distance		m	270	230	180	50	180
Minimum Hor- izontal Curve Radius	SE = 4%	m	215	145	95	20 ⁽³⁾	95
	SE = 6%	m	195	135	85	18 ⁽³⁾	85
	SE = 8%	m	175	120	80	17 ⁽³⁾	-
	SE=10%	m	160	110	75	16 ⁽³⁾	-
Max. Gradient (desirable)		%	4	6	8	8	7
Max. Gradient (absolute)		%	6	8	10 ⁽⁴⁾ ⁽⁵⁾	10 ⁽⁴⁾ ⁽⁵⁾	9
Minimum Gradient		%	0.5 ⁷	0.5	0.5	0.5	0.5
Min. Crest Vertical Curve ⁽⁶⁾		K	30	17	10	2	10
Min. Sag Vertical Curve		K	12	9	7	2	7
Normal Cross-fall		%	3	3	3	3	3
Shoulder Cross-fall		%	4	4	3	3	4

Notes:

- 1) If there are more than 80 Large Heavy Vehicles per day then LVSR5 should be used.
- 2) If the number of Large Heavy Vehicles is >40 per day then this should be increased to 7.0m.
- 3) On hairpin stacks the minimum radius may be reduced to a minimum of 15m.
- 4) Length not to exceed 200m and relief gradients required (< 6% for minimum of 200m).
- 5) If the number of Large Heavy Vehicles <10 this can be increased to 12%.
- 6) These values are based on an object height of 0.2m. Use of a different sized object requires approval.
- 7) In some circumstances in very flat terrain this can be reduced to 0.3%

Table 3-9 Geometric Parameters for Design Class Paved LVSR1⁽¹⁾ (AADT 25-150)

Design Element		Unit	Flat	Rolling	Mountain	Escarp't	Urban Peri-Urban
Design Speed		km/hr	70	60	50	25	50
Width of Carriageway		m	5.5	5.5	4.5	4.5	6.0
Width of Shoulders ^{(2) (3)}		m	Table 35 and Table 36				
Minimum Stopping Sight Distance	g = 0%	m	110	85	65	25	65
	g = 5%	m	120	90	70	25	70
	g = 10%	m	140	105	75	25	75
Min. Passing Sight Distance		m	275	225	175	60	175
Minimum Horizontal Curve Radius	SE = 4%	m	215	145	95	20 ⁽³⁾	95
	SE = 6%	m	195	135	85	18 ⁽³⁾	85
	SE = 8%	m	175	120	80	17 ⁽³⁾	-
Max. Gradient (desirable)		%	6	7	10 ⁽⁴⁾	10 ⁽⁴⁾	7
Max. Gradient (absolute)		%	8	9	12 ⁽⁴⁾	12 ⁽⁴⁾	9
Minimum Gradient		%	0.5 ⁶	0.5	0.5	0.5	0.5
Min. Crest Vertical Curve ⁽⁵⁾		K	30	17	10	2	10
Min. Sag Vertical Curve		K	12	9	7	2	7
Normal Cross-fall		%	3	3	3	3	3
Shoulder Cross-fall		%	4	4	3	3	4

Notes:

- 1) If there are more than 30 Large Heavy Vehicles, then LVSR4 should be used (Table 3.1.6).
- 2) On hairpin stacks the minimum radius may be reduced to a minimum of 15m.
- 3) In urban and peri-urban areas parking lanes and footpaths may be required (Table 3.1.5)
- 4) Length not to exceed 200m and relief gradients required (< 6% for minimum of 200m).
- 5) These values are based on an object height of 0.2m. Use of a different sized object requires approval.
- 6) In some circumstances in very flat terrain this can be reduced to 0.3%.

4.0 Drainage Design

4.1 Introduction

This chapter presents information relevant to the design or sizing of road drainage structures. The chapter presents information relating to ensuring that the pavement structure possesses good surface and subsurface drainage.

4.2 Hydrology

The following methods are very useful for estimating maximum flow in a watercourse:

- i. Direct observation of the size of watercourse, erosion and debris on the banks, history and local knowledge;
- ii. The Rational Method; and
- iii. The Soil Conservation Service (SCS) method.

Greater reliability can be achieved by estimating flow using all of the above methods and using the more conservative result.

4.2.1 Direct observation method

This method offers a practical approach to the estimation of flow. It is particularly useful where data is non-existent or not readily available. The cross-sectional area of the water course is measured and a cross-sectional area of apertures of the structure provided that is equal to that of the water course. If the return period of the maximum flow is much longer than that for which the structure is being designed, the typical high water level can be estimated from lateral erosion on the banks or debris caught in the branches of trees. The cross-sectional area of the water course to this level is calculated and a structure provided with cross-sectional area of apertures equal to this area. Local residents may offer useful recollections about the water flow levels that were observed during certain floods.

In conjunction with this approach, structures in nearby areas that have evidently performed well without collapse and overtopping may be replicated. This is especially useful for culvert design.

4.2.2 Rational method

The flow of water in a channel, q , is calculated from the following equation.

$$Q = 0.278 \times C \times I \times A \text{ (m}^3\text{/s)} \quad \text{Equation 41}$$

Where:

- C = the catchment runoff coefficient
 I = the intensity of the rainfall (mm/hour)
 A = the area of the catchment (km²)

The Catchment Runoff Coefficient, C , is obtained from Table 4-1.

Table 4-1 Runoff Coefficient for Humid Catchment

Average Ground Slope	Soil Permeability			
	Very low (rock & hard clay)	Low (clay loam)	Medium (sandy loam)	High (sand & gravel)
Flat 0-1%	0.55	0.40	0.20	0.05
Gentle 1-4%	0.75	0.55	0.35	0.20
Rolling 4-10%	0.85	0.65	0.45	0.30
Steep >10%	0.95	0.75	0.55	0.40

The intensity of rainfall (I) is obtained from the Intensity-Duration-Frequency charts or can be calculated from daily rainfall assuming 3 hour storm duration. In many situations the charts will not be available because rain stations often measure only the rainfall in 24 hours. The SCS method (see below) is then a better option. The storm duration is equal to the Time of Concentration (T_c). T_c is the time taken for water to flow from the farthest extremity of the catchment to the crossing site.

$$T_c = \text{Distance from farthest extremity (m)} / \text{Velocity of flow (m/s)} \quad \text{Equation 42}$$

The velocity of flow depends on the catchment characteristics and slope of the watercourse. It is estimated from Figure 4-1.

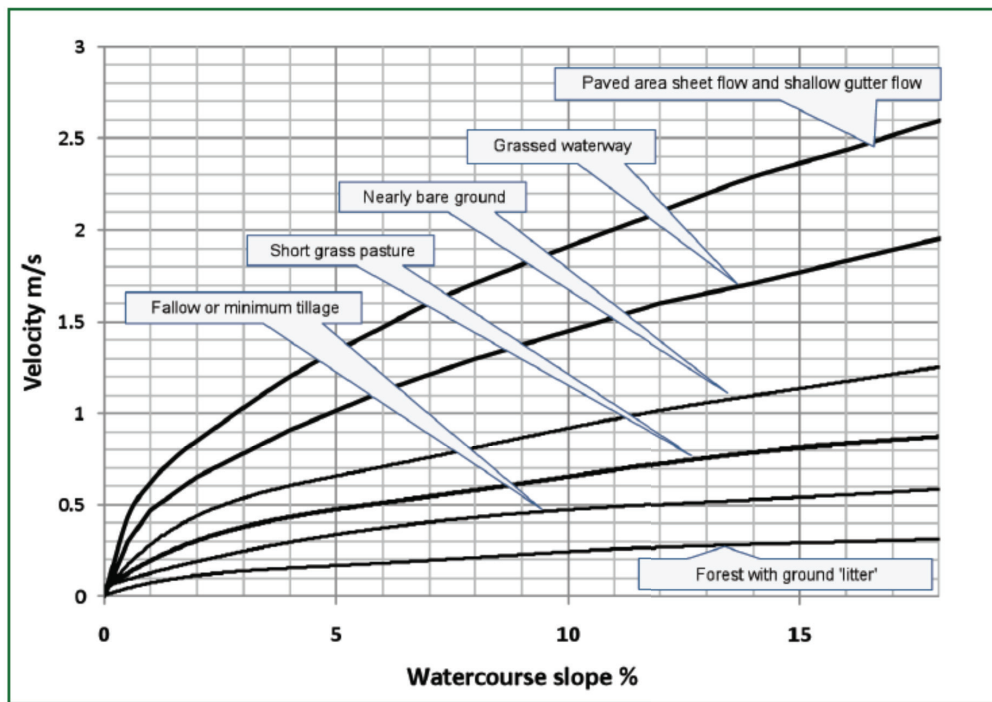


Figure 4-1 Velocity of Flow for Varying Surface Cover

The storm design return period is taken from Table 4-2. If the route is of strategic importance, or if the alternative route in the event of a drainage failure is more than an additional 75km or if there is no alternative route, Table 4-3 should be used.

Table 4-2 Storm Design Return Period (years)

Drainage Structure	Geometric Design Standard		
	LVS R3	LVS R2	LVS R1
Gutters and inlets	2	2	2
Side ditches	10	5	5
Ford	10	5	5
Drift	10	5	5
Culvert diameter <2m	15	10	10
Large culvert diameter >2m	25	15	10
Gabion abutment bridge	25	20	15
Short span bridge <10m	25	25	15
Masonry arch bridge	50	25	25
Medium span bridge (15-50m)	50	50	25
Long span bridge >50m	100	100	50

Table 4-3 Storm Design Return Period (years) for Severe Risk Situations

Drainage Structure	Geometric Design Standard		
	LVSr3	LVSr2	LVSr1
Gutters and inlets	5	5	5
Side ditches	15	10	10
Ford	15	10	10
Drift	15	10	10
Culvert diameter <2m	25	20	20
Large culvert diameter >2m	50	25	20
Gabion abutment bridge	50	25	20
Short span bridge <10m	50	50	25
Masonry arch bridge	50	50	25
Medium span bridge (15-50m)	100	100	50
Long span bridge >50m	100	100	100

The area of the drainage catchment (A) should be estimated from topographical maps or through the use of aerial photographs.

In the Rational Method it is assumed that the intensity of the rainfall is the same over the entire catchment area. The consequence of applying the method to large catchments is an over-estimate of the flow and therefore a conservative design.

A simple modification can be made to take into account the spatial variation of rainfall intensity across a larger catchment. The effective area of the catchment is reduced by multiplying by the areal reduction factor (ARL) given by the following equation:

$$ARL = 1 - 0.04 \times t^{1/3} \times A^{1/2} \quad \text{Equation 43}$$

Where,

t = storm duration in hours

A = catchment area in km²

4.2.3 Soil Conservation Service (SCS) method

The United States SCS method for calculating rates of runoff requires much of the same basic data as the Rational Method namely: catchment area, a runoff factor, time of concentration and rainfall. However the SCS method also considers the time distribution of the rainfall, the initial rainfall losses to interception and storage, and an infiltration rate that decreases during the course of a storm. It is therefore potentially more accurate than the Rational Method and is applicable when the catchment area is larger than 50 hectares.

4.2.3.1 Catchment area

The catchment area is determined from topographic maps and field surveys. For large catchment areas it might be necessary to divide the area into sub-catchment areas to account for major land use changes.

4.2.3.2 Rainfall

The SCS method is based on a 24-hour storm event. The characteristics of storms are defined in terms of the relationship between the percentages of the total storm rainfall that has fallen as a function of time. Three basic types of storm are defined for three levels of maximum intensity, Type I being the least intense and Type III being the most intense. Type III should be used in Uganda.

A relationship between accumulated rainfall and accumulated runoff was derived by SCS for numerous hydrologic and vegetative cover conditions. The storm data included total amount of rainfall in a calendar day but not its distribution with respect to time. The SCS runoff equation is therefore a method of estimating direct runoff from 24-hour or 1-day storm rainfall. The equation is:

$$Q = (P - I_a)^2 / [(P - I_a) + S] \quad \text{Equation 44}$$

Where:

Q = accumulated direct runoff, mm

P = accumulated rainfall (i.e. the potential maximum runoff), mm

I_a = initial abstraction including surface storage, interception, and infiltration prior to runoff, mm

S = potential maximum retention, mm

S is related to the soil and cover conditions of the catchment area through the Curve Numbers, CN, described below.

$$S = 25.4(1000/CN - 10) \quad \text{Equation 45}$$

The relationship between I_a and S was found to be;

$$I_a = 0.2S = 50.8(100/CN - 1) \quad \text{Equation 46}$$

Substituting into Equation 4-4:

$$Q = [P - 50.8(100/CN - 1)]^2 / [P + 203.2(100/CN - 1)] \quad \text{Equation 47}$$

Figure 4-2 shows a graphical solution which enables Q , the direct runoff from a storm, to be obtained if the total rainfall and catchment area curve number are known.

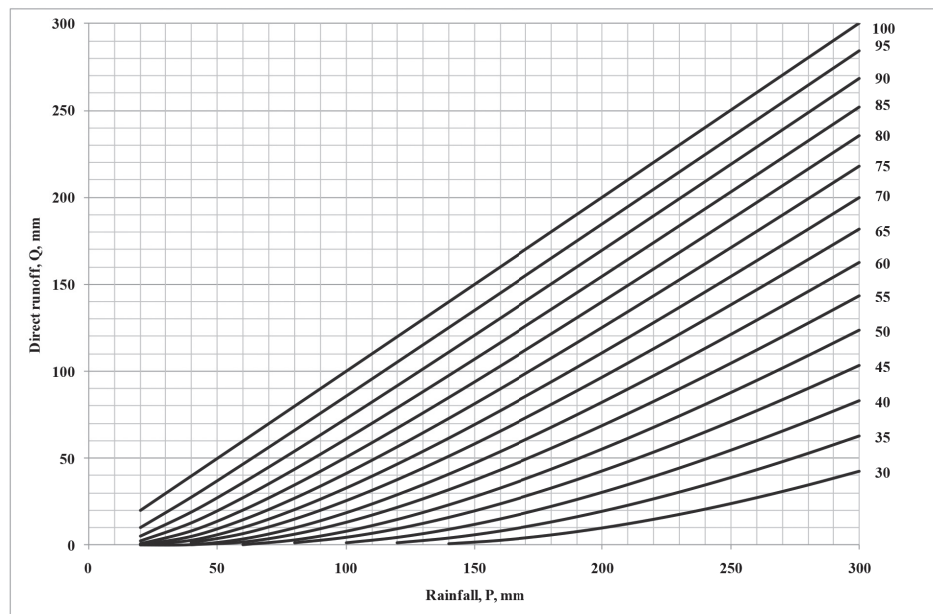


Figure 4-2 Relationship between Precipitation, Direct Runoff and CN

4.2.3.3 Runoff and Curve Numbers

The physical catchment area characteristics affecting the relationship between rainfall and runoff (i.e. the CN values) are land use, land treatment, soil types, and land slope. Land use is the catchment area cover and it includes agricultural characteristics, type of vegetation, water surfaces, roads, roofs, etc. Land treatment applies mainly to agricultural land use, and it includes mechanical

practices such as contouring or terracing and management practices such as rotation of crops. The SCS method uses a combination of soil conditions and land-use to assign a runoff factor to an area. These runoff factors or curve numbers (CN), indicate the runoff potential of an area. The higher the CN, the higher is the runoff potential.

Soils are divided into four hydrologic groups (Groups A, B, C and D) based on infiltration rates (Table 4-4).

Runoff curve numbers also vary with the antecedent soil moisture conditions, defined as the amount of rainfall occurring in a selected period preceding a given storm. In general, the greater the antecedent rainfall, the more direct runoff there is from a given storm. A five-day period is used as the minimum for estimating antecedent moisture conditions.

Table 4-4 Hydrological Characteristics of Soil Groups

Soil Group	General Description	
A	Well drained, sandy	High infiltration, low runoff
B	Sandy loam, low plasticity	Moderate infiltration
C	Clay loam, medium plasticity	Slow infiltration
D	Highly plastic clay	Very slow infiltration, high run off

Table 4-5 gives runoff curve numbers for various land uses. This Table is based on an average antecedent moisture condition (i.e. soils that are neither very wet nor very dry when the design storm begins). Table 46 gives conversion factors to convert average curve numbers to wet and dry curve numbers.

Table 4-5 Runoff Curve Numbers (CN)

Land use			A	B	C	D
Cultivated land	Without conservation treatment		72	81	88	91
	With conservation treatment		62	71	78	81
Pasture land	Poor condition		68	79	86	89
	Good condition		39	61	74	80
Meadow			30	58	71	78
Wood or forest	Thin stand, poor cover, no mulch		45	66	77	83
	Good cover		25	55	70	77
Open spaces, lawns, parks	Good condition, grass cover >75% of area		39	61	74	80
	Fair condition, grass on 50-75%		49	69	79	84
Urban districts	Commercial and business areas, 85% impervious		89	92	94	95
	Industrial districts, 70% impervious		81	88	91	93
Residential	Average lot size	Average % impervious				
	< 0.05 hectares	65	77	85	90	92
	0.1 hectares	38	61	75	83	87
	0.2 hectares	25	54	70	80	85
	0.4 hectares	20	51	68	79	84
	0.8 hectares	12	46	65	77	82
Paved roads with curbs, storm drains, paved parking areas, roofs.			98	98	98	98
Gravel roads			76	85	89	91
Earth roads			72	82	87	89
Open water			0	0	0	0

Table 4-6 Conversion from Average to Wet and Dry Antecedent Moisture Conditions

CN values		
Average conditions	Dry	Wet
95	87	98
90	78	96
85	70	94
80	63	91
75	57	88
70	51	85
65	45	82
60	40	78
55	35	74
50	31	70
45	26	65
40	22	60
35	18	55
30	15	50

4.2.3.4 Time of concentration

The next step in the SCS Method is to determine the Time of Concentration. This is the time it takes water to flow from the edge of the catchment area to the point of interest. It is a combination of three values:

- i. Sheet flow;
- ii. Shallow concentrated flow; and
- iii. Open channel flow.

The type that occurs is a function of the conveyance system and is determined by field inspection. It is often a combination of these so that the total travel time is the sum of the time taken for the water to pass through all of the segments of the catchment.

Travel time is the ratio of flow length to flow velocity:

$$T = L/(3600V) \quad \text{Equation 48}$$

Where:

- T = travel time, hours
 L = flow length, m
 V = average velocity, m/s
 3600 = conversion factor from seconds to hours

Sheet flow

Sheet flow is flow over plane surfaces. It usually occurs in the headwater of streams. With sheet flow, the friction value (Manning's n) is an effective roughness coefficient that includes the effect of raindrop impact; drag over the plane surface; obstacles such as litter, crop ridges, and rocks; and erosion and transportation of sediment. These n values are for very shallow flow depths of about 0.03m or so. Table 4-7 gives Manning's n values for sheet flow for various surface conditions.

Table 4-7 Manning's Roughness Coefficients for Sheet Flow

Surface	n ¹
Smooth surfaces: concrete, asphalt, gravel or bare soil	0.011
Fallow (no residue)	0.05
Cultivated soils	
Residue cover < 20%	0.06
Residue cover > 20%	0.17
Short grass	0.15
Dense grass	0.24
Range	0.13
Woods ¹	
Light underbrush	0.4
Dense underbrush	0.8
Note 1 Consider cover to a height of 30mm. This is the only part of the cover that will affect sheet flow	

For sheet flow of less than 100 metres Manning's kinematic solution should be used to compute the travel time T,

$$T = [0.091 (n.L)^{0.8} / (P_2)^{0.5} S^{0.4}] \quad \text{Equation 49}$$

Where:

T	=	travel time, hour
n	=	Manning's roughness coefficient (Table 47)
L	=	flow length, m
P ₂	=	2-year, 24-hour rainfall, mm
S	=	slope of hydraulic grade line (land slope), m/m

Shallow Concentrated Flow

After a maximum of 100 metres, sheet flow usually becomes shallow concentrated flow. The average velocity for this flow can be determined from the following equations in which average velocity is a function of watercourse slope and type of channel.

Unpaved	V	=	4.918 (S) ^{0.5}	
Paved	V	=	6.196 (S) ^{0.5}	Equation 410

Where:

V = average velocity, m/s

S = slope of hydraulic grade line (watercourse slope), m/m

After determining average velocity, the travel time for the shallow concentrated flow segment is calculated from Equation 4-9.

Open Channel flow

Open channels are assumed to begin where surveyed cross section information has been obtained, where channels are visible on aerial photographs, or where blue lines (indicating streams) appear on topographic maps (1:50,000). Average flow velocity is usually determined for bank-full elevation. Manning's equation or water surface profile information can be used to estimate average flow velocity. When the channel section and roughness coefficient (Manning's n) are available, then the velocity can be computed using the Manning Equation:

$$V = (R^{2/3} \cdot S^{1/2})/n \quad \text{Equation 411}$$

Where:

V	=	average velocity, m/s
R	=	hydraulic radius, m (equal to a/P_w)
a	=	cross sectional flow area, m ²
P_w	=	wetted perimeter, m
S	=	slope of the hydraulic grade line, m/m
n	=	Manning's roughness coefficient (Table 47).

After the average velocity is computed, the travel time for the segment can be calculated using Equation 4-9.

Reservoir or Lake

Sometimes it is necessary to compute a time of concentration for a catchment area having a relatively large body of water in the flow path. The travel time is computed using the equation:

$$V_w = (g \cdot D_m)^{0.5} \quad \text{Equation 412}$$

Where:

V_w	=	the wave velocity across the water, m/s
g	=	9.81 m/s ²
D_m	=	mean depth of lake or reservoir, m

This equation only deals with the travel time across the lake, not the time at the inflow or outflow channels. The times for these are generally very much longer and must be added to the travel time across the lake. Equation 4-11 can be used for swamps with much open water, but where the vegetation or debris is relatively thick (less than about 25 percent open water), Manning's equation is more appropriate.

Steps in the SCS procedure

The steps in using the SCS method are as follows:

- i. Determine the catchment area, A, and its soil and land use characteristics;
- ii. Determine the 'curve runoff number, CN, from Table 45 and any adjustment based on the likely antecedent soil moisture conditions (Table 46);
- iii. Calculate the value of I_a from Equation 4-6;
- iv. Choose the appropriate design storm recurrence frequency. This is based on the class of road and the drainage structure being designed;
- v. For the recurrence frequency chosen, determine the 24-hour rainfall (P) for the appropriate rainfall region from Figure 43;
- vi. Determine the direct runoff (Q) for the rainfall (P) and curve number (CN) obtained in steps iii and v from Figure 42;
- vii. The catchment must be divided into uniform areas for the purpose of determining the Time of Concentration, T_c . The flow lengths for sheet flow, shallow concentrated flow, and channel flow must be determined and the equations in Section 4.2.2 and Section 4.2.3 are used to calculate the total time of concentration;

At this stage the following data have been obtained,

I_a – the initial abstraction based on the curve number CN (step iv)

P – the design storm precipitation (step vi)

Q – the accumulated direct runoff (step vii)

T_c – the Time of Concentration (step viii);

viii. The next step is to determine the unit peak discharge, Q_u and this is done using Figure 44, the value of I_a/P and the Time of Concentration, and

ix. The final step is to compute the actual peak discharge from the unit value as follows.

Design peak discharge = $Q_u \times Q \times A$

Where Q is in mm and A is in units of 100 hectares

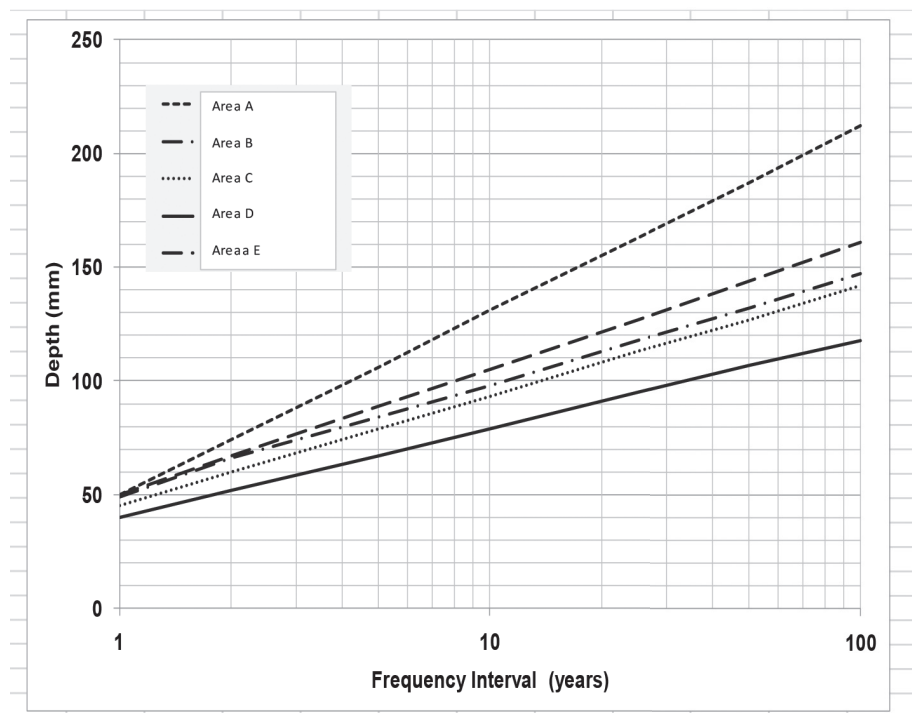


Figure 4-3 Recurrence Interval and 2-4 hour Rainfall Depth for Different Regions

Table 4-8 24 hour Precipitation for Flood Return Periods

Region Represented by graphs on Figure 43	Frequency Interval (years)					
	2	5	10	25	50	100
	24-hour depth in mm					
Area A	74	106	131	163	187	211
Area B	65	84	98	118	132	147
Area C	60	79	93	113	127	142
Area D	52	67	79	95	107	118
Area E	67	89	105	127	144	161

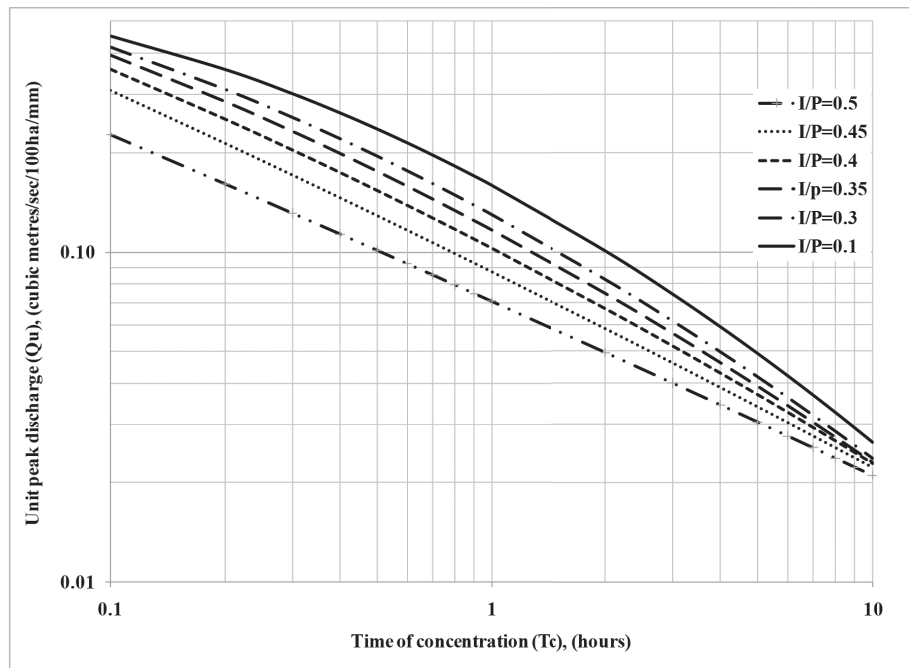


Figure 4-4 Unit Peak Discharge (Type II Rainfall)

4.3 Hydraulics and Design of Drainage Structures

4.3.1 Design of culverts

Two methods of culvert design namely, monograph and replicating successful practice are used as described below. To minimise the effect of potential silting and deposition of debris in the culvert, the slope/fall should be 2 – 5 %.

All pipes should have a minimum diameter or a minimum height of 600mm to ensure that they can be manually cleaned.

4.3.1.1 Monograph method for culvert sizing

The water flow through the culvert should be estimated by any one of the methods in Section 4.2.

The required size of a culvert opening is estimated using the monographs in Figure 4-5 to Figure 47. These figures apply to culverts with inlet control where there is no restriction to the downstream flow of the water.

In flat terrain, where there is a high risk of silting, a factor of safety of 2 should be allowed in the design of the culvert.

4.3.1.2 Replicating successful practice

If a high proportion of structures along a road or in a region have been in operation for a number of years without overtopping, it is reasonable to assume that the relationship between catchment area, catchment characteristics, rainfall intensity and maximum water flow used in their design is valid. The design of new culverts can be based on simply the catchment area using the same relationships.

4.3.2 Design of drifts and fords

The following criteria should be considered when designing drifts:

- The level of the drift should be as close as possible to the existing river bed level.
- The normal depth of water should be a maximum of 300 mm and the maximum 5 year flow should be 6m³/second on the drift to allow traffic to pass.

- Approach ramps should have a maximum gradient of 10% (7% for roads with large numbers of heavy trucks).

The engineer has the responsibility of determining the dimensions applicable to a particular site.

4.3.3 Side drains

Side drains can be constructed in three forms: V-shaped, rectangular or trapezoidal. The trapezoidal cross-section facilitates maintenance and improves traffic safety. Trapezoidal drains can be easily constructed and maintained by hand. The minimum recommended width of bottom of a trapezoidal side drain is 300mm and the minimum recommended longitudinal gradient is 2%.

Design volumes of run-off in side drains and other channels are estimated using the Rational Method. The cross sectional area of the drain must be sufficient to accommodate the expected flow of water, Q , where:

$$Q = AV$$

The flow velocity is calculated from the Manning equation.

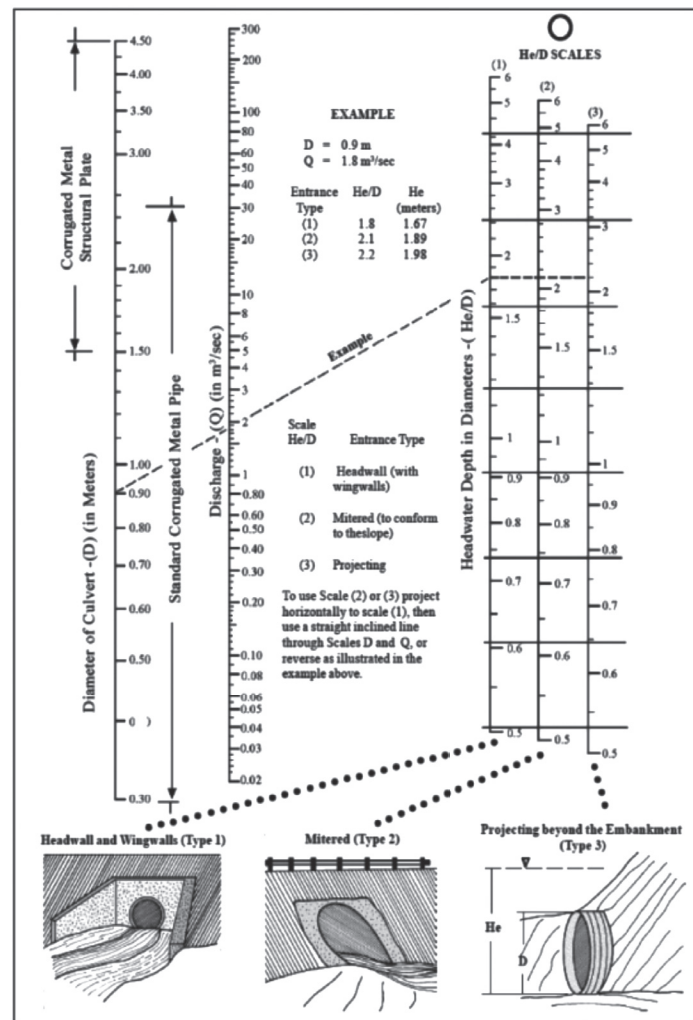


Figure 4-5 Headwater Depth and Capacity for Corrugated Metal Pipe Culverts with Inlet Control (Adapted from FHWA, 1998)

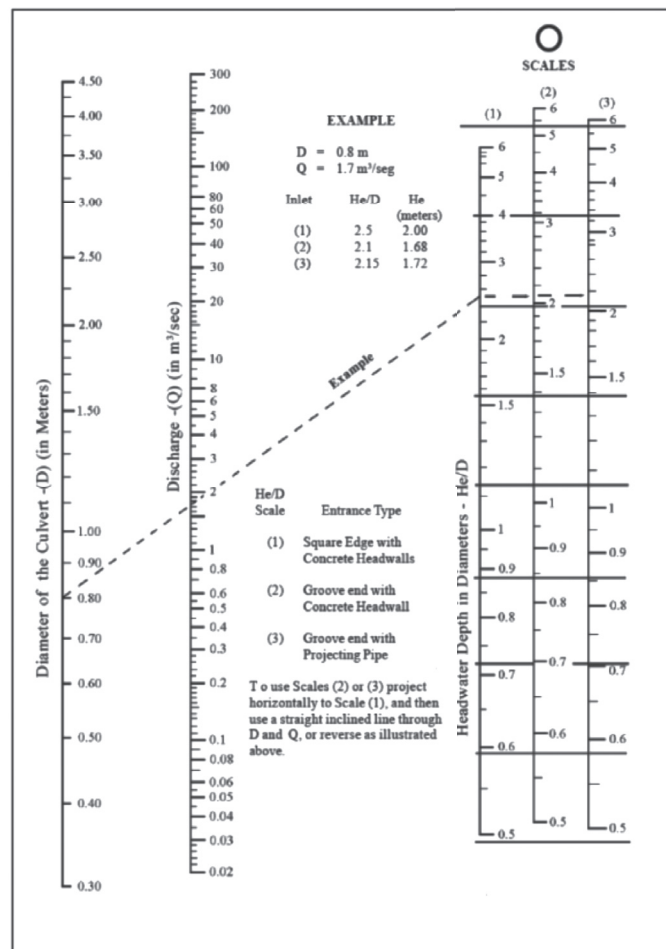


Figure 4-6 Headwater Depth and Capacity for Concrete Pipe Culverts with Inlet Control (Adapted from FHWA, 1998)

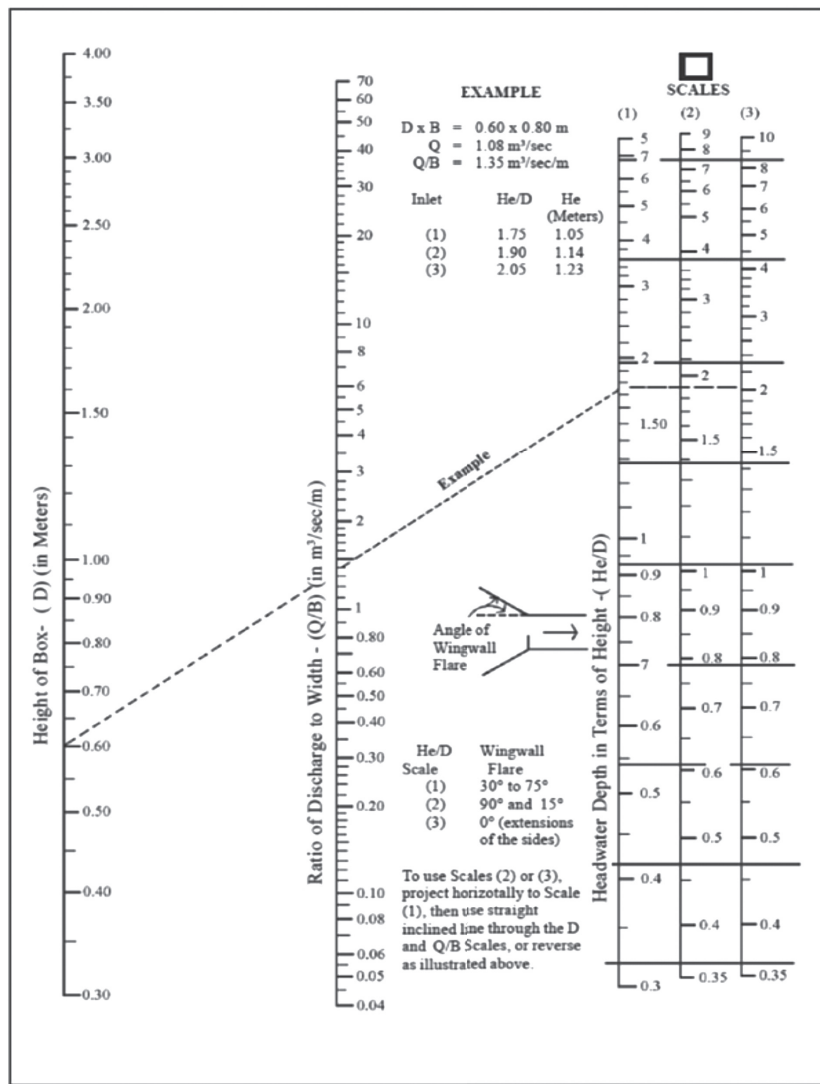


Figure 4-7 Headwater Depth and Capacity for Concrete Box Culverts with Inlet Control (Adapted from FHWA, 1998)

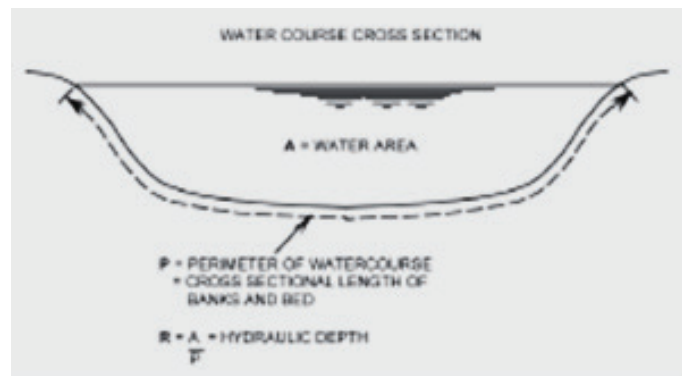


Figure 4-8 Manning's Formula

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

Equation 413

Where:

V = velocity in m/s

A= the cross-sectional area of water (m²)

R = hydraulic depth (the area for the stream flow divided by the wetted perimeter)

S = hydraulic gradient (the slope of the drain or watercourse)

n = roughness coefficient (Table 4-9)

Table 4-9 Roughness Coefficient (n) for Drains

Material in the drain	Roughness coefficient
Sand, loam, fine gravel, volcanic ash	0.022
Stiff clay	0.020
Course gravel	0.025
Conglomerate, hard shale, soft rock	0.040
Hard rock	0.040
Masonry	0.025
Concrete	0.017

Erosion control in the side drain

Limiting flow velocities to prevent scour in excavated drains are given in Table 4-10.

Drain erosion is controlled by building scour checks or lining the drain. The distance between scour checks depends on the road gradient and the erosion potential of the soils. Table 411 shows recommended values. In areas that evidently have erodible soils, it is recommended that the drains are lined at gradients above 3%. The preferred shape of lined side drains is trapezoidal.

Measures must be taken to control erosion downstream of the drain outlet.

Dry stone pitching for drain lining is usually only suitable where the discharge is lower than 1 m/sec per metre width, and where sediment load is relatively fine-grained.

Table 4-10 Permissible Flow Velocities (m/sec) in Excavated Side Drains

Soil type	Clear water	Water carrying fine silt	Water carrying sand and fine gravel
Fine sand	0.45	0.75	0.45
Sandy loam	0.55	0.75	0.6
Silty loam	0.6	0.9	0.6
'Good' loam	0.75	1.05	0.7
Lined with established grass on good soil	1.7	1.7	1.7
Lined with bunched grasses (exposed soil between plants)	1.1	1.1	1.1
Volcanic ash	0.75	1.05	0.6
Fine gravel	0.75	1.5	1.15
Stiff clay	1.15	1.5	0.9
Graded loam to cobbles	1.15	1.5	1.5
Graded silt to cobbles	1.2	1.7	1.5
Alluvial silts (non-colloidal)	0.6	1.05	0.6
Alluvial silts (colloidal)	1.15	1.50	0.9
Coarse gravel	1.2	1.85	2.0
Cobbles and shingles	1.5	1.7	2.0
Shales	1.85	1.85	1.5
Rock	Negligible scour at all velocities		

Table 4-11 Spacing between Scour Checks

Road gradient (%)	Scour check interval (metres)
3	Not required
4	17
5	13
6	10
7	8
8	7
9	6
10	5
12	4

4.3.3.1 Mitre drains or turnouts

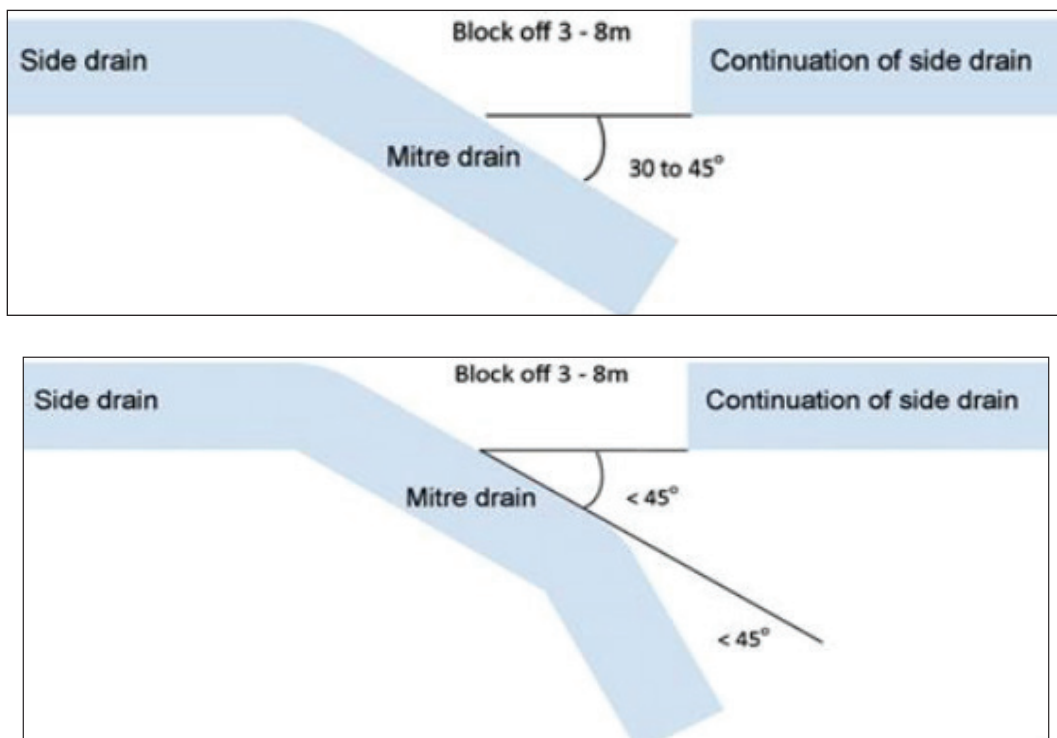
To minimise erosion in the side drains, water drains should be discharged as frequently as possible. Where topographical and land features permit, it is preferable to discharge the water on the same side of the road as the drain to minimise culvert costs. A turnout or mitre drain is used to lead the water onto adjacent land. Table 412 shows the maximum spacing of mitre drains related to gradient and Figure 4-9 shows the layout of the mitre drains.

Table 4-12 Maximum Spacing of Mitre Drains

Road gradient (%)	Maximum mitre drain interval (m)
12	40
10	80
8	120 ¹
6	150 ¹
4	200 ¹
2	80 ²
<2	50 ²

Notes

- 1 A maximum of 100m is preferred but not essential.
- 2 At low gradients silting becomes a problem.

**Figure 4-9 Schematic Layout of Mitre/Turnout Drains**

A block-off is required to ensure that water flows out of the side drain into the mitre drain. The angle between the mitre drain and the side drain should preferably be 30 degrees, but not greater than 45 degrees.

The desirable slope of the mitre drains is 2-5%. The drain should lead gradually across the land, getting shallower and shallower. Stones may need to be laid at the end of the drain to help prevent erosion.

In flat terrain, a small gradient of 1% or even 0.5% may be necessary to discharge water, or to avoid very long drains.

4.3.4 Water crossings and associated structures

The sizing of openings of water crossing structures is done by estimating the flow by either the observation method or the rational method or SCS method as discussed in preceding sections above. For more complex bridges, please refer to the Small Bridges Manual which can be downloaded from the gtkp website.

4.3.5 Design for climate resilience

Uganda's transport systems and other infrastructure continue to be built without taking predicted climate change patterns into account. Climate-related hazards and predicted impacts of climate change threatens vital transport infrastructure such as roads, bridges and other road furniture. The economic cost of the impacts of climate change on infrastructure damage, repairs and reconstructions, though difficult to estimate, is very high hence a need to integrate Climate change issues. The following should be considered while planning and construction of roads:

- i. Integrate climate change issues into the existing framework for road planning and construction especially condition assessment guidelines and work methodology;
- ii. Comply with climate change-resilient standards for infrastructure planning and development through monitoring and reporting systems;
- iii. Promote water catchment protection including road water harvesting in transport infrastructure development and maintenance; and
- iv. Greening of road corridors.

4.3.6 Other drainage considerations

4.3.6.1 Wet lands

Special drainage or construction methods are needed if wet areas must be crossed. An embankment is normally required. The embankment should include multiple drainage pipes or coarse permeable rock-fill to keep the flow dispersed. Subgrade reinforcement with coarse permeable rock, filter layers and geotextiles may also be required. The objective is to maintain the natural groundwater level and flow pattern dispersed across the wet land and, at the same time, provide a stable, dry roadway surface.

Subsurface drainage, through use of under-drains, interceptor trenches or aggregate filter blankets, is used in localized wet or spring areas to remove the groundwater and keep the roadway subgrade dry. In localised areas subsurface drainage is often more cost-effective than adding a thick structural section to the road. In extensive swamp or wet areas subsurface drainage is often less effective than raising the roadway platform or providing a thick aggregate layer to support the road pavement.

4.3.6.2 Subsoil drains

Longitudinal subsoil drains can be used to locally lower a water table. They normally consist of porous concrete, open jointed or perforated pipe laid in a trench and backfilled with a free draining material such as graded crushed stone or gravel. The pipe size should not be less than 15cm internal diameter. The trench should be at least 60cm wide and 1.5m deeper than the formation level of the road.

4.3.6.3 Filters

A filter is a transitional layer of small gravel or geotextile placed between a structure, such as riprap or gabions, and the underlying soil. Its purpose is to prevent the movement of soil behind the structure or into under-drains. Filters allow groundwater to drain from the soil without building up pressure. A sand or gravel filter layer is typically about 150 to 300 mm thick. In some applications, two filter layers may be needed between fine soil and very large rock. Geotextiles are commonly used to provide filter zones between materials of different size and gradation. The geotextile can be a woven monofilament or a needle punched non-woven geotextile, but it must be permeable. The geotextile should have an apparent opening size of 0.25 to 0.5mm. A 200g/m² needle-punched non-woven geotextile is commonly used for soil filtration and separation applications.

4.3.6.4 Interceptor, cut-off or catch-water drains

These drains are constructed to prevent water flowing into vulnerable locations by 'intercepting', 'cutting off' or 'catching' the water flow and diverting it to a safe point of discharge, usually a natural watercourse.

Interceptor drains above cut faces should have a gradient of 2% on their full length and should be at least 3 to 5 m from the cut face. If steeper gradients in the drain are unavoidable then scour checks should be installed or the drain should be lined. The drain should also be lined where seepage will weaken the cut slope. Alternatively the drain should be replaced by a vegetated earth bund.

Interceptor drains should be 60 cm wide, 40 cm (minimum) deep with sides back-sloped at 3:1 (vertical: horizontal).

4.3.6.5 Chutes

Chutes are structures intended to convey a concentration of water down a slope that, without such protection, would be subject to scour. Since flow velocities are very high, stilling basins are required to prevent downstream erosion. The entrance of the chute needs to be designed to ensure that water is deflected from the side drain into the chute, particularly where the road is on a steep grade. On embankments it may be necessary to lead water to the top of chutes using kerbing.

4.4 Design Standards for Road Drainage

Table 4-11, Table 4-12, Table 4-13, Table 4-14 and Table 4-15 show the design standards that arise from the discussions in all sections of this chapter. Before applying these standards, the designer must first understand the rationale behind these figures by studying the whole manual.

Table 4-13 Slope Dimensions for Cross Sections and Side Drains (Vertical : Horizontal)

Material Type	Height of slope (m)	Side Slope		Back slope
		Cut	Fill	
Earth/Gravel	0.0-1.0	1:4	1:4	1:3
	1.0-2.0	1:3	1:3	1:2
	>2.0	1:2	1:2	1:1.5
Expansive Clays	0-2.0	n/a	1:6	
	>2.0	n/a	1:4	
Rock	Any height	Dependent on costs		

Notes:

1. Certain slopes may be unstable at slopes 1:2. Geotechnical advice required.

Table 4-14 Minimum Height between Road Crown and Invert Level of Drain

Road Class	Climate	
	Wet (N<4)	Dry (N>4)
	H _{min} (mm)	H _{min} (mm)
LVSR-1	350	250
LVSR-2	400	450
LVSR-3	500	300

Table 4-15 Other Standards

Parameter	Minimum Dimension (mm)
Ditch/Side drain width at the bottom (trapezoidal drains)	300
Culvert Internal Diameter or height of other structure	900

5.0 Materials

5.1 Introduction

Materials required for the construction of roads are getting increasingly scarce. As a result of this, it is imperative that whatever material extracted must be used judiciously. Naturally occurring soils and gravels usually meet the requirements for low volume roads. It is only in exceptional circumstances that the more expensive crushed stone materials should be used for construction of pavement layers.

Many naturally occurring materials do not meet conventional specifications. However, they have been used successfully in situations of lowly trafficked roads, favourable climatic conditions and where good and appropriate construction techniques were used.

5.2 Use of Naturally Occurring Materials

Naturally occurring materials of varying properties may be found locally close to the project areas. These materials may in many cases not meet conventional standard specifications for heavily trafficked roads. Such materials may however be used within a framework of standards and specifications which have been developed and are appropriate for low volume roads. However, their use demands careful attention to three factors:

- i. There must be good internal and external drainage of the pavement;
- ii. The construction quality control must be rigorous with serious attention paid to compaction; and
- iii. There must be a realistic commitment to appropriate maintenance strategies.

Examples of common naturally occurring materials used for the construction of road bases and sub-bases are shown in Table 51.

Table 5-1 Common Examples of Natural Base and Sub-base Materials

Type of gravel	MaterialGeologicalCharacteristics	General Engineering Properties
Laterite gravel	Being a precipitate laterites mature with age. They start off as small nodules which then grow in size and harden with time. Very mature laterites coagulate into large lumps or boulders which are hard to break during construction	The behaviour of laterite gravel is largely dependent on the parent soil in which it is formed. Laterites that form in clayey soils tend to be plastic and this is very common. This is the major reason why laterites have not been widely recommended for road construction as the high plasticity leads to lower CBRs and most therefore do not meet the minimum standard specifications of road base of not less than 80% CBR
Calcrete gravel	Calcrete is formed in the same manner as laterite i.e. through the precipitation of calcium oxides and carbonates forming nodules which grow and harden over time. Very mature calcrete is found with high content of coagulants which are hard to break even with use of machines	Mature calcrete can give an ACV of about 25 which is good enough for use as surfacing aggregate. The properties of calcrete are dependent on the host material in which it is deposited. If it is deposited in clay plasticity is usually high and that which is deposited in sand soils tends to have low plasticity. The maturity of the calcrete determines the strength and how and where it can be used in the road structure.

Type of gravel	MaterialGeologicalCharacteristics	General Engineering Properties
Quartzitic gravel	Quartz is formed from decomposition of igneous rocks which have a large proportion of quartzite. In most cases the quartzite is mixed with feldspar but in road construction it's all considered as quartzite.	Quartzite gravel is a very important material in road construction. Quartzite is a very hard material and it is usually angular. This is what gives quartzitic gravel an edge over the other forms of gravel in terms of its performance in service. Quartz gravel is usually well graded but it depends on the composition of the parent rock. Quartz gravel may have a high content of silt which increases the plasticity of the whole gravel matrix. Most quartz gravels have low plasticity if not non-plastic. The CBR is usually high and this makes quartzitic gravel suitable natural material for construction of bases usually without the need for stabilisation
Decomposed granite	Decomposed granite is a product of the decomposition of granitic rocks with a low content of quartz or silica. The strength of individual particles is usually low and sometime can be broken with fingers. They are usually greyish or pale in colour and generally fine grained	Decomposed granite gravels tend to have low plasticity and regardless of the low particle strength the material can be used for the construction of bases on low volume roads or sub-bases on high volume roads without treatment. However it may be necessary to consider stabilisation with cement for use on road base for higher volume roads.
Basaltic gravel	Basaltic gravels are formed from the decomposition of basalt rocks. Basalt is an igneous rock which is usually formed from the magma that is extruded to the surface and it is fine grained. The decomposition of the basalts usually results in the formation of coarse gravels which are gap graded and plastic.	The engineering properties of basalt gravels tend to show high variability and need close examination and monitoring. High coarseness usually leads to high bearing capacity but when plasticity is high the bearing capacity becomes highly compromised. Basalt gravels also have a tendency to degenerate in service depending on a number of factors including the moisture regime and traffic loading. Such degeneration can lead to sharp increases in plasticity of the gravel layer in-service leading to significant reduction in strength which in turn may lead to premature failures
Ryolite gravel	Ryolites are formed over short periods of time as a result of violent volcanic activities. They are basically a deposition of stone and volcanic ash from the explosive extrusions. In some instances they have the appearance of limestone gravel and one can be mistaken for the other. Acid test can be used to distinguish the two with the limestone gravel giving effervescence from the chemical reaction between the acid and the basic material.	Ryolite gravels can be found in many different forms but with the appropriate work norms it is possible to build good natural road bases with it. High content of volcanic ash makes construction difficult as a result of the powdery nature of the ash. Screening is usually necessary due to high contents of reject material which may include large boulders up to 300mm in diameter.
Cinders/red ash	Cinder gravels are common in the rift valley and they are formed from violent volcanic activities with a high content of gases. They are found in many different forms. When the gas content is very high stones that are formed from rapidly expanding and cooling lava tend to have a high content of voids and as a result the aggregate tends to be very light and weak even to the extent of floating on water. When the gas content is low the content of basaltic stone would be high leading to strong gravel.	The plasticity of cinder gravels is usually low and most gravels are non-plastic. The grading is also variable and the gravel may be coarse with large boulders or fine grained. The most important parameters are the aggregate strength, the density (the maximum dry density) which can be very low and the plasticity. Cinder gravels which a high content of basaltic material tend to be plastic and other forms tend to have low plasticity. Cinder gravels with low particle strength tend to degenerate during construction and care should be taken to prevent excessive degeneration as this compromises the overall bearing capacity of the base layer. Modification of the material may be necessary to enhance performance of the road base in-service

5.3 Sampling and Selection of Materials

5.3.1 Subgrade testing and classification

Subgrade investigations shall be conducted using a Dynamic Cone Penetrometer (DCP) and trial pit excavations. The DCP testing shall be conducted at intervals of 100 – 500m. Trial pits shall be excavated at intervals of 500m along the road section. However, to understand the subgrade properties more, sampling and laboratory testing are necessary. Sampling should be conducted for subgrade materials for all identified sections.

The depth of material sampled for subgrade testing should lie between 700 and 800mm below the would-be completed road surface. These depths ensure that there are no weaker underlying materials that would negatively affect the pavement performance.

Subgrades are classified on the basis of soaked CBR tests on samples compacted to 95% of the maximum dry density determined by the BS Light Compaction method. The CBRs obtained from this test are used to classify all the subgrade. A minimum of 10 samples are required for each uniform section. The subgrade design CBR for each uniform section is determined by using the percentiles value as shown in Table 5-2.

Table 5-2 Percentiles for Subgrade Classification

Traffic Class	Design CBR
0.5 – 1.0 MESA	Lower 10 th percentile
0.1 – 0.5 MESA	Lower 15 th percentile
< 0.1 MESA	Lower 30 th percentile

Using the values obtained from the testing and the percentiles, the subgrade may be classified as shown in Table 5-3.

Table 5-3 Subgrade Classification

Subgrade Class	S2	S3	S4	S5	S6
Strength (CBR)	3-4	5-8	9-14	15-29	30+

The use of Class S2 soils and any weaker soils as direct support for the pavement should be avoided as much as possible. Wherever practicable, such relatively poor soils should be excavated and replaced, or covered with an improved subgrade of minimum soaked CBR 15%.

5.3.2 Material depth

The concept of “material depth” is used to denote the depth below the finished level of the road to which soil characteristics have a significant effect on pavement behaviour and which the nominal subgrade strength selected for design should be maintained throughout.

Table 54 specifies typical material depths used for determining the design CBR of the subgrade. Note that this depth may be insufficient in certain special cases where “problem” soils occur.

Table 5-4 Material Depth by Road Category

Road Category	Material Depth (mm)
LVSR2 and LVSR3	800 /400
LVSR1	700/ 300

The minimum depths indicated in Table 54 are not depths to which re-compaction and reworking is necessarily required. Rather, they are the depths to which the Engineer should confirm that the nominal subgrade strength is available. In general and only for existing roads, unnecessary working of the subgrade should be avoided and limited to rolling prior to constructing overlying pavement layers. This is based on the presumption that the subgrade under an existing road

would have gained significant strength over long periods of time through consolidation by traffic. For constructed roads on virgin ground, the subgrade should be reworked. The design subgrade strength should be that which is obtained from the laboratory after compaction at 95% of maximum dry density or BS Light Compaction.

5.3.3 Materials prospecting

Materials prospecting is both an art and a science which involves searching for clues for the occurrence of useful materials and then excavating to see whether they exist. Materials prospecting can be very complicated and often take several months or sometimes years. However, for most low volume roads, local residents and the road maintenance department often have information relating to the occurrence of road building materials.

Identifying features that indicate the presence of gravel from interpretation of maps, vegetation, outcrops, burrows and other information is a key activity in prospecting. Prospecting involves a desk study followed by field surveys and trial pitting.

Information about gravels in the landscape typically comes from five main sources:

- i. Geological information from geological maps and reports;
- ii. Soils information from agricultural soils maps and reports;
- iii. Botanical indicators (such as certain weed types are often synonymous with the occurrence of certain gravels);
- iv. Landscape information from topographical maps, aerial photos and satellite imagery; and
- v. Other local information (e.g. existing borrow pits).

The aggregated analysis of all this information allows for the prospective areas of occurrence to be significantly narrowed down.

5.3.4 Borrow Pit Materials

The maintenance department and most often local residents will know locations where materials may be extracted. It is important to mark by GPS, locations of potential sources of materials. Samples should then be taken from these locations for laboratory testing.

Existing gravel wearing courses on the road usually have high plasticity and low CBRs, however, they can be used satisfactorily as sub-base in the new pavement. A base of better quality borrow materials will have to be constructed.

The material for the base can often be obtained from borrow pits as as-dug material or modified through screening, blending, stabilisation or other methods. Borrow pit materials should generally be classified in terms of the parameters defined in the Table 5-5.

Table 5-5 Material Classes and Specifications

Material Code	Material Type	Material Specifications
G80	Natural Gravel / Crushed Stone	Min. CBR: 80% @ 98% AASHTO T180 and 4 days soaking Max. Swell: 0.5% Max. Size and grading: Max size 37.5mm, grading as specified PI: < 6 or as otherwise specified (material specific)
G65	Natural Gravel	Min. CBR: 65% @ 98% AASHTO T180 and 4 days soaking Max. Swell: 1% Max. Size and grading: Max size 37.5mm, grading as specified PI: < 10 or as otherwise specified (material specific)
G55	Natural Gravel	Min. CBR: 55% @ 98% AASHTO T180 and 4 days soaking Max. Swell: 0.2% Max. Size and grading: Max size 37.5mm, grading as specified PI: < 12 or as otherwise specified (material specific)
G45	Natural Gravel	Min. CBR: 45% @ 95% AASHTO T180 and 4 days soaking Max. Swell: 0.5 % Max. Size and grading: Max size 37.5mm, grading as specified PI: < 14 or as otherwise specified (material specific)
G30	Natural Gravel / soil	Min. CBR: 30% @ 95% AASHTO T180 & highest anticipated moisture content Max. Swell: 1.0% 1.5% @ 100% AASHTO T180 Max. Size and grading: Max size 63mm or 2/3 layer thickness PI: < 16 or as otherwise specified (material specific)
G25	Natural Gravel / soil	Min. CBR: 25% @ 95% AASHTO T180 & highest anticipated moisture content Max. Swell: 1.0% 1.5% @ 100% AASHTO T180 Max. Size and grading: Max size 63mm or 2/3 layer thickness PI: < 16 or as otherwise specified (material specific)
G15	Natural Gravel / soil	Min. CBR: 15% @ 95% AASHTO T180 & highest anticipated moisture content Max. Swell: 1.5% 1.5% @ 100% AASHTO T180 Max. Size: 2/3 of layer thickness PI: < 16 or 3GM + 10 or as otherwise specified (material specific)

The grading envelopes to be used for road base are shown in Table 5-6. Envelope A varies depending whether the nominal maximum particle size is 37.5mm, 20mm or 10mm. A requirement of five to ten per cent retained on successive sieves may be specified at higher traffic (>0.3 mesa) to prevent excessive loss in stability. Envelope C extends the upper limit of envelope B to allow the use of sandy materials, but its use is not permitted in wet climates. Envelope D is similar to gravel wearing course specification, and is used for very low traffic volumes. The grading is specified only in terms of the grading modulus (GM).

Table 5-6 Particle Size Distribution for Natural Gravel Road bases

Test Sieve Size	Per cent by mass of total aggregate passing test sieve				
	Envelope A Nominal maximum particle size			Envelope B	Envelope C
	37.5mm	20mm	10mm		
50 mm	100			100	
37.5 mm	80-100	100		80-100	
20 mm	55-95	80-100	100	55-100	
10 mm	40-80	55-85	60-100	40-100	
5 mm	30-65	30-65	45-80	30-80	
2.36 mm	20-50	20-50	35-75	20-70	20-100
1.18 mm	-	-	-	-	-
425 µm	8-30	12-30	12-45	8-45	8-80
300 µm	-	-	-	-	-
75 µm	5-20	5-20	5-20	5-20	5-30
Envelope D					
1.65 < GM < 2.65					

5.3.4.1 Strength and Plasticity

The strength and plasticity requirements vary depending on the traffic level and climate as outlined in Table 57. The soaked CBR test is used to specify the minimum roadbase material strength.

A maximum plasticity index of 6 has been retained for higher traffic levels (> 1 MESA) and also on weaker subgrades. For designs in arid conditions, the plasticity modulus for each traffic and subgrade class can be increased depending on the crown height and whether unsealed or sealed shoulders are used as described herein.

Table 5-7 Plasticity Requirements for Natural Gravel Road Base Materials

SubgradeClass	Material Property	Upper Limit of Design Traffic Class						
		0.01M	0.05M	0.1M	0.3M	0.5M	1M	3M
	I _p	<12	<12	<9	<6	<6	<6	<6
S2	PM	400	250	150	120	90	90	90
	Grading	B	B	B	A	A	A	A
	I _p	<15	<12	<12	<9	<6	<6	<6
S3	PM	550	320	250	180	90	90	90
	Grading	C ⁽¹⁾	B	B	B	A	A	A
	I _p	Note (2)	<15	<12	<12	<9	<9	<6
S4	PM	800	450	320	300	200	90	90
	Grading	D ⁽³⁾	B	B	B	B	A	A
	I _p	Note (2)	<15	<15	<12	<12	<9	<6
S5	PM	n/s	550	400	350	250	150	90
	Grading	D ⁽³⁾	C ⁽¹⁾	B	B	B	A	A
	I _p	Note (2)	<18	<15	<15	<12	<9	<6
S6	PM	n/s	650	550	500	300	180	90
	Grading	D ⁽³⁾	C ⁽¹⁾	C ⁽¹⁾	B	B	A	A

Notes:

Grading 'C' is not permitted in wet environments or climates (N<4); grading 'B' is the minimum requirement

Maximum I_p = 8 x GM

Grading 'D' is based on the grading modulus 1.65 < GM < 2.65

All base materials are natural gravels

Subgrades are non-expansive

Separate notes are provided covering the use of laterites, calcretes (N>4) and weathered basalts

I_p Plasticity index

PM Plasticity modulus

n/s Not specified

5.3.4.2 Lateritic Gravels for Road Bases

The requirements for selection and use of lateritic gravels for bases (Table 5-8) are slightly different to those given for other natural gravels (Table 5-7). A maximum plasticity index of 6 has been specified for higher traffic levels and weak subgrades. For design traffic levels greater than 0.3 MESA, a requirement is set that the liquid limit should be less than 30. Below this traffic level, this requirement is relaxed to a liquid limit of less than 35. Where sealed shoulders over one metre wide are specified in the design, the maximum plasticity modulus may be increased by 40 per cent. A minimum field compacted dry density of 2.0 Mg/m³ is required for these materials.

Table 5-8 Requirements for Lateritic Gravel Roadbase Materials

Subgrade CBR	Material Property	Design Traffic Class (MESA)					
		0.01	0.05	0.1	0.3	0.5	1.0
S2	I _p	<15	<15	<12	<9	<9	<6
	PM	<400	<250	<150	<150	<120	<90
	GE	B	B	B	A	A	A
S3	I _p	<18	<15	<15	<12	<9	<6
	PM	<550	<320	<250	<180	<120	<90
	GE	B	B	B	B	A	A
S4	I _p	<20 ⁽¹⁾	<18	<15	<15	<9	<9
	PM	(800	<450	<320	<300	<200	<90
	GE	GM 1.6-2.6	B	B	B	B	A
S5	I _p	<25 ⁽¹⁾	<20	<18	<15	<12	<9
	PM	n/s	<550	<400	<350	<250	<150
	GE	GM 1.6-2.6	B	B	B	B	B
S6	I _p	<25 ⁽¹⁾	<20	<20	<18	<15	<12
	PM	n/s	<650	<550	<400	<300	<180
	GE	GM 1.6-2.6	B	B	B	B	A
Notes: Ip maximum = 8 x GM n/s = not specified Unsealed shoulders are assumed (1) The high PI is only permissible for pedogenic materials like laterite.				Ip = plasticity index PM = plasticity modulus GE = grading envelope GM = grading modulus			

The following should also be noted:

- Further modification to the limits can be made if the shoulders are sealed
- The compaction requirement for the soaked CBR test is 100 per cent BS 4.5 kg rammer or equivalent compaction with a minimum soaking time of four days or until zero swell is recorded; this is a relaxation of the soaked CBR requirement for natural gravel base materials given in Charts 1 and 2

5.3.4.3 Sub-bases

In order for the sub-base to provide the required bearing capacity and support to the upper pavement layers, the particle size distribution shown in Table 5-9 and the Atterberg limits shown in Table 5-10 should be met.

Table 5-9 Sub-base Grading Requirements

Sieve Size (mm)	Per cent by mass of total aggregate passing test sieve
50	100
37.5	80-100
20	60-80
5	30-100
1.18	17-75
0.3	9-50
0.075	5-25

Table 5-10 Atterberg Limits for Granular Sub-bases

Climate	Liquid Limit	Plasticity Index	Linear Shrinkage
Moist tropical and wet tropical (N<4)	<35	<6	<3
Seasonally wet tropical (N<4)	<45	<16	<6
Arid and semi-arid (N>4)	<55	<20	<10

5.3.4.4 Blending Materials for Road Bases

The purpose of blending is to improve the properties of one or more materials. If the materials already possess the desirable properties, there is no need for blending unless significant cost reductions can be achieved. The materials to be blended must complement each other, for example non-plastic sand with highly plastic laterite.

The properties of the final blended material must meet the requirements of the design catalogue for granular materials for the layer in which it is intended for use. The blend ratio that meets these requirements in the most economical way should be used.

5.3.5 Cement and Lime Stabilised Road Bases and Sub-bases

Suitability of materials for stabilization with either cement or lime can be determined from Table 5-11.

Table 5-11 Selection of Stabilisation Methods

Type of Stabilisation	Soil Properties					
	> 25% Passing the 0.075mm sieve			< 25% passing the 0.075mm sieve		
	PI ≤ 10	10<PI≤20	PI > 20	PI ≤ 6 PP ≤ 60	PI ≤ 10	PI > 10
Cement	Yes	Yes	Marginal	Yes	Yes	Yes
Lime	Marginal	Yes	Yes	No	Marginal	Yes

The most common type of cement used for stabilisation is Ordinary Portland Cement and hydrated lime is preferred for lime stabilisation. The properties of the final stabilised material must meet the requirements shown in Table 5-12 for the layer in which it is intended for use and the design catalogue for stabilised materials. The amount of cement used should be greater than the quantity of initial consumption of cement (ICC) and for lime greater than initial consumption of lime (ICL) determined in the laboratory. During construction, from the point the stabiliser comes into contact with the material to completion of compaction should be less than 3 hours for cement and 8 hours for lime.

Table 5-12 Minimum Characteristics of Stabilised Materials for Road Bases and Sub-bases

Material Class	Description	Unconfined Compressive Strength or CBR
CB2	Stabilized Road base	1.5-3.0 MPa
G30 or CS	Granular Sub-base Stabilised Sub-base	Min CBR 30% 0.7-1.5 MPa
G10	Selected Subgrade Fill	Min CBR 10% Mod. AASHTO at highest anticipated moisture content

5.4 Aggregates for Surfacing

For the purpose of construction of low volume roads of short lengths, it is often not cost effective to set up a quarry plant to produce surfacing aggregates. Naturally occurring aggregates may be available locally that can be screened and used in the construction of surfacings. In many cases, these aggregates may be screened by labour-based methods if well-coordinated and planned.

Natural deposits of sand are usually readily available, although care has to be taken to ensure they do not contain adverse quantity of clay and organic materials.

In the case where natural aggregates are not available, nearby commercial quarries should be sought to provide aggregates that are suitable for surfacings. The detailed specifications for aggregates for each type of surfacing are discussed under the surfacing design chapter.

In all cases, the aggregates should generally be of the following characteristics:

- i. The materials should be strong, durable and sound;
- ii. Free of deleterious materials;
- iii. Not susceptible to excessive polishing action of traffic;
- iv. Clean and generally free of dust (with the exception of Otta Seals that may contain a limited quantity); and
- v. Cubical in shape, not rounded or flaky.

It is usually difficult to obtain aggregates that meet all of the requirements outlined above and therefore a compromise has to be made from time to time. Engineering judgement must be exercised in these situations. Table 5-13 shows examples of materials that are commonly used for surfacing of low volume roads.

Table 5-13 Types of Aggregates for Surfacing

Materials for surfacing	Description
	Calcrete graded aggregate for Otta seal surfacing.
	Fine quartz graded aggregate for Otta seal surfacing sieved from a quartzitic gravel
	River sand for sand seal. The sand has to be coarse, clean and dry or slightly moist.
	Coarse hand or machine crushed aggregate (20-40mm) for penetration macadam
	Fine aggregate/grit for a grit seal
	Stone paving with concrete screed topping
	Crushed stone (plant or hand crushed)

5.5 Testing of Materials

Laboratory test methods should be consistent in order to avoid problems that may be encountered by the use of different standards. In Uganda, the predominant test methods are the British Standards (BS) test methods although some few tests may be specified under the AASHTO or ASTM test methods. The list of tests that must be carried out as a minimum is shown in Table 5-14. These methods generally differ in many aspects and results from one cannot simply be used to substitute another. For example, the use of the BS Liquid Limit test device will in most cases lead to Plasticity Index 4 Units higher than those using an ASTM Liquid Limit device.

Table 5-14 Mandatory Materials Tests

Test Type	Test Designation
Subgrade, Sub-base, Base	
Liquid Limit	BS 1377-2:1990
Plastic Limit	BS 1377-2:1990
Maximum Dry Density and OMC	BS 1377-4:1990 or Equivalent (AASHTO T180-01)
California Bearing Ratio (CBR)	BS 1377-4:1990 or Equivalent (AASHTO T193-99)
Particle Size Distribution	BS 1377-2:1990
Unconfined Compressive Strength (Lime or Cement Stabilised Materials)	BS 1924-2:1990
Surfacing Aggregates	
Loose Density	BS EN 1097-3:1998
Particle Size Distribution	BS 812: Part 103.1:1985
Average Least Dimension	BS EN 933-3:1997
Flakiness Index	BS EN 933-3:1997
Aggregate Crushing Value	BS 812: Part 110:1990
Ten Percent Fines Value	BS 812: Part 110:1990
Water Absorption	BS EN 1097-3:1998

5.6 Materials Specifications

Materials specifications are dependent on the traffic levels and the climatic region of the road being designed. The designer is advised to refer to preceding subsections of this chapter and the chapter on pavement design.

6.0 Pavement Design

6.1 Introduction

The main objective of pavement design is to provide an economic, adequate structure in terms of material types and thicknesses that can withstand the expected traffic loading over the required design life of the pavement without deteriorating below a pre-determined level of service.

The types of materials and the thickness of the layers are dependent upon the design traffic, sub-grade strength and the environmental conditions that affect the performance of the road. For this reason, it is common practice to have different pavement design sections on a given road.

To upgrade a road to a sealed standard cost effectively, use must be made of the existing pavement layers. As a result of undergoing many years of trafficking, coupled with climatic wetting and drying cycles, the existing unsealed pavement would have achieved a significant degree of consolidation. If undisturbed, the layers can be incorporated into the new pavement to provide significant strength, often as sub-bases and sometimes as base course.

6.2 Traffic

6.2.1 Classification

The deterioration of sealed roads is caused partly by traffic, resulting from both the magnitude and frequency of the individual axle loads. Thus, pavement design requires information on the total number of commercial vehicles which will use the road and their axle loads.

Traffic is measured in numbers expressed as the Annual Average Daily Traffic (AADT). The AADT is defined as the total annual traffic summed for both directions and divided by 365. It is usually obtained by recording actual traffic volumes over a much shorter period from which the AADT is then estimated. In many cases, it is not possible to obtain AADT data in which case, the Average Daily Traffic (ADT) is measured. The ADT is the traffic measured (counted) for a minimum of 7 days and expressed as a daily average.

During traffic counts the vehicles are categorised into classes as shown in Table 6-1. The traffic classification in Uganda is given in Table 6-2.

Traffic engineering calculations should be carried in order to estimate the cumulative traffic over the design life of the road. Alternatively, traffic information might be available from an economic evaluation carried out to justify the road.

Table 6-1 General Vehicle Classification

Class	Type	Axles	Description
A	Car	2	Passenger cars and taxis
B	Pick-up/4-wheel drive	2	Pick-up, minibus, Land Rovers, Land Cruisers
C	Small bus	2	≤ 27 seats
D	Bus/coach	2	> 27 seats
E	Small truck	2	> 3.5 tonnes (unladen weight)
F	Medium Truck	2 or 3	>3.5 – 7.5 tonnes
G	Large Truck	> 4	>7.5 tonnes
H	Tractor	2	*
I	2-axled trailer		>7.5 tonnes
J	3-axled trailer	3	*
K	4-axled trailer	4	*
L	Motorcycles and motor cycle taxis		*
M	Bicycles		*
N	Pedestrians		*
O	Other NMT		*

*No need for definition

Table 6-2 MoWT Vehicle Classification in Uganda

Vehicle type	Designation	Axles
Salon cars and taxis		2
Light goods vans, pickups and 4WDs		2
Minibuses and Matatus		2
Buses		3
Lorries, Single unit and tractors		3
Truck Trailer and Semi-trailer		>3
Motorcycles		
Bicycles		
Carts		

6.2.2 Traffic Surveys

The initial traffic volume for each vehicle class is determined on the basis of appropriate traffic surveys to establish the traffic volume by each traffic class in terms of the Annual Average Daily Traffic (AADT) at the time of opening the road. The two most commonly used types of traffic surveys for LVRs are:

- i. Automatic Traffic Survey (traffic counters with inductive loops and weighing-in-motion (WIM) sensors; and
- ii. Manual Traffic Survey.

The objective of undertaking either of the above types of traffic surveys is essentially to obtain an estimate of the AADT using the road, disaggregated by vehicle type. Prediction of such traffic is notoriously imprecise, especially where the roads serve a predominantly developmental or social function. Thus, the timing, frequency and duration of traffic surveys should be given very careful consideration in terms of striking a balance between cost and accuracy. Errors in estimating traffic for LVSs can be reduced by conforming to the following rules:

- a) Count for seven consecutive days;
- b) On some days count for a full 24 hours, preferably with one 24-hour count on a weekday and one during a weekend; on other days, 16-hour counts (typically 06.00 – 22.00 hours) should be made and extrapolated to 24 hour counts using a previously established 16:24 hour ratio;
- c) Avoid counting at times when road travel activity increases abnormally; for example, just after the payment of wages and salaries, or at harvest time, public holidays or any other occasion when traffic is abnormally high or low; and
- d) If possible, repeat the seven-day counts several times throughout the year.

Care should be exercised in selecting appropriate locations for conducting the traffic counts to ensure a true reflection of the traffic using the road and to avoid under- or over-counting.

If any junctions occur along the road length, counts should also be conducted before and after the junctions.

6.2.3 Traffic Growth

Future traffic falls into the following three categories:

- i. Normal traffic - Traffic which would pass along the existing road or track even if no new pavement were provided. This increases naturally by virtue of normal economic growth;
- ii. Generated traffic - Additional traffic which occurs in response to the provision or improvement of the road. This is traffic generated by any extra economic growth resulting from the road; and
- iii. Diverted traffic - Traffic that changes from another route (or mode of transport) to the project road because of the improved pavement, but still travels between the same origin and destination.

The AADT in both directions in the first year of analysis consists of the current traffic plus an estimate of the diverted traffic. If the total traffic is denoted by $AADT_0$ and the general growth rate is i per cent per annum, then the traffic in any subsequent year, x , is given by the following equation:

6.2.4 Equivalent Standard Axles (ESA) for Each Vehicle Class

The of Equivalent Factor (EF) of an axle for a traffic category or class (m) is related to the axle load as follows:

$$EF_{(m)} = (P/8160)^n \text{ (for loads in kg)}$$

or $EF_{(m)} = (P/80)^n \text{ (for loads in kN)}$

Where:

$EF_{(m)}$ = number of equivalent standard axles (ESAs)

P = axle load (in kg or kN)

n = damage exponent (n lies between 2.5 and 4.5 – a value of 4 is used for LVRs)

The sum of the individual $EF_{(m)}$ or E80 values for each axle of the vehicle gives the equivalence factor for the vehicle as a whole, $EF_{(m)}$. Guidance on the likely average $EF_{(m)}$ for different vehicle classes derived from historical data is given in Table 63. This data can be used where traffic counts have been conducted but axle load data is not available. However, data from any recent axle load survey on the road in question or a similar road in the vicinity is better than using countrywide averages.

Table 6-3 Average Equivalency Factors for Different Vehicle Types

Class	Type	No of axles	Average ESAs per vehicle - loaded	Average ESAs per vehicle - half loaded ⁽¹⁾
A	Car	2	-	-
B	Pick-up/4-wheel drive	2	-	-
C	Small bus	2	0.3	0.15
D	Bus/coach	2	2.0	1.0
E	Small truck	2	1.5	0.7
F	Medium Truck	2	5	2.5
G	Large Truck 2 Axle	2	10	5
H	3 – axle truck	3	12	3.5
I	4 – Axle Truck	4	15	7.5
J	5 - axle	5	17	8.5
K	6- Axle	6	17	8.5
L	Tractor	3	2.5	1.5
M	2-axled trailer	2	10	5
N	3-axled trailer	3	12	6
O	4-axled trailer	4	12	6

Note: -It is common to find that vehicles have no back load hence half the vehicles are likely to be empty, or nearly so.

-Vehicle Category A and B have negligible damaging effect on the pavement.

The cumulative ESAs over the design period for each vehicle class is obtained by multiplying EF by the cumulative traffic, $T_{(m)}$. The total number of cumulative standard axles for all vehicle classes is then obtained by adding together the values of $EF_{(m)} \times T_{(m)}$ for all the classes.

In some cases, there will be distinct differences in each direction and separate vehicle damage factors for each direction should be derived. The higher of the two directional values should be used for design.

On narrow roads, the traffic tends to be more channelized than on wider two-lane roads. In such cases the effective traffic loading is greater than that for a wider road and the design traffic loading (Esa) is calculated using the relationships given in Table 6-4.

Table 6-4 Factors for Design Traffic Loading

Cross Section	Paved width	Corrected Design Traffic Loading (Esa)	Explanatory Notes
Single carriageway	< 3.5 m	Double the sum of Esa in both directions	The driving pattern on this cross-section is much channelized.
	Min. 3.5 m but less than 4.5.m	The sum of Esa in both directions	Traffic in both directions uses the same lane
	Min. 4.5 m but less than 6 m	80% of the Esa in both directions	To allow for overlap in the centre section of the road
	6 m or wider	Total Esa in the heaviest loaded direction	Minimal traffic overlap in the centre section of the road.
More than one lane in each direction		90% of the total Esa in the studied direction	The majority of vehicles use one lane in each direction.

Construction traffic can also be a significant proportion of total traffic on LVRs (sometimes 20 – 40 % of total traffic) and should be taken into account in the design of the pavement.

For very low volume roads (traffic <25 vpd), a detailed traffic analysis is seldom warranted because environmental rather than traffic loading factors generally determine the performance of roads.

6.2.5 Computation of the design MESA

For structural pavement design the cumulative traffic loading of each of the motorised vehicle classes over the design life of the road in one direction is required.

Basic traffic engineering methods may be used to determine the cumulative traffic loading for the life of the road or performance period. For a given class, m , the equation below is used to calculate the design cumulative traffic (T_m) for the design life (n).

$$T_m = \sum_0^n T_i = (AADT_0 \times 365) \times \left(1 + \frac{r}{100}\right)^n$$

Where:

- n = Design life of the road in years from the day it's opened to traffic
- T_m = Design traffic for class (m) for the design life of the road
- $AADT_0$ = Current annual average daily traffic
- r = Annual traffic growth rate (%)

Generally, a life-cycle of 20 years is assumed but this can be varied in accordance with country standards. For low volume roads, the design life can be 15 years from the year of opening to traffic.

The cumulative traffic for each class of vehicle is multiplied by the average number of equivalent standard axles of vehicles in that class to calculate the cumulative total number of equivalent standard axles over the life of the road (in MESAs).

6.2.6 Pavement Design Classes

The pavement design classes for structural design are shown in Table 65. They are defined in the form of ranges of traffic loading classes, TLC.

Table 6-5 Pavement Design Classes

Class	MESA
TLC1	< 0.01
TLC2	0.01-0.1
TLC3	0.1 – 0.3
TLC4	0.3 – 0.5
TLC5	0.5 – 1.0

6.3 Design Methods

The surfacing of many types of low volume paved roads is a thin flexible bituminous layer designed to produce a durable and waterproof seal. The structural design of such roads is identical because the surfacing adds no structural strength, or very little.

6.3.1 Catalogue Method

The catalogue method is the most convenient and, indeed, the most common method of design. For each type of structure, designs have been produced based on the range of subgrade strengths and the range of traffic loading levels. The designer has to estimate or measure the subgrade strength and traffic level, choose the preferred pavement structure type and then look up the design in the appropriate catalogue. The subgrade class is determined as discussed in Section 5.3.1 and the design traffic class as discussed in Section 6.2.6.

The road bases can be either unbound granular material or a stabilised material. Table 5-5 shows the material types for the various unbound structural layers. Section 5.3 also includes various different bound materials for road bases and materials for sub-bases. Generally, sub-bases are unbound and comprise G30 or G25 materials. Both are suitable but G30 is preferred.

The design charts for roads with thin bituminous road surfaces are shown in Table 6-6 and Table 6-7. The use of the charts is described below.

Table 6-6 Bituminous Pavement Design Chart 1 (N<4)

SG CBR	TLC1	TLC2	TLC3	TLC4	TLC5
	<0.01	0.01-0.1	0.1-0.3	0.3-0.5	0.5-1.0
S2 (3-4%)	150 G60	150 G60	150 G60	175 G80	200 G80
	150 G15	125 G30	150 G30	175 G30	175 G30
		150 G15	175 G15	175 G15	200 G15
S3 (5-7%)	125 G65	150 G60	150 G60	175 G60	200 G80
	150 G15	100 G30	150 G30	150 G30	150 G30
		100 G15	150 G15	150 G15	150 G15
S4 (8-14%)	175 G40	150 G60	150 G60	175 G60	200 G80
		120 G30	200 G30	200 G30	200 G30
S5 (15-29%)	175 G40	125 G60	150 G60	175 G60	175 G80
		125 G30	150 G30	150 G30	150 G30
S6 (>30%)	150 G40	150 G60	175 G60	175 G60	200 G80

Table 6-7 Bituminous Pavement Design Chart 2 (N>4)

SG CBR	TLC1	TLC2	TLC3	TLC4	TLC5
	<0.01	0.01-0.1	0.1-0.3	0.3-0.5	0.5-1.0
S2 (3-4%)	150 G40	150 G60	150 G80	175 G80	200 G80
	150 G15	120 G30	150 G30	150 G30	175 G30
		120 G15	150 G15	150 G15	175 G15
S3 (5-7%)	125 G40	150 G60	175 G60	200 G60	200 G60
	125 G15	150 G30	175 G30	200 G30	250 G30
S4 (8-14%)	150 G40	150 G40	150 G60	175 G60	175 G60
		100 G30	150 G30	175 G30	200 G30
S5 (15-29%)	150 G40	175 G60	175 G60	175 G60	175 G60
S6 (>30%)	150 G40	150 G40	150 G60	150 G60	175 G60

6.3.1.1 Climatic Zones Where $N < 4$ (wet)

- Where the total sealed surface width is 8 m or less, Pavement Design Chart 1 (Table 6-6) should be used. No adjustments to the road base material requirements;
- Where the total sealed surface is 8 m or more, Pavement Design Chart 2 (Table 6-7) should be used. The limit on the plasticity modulus of the road base may be increased by 20 per cent; and
- Where the total sealed surface is less than 8 m but the pavement is on an embankment in excess of 1.2 m in height, Pavement Design Chart 2 (Table 6-7) should be used. The limit on the plasticity modulus of the road base may be increased by 20 per cent.

If the engineer deems that other risk factors (e.g. poor maintenance and/or construction quality) are too high, then Pavement Design Chart 1 should be used.

6.3.1.2 Climatic Zones Where $N > 4$ (dry)

Pavement Design Chart 2, Table 67 should be used:

- Where the total sealed surface is less than 8 metres, the limit on the plasticity modulus of the road base may be increased by 40%; and
- Where the total sealed surface is over 8 metres and when the pavement is on an embankment in excess of 1.2 metres in height, the plasticity modulus of the road base may be increased by up to 40% and the plasticity index by 3 units.

Once the quality of the available materials and haul distances are known, the flow chart shown in Figure 61 and the design charts can be used to review the most economical cross-section and pavement.

When the project is located close to the border between the two climatic zones, the lower N-value should be used to reduce risks.

When the design is close to the borderline between two traffic design classes, and in the absence of more reliable data, the next highest design class should be used.

It may be more economical to use a wider cross-section in the seasonal tropical and wet climate zone and then use Pavement Design Chart 2 rather than to design a narrow cross-section and a pavement using Pavement Design Chart 1.

The design charts do not cater for weak sub-grades (CBR < 3%) and other problem soils.

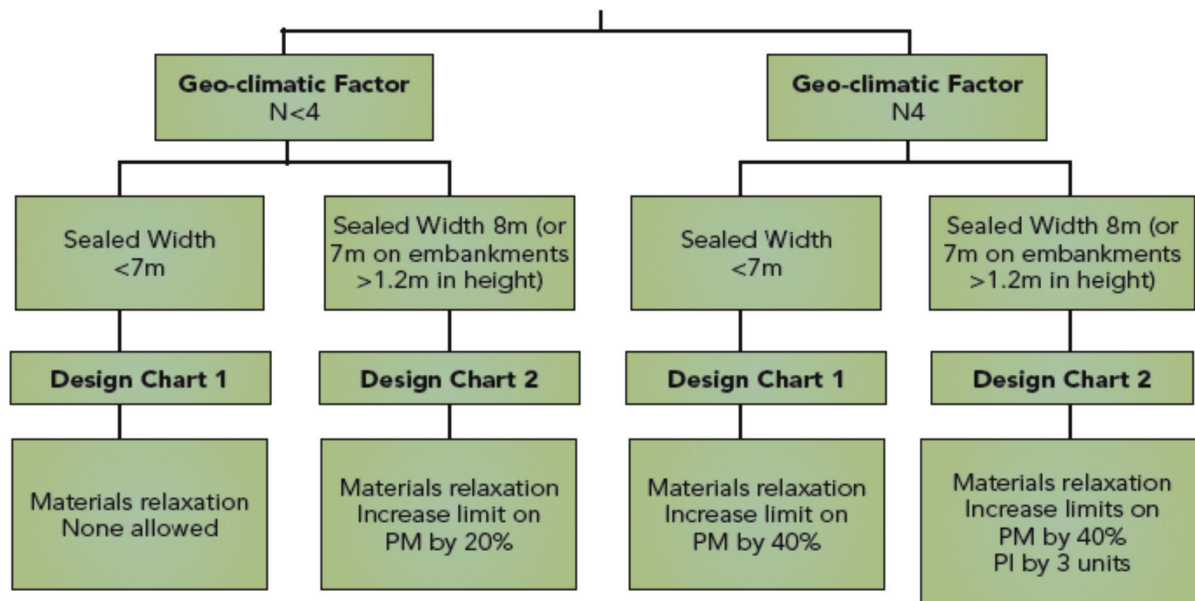


Figure 6-1 Pavement Design Flow Chart

6.3.2 DCP-CBR/SN Method

It is now widely accepted that the DCP is a tool that can be used for evaluation of pavement strength. Data from the DCP analysis can be used to assign pavement layer thicknesses and strength in terms of both the California Bearing Ratio (CBR) and Structural Number (SN).

There are currently different methods by which DCP data is used for analysis and design of pavements for low volume roads. Here we present the DCP-CBR/SN method which can easily be used with a CBR based catalogue or with the AASHTO SN Method.

The designer would normally make appropriate use of a dynamic cone penetrometer (DCP) to obtain much of the required design information, particularly a longitudinal profile of in situ strengths of the pavement layers of the existing road in terms of DN values (penetration per blow in mm/blow). The DN number is normally converted to CBR so that a diagram of CBR versus depth is obtained. The equation used to do this is based on the BS method of CBR testing and the equation is:

$$\text{Log}_{10} \text{CBR} = 2.48 - 1.057 \text{ Log}_{10} \text{DN} \quad \text{Equation 61}$$

In order to make optimum use of the existing layers the method makes use of the structural number concept (AASHTO 1993). Using this method, the difference between the structural number of the existing road and that required for the upgraded road, obtained from the catalogue of structures, defines the additional requirements for upgrading, rehabilitation or reconstruction. The flow diagram for upgrading an existing road is shown in Figure 13-1. It should be noted that the required designs are determined for every DCP test point. No averaging or percentiles are computed at this stage. Only when the final designs are completed is a percentile of the required strengthening requirements selected for implementation. This is the most precise method of analysis because there are many very non-linear relationships in pavement theory hence basic averaging at any stage is statistically incorrect and can lead to incorrect design.

For details on how to use the DCP and the UKDCP software, the reader is referred to TRL publication "Dynamic Cone Penetrometer Tests and Analysis, by Colin Jones. PR/INT/277/04" which may be obtained from www.transport-links.org for instructions on how to use the DCP and to <http://www.transport-links.org/ukdcp/index.htm> for a software for analysis of the data.

6.3.2.1 Design Approach

The DCP survey provides the thicknesses and in situ strengths of the layers of the existing road along the entire alignment. The analysis of the DCP data, preferably using the DCP program, provides the overall strength of the pavement at each test point based on the structural number approach. The flow diagram is shown in Figure 6-2.

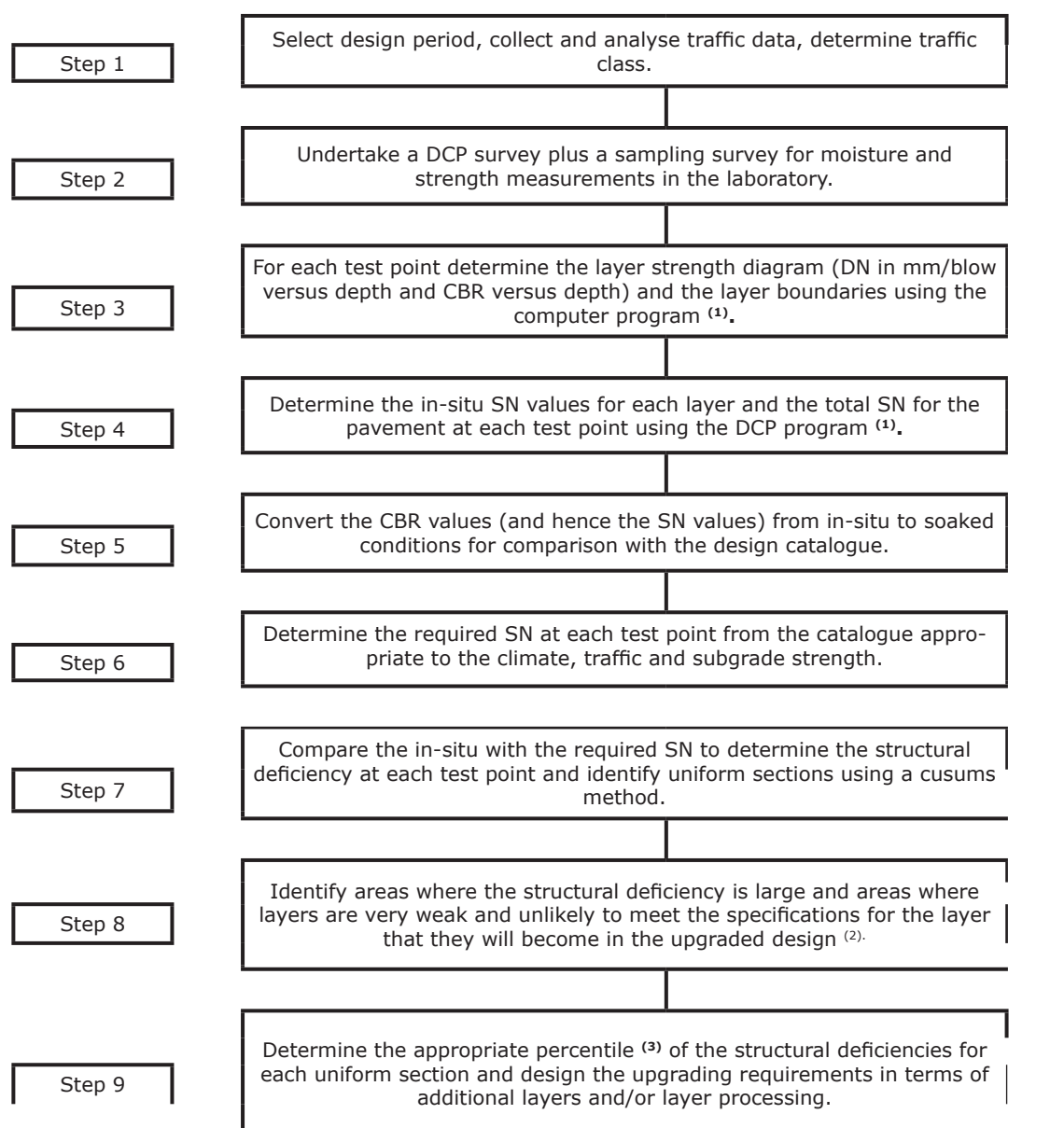
The structural number is essentially a measure of the total thickness of the road pavement weighted according to the 'strength' of each layer and calculated as follows:

$$SN = 0.0394 \sum m_i \cdot a_i \cdot h_i \quad \text{Equation 62}$$

Where:

- SN = structural number of the pavement
- a_i = strength coefficient of the i^{th} layer
- h_i = thickness of the i^{th} layer, in millimetres
- m_i = 'drainage' coefficients that modify the layer strength coefficients of unbound materials if drainage is poor and/or climate is favourable or severe.

The summation is for the number of pavement layers, n .



- Notes
- 1 These calculations can be done by hand using a spreadsheet but the DCP program makes this very straightforward.
 - 2 Areas with material that does not meet the specifications for the layers that they will normally become in the upgraded design will require some form of reconstruction (or additional strengthening layers to increase the depth of the inadequate layers in the new pavement).
 - 3 Depends primarily on traffic level.

Figure 6-2 DCP-CBR method - Flow Diagram for Upgrading an Existing Road

The individual layer strength coefficients are determined from the normal tests that are used to define the strength of the material in question e.g. CBR for granular materials, UCS for cemented materials etc. Table 68 shows typical values.

Table 6-8 Pavement Layer Strength Coefficients

Layer Type	Condition	Coefficient
Surface treatment		$a_i = 0.2$
	Default	$a_i = (29.14 \text{ CBR} - 0.1977 \text{ CBR}^2 + 0.00045 \text{ CBR}^3) 10^{-4}$
	CBR > 100%	$a_2 = 0.145$
	CBR = 100%	$a_2 = 0.14$
	CBR = 80%	$a_2 = 0.135$
	With a stabilised layer underneath	
	With an unbound granular layer underneath	$a_2 = 0.13$
	CBR = 65%	$a_2 = 0.12$
	CBR = 55%	$a_2 = 0.107$
	CBR = 45%	$a_2 = 0.1$
Bitumen treated gravels and sands	Marshall stability = 2.5 MN	$a_i = 0.135$
	Marshall stability = 5.0 MN	$a_i = 0.185$
	Marshall stability = 7.5 MN	$a_i = 0.23$
Cemented ³	Equation	$a_i = 0.075 + 0.039 \text{ UCS} - 0.00088(\text{UCS})^2$
	CB 1 (UCS = 3.0 – 6.0 MPa)	$a_i = 0.18$
	CB 2 (UCS = 1.5 – 3.0 MPa)	$a_i = 0.13$
Granular unbound sub-bases	Equation	$a_j = -0.075 + 0.184(\log_{10} \text{ CBR}) - 0.0444(\log_{10} \text{ CBR})^2$
	CBR = 40%	$a_3 = 0.11$
	CBR = 30%	$a_3 = 0.1$
	CBR = 20%	$a_3 = 0.09$
	CBR = 15%	$a_3 = 0.08$
	CBR = 10%	$a_3 = 0.065$
Cemented	(UCS = 0.7 – 1.5 MPa)	$a_i = 0.1$

Notes: 1. Unconfined Compressive Strength (UCS) is quoted in MPa at 14 days.

2. Used for low volume roads (see LVR manual)

The drainage coefficients are effectively calibration factors for the moisture regime experienced by the road and are therefore related to both climate and drainage. Values range from 0.7 for extremely poor conditions up to 1.3 for very good conditions, but the usual working range is 0.9 to 1.1. Where $N < 4$ (wet conditions) a value of m_i of 0.9 will provide a suitable safety factor. However, for a well-designed road the effects of its moisture regime or climate on the strength of the road are primarily manifest in the strength of the subgrade and so a value of 1.0 should be used for the pavement layers for relatively 'normal' conditions and a value of 1.1 for very dry conditions.

6.3.2.2 Design Procedure

To design the upgrading or rehabilitation it is first necessary to measure the structural number at each test point as indicated in Figure 6-2. The calculation of SN for design purposes is the AASHTO method which is based on the value of the soaked CBR of the layers. To convert from the in-situ values to the soaked values requires a measurement of the in-situ moisture condition, expressed as the ratio of in situ moisture content divided by the optimum moisture content, and the use of Figure 6.3. The in-situ moisture condition is obtained from the samples collected for laboratory analysis during the DCP survey. A minimum of three samples per kilometre is recommended. The relationship between soaked and in situ strength (CBR) depends on the characteristics of the materials. However, for the level of accuracy required, Figure 6-3, which is based on extensive research, is adequate.

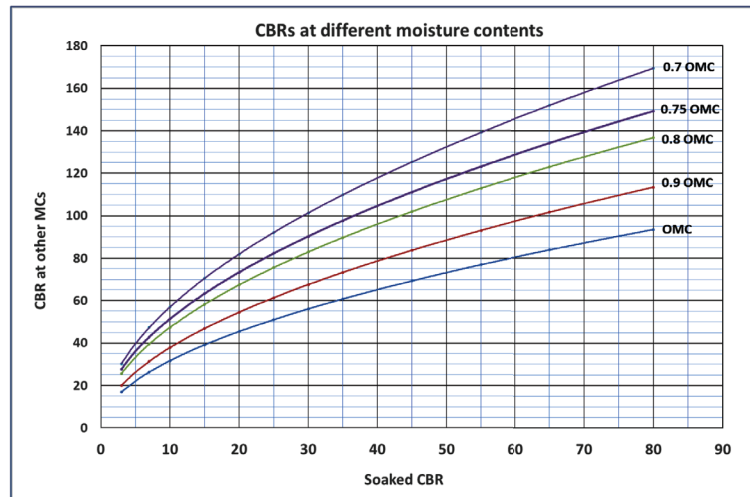


Figure 6-3 CBR at Different Moisture Contents

6.3.2.3 Modified Structural Number

The effect of different subgrades can also be included in the structural number approach. The subgrade contribution is defined as follows:

$$\text{SNC} = \text{SN} + 3.51 (\log_{10} \text{CBR}_s) - 0.85 (\log_{10} \text{CBR}_s)^2 - 1.43 \quad \text{Equation 63}$$

where

SNC = Modified structural number of the pavement

CBR_s = In-situ CBR of the subgrade

The modified structural number (SNC) has been used extensively and forms the basis for defining pavement strength in many pavement performance models. It should be used to identify the overall strength of each DCP test point in the old road if the subgrade is particularly variable.

6.3.2.4 Target Structural Numbers

When designing upgrading or rehabilitation it is necessary to determine the existing effective SN as described above at each test point and the required SN to carry the new design traffic. Table 6-9 to Table 6-10 show the target values of SN and SNC for different subgrade conditions and for different traffic levels calculated from the design charts for roads with a thin bituminous surfacing. The difference between the required structural number and the existing structural number is the deficiency that needs to be corrected.

Table 6-9 Structural Numbers for Bituminous Pavement Design Chart 1 (Wet: N<4)

	TLC1	TLC2	TLC3	TLC4	TLC5
	< 0.01	0.01 – 0.1	0.1 – 0.3	0.3 – 0.5	0.5 – 1.0
S1 (<3%)	Special subgrade treatment required				
S2 (3-4%)	1.05	1.7	1.95	2.05	2.30
S3 (5-7%)	0.95	1.45	1.70	1.85	2.1
S4 (8-14%)	0.8	1.25	1.55	1.65	1.85
S5 (15-29%)	0.7	1.0	1.25	1.35	1.5
S6 (>30%)	0.6	0.7	0.85	0.95	1.0

Note, these values exclude a contribution from the surfacing

Table 6-10 Structural Numbers for Bituminous Pavement Design Chart 2 (Dry: N>4)

Subgrade CBR	TLC1	TLC2	TLC3	TLC4	TLC5
	< 0.01	0.01-0.1	0.1-0.3	0.3-0.5	0.5-1.0
S1 (<3%)	Special subgrade treatment required				
S2 (3-4%)	1.05	1.55	1.80	2.0	2.15
S3 (5-7%)	0.9	1.35	1.55	1.70	1.95
S4 (8-14%)	0.7	1.05	1.35	1.45	1.6
S5 (15-29%)	0.6	0.85	1.05	1.1	1.3
S6 (>30%)	0.6	0.7	0.75	0.85	0.9

Note, these values exclude a contribution from the surfacing

Table 6-11 Required Modified Structural Numbers Chart 1 (Wet: N<4)

Subgrade CBR	TLC1	TLC2	TLC3	TLC4	TLC5
S2	1.1	1.75	2.0	2.1	2.35
S3	1.55	2.05	2.35	2.45	2.7
S4	1.85	2.25	2.6	2.7	2.9
S5	2.2	2.55	2.75	2.9	3.05
S6	2.5	2.6	2.75	2.85	2.9

Table 6-12 Required Modified Structural Numbers Chart 2 (Dry: N>4)

Subgrade CBR	TLC1	TLC2	TLC3	TLC4	TLC5
S2	1.1	1.6	1.85	2.05	2.2
S3	1.5	1.95	2.15	2.35	2.55
S4	1.75	2.1	2.4	2.5	2.65
S5	2.1	2.35	2.55	2.65	2.8
S6	2.5	2.6	2.65	2.75	2.8

The final step is to determine uniform sections based on the strengthening requirements using a cum-sums method. For each uniform section, the following percentiles of the strengthening requirements should be used:

- i. Median for TLC1 and TLC2;
- ii. Upper 75 percentile for TLC3; and
- iii. Upper 90 percentile for TLC4 and TLC5.

However, when the strengthening requirements are large it may be more cost effective to carry out some reconstruction and, conversely if they are small, maintenance may be all that is required. Table 6-13 is a guide to the treatments.

Table 6-13 Structural Deficiency Criteria

Structural deficiency based on appropriate percentiles	Action	Notes
0.2 or negative	Maintain with a surface treatment (e.g. a surface dressing)	A thin granular overlay can be used to correct other road defects.
0.2 – 1.2	New granular layer. The existing layers must be checked for quality (sub-base or road base). The minimum thickness of new roadbase should be 50mm.	Some localised remedial works can be expected. A surface treatment is required.
1.2 – 1.8	The existing road base is likely to be only of sub-base quality and should be checked. Additional sub-base and a new road base are required.	Some localised remedial works will be needed. A surface treatment is required.
> 1.8	The existing layers are likely to be less than sub-base quality, hence a new sub-base and road base are required. Chemically stabilising existing material should be considered.	Localised remedial treatment and a surface treatment are required.

6.4 Design Example

- 1) An old road needs to be upgraded to carry a cumulative design traffic loading of 0.3 MESA.
- 2) A DCP survey is carried out in the dry season. The data are analysed using the TRL UKDCP software or WinDCP software.
- 3) Samples of each layer of the pavement are taken for laboratory testing.
- 4) The climate zone is moderate to dry hence design Chart 2 (Table 67 and Table 610) are to be used.

The following analysis is carried out for each DCP test point.

Step 1 A typical DCP result is shown in Figure 64. The program identifies the layer boundaries automatically and outputs, for each layer, the thickness, average mm/blow, and CBR. Usually more than one sub-base layer is identified.

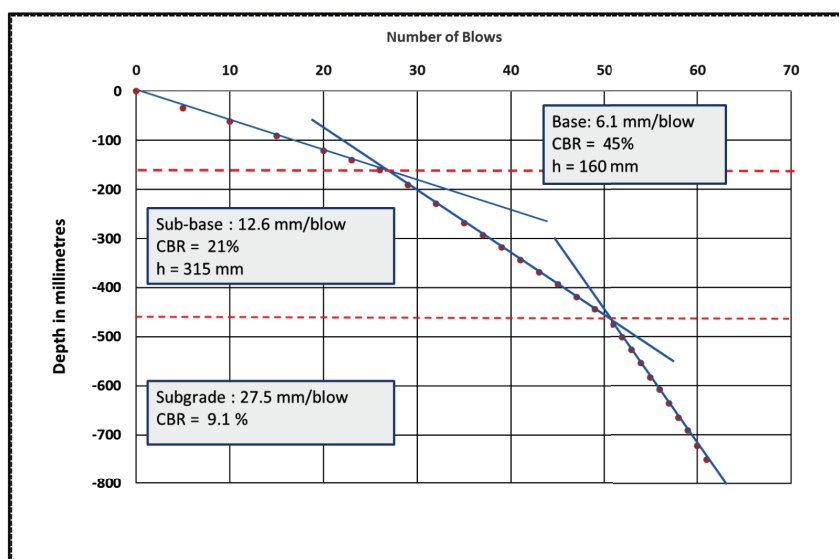


Figure 6-4 Typical DCP Test Result

Step 2 The user must define each layer as road base, sub-base, or subgrade. The program then calculates the contribution of each layer to the overall structural number. The strength coefficients are calculated automatically or Table 6-8 can be used.

Table 6-14 Initial CBRs and Strength Coefficients

Layer No	CBR %	Thickness (mm)	Depth (mm)	Position	Strength Coefficient	SN
1	45	160	160	Road base	0.10	0.63
2	21	315	475	Sub-base	0.09	1.12
	9.1	-		Subgrade		
Total		475				1.75

Step 3 These SNs are the values obtained at the in-situ conditions. For evaluation and design purposes the SN of each layer in the soaked condition is required. The user must estimate the soaked CBR values from the in-situ moisture contents measured in laboratory tests of the samples. This is done using Figure 6-3. This conversion cannot be exact because the relationships shown in the Figure depend on various material properties such as PI, hence a high level of precision is not possible, and nor is it necessary. In this example, the in-situ conditions are not extreme (in terms of wet or dry). An average in situ moisture content of OMC was obtained and used with Figure 6-3 to convert the CBRs to soaked conditions. The revised CBRs are shown in Table 6-15.

Table 6-15 Revised CBR and SN Corrected for Moisture Conditions

Layer No	CBR (%)	Thickness (mm)	Position	Revised CBR (%)	Revised strength coefficient	Revised SN
1	45	160	Road base	20	0.09 ⁽¹⁾	0.57
2	21	315	Sub-base	5	0.03	0.37
	9.1	-	Subgrade			
Total		475				0.94

Note 1 If strengthening is required this layer will become sub-base

Step 4 Using the revised CBR values, the revised SN is calculated for each layer and then summed to give the total value. This is 0.94.

Step 5 The strength of the sub-grade must also be adjusted to give an estimate of the soaked value. However, it is only necessary to identify the subgrade class. For low values of CBR, if the in-situ moisture regime is OMC the soaked value is typically half to one third of the in-situ value. If the moisture regime is dry (0.75*OMC) then the soaked value is one third to one quarter of the in-situ value (Figure 6-3). The subgrade class is therefore S2 (3 - 5% CBR).

Step 6 With the subgrade strength and the existing SN for the example DCP test point, Design Chart 2 (Table 610) is used to determine the SN required for the new road to carry the design traffic on the design subgrade. The subgrade is S2 and the required SN for 0.3 MESA is 1.8. The difference between the required SN and the existing SN (Δ SN) is

$$\Delta \text{SN} = 1.8 - 0.94 = 0.86$$

The above calculation is carried out for every test point remembering that the subgrade may not be S2 for all the DCP test points.

Step 7 The road displays a range of Δ SNs along its length. It is at this point that the final choice of uniform sections are identified based on a cum sums analysis of Δ SN. Figure 65 shows the cum sums analysis. Uniform sections are identified as shown in

Table 616.

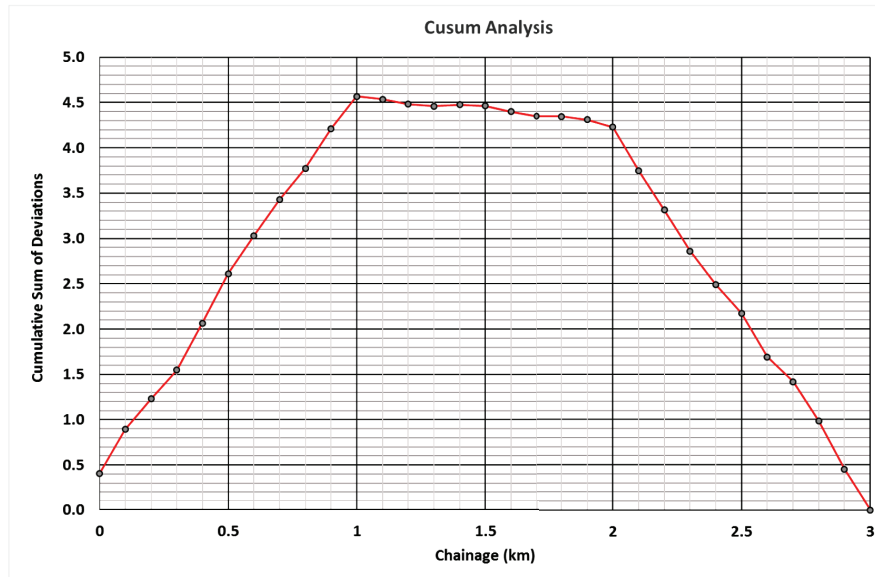


Figure 6-5 Cusum Analysis to Identify Uniform Sections

Table 6-16 Uniform Sections and Design Options

Section	Chainage	Median Δ SN required	75 th Percentile of the Δ SN ⁽¹⁾	Options
1	0.0 – 1.0	0.86	0.92	175mm of good quality graded stone aggregate
2	1.0 – 2.0	0.35	0.41	75 mm of good quality graded stone aggregate
3	2.0 – 3.0	-0.04	0.08	Just a maintenance reseal

Notes 1 The individual Δ SN values are not shown here

Step 8 For each uniform section there will still be a range of values of Δ SN but this will be much smaller now that uniform sections have been defined. The next step is to choose the appropriate percentile of those ranges for design. The percentile depends on the reliability required; the recommended values are:

- Median for LVSR 1 (0.01 MESAs max) and LVSR 2 (0.1 MESAs max).
- Upper 75 percentile for LVSR 3 (0.3 MESAs max).
- Upper 90 percentile for LVSR 4 (0.5 MESAs) and LVSR 4 (1.0 MESAs max).

Thus, for LVSR 3 (0.3 MESAs max) a 75th percentile is recommended.

Table 616 shows the median and the 75th percentiles of the Δ SNs for the three uniform sections of road. If the preferred design solution is to add an additional road base layer of good quality granular material the thickness h (mm) required is calculated as follows.

$$h = 25.4 \times \Delta\text{SN}/(\text{Strength coefficient}) = 25.4 \times 0.92/0.135 = 173\text{mm}$$

The same procedure is repeated for all the uniform sections. The full DCP data are not shown but the results of the analysis are summarised in Table 6-16.

6.5 Design of Roads with Stabilised Road bases

6.5.1 Design of Roads with Cement or Lime Stabilised Road Bases

Table 617 shows the structural designs for roads with chemically stabilized road base.

Table 6-17 Design Catalogue for Chemically Stabilised Materials

SG CBR	TLC1	TLC2	TLC3	TLC4	TLC5
	< 01	0.01-0.1	0.1-0.3	0.3-0.5	0.5-1.0
S1 (2-3%)		125 CB2	150 CB2	150 CB2	175 CB2
		150 G30	150 G30	175 CB2	175 G30
		250 G15	300 G15	300 G15	300 G15
S2 (3-4%)		125 CB2	150 CB2	150 CB2	175 CB2
		125 G30	150 G30	175 G30	175 G30
		150 G15	175 G15	175 G15	200 G15
S3 (5-7%)		125 CB2	150 CB2	150 CB2	175 CB2
		100 G30	125 G30	150 G30	150 G30
		100 G15	125 G15	150 G15	150 G15
S4 (8-14%)		125 CB2	150 CB2	150 CB2	175 CB2
		150 G30	175 G30	200 G30	200 G30
S5 (15-29%)		125 CB2	150 CB2	150 CB2	175 CB2
		100 G30	125 G30	150 G30	150 G30
S6 (>30%)		125 CB2	150 CB2	150 CB2	175 CB2

6.5.2 Emulsion Stabilised Road Bases

Table 6-18 shows the structural designs for roads with road bases stabilized with bitumen emulsion.

Table 6-18 Design Catalogue for Emulsion Stabilised Bases

Test	BSM1 (>6M ESAs)	BSM2 (<6M ESAs)	BSM3 (<1M ESAs)
P0.075			0 – 20
GM		1.2 – 2.5	0.75 – 2.7
CBR	>80	25 – 80	10 – 25
DCP (mm/blow)	<3.7	3.7 – 9.1	9.1 – 19.0
FWD (MPa)	>300	150 – 300	70 – 150
ITS dry (kPa)	>225	175 – 225	125 – 175
ITS wet (kPa)	>100	75 – 100	50 – 75
UCS (kPa)	1200 – 3500	700 – 1200	450 – 700

6.6 Design of Roads with Semi-Structural Surfacing (SSS)

Several surfacings that are used for LVRs provide a structural component to the pavement because of their thickness and stiffness. Such surfacings are relatively flexible so their structural contribution is small but if they are thick enough the contribution is worth taking into account and are usually 35-50 mm thick.

6.6.1 Cold Mix

Cold mix is a mixture of aggregate and emulsion and is a good alternative in terms of application technology. It can be applied by both machine (pavers) and labour based technology. The design

of cold mix is described in Section 3.6. Cold mix behaves in the same manner as hot mix asphalt and contributes significantly to structural strength depending on the thickness. The design specifications of pavements with cold mix asphalt are given in Table 6-19.

6.6.2 Penetration Macadam

Penetration Macadam is a series of layers of stone bound by bitumen emulsion designed as explained in Section 3.6 and is usually 40 and 50mm thick and suitable structural designs are shown in Table 6-19 where the use of a semi structural surface has reduced road base thicknesses by 50mm. This table is the equivalent of Table 6-6. A similar chart for dry areas can be compiled.

Table 6-19 Design of Roads with Semi-structural Surfacing (SSS)

SG CBR	TLC1	TLC2	TLC3	TLC4	TLC5
	< 01	0.01-0.1	0.1-0.3	0.3-0.5	0.5-1.0
S2 (3-4%)		40 SSS	40 SSS	40 SSS	40 SSS
		100 G65	100 G65	125 G80	150 G80
		125 G30	150 G30	175 G30	175 G30
		150 G15	175 G15	175 G15	200 G15
S3 (5-7%)		40 SSS	40 SSS	40 SSS	40 SSS
		100 G65	100 G65	150 G65	150 G80
		100 G30	150 G30	150 G30	150 G30
		100G15	150 G15	150 G15	150 G15
S4(8-14%)		40 SSS	40 SSS	40 SSS	40 SSS
		100 G65	100 G65	125 G65	150 G800
		125 G30	200 G30	200 G30	200 G30
S5(15-29%)		40 SSS	40 SSS	40 SSS	40 SSS
		100 G65	100 G65	125 G65	125 G80
		100 G30	150 G30	150 G30	150 G30
S6(>30%)		40 SSS	40 SSS	40 SSS	40 SSS
		100 G65	125 G65	125 G65	150 G80

6.6.3 Thick Otta Seals

A double Otta seal with a total thickness of 40mm is also a semi-structural surface and therefore Table 6-19 should be used for the structural designs of such roads.

7.0 Design of Surfacing

7.1 Introduction

Surfacing is the application of a bituminous seal or a layer of discrete elements which provides a good riding surface on a road while also protecting pavement layers below from weather elements and wear and tear from traffic. Figure 7-1 gives an illustration of typical bituminous surfacings.

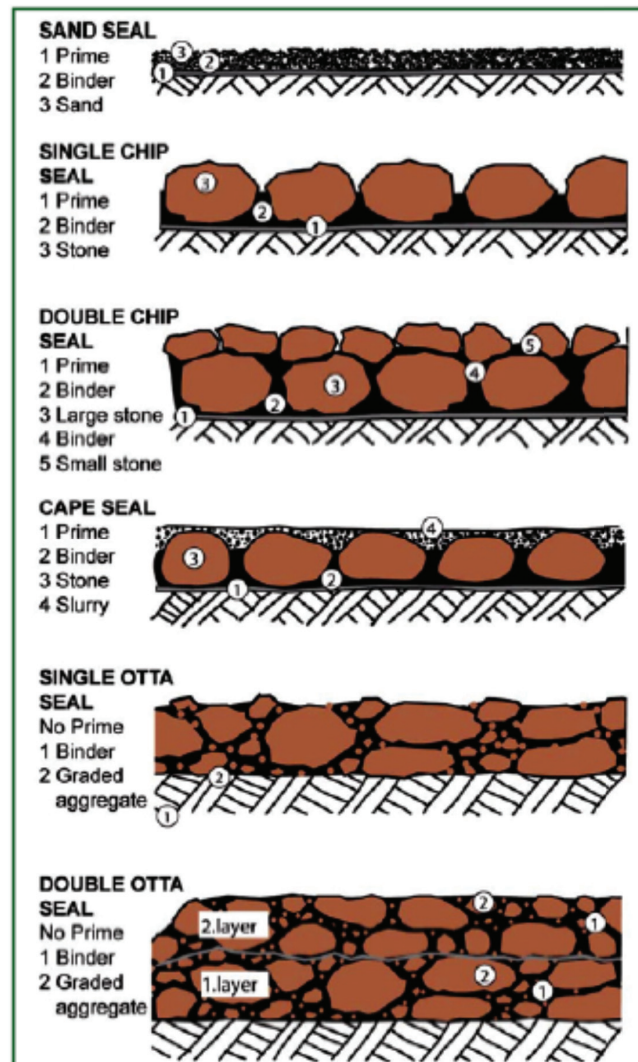


Figure 7-1 Typical Bituminous Surfacing

A bituminous surfacing is a layer of aggregate bound by bitumen. It provides a seal or impervious layer which prevents water from reaching the base. It also provides a resilient wearing and smooth riding surface.

There are many different types of surfacings and factors determining their suitability. Table 7-1 shows different surfacing types and their appropriateness for use in sealing low volume roads.

Table 7-1 Appropriateness of Surfacing Options for Low Volume Roads

Parameter	Degree	Type of Surfacing						
		SS	SIS	SCS	DCS	CS	SOS+SS	DOS
Service life Required ¹	Short							
	Medium							
	Long							
Traffic level	Light							
	Medium							
	Heavy							
Impact of traffic turning action	Low							
	Medium							
	High							
Gradient	Mild							
	Moderate							
	Steep							
Material quality	Poor							
	Moderate							
	Good							
Pavement and base quality	Poor							
	Moderate							
	Good							
Suitability for labour-based methods								
Contractor experience/capability	Low							
	Moderate							
	High							
Maintenance capability	Low							
	Moderate							
	High							

Key:

SS = sand seal, SIS = Slurry Seal, SCS = single chip seal, DCS = double chip seal, CS = Cape seal, SOS+SS = Single Otta seal + sand seal, DOS = double Otta seal

	Suitable/Preferred		Less suitable/not preferred		Not suitable/not applicable
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The durability of the different surfacing options is given in Table 7-2. These values are only indicative and the durability is dependent on traffic, environment, quality of surfacing and maintenance regime.

Table 7-2 Estimated Service Life of Low Cost Seals

Type of Surfacing	Typical Service Life (years)
Single sand seal	2-3
Double sand seal	3-6
Slurry seal	2-4
Single surface dressing	3-5
Double surface dressing	7-12
Single Otta seal	6-10
Single Otta seal plus sand seal	8-12
Cape seal (13mm + slurry)	6-10
Cape seal (19mm + double slurry)	8-14
Double Otta seal	12-16
Penetration macadam	12-20

Other factors influencing the choice of surfacing options are given in Table 7-3.

Table 7-3 Factors Influencing Choice of Surfacing Options for LVSRs

Surfacing	Applicability
Surface dressing	a) Works for both low volume and high volume roads b) Cheap c) Life cycle on low-volume road (SSD – approx. 3 to 5 years, DSD – approx. 7-12 years) d) Quick to construct
Otta Seal	a) Not suitable for very low volume roads because it needs traffic action to cure properly b) More expensive than DSD c) Difficult to construct – requires a prolonged period of rolling and curing where traffic is low approx. 2 weeks d) Long life cycle – approx. 8-12 years for single Otta seal +sand seal or emulsion based Otta Seals or double Otta seals e) Low maintenance f) Uses locally available natural gravels
Penetration macadam	a) Relatively more expensive than DSD b) Less stringent on quality of stone c) Long life 12-20 years on low-volume roads d) Rougher surface e) Usable on weak bases e.g. sand bases f) Low maintenance
Slurry seal	a) Slightly more expensive than DSD b) Slow to construct with labour but fast with plant mix c) Thick slurry (15mm) is durable (approx. 7years) and strong, thin slurry (5mm) is much less durable (approx. 4 years) d) Useful in sandy areas where there are no materials for other surfacing options
Sand seal	a) Single sand seal is not durable (approx. 2-3 years), double sand seal is more durable (approx. 3 – 6 years), thick sand seal (cold sand asphalt approx. 5-7 years) b) Useful in sandy areas where there are no materials for other surfacing options c) Cheap d) Easy and quick to construct
Stone paving	a) Suitable for short interventions b) Slow to construct c) Low cost where stone is readily available d) Low maintenance e) Low-technology for construction (constructed with labour) f) Cheaper alternative to construction of concrete slabs g) Most useful on ramps and swamps

For design purposes, the surfacings can be classified by their structural functionality and composition as presented in Table 7-4.

Table 7-4 Structural Functionality of Surfacings

	Structural Contribution	Design Considerations
1	Non-structural	These are thin surfacings which provide insignificant contribution to the structural strength of the pavement. Such surfacings include sand seals, single and double surface dressings, single surface dressings with sand seals single Otta seals, etc.
2	Semi-structural	These are thicker surfacings which provide a moderate contribution to the structural strength of the pavement and they include penetration macadam, thin cold mix asphalt, cold sand asphalt, etc.

The design of the individual surfacings takes into account basic properties of materials, Table 7-5.

Table 7-5 Material Properties and their Significance in Design of Surfacing

Parameter	Description	Significance in low volume roads design
Aggregate:		
a. Grading	This is the particle size distribution and it is critical for the strength and stability of the surfacing. Surface dressings require single size aggregate. Graded aggregate is used for Otta seals or graded aggregate seals. Fine aggregate is used for sand and grit seals and large aggregate is used for penetration macadam.	This parameter always has a significant influence on the performance of the surfacing and should always be specified.
b. Strength	Aggregate strength is the major contributor to the resilience of the surfacings. Different rocks have different competencies and aggregate strength contributes significantly to the durability of the surfacing. The strength is measured in the form of resistance to crushing, abrasion and polishing.	This parameter is very significant and the specifications should be adhered to. Appropriate aggregate strength parameters have been developed for low volume roads. The aggregate strength specifications are less stringent for Otta seals or graded aggregate seals.
c. Shape	Shape is measured in the form of flakiness or flakiness index. Elongation index is less commonly used. Flakiness is a measure of how flat and elongated the aggregate is.	Flakiness index is not very significant. The effect of high flakiness index is in the adjustment of the binder content at design stage. Flaky aggregate requires more binder in the surfacing.
d. Content	The content of aggregate is usually determined in the form of proportion by mass. However, when simpler technologies are used such as labour based technology the proportions can be converted to volume for easy of application.	This parameter is critically important in low volume roads design and should be clearly specified including tolerances.
Binder		
a. Type	There are many different types of binders and the design of surfacings should take into account binder types and their appropriateness for the environment and construction technology. Emulsions are the most suitable for labour based operations and there are 2 types i.e. cationic and anionic emulsions and the choice depends on the type of aggregate i.e. acidic and basic respectively. Hard binders such as 80/100 pen are the most appropriate for hot climates and 150/200pen made from cutting back 80/100 pen with diesel is highly durable.	This parameter is important for low volume roads. The choice of binders in the design process is critical because the cost component of the binder is usually very significant i.e. 40-60% of the total cost of surfacing. Technology choice is also a determinant factor. Emulsions are more appropriate for labour based operations because surfacing can be applied at ambient temperature
b. Quality	The quality of binder is usually specified by the manufacturers. The parameters which define the quality of binder include penetration, softening point and viscosity among many other less critical parameters. The specification of a bitumen determines its binding properties and ultimately the durability of the binder and surfacing	The quality of the binder is critical in the delivery of a good quality surfacing. Most of the low cost seals on low volume roads are thin and therefore more vulnerable to degradation by UV radiation and oxidation. The quality of binder and surfacing design can be key to reducing the rate of deteriorations leading to longer life.

Parameter	Description	Significance in low volume roads design
c. Content	Bitumen content is measured in percentage by weight of dry aggregate. This is much more important in the preparation of bituminous mixes like cold and hot mix asphalt concrete. The binder content in spray and chip type of surfacings is expressed in the form of application rates i.e. litres/m² or kg/m². In emulsions and other dilute binders bitumen content may refer to the residual bitumen in the binder mix. Emulsions are a mixture of water and bitumen and an emulsifying agent. It is important to specify the binder application rate of content taking into account the actual residual because 30 to 40% is water.	Bitumen content is one of the most important aspects of surfacing design. There is very little leeway for error in this regard. It is important to accurately determine the binder content in the surfacing and also specify the tolerances. Usually the tolerance in which should be specified is $\pm 5\%$.
d. Durability	Durability is specified indirectly using viscosity. This is a measurable parameter. This can also be more accurately determined using a rotating plate oven test which is an aging test.	For low volume roads design specifications this parameter is optional but this does not diminish its importance.

7.2 Prime Coat

Prime coat is essentially an application of low viscosity binder on a road base before the application of surfacing. Typical primes are:

- **Bitumen primes:** Low viscosity, medium curing cutback bitumen such as MC-30, MC-70, or in rare circumstances, MC-250, can be used for prime coats.
- **Emulsion primes:** Bitumen emulsion primes are not suitable for priming stabilised bases as they tend to form a skin on the road surface and do not penetrate easily into the base. However, where emulsions are to be used for priming, they should be further diluted by adding water at a volumetric ratio of 1:1 to minimise the effect of forming the film.
- **Tar primes:** Low-viscosity tar primes such as 3/12 EVT are suitable for priming road surfaces but are no longer in common use because of their carcinogenic properties which are potentially harmful to humans and the environment.

The prime coat has the following main functions:

- To provide a good bond between the surfacing and the base course. Being a low viscosity binder, it penetrates into the base upon application and this creates a crust on the upper part of the base which is rich in binder. This has the effect of aiding bonding between the bituminous surfacing and the base. Priming makes the surface of the base impermeable and prevents ingress of water which further enhances the strength of the surface of the base; and
- To provide a hard surface on which to construct the surfacing. Low viscosity of the prime is a result of dilution with high contents of solvents which are essentially volatiles. This aid rapid hardening of the prime and the surface of the based as the volatiles evaporate.

Priming is effective when the base is semi-porous and semi-permeable. When the road base has very low permeability such as cement and emulsion stabilised bases, normal priming is not recommended and fog sprays are applied instead. Fog spray is a very light application of low viscosity binders such as emulsion diluted 50:50 with water.

Table 76 gives the most commonly known primers and their application rates.

Table 7-6 Types and Application Rates of Prime

Pavement Surface	Prime	
	Grade	Rate of application (l/m ²)
Tightly bonded (light primer)	MC 70	0.6-0.7
Medium porosity (medium primer)	MC30/MC70	0.7-0.8
Porous (heavy primer)	MC30	0.8-1.1

7.3 Single and Double Surface Dressing

Surface dressing also called chip seals (single or double) consists of a spray(s) of bituminous binder followed by the application of a layer(s) of aggregate (stone chippings). The binder acts as a waterproofing seal preventing entry of surface water into the road structure while the chippings protect this film from damage by vehicle tyres.

Chip Seals can be used for a number of purposes, including:

- i. New construction (normally double surface dressings only);
- ii. Temporary by-passes or diversions (normally single surface dressings);
- iii. Maintenance resealing (normally single surface dressing);
- iv. First layer of a Cape Seal; and
- v. First layer of a single surface dressing with sand capping.

The design methods for both single and double chip seals are presented in Overseas Road Note 3 (2nd edition): A guide to surface dressing in tropical and sub-tropical countries. The design is based on the concept of partially filling the voids in the covering aggregate. This is controlled by the natural orientation of the chippings as they lie on the road surface with their 'least dimension' in the vertical direction. Thus, the Average Least Dimension (ALD) of the chippings is the parameter that mainly determines how much bitumen is required. Corrections to the spray rate need to be subsequently carried out to take account of site conditions as described in the guide. These conditions include traffic level, hardness of existing road surface (controlling embedment of the chippings), shape and condition of chippings, downhill or uphill road gradient, grade of bitumen, and climate.

Typical constituents for chip seals are:

- i. Binder: The bituminous binder can consist of any of the following:
 - a) Emulsion – Spray grade anionic (A_n60) and cationic (K_n65 or K_n70);
 - b) 80/100 or 150/200 penetration grade bitumen preferably cutback with diesel;
 - c) MC 3000 grade cutback bitumen;
 - d) Modified binders (polymer modified and bitumen rubber); and
 - e) Foamed bitumen.
- ii. Aggregate: The aggregate for a Chip Seal shall be durable and free from organic matter or any other contamination. Typical grading requirements for Chip Seals are given in Table 7-7.

Table 7-7 Aggregate Requirements for Bituminous Chip Seals

Test Sieve (mm)	Nominal Size of Aggregates (mm)			
	20	14	10	7
25	100	-	-	-
20	85-100	100	-	-
14	0-30	85-100	100	-
10	0-5	0-30	85-100	100
6.3	-	0-5	0-30	85-100
5	-	-	0-5	0-40
2.36	-	-	-	0-5
0.425 (fines)	< 0.5	<1.0	< 1.0	< 2
0.075 (dust)	< 0.3	< 0.5	< 0.3	< 1
Max. Flakiness Index	20	25	25	30
TFV (dry)	AADT >1000vpd Min =120kN		AADT <1000vpd Min =80kN	
TFV (wet)	Minimum 75% of dry 10% FACT			

For planning purposes, typical binder and aggregate application rates for single bituminous Chip Seals are given in Table 7-8

Table 7-8 Specifications for Total Binder Application Rates for DSD

Traffic (vehicles/lane/day)	Total Binder Application Rate (l/m ²)	
	20/10 mm	14/7 mm
<25	2.6	2.25
25-75	2.5	2.15
>75	2.35	2.05

These apply in situations where the surfacing stone meets conventional specifications as in table 7-7. In situations where the materials are marginal the following design procedures should be mandatory.

7.3.1 Design Procedure for Surface Dressing

Laboratory tests

- Sample surfacing stone from the selected quarry or quarries and carry out laboratory tests. Compare the laboratory tests with the specifications given in Table 7-9.

Table 7-9 Specification for Surfacing Aggregates

Tests	Specifications
Grading	Table 7-9. The stone should be single size. Graded surfacing stone may be used on low-volume roads and the 50 percentile value read from the grading curves should be used instead of the nominal maximum size.
Aggregate crushing value (ACV)	≤ 25 (≤ 36 is permissible for low-volume roads)
TFV (alternative to ACV)	Not less than 80kN
Flakiness	< 36 for 7mm stone < 25 for 14mm and 20mm stone (< 30 is permissible for very low-volume roads)
Elongation	≤ 40 (but not critical for low volume roads)

ii. Calculate application rates.

- a. Determine the Average Least Dimension (ALD) from the chart (Figure 7-2) using the value of the measured flakiness index and the nominal size value obtained from the sieve analysis. The intercept of ALD line and a straight line drawn from the aggregate size scale to the flakiness scale should be read as the ALD value of the surfacing stone. Alternatively place a random sample of 100 stones on a flat table. The ALD is their vertical height measure from their most stable face. Measure this for each stone with callipers and take the average;

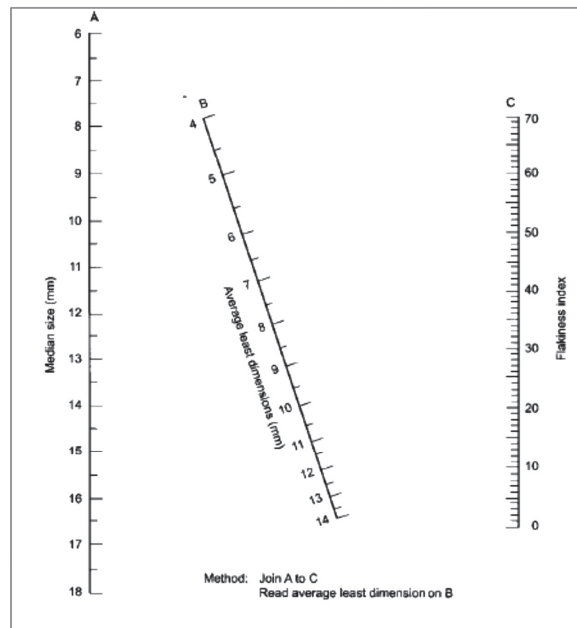


Figure 7-2 Determination of Average Least Dimension (ALD)

- b. Determine the weighting factor, the sum of the individual factors given in Table 7-10;
- c. Determine the binder application rate using the following formula

$$R = 0.625 + (0.023 \times F) + [0.0375 + (0.0011 \times F)] \times \text{ALD}$$

Where F = Overall weighting factor
 ALD = Average least dimension
 R = Rate of binder application in kg/m²

This formula is correct for MC3000 grade bitumen. Adjustment factors for different cut-back bitumens, penetration grade bitumens and emulsions are required (see TRL's ORN 3 for a more detailed description);

- d. Calculate the application rate for the chippings or surfacing stone,

A rough estimate of the application rate for the chippings can be obtained using the following formula assuming the density of loose aggregate to be approximately 1.35 kg/litre.

$$\text{Chipping application rate} = 1.364 \times \text{ALD}$$

The weighting factor, F, is obtained from Table 7-10.

Table 7-10 Determination of Weighting Factor (F)

Description	Factor (F)
Traffic vehicles/lane/day	
Very light (0 – 20)	+8
Light (20 – 100)	+4
Medium light (100 – 250)	+2
Medium (250 – 500)	0
Existing surface	
Untreated or primed	+6
Very lean bituminous	+4
Lean bituminous	0
Average bituminous	-1
Very rich bituminous	-3
Climatic conditions	
Wet and cold	+2
Tropical (wet and hot)	+1
Temperate	0
Semi-arid (hot and dry)	-1
Arid (very hot and very dry)	-2
Type of chippings	
Round/dusty	+2
Cubical	0
Flaky	-2
Pre-coated	-2

Conversions from hot spray rates in volume (litres) to tonnes for payment purposes must be made for the bitumen density at a spraying temperature of 180°C. For planning purposes, a hot density of 0.90 kg/l should be used until reliable data for the particular bitumen is available.

7.3.2 Surface Dressing using Emulsions

Specifications for the single and double surface dressing using emulsion are given in Table 7-11.

Table 7-11 General Specifications for Surface Dressing using Emulsions

Surfacing type	Aggregate	Binder type	Application rate of aggregate	Application rate of binder
Single surface dressing	Acidic aggregate: granite, quartz, dolerite, etc.	Cationic emulsions K1-60, K2-60 and K3-60, or K3-70	14mm aggregate – 12L/m ²	1 st application before spreading aggregate 0.8 – 1.0L/m ² 2 nd application after spreading aggregate – 1.2 – 1.4L/m ²
	or Basic aggregate: calcrete, limestone, etc.	or Anionic emulsions K1-60, K2-60 and K3-60, or K3-70	10mm aggregate – 8L/m ²	1 st application before spreading aggregate – 0.8 – 1.0L/m ² 2 nd application after spreading aggregate – 1.2 – 1.4L/m ²
Double surface dressing	Acidic aggregate: granite, quartz, dolerite, etc.	Cationic emulsions K1-60, K2-60 and K3-60, or K3-70	Tack coat: 20mm aggregate – 14L/m ²	1 st application before spreading 19mm aggregate – 1.2 – 1.4L/m ²
	Basic aggregate: calcrete, limestone,	Anionic emulsions K1-60, K2-60 and K3-60, or K3-70	Seal Coat: 14mm aggregate – 12L/m ² or 10mm aggregate – 8L/m ²	Application after spreading 14mm or 10mm aggregate – 2.0 – 2.2L/m ²

Note:

The construction procedure involves the application of binder on primed base followed application of 20mm stone for tack coat. After rolling the tack coat, 14mm or 10mm stone is applied and this is followed by the application of emulsion for the seal coat.

Where necessary the design procedure given in Section 7.3.1 should be followed and the application rate of binder shall be based on residual binder when the emulsion breaks.

7.4 Single Surface Dressing with Sand Capping

Single surface dressing is designed in the manner and specifications given above (Section 7.3) for both hot and emulsion binders. The application of sand capping improves the seal through a dense mix of sand and binder on top of the surface dressing. This further reduces permeability and penetration of UV light into the surfacing and also minimising the loss of volatiles. This keeps the binder soft for longer which in turn prolongs the life of the binder and surfacing.

Specifications for single surface dressing with sand capping are given in the Table 7-12.

Table 7-12 Specifications for Single Surface Dressing with Sand Capping

Type of surfacing	Aggregate	Type of binder	Application rate of aggregate	Application rate of binder
Single surface dressing	14mm for machine crushed stone	Emulsion K1-60, K2-60 and K3-60	12- 14L/m ²	Application before spreading aggregate -0.8-1.0L/m ²
	10-14mm for hand crushed stone		13-15L/m ²	
	14mm for machine crushed stone	MC3000 or 150/200 pen	12-14L/m ²	Application before spreading aggregate -0.8-1.0L/m ²
	10-14mm for hand crushed stone		13-15L/m ²	
Sand capping	Nominal maximum size of 5 - 7mm	Emulsion K1-60 K2-60 and K3-60	7-9L/m ²	Application on layer of SSD before application of sand – 1.6-1.8L/m ²
	Nominal maximum size of 2-3mm	MC3000 or 150/200 pen		Application on layer of SSD before application of sand – 1.6-1.8L/m ²

7.5 Otta Seals/Graded Aggregate Seals with Hot Binders

An Otta Seal is a bituminous surfacing comprising of a mixture of graded aggregates ranging from natural gravel to crushed rock with relatively soft (low viscosity) binder, with or without a sand cover seal. Otta seals can be single or double (two layers). Otta Seals can be used for a variety of purposes:

- New construction: Single Otta seal with sand seal or double Otta Seals is recommended;
- Temporary seal: Normally single Otta seal for diversions, haul roads, temporary accesses, etc will be appropriate; and
- Maintenance reseals: A single Otta seal will normally be used.

The Otta seal is applied to a previously constructed and approved primed surface in accordance with specifications and in conformity with the lines and typical cross sections shown on the drawings, or as directed by the Engineer.

The design of an Otta Seal relies on an empirical approach in terms of the selection of both an appropriate type of binder and an aggregate application rate. Full details of the design methods are given in the *Botswana Guideline No. 1: The Design, Construction and maintenance of Otta Seals (1999)*.

Bitumen

The choice of binder in relation to traffic and aggregate grading is given in Table 713.

Table 7-13 Choice of Otta Seal Binder in Relation to Traffic and Grading

AADT at time of construction	Type of bitumen					
	Open Grading	Medium Grading	Dense Grading			
>1000	N/A	150/200 pen grade	M	C	3	0 0 0
			MC800 in cold weather			
100-1000	150/200 pen grade	150/200 pen grade in cold weather	M	C	3	0 0 0
			MC800 in cold weather			
<100	150/200 pen grade	MC 3000	MC 800			

In the shaded cells of Table 7 – 13 are the preferred binders in relation to traffic.

- The binder should consist of cut-back bitumen. It should be free from water, show no separation prior to use, and conform to all the requirements for Grade MC-3000 medium-curing cutback bitumen as specified in AASHTO M81. 150/200 penetration grade bitumen or MC 3000 cutback grade bitumen should be used in warm weather. In cold weather, when night temperatures are likely to fall below 10°C, MC 800 cutback grade bitumen may be used or alternatively 150/200 penetration grade bitumen may be cutback with power paraffin to the appropriate viscosity range as directed by the Engineer;
- Alternative types of bituminous materials may be used, where the Contractor can prove its satisfactory use on similar projects and demonstrate its performance to the Engineer's approval;
- Each consignment brought to site should be accompanied by the manufacturer's certificate and a test report stating the shipment number, date of shipment, net weight, and results of the tests specified;
- Certificates and samples for each consignment of bitumen materials should be submitted to the Engineer at least 28 days before the Contractor intends to use the materials; and
- No bitumen material, other than that represented by certificates, test results and samples submitted to the Engineer, should be used.

Table 7-14 gives the recommended hot spray rates for primed base courses.

Table 7-14 Nominal Binder Application Rates for Otta Seal

Type of Otta Seal	Open	Medium	Dense	
			AADT < 100	AADT > 100
Double				
1 st layer	1.7	1.8	1.8	1.7
2 nd layer	1.6	1.4	2.0	1.9
Single with sand cover				
1 st layer	1.7	1.8	2.0	1.9
Fine sand	0.8	0.7		0.9
Crusher dust/Coarse river sand	0.9	0.8		0.7
Single	1.8	1.9	2.1	2.0
Maintenance reseal	1.7	1.8	2.0	1.8

The following points should be noted with regard to the binder application rates:

- Hot spray rates lower than 1.6 l/m² should not be allowed;
- Binder for the sand capping shall be MC 3000 for crusher dust or coarse river sand and MC 800 for fine sand; and
- Where the aggregate has a water absorbency of more than 2%, the hot spray rate should be increased by 0.3 l/m².

Bitumen Additives

Bitumen additives can be used, if and when directed by the Engineer. Prior approval must be obtained from the Engineer regarding the type and purpose of additive to be used.

Aggregates

- i. Aggregates should be screened natural gravel, crushed rock, crushed stone or crushed gravel of uniform quality;
- ii. The cover material requirements are less stringent than for a double chip seal and should be graded. Aggregates should be hard, durable, and rounded or cubical in shape. Some cut faces in gravel material would be beneficial. As-dug gravels should be screened to remove oversized material and excessive fines. Aggregates should be free from harmful, deleterious or objectionable material such as organic matter, and from excess of flat and elongated pieces, and be dry;
- iii. The preferred maximum particle size is 16mm, but 19mm can be accepted in the first seal where a double seal is to be constructed;
- iv. The amount of fines ($< 0.075\text{mm}$) should not exceed 10% (or an absolute maximum of 13%);
- v. Aggregates should be of such a nature that when thoroughly coated with bitumen material proposed for the work, the coating will not be removed upon contact with water;
- vi. The aggregate should meet the grading requirements in Table 7-15 and the specifications in Table 7-16;
- vii. If satisfactory tests and service records of the aggregates are not available, the Engineer may require that other tests be carried out to determine the acceptability for Otta seal;
- viii. The grading to be used is indicated on the drawings, or as directed by the Engineer;
- ix. Sources of aggregate materials should be selected by the Contractor and be subject to acceptance by the Engineer;
- x. Acceptance of source should not mean approval of materials from that source; and
- xi. Use of aggregates for surface treatment is subject to testing in conformity with these Specifications and approval by the Engineer.

Table 7-15 Grading Specifications for Otta Seal Aggregate

AASHTO Sieve (mm)	Overall requirement	Preferred Open Grading AADT < 100	Preferred Medium Grading 100 < AADT < 1,000	Fine Grained Otta seal for LVRs (nominal max. size 13 mm)
	% passing	% passing	% passing	% passing
25	100	100	100	
19	100	100	100	
16	80 - 100	80 - 100	84 - 100	100
13.2	52 - 100	52 - 82	68 - 94	55 - 100
9.5	36 - 98	36 - 58	44 - 73	33 - 70
6.7	20 - 80	20 - 40	29 - 54	20 - 52
4.7	10 - 70	10 - 30	19 - 42	12 - 40
2.0	0 - 48	0 - 8	3 - 18	5 - 25
1.18	0 - 38	0 - 5	1 - 14	2 - 18
0.425	0 - 25	0 - 2	0 - 6	1 - 15
0.075	0 - 10	0 - 1	0 - 2	0 - 13

Table 7-16 Specifications for Otta Seal Aggregate

Test	Requirement
Los Angeles Abrasion	Max 35%
Sodium Sulphate Soundness	Max 10%
Magnesium Sulphate Soundness	Max 12%
Flakiness Index	Max 30%
Wet and dry strength ratio	Min 60%
Water Absorption	Max 1%

Table 7-17 gives the aggregate application rates for Otta Seals.

Table 7-17 Nominal Otta Seal Aggregate Application Rates

Type of Seal	Aggregate Application Rates (m ² /m ³)		
	Open Grading	Medium Grading	Dense Grading
Otta Seals	63-77	63-77	50-63
Sand Cover Seals	83-100		

The following points should be noted with regard to the aggregate application rates:

- i. Sufficient amounts of aggregate should be applied to ensure that there is some surplus material during rolling (to prevent binder pick-up) and through the initial curing period of the seal; and
- ii. Aggregate embedment will normally take about 3 – 6 weeks to be achieved where crushed rock is used, after which any excess aggregate can be swept off. Where natural gravel is used the initial curing period will be considerably longer (typically 6 – 10 weeks).

7.6 Emulsion Based Otta Seals/ Graded Aggregate Seals

Emulsion based Otta seals or graded aggregate seals involve the application of a layer or layers of graded aggregate bound by bitumen emulsion sprayed in split application. All types of emulsion i.e. K1-60, K2-60 and K3-60 can be used. However, if the aggregate has too much fines or dust, K1-60 should not be used because it will set before it penetrates through the layer of aggregate.

The aggregate may be crushed stone (machine or hand crushed) or natural aggregate sieved from naturally occurring gravels. Natural aggregate includes:

- i. Quartzite – this type of aggregate consists mainly of quartz with small quantities of feldspar and other stones. Quartz is very hard and durable and it is mostly found in rounded and cubical shapes with small quantities of flat or flaky stones. Natural quartz aggregate may have significant sand or silt fractions and the fines should be minimised by sieving the aggregate on 5mm sieves;
- ii. Laterite/murram aggregate – the aggregate is derived from sieving laterite or murram gravel. The aggregate is usually rounded and nodular. Coagulated laterite should be crushed or removed;
- iii. Calcrete/limestone – the aggregate is usually found in the form of nodules with rounded shapes. When partially machine or hand crushed limestone aggregate tends to be flaky; and
- iv. Basaltic, cinder, granitic and other aggregate – the aggregate tends to contain significant fractions of weak to very weak aggregate.

The content of fines is usually high and to cater for the high dust content a 10% increase in the application rate of binder is required.

The aggregate should be assessed through aggregate quality tests such as Aggregate Crushing Value (ACV), Aggregate Impact Value (AIV) and TFV.

Table 7-18 shows the specifications for emulsion based graded aggregate seals/ Otta seals.

Table 7-18 Specifications for Emulsion Based Graded Aggregate Seals / Otta Seals

Type of surfacing	Aggregate	Type of binder	Application rate of aggregate	Application rate of binder
Emulsion based single graded aggregate seal/ single Otta seal	Nominal maximum size of stone - 13mm	Emulsion K2-60 and K3-60 (K1-60 possible but not recommended)	16L/m ²	1 st application before spreading aggregate – 0.8L/m ² 2 nd application after spreading aggregate – 1.8L/m ² 3 rd application 2 nd layer of aggregates – 1.0L/m ²
Emulsion based double graded aggregate seal/ double Otta seal	1 st layer: Nominal maximum size of stone - 19mm	Emulsion K2-60 and K3-60 (K1-60 possible but not recommended)	16L/m ²	1 st application before spreading 19mm (max) aggregate for 1 st layer – 0.8L/m ² 3 rd application after 2 nd layer of aggregates – 1.0 L/m ²
	2 nd layer: Nominal maximum size of stone - 13mm		13L/m ²	1 st application before spreading aggregate – 1.6L/m ² 2 nd application after spreading aggregate for 2 nd layer – 1.6L/m ² 3 rd application after 2 nd layer of aggregates – 1.0 L/m ²

7.7 Cold Mix Asphalt

Cold mix asphalt is a mixture of aggregate and emulsion. It is easy to produce, aesthetically appealing and durable. Thin cold mix is semi-structural and thick cold mix is structural. Graded aggregate is used in the production of cold mix asphalt. The aggregate may be machine or hand crushed and fines may be added to make the asphalt more stable and dense.

The emulsion K2-60 and K3-60 should be used in the production of cold mix. However slow setting K3-60 is the most preferred because it allows adequate time for mixing before it starts to break and set. The asphalt can be mixed mechanically using concrete mixers or mixing plants and manually by labour.

Plant mixing and use of concrete mixers is not feasible when K1-60 is used because it breaks and sets rapidly and clogs up machines and only in-situ mixing by labour is possible. It is very difficult to mix the asphalt once the emulsion has started breaking and K1-60 is therefore not desirable for labour based operations.

The specifications for cold mix are given in Table 7-19.

Table 7-19 Specifications for Cold Mix Asphalt

BS Sieve size (mm)	Coarse	Fine
20	100	
14	80 – 100	100
5	54 – 72	62 – 80
2.36	42 – 58	44 – 60
1.18	34 – 48	36 – 50
0.6	26 – 38	28 – 40
0.3	18 – 28	20 – 30
0.15	12 – 20	12 – 20
0.075	6 – 12	6 – 12
Bitumen Content (80/100) by weight of aggregate	5 – 7%	5.5 – 7.4%
Thickness (mm)	25 – 35	20 – 30

7.8 Sand Seals

Sand seal consists of a spray of bitumen followed by the application of a coarse, clean river/ lake sand or quarry sand as aggregate. This surfacing is used on low-volume roads, especially in drier regions, but can also be used for maintenance resealing, or for temporary by-passes or diversions.

For new construction, two layers are usually specified as single layers tend not to be durable. There is an extended curing period (typically 8 – 12 weeks) between the first and second seal applications to ensure complete loss of volatiles from the first seal and thus minimise the risk of bleeding. The period of curing is also dependent on the volume of traffic.

There are no formal methods for the design of sand seals.

Typical constituents for sand seals are:

- i. Binder - The following grades of binder are used:
 - a. MC-800 cut-back bitumen
 - b. MC-3000 cut-back bitumen
 - c. Spray-grade emulsion (60 or 65% or 70% of net bitumen)
 - d. 150/200 penetration grade bitumen
- ii. Aggregate: The grading of the sand may vary, but the grading requirements of Table 7-20 must be recommended.

Table 7-20 Grading of Sand for use in Sand Seal

Sieve size (mm)	Percentage by mass passing through sieve (river Sand)	Crusher dust
10	100	100
5	85 – 100	85 – 100
1.18	20 – 60	20 – 80
0.425	0 – 30	-
0.3	0-15	-
0.15	0-2	0 – 30
Sand equivalent (%) 35 minimum		

For planning or tender purposes, typical binder and aggregate application rates for sand seals are shown in Table 7-21.

Table 7-21 Binder and Aggregate Application Rates for Sand Seals

Application	Hot spray rates of MC3000 cut back bitumen (l/m ²)	Aggregate application rate (m ³ /m ²)
Double sand seal used as a permanent seal	1.2 – 1.4 per layer	0.01 – 0.012 per layer
Single sand seal used as a cover over an Otta seal or chip seal 0.8 – 1.0		0.01 – 0.012
Single seal used as a maintenance remedy on an existing road	1.0 – 1.2	0.01 – 0.012

7.9 Cape Seal

A Cape Seal is a combination of a single chip seal with a slurry seal on top. The design is similar to that for a Chip Seal and Slurry Seal as described above.

Typical constituents for Cape Seals are:

- Binder - A variety of binder types may be used for constructing a Cape Seal; and
- Aggregate - The same requirements are required as for Chip Seals and Slurry Seals.

For planning purposes, typical binder and aggregate application rates for single Chip seals are as shown in Table 7.22.

Table 7-22 Binder and Aggregate Application Rates for a Cape Seal

Nominal size of aggregate (mm)	Nominal rates of application for tendering purposes	
	Binder (litres of net cold bitumen per m ²)	Aggregate (m ² /m ³)
14	0.6	110
20	1.1	75

7.10 Penetration Macadam

Penetration macadam consists of layers of broken or crushed stones of up to 63mm, interspersed with applications of heated or cold bitumen to grout and seal the surface. It is laid as a waterproof surface on a previously prepared road base. The effect is to achieve a matrix of keyed stones grouted and sealed with bitumen to a depth of about 50 – 63mm.

The process requires only a basic bitumen heater-distributor and an 8-10 tonne deadweight roller. Labour based methods can be used to place and spread the materials.

Applications of bitumen are high at 7 – 9 kg/m². Other types of bitumen surface can achieve a durable waterproof surface with lower bitumen application rates.

Stone Aggregate

The aggregate may be machine-crushed or hand-broken fresh material that may be quarried rock, natural granular material such as rocks, gravel or boulders. The material shall be single sized, separated by screening. After crushing/breaking, the material should be angular in shape. Flat chippings should not be more than 10% by weight. These are defined as chippings with length + width being more than 6 times thickness.

Three stone applications are made:

- First application: 40/60mm, or 50/70mm aggregate;
- Second application: 10/20mm aggregate; and
- Third application: 5/10mm aggregate.

For low volume roads, the configuration is slightly different. The conventional penetration macadam surfacing consisting of three layers is expensive because the amount of bitumen required is high. Two layers are sufficient for low volume roads.

Table 7-23 shows the specifications for penetration macadam for low volume roads.

Table 7-23 Specifications for Penetration Macadam

Type of surfacing	Aggregate	Type of binder	Application rate of aggregate	Application rate of binder
Penetration Macadam	First layer: 20-40mm stone	Emulsion K2-60 and K3-60	26L/m ²	1 st application before spreading aggregate 20-40mm -0.8-1.0L/m ²
	Second layer: 6-14mm stone		16L/m ²	2 nd application after spreading aggregate for 1 st and 2 nd layer 2.0-2.5L/m ²
	First layer: 20-40mm stone	MC3000 or 150/200 pen	26L/m ²	1st application before spreading aggregate – 1.0-1.2L/m ²
	Second layer: 6-14mm stone		16L/m ²	2nd application after spreading aggregate for 1st layer 2.0-2.5L/m ²

7.11 General Guide for Selection of Surfacing Options Based on AADT

The sections above have provided the necessary information on design guidelines and standards for different surfacing options. The final choice of the most appropriate surfacing option for a particular road or section of roads is dependent on the following major aspects:

- Available materials: Low cost surfacing technology is premised on reduction of cost through use of locally available materials and this is a major influencing factor for the choice of surfacing for a particular project; and
- Appropriateness of the surfacing for the traffic level (AADT): Not all surfacings work for all the three categories of traffic (i.e. LVSR 1, 2 and 3). The appropriateness depends on the strength and durability which then influence the service life of the surfacing and maintenance demand.

Table 7-24 gives general guidance on the appropriateness of the different surfacing options for the different categories of low volume roads.

Table 7-24 Guide for Selection of Surfacing Options for LVSR Categories

Surfacing option	LVSR1 25<AADT≤150	LVSR2 150<AADT≤300	LVSR3 300<AADT≤1000
Double sand seal	X		
Double grit seal	X		
Single surface dressing	X		
Single surface dressing with sand capping	X	X	X
Double surface dressing	X	X	X
Single Otta seal	X	X	
Single Otta seal with sand capping	X	X	X
Single Otta seal with slurry capping	X	X	X
Cape seal	X	X	X
Penetration macadam	X	X	X
Cold mix asphalt	X	X	X
Emulsion based single Otta seal	X	X	
Emulsion based double Otta seal	X	X	X

8.0 Environmental and Social Safeguards

8.1 Introduction

Roads are generally constructed to improve the economic and social welfare of people using them or those they serve. The development of the LVSRs whether newly constructed or maintained inevitably brings changes and/or modifications to the natural environment within the project areas. This is in form of environmental or social impacts which may be positive or negative.

In the past, the development of roads mainly focused on the potential social benefits resulting from the provision of improved road facilities. However, there are environmental and social problems that result from improving the roads, which may receive little or no attention largely because they are considered unimportant or heavily affect the cost of the project. This section considers environmental and social issues that need to be addressed as a result of providing LVSRs. The Social Safeguards are mainly guided by the MoWT guidelines and policy statements for HIV/AIDs, Gender, Occupational Health and Safety and Persons with Disabilities (PWDs) of 2008.

The Environmental Safeguards on the other hand, are guided by the National Environment Policy (1994), Climate Change Policy (2015) and the National Environment Act (2000).

8.2 Environmental Safeguards

The planning and construction of LVSR shall be in compliance with the Statutory Regulations in force in Uganda regarding environmental management and climate change considerations. These shall be incorporated into the design, tender and contract documents. Also works executed under force account schemes shall conform. To ensure compliance with all environmental safeguards, the implementing road entities, consultants and contractors shall liaise with the responsible national and local environmental management authorities. This shall be achieved through preparation and submission of the relevant environmental management plans to the relevant authorities, which shall in turn review and approve them. Many environmental safeguards are highlighted in several documents including the Uganda Specifications for Roads and Bridges 2005 among others. These in addition to others are further outlined below.

8.2.1 Land Preservation and Re-vegetation

The construction of LVSR may lead to destruction of the natural landscape of a given area. This results from construction operations such as clearing natural vegetation, borrowing and/or extraction of natural material. The extent of destruction varies from one activity to another. These have to be reinstated or re-vegetated in order to preserve the natural landscape. Activities of particular concern in this regard include construction of new roads or detours, borrowing material, quarrying and location of construction facilities like site camps.

8.2.1.1 Construction Roads

The construction of main roads will often require establishment of detours and construction roads to be used by the construction equipment. The location, alignment, and grading of these roads shall be subject to approval of the supervising Engineer. When no longer required by the Contractor, construction roads shall, if not required, be restored to the original contour and made impassable to vehicular traffic. The surfaces of such construction roads shall be scarified as needed to provide a condition which will facilitate natural revegetation, provide for proper drainage, and prevent erosion.

8.2.1.2 Borrow Pits and Quarries

Materials for construction of pavement layers and bituminous surfacings are obtained from borrow-pits and quarries respectively. Extraction of materials if not well managed causes destruction of the natural landscape and often leads to several impacts including: permanent loss of natural resources, modification of the natural drainage, increased soil erosion and siltation of watercourses by disturbance of soil and contamination of water and soil by fuel and oil spillage among others.

Extraction of materials from quarries and borrow areas must be well managed. Provisions should be made in the tender and contract documents for their management. On completion of extraction of the materials, all quarry sites, borrow pits and areas used for the disposal or storage of surplus materials shall be reinstated by landscaping including the replacement and spreading of topsoil as directed by and to the satisfaction of the Engineer. Steep quarry walls shall be fenced to protect livestock and people falling in the quarry.

8.2.1.3 Location of Construction Facilities

Where they are required, the Contractor's workshops, office, and yard area shall be located and arranged in a manner to preserve trees and vegetation to the maximum practicable extent. On abandonment, all temporary buildings, including concrete footings and slabs, and all construction materials, equipment and debris shall be removed from the site. The area shall be re-graded, as required, so that all surfaces drain naturally, blend with the natural terrain, and are left in a condition that will facilitate natural re-vegetation, provide for proper drainage, and prevent erosion.

8.2.2 Temporary Soil Erosion Control

Road construction works will often lead to destabilization of existing soil stability and may cause changes in natural drainage patterns of a given area, often leading to soil erosion during construction of the roads. Temporary control measures have to be provided for in the design and contract documents to control soil erosion and water pollution. These may be in different forms as may be determined by the design consultants. Examples of control measures may be by use of silt fences, brush barriers, dams, sediment basins, netting, gravel, mulches, grasses, slope drains, and other erosion control devices or methods. Appropriate control measures will be required to avoid emission of high concentration of sediments into wetlands, swampy areas and other particular sensitive areas.

The temporary erosion control provisions shall be coordinated with permanent erosion control features to assure economical, effective and continuous erosion control throughout the period of the Works.

8.2.3 Preservation of Trees and Shrubbery

All trees and shrubbery which are not specifically required to be cleared or removed for construction purposes shall be preserved and shall be protected from any damage that may be caused by the Contractor's construction operations. The removal of trees or shrubs will be permitted only after prior approval by the supervising Engineer. Existing trees not approved for removal by the Engineer shall be preserved and adequately protected from either direct or indirect damage by the Contractor's operations. Where trees have to be inevitably removed to ease the construction, new ones shall be planted as replacements. This should be provided for in the design, tender and contract documents.

8.2.4 Prevention of Water Pollution

Road construction activities may often lead to pollution of surrounding watercourses, lakes, and underground water sources. This is often though accidental fuel and oil spillage, generation of dust during processing, loading and transportation of materials as well construction of access trucks, refuse, garbage, cement, concrete, sanitary waste, industrial waste and radioactive substances

among others. Provisions have to be made in the design, tender and contract documents for prevention of contamination and pollution of the water sources. These shall be in compliance with all applicable Ugandan laws, orders, regulations, and water quality standards concerning the control and abatement of water pollution.

8.3 Climate Change Considerations

Roads continue to be built without taking predicted climate change patterns into account. Climate-related hazards and predicted impacts of climate change threaten roads, bridges and other road furniture. The economic cost of the impacts of climate change on road damage, repairs and reconstructions, though difficult to estimate, is very high hence a need to integrate Climate change issues.

The following should be considered while planning and construction of LVSR roads:

- v. Integrate climate change issues into the existing framework for LVSR planning and construction especially condition assessment guidelines and work methodology;
- vi. Comply with climate change–resilient standards for LVSR planning and development through monitoring and reporting systems;
- vii. Promote water catchment protection including road water harvesting during LVSR construction and maintenance; and
- viii. Greening of road corridors.

8.4 Social Safeguards

The design of LVSRs must address several social safeguards resulting from the construction of the roads through provision of measures and methods to:

- i. Ensure that Occupational Health and Safety (OHS) standards are adhered to during project implementation;
- ii. Prevent the spread of HIV/AIDS and sexually transmitted diseases (STDs);
- iii. Ensure equitable employment opportunities for men and women as well as addressing their specific gender needs; and
- iv. Ensure that there is accessibility for people with disabilities and elderly persons.

8.4.1 Occupation Health and Safety (OHS)

According to the Uganda MoWT Road Sub-sector guidelines for Mainstreaming OHS (2008), the design, tender and contract documents for the Consultants and Contractors involved in the implementation of the LVSR shall have provisions with the relevant clauses and conditions to ensure OHS is adhered to. Also works executed under force account schemes shall conform. The following documents and required provisions will be provided:

- i. Terms of Reference (ToR) shall:
 - Clearly specify the need for a OHS expert on the Consultant’s team;
 - Explicitly incorporate the objectives and expected outputs with respect to OHS; and
 - Provide for utilization of methods to identify hazards, risks and mitigation measures during implementation stages
- ii. Prequalification and bidding documents: These should indicate a need for Consultants/ Contractors to have capacity to address the OHS concerns for the project;
- iii. Certificate of completion of works: This should only be awarded after ensuring compliance with the OHS provisions;

- iv. Bills of Quantities (BoQs): These should provide bill items relating to Mobilization, Awareness creation, enabling a safe and healthy working environment and participatory monitoring; and
- v. Contract documents for Contractors: These shall provide for:
 - Equitable mobilization and recruitment of workers
 - Training skills for road workers
 - Compensation schemes for accident victims
 - Protective gear including- high visibility vests, protective boots, gloves and dust masks among others.
 - Emergency facilities at construction sites to provide first aid services.

Vi) Work plans and method statements. These shall include measures of addressing OHS.

8.4.2 HIV/AIDS and STD prevention

The road sub-sector just like any other is greatly affected by HIV/AIDS and Sexually transmitted diseases (STDs). Workers in the road subsector are extremely vulnerable and at a risk of acquiring HIV infection due to their nature of work which involves movements of people from one place to another while leaving behind their families. This not only deprives them of social support but may lead to engaging in risky sexual behavior. The strenuous work may quickly lead to the onset of AIDS, which ultimately affects productivity within the road sector. It is therefore important to provide guidance and provisions within the design and tender documents on how best to create awareness, enabling work environment, prevention and control of HIV/AIDS and other STDs.

In line with the Uganda MoWT guidelines for mainstreaming HIV/AIDS into the road sub-sector 2008 and Specifications for Roads and Bridges 2005, the design of LVSRs shall have provisions with the relevant clauses and conditions incorporated into the Terms of Reference (ToR) and all contracting documents shall ensure that HIV/AIDS and STDs prevention is achieved especially during the implementation stages. Also works executed under force account schemes shall conform. The following documents and required provisions will be provided:

- i. Terms of Reference (ToR): These shall have clauses requiring the Contractors to have an HIV/AIDS Management Plan that shall include details of the measures proposed to combat the spread of HIV/AIDS and sexually transmitted diseases (STD) between staff, labour and the local community. The plan shall also outline workplace policies and programmes for employees living with HIV/AIDS, information and awareness campaigns and effective screening and counseling policies for STD and HIV/AIDS cases of his/her project staff.
- ii. Bills of Quantities (BoQs): These should provide bill items covering costs related to information and awareness campaigns and effective screening and counseling services for STD and HIV/AIDS cases of project staff.

8.4.3 Gender issues

According to the Uganda MoWT guidelines for Mainstreaming Gender into Road Sub-sector, the design of all roads shall be gender responsive. This is aimed at ensuring that equitable employment opportunities are accorded to both men and women as well as achieving other gender related objectives. This is a responsibility of the implementing entities as well as the private consulting and contracting firms. In line with this, the design of LVSRs shall have provisions with the relevant clauses and conditions incorporated Terms of Reference and all contracting documents ensure Gender responsiveness is achieved. The following documents and required provisions will be provided:

- i. Terms of Reference (ToR): These shall:
 - Clearly specify the need for a gender-sensitive consultant's team or the need for a gender specialist on the team;
 - Explicitly incorporate gender objectives and expected outputs; and
 - Provide for utilization of gender focused methods such as, analysis and interpretation of data.
- ii. Prequalification and bidding documents: These should indicate a preference for gender sensitive Consultants or Contractors and other service providers.
- iii. Bills of Quantities (BoQs): These should provide bill items covering costs related to mobilization, gender-awareness creation, enabling working environment and participatory monitoring.
- iv. Contract documents for Contractors: These shall provide for:
 - Equitable employment for both men and women or a 30% quota for women for labour-based works;
 - Equitable mobilization and recruitment of workers;
 - A gender sensitive human resource policy;
 - Equal payment to men and women for similar work and that payment of wages is made to the workers and not to representatives;
 - Flexible working hours as much as possible to take account of multiple roles of women and cultural norms;
 - Representation for both men and women in any information and consultative meetings held at the site and that gender and social issues are raised and analyzed; and
 - Contractor's reporting format that provides for the recording of sex or gender disaggregated data.

PART B: WORK STANDARDS, GUIDELINES AND CONSTRUCTION MANUAL FOR LOW VOLUME SEALED ROADS

9.0 Work Standards and Guidelines

Introduction

This part of the manual covers standards and guidelines for the execution of road works. Techniques in this field have been developed over a long time and the technology is developing steadily with new innovation developed recently replacing some of the old approaches.

It also covers many aspects of road construction and the works are categorised into Series of activities. Each Series has a number of activity codes which are basically stand-alone but do make references to other codes for specific operations.

Each activity code consists of a number of aspects of construction which must be followed by the practitioners in the execution of the particular activity as follows:

- i. The code number and activity title such as 'Construction of single surface dressing';
- ii. The Application of the code which is the description of the code and its purpose and where and how it can be applied;
- iii. The tasks assigned for the accomplishment of the activity;
- iv. The standards and guidelines which should be followed and satisfied by the practitioners;
- v. Materials description, guidelines and handling.
- vi. Measurement, approval of works and payment.

There are guidelines, diagrammatic and photo illustrations of the activities and expected products which will assist the contractor or force account units

The activity codes are used as follows:

- i. Depending on the requirement of the project appropriate activity codes and the references can be picked up and placed in bidding documents and this will help in the efficient procurement of works;
- ii. The activity codes can also be picked up and placed in the contract documents. This will ensure uniformity in the documentations and that the works are correctly presented without ambiguity; and
- iii. The standards in the code provide a check list of standards to be met in the execution of the works.

This is the new approach in presenting construction manuals.

9.4 Series 100 Preliminaries and General

Introduction

Preliminaries and general items involve all activities relating to establishment and provisions for the site, demobilisation and compliance with General and Specific Conditions of Contract.

Establishment and provisions on site

This includes construction of contractor's camp and site including accommodation for key staff on site, abolition and sanitation facilities like toilets, bath rooms, storage facilities and workshops for equipment and recreational facilities.

Mobilisation

The contractor should mobilise all the necessary resources such as personnel, equipment and tools as is stipulated in the contract. Mobilisation may also include materials but this depends on the requirements of the contract.

Compliance with General and Specific Conditions of Contract

The contract is required to ensure the works and site personnel against injury and damage. This includes situations where unforeseen damage occurs to works before they are handed over to the client. Such phenomena may include storm damage and other natural calamities. Workers may be injured on site while on duty and the responsibility is of contractor and of the workers to ensure safe on site. In the event that workers get injured on site it becomes the contractor's liability.

There may also be conditions regarding policies on employment such as meeting the proportions of female employees or casual labourers and employment of local labour, and provisions can be made in this regard.

Other obligations may relate to infection and disease control such as HIV/AIDS programmes and so on.

In previous war zones there could be problems of land mines and the investigation and demining processes should be covered under preliminaries and general. This is a unique activity but should be considered when there is risk of encountering land mines. Land mines can last for a long time and usually remain live and they can cause damage, injury or death to workers.

Clearance of site can also be considered as part of the preparatory work before the commencement of earthworks.

Demobilisation

Once the construction activities have been completed the contractor is expected to demobilise from site and this includes the removal of the contractor's camp and facilities, equipment and contractor's staff and casual labourers. The land or site should be reinstated and all deleterious materials should be removed and disposed of safely.

THIS SERIES SHALL BE READ IN CONJUNCTION WITH THE GENERAL SPECIFICATIONS FOR ROADS AND BRIDGES – SERIES 1000.

Activity: Mobilisation		Code: 110
Reference Code		
Application	This code is used for activities involving construction of the Contractor's worksite, and mobilization of equipment and labour.	
Tasks	• Construction of site accommodation for the Contractor's <i>staff</i>. • Latrines, bathrooms and social infrastructure. • Provision of drinking water for the Contractor's <i>Staff</i> and other workers on the site. • Provision of hygiene and security equipment for the Contractor's <i>Staff</i> and other workers on site. • Construction of storeroom(s), office(s), workshop and laboratory. • Provision of tools and equipment. • Transport at the worksite and the necessary resources. • Recruitment of the casual workers in coordination with the traditional, administrative authorities and the community. • Any changes in the worksite that may be necessary during the project execution. • Maintenance of the camp installations and facilities during the execution of the works.	
Standards	• The camping place must be approved by the Engineer and the Local Authorities. • The following shall be constructed: 1) Conventional houses to be used by the Contractor's <i>staff</i>. 2) Bath rooms and improved latrines – 1 latrine for 6 workers that are living in the camp. 3) Shed for a social place. 4) Offices (with chairs and tables), workshop and storerooms for the tools, materials and parts. 5) Place for storage of fuel, at a distance not less than 30m from the houses. • Placing of a water tank with a minimum of 1000 litre capacity. • The contractor's <i>staff</i> should be accommodated while working in the worksite. • The mobilization on site of the equipment listed in the contract. • Procurement of tools. • Recruitment should be done in coordination with the District Administrator and the Traditional Authorities of the area to ensure a transparent procedure. • At least 60% of the labour force shall be recruited from the local community.	
Materials	All material and equipment should be approved by the Engineer. Generally, the construction of the camp may be carried out using local materials.	
Measurement	• Lump sum. • Camping place constructed and working according to the above defined standards including the availability of equipment and minimum required <i>staff</i> as indicated in the contract. • Measurements shall be taken only in the construction of the first camp. Any changes in the camping shall be priced by the Contractor and must be included in his unitary prices.	
Payment	• The first instalment, 50% of the lump sum, will be paid in the first payment certificate after the Contractor has met all his obligations under this Section and has made a substantial start with construction in accordance with the approved programme. • The second instalment, 50% of the lump sum, will be paid when the value of the work done reaches one half of the tendered amount, excluding contingencies and price adjustments in terms of the General Conditions of Contract. • Before any payment is made under this code, the Contractor shall satisfy the Engineer that he has provided camps and construction plant of good quality on the site, the value of which exceeds that of the first instalment.	

Activity: Demobilisation		Code: 111
Reference Code		
Application	This code is used in activities involving demolition of Contractor's site camp and the removal of equipment and laying off the temporary <i>staff</i> .	
Enclosed tasks	• Demolition of accommodation, latrines, bathrooms and social infrastructures that were constructed for the Contractor's <i>staff</i> on site. • Demolition of warehouses, offices and workshops and removal of tools, equipment, scrape, waste and contaminated soil. • Transferring or laying-off temporary <i>staff</i>. • Cleaning and restoration of the camping area.	
Standards	• The following should be remove completely from camping site: 1) All the material and scrape that resulted from the camp and site demolition. 2) Any waste or soil contaminated by petrol, fuel, tar or any other chemical material. • The camp area should be cleaned and replaced to its original condition to fulfill the environmental requirements. • The transfer or laying-off of workers should be done in conformity with the District Administrator and the Traditional Authorities of the area.	
Materials	Not Applicable	
Measurement	• Lump sum • Camp removed according to the standards mentioned above. • The measurements are taken only in the first camp removal, followed by the <i>staff</i> and equipment demobilisation after provisional reception of the works by the client. MAE.	

Activity: : Insurances: Against civil duty Against work accidents		Code: 120 121
Reference Code		
Application	These codes are used to pay for the insurance policy that the Contractor has to buy to comply with the General and Specific Conditions of Contract, particularly in respect of the insurances and sureties required and his general obligations to the public and the Employer. He/she shall comply with all the regulations of statutory bodies.	
Tasks	All expenses linked to the insurances policy payment and any extensions if necessary.	
Standards	• The insurances must include all risks covered in the General and Special Conditions of the contract (contract data). • The insurance company must be registered and operating in Uganda. • The insurances values should be in conformity with contract data. • The insurances must be valid during the period covered by the General and Special Conditions of Contract. The extension of the insurance policy should be refunded according to the initial value and the extension time, unless the extension reasons are primarily the Contractor's responsibility.	
Materials	Not Applicable	
Measurement	• Monthly value. • Payable on the issuance of the insurance policy, valid during the work duration, and issued receipts. MAE.	

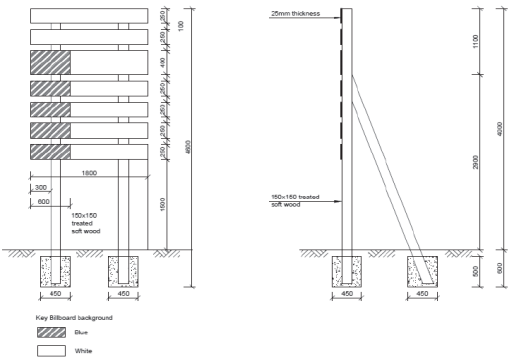
Activity:	Demining: Verification of the mining level Demining of the worksite	Code: 130 131
Reference Code		
Application	It is defined as contractor's responsibility to guarantee conditions of safety for the <i>labour</i> during the execution of the works. This code is used for all the activities necessary to comply with this obligation related to explosive devices and may be applicable in previous conflict zones.	
Enclosed tasks	- Consulting the local communities and authorities, the police and military departments for the determination of the risk levels existing in the work area. - Capacity of the <i>staff</i> in dealing with explosive devices and care to avoid accidents. - Demining activities necessary before and during the work execution. - Demining of work areas indicated as high level risk areas. - Dealing with suspect explosive devices found on site, during the execution.	
Standards	- The Contractor has to show that he has done all necessary enquiries and surveys to determine the risk of mines and other explosion devices around the worksite; - Each <i>staff</i> member should be prepared in mines risk skills and how to avoid them. - Any area used by the <i>staff</i> members during the work execution should be away from the high level mining risks. - A free mines risk area is defined as: 1) An area that has been de-mined by a recognized company or entity. 2) An area where local authorities and population have the information of no risks of mines. - In case there are some areas in the worksite which are not notable as "free of high mines risk" the Contractor has to contact the authorities for the demining procedure. - Any area recognized as <i>high mines risk</i> has to be marked to prevent staff and local people from stepping in. - The entity that works on demining should have the approval of the National Program of Demining. - The procedures for the demining work must always follow the recommendations of the National Program of Demining. - The method used in the site must be appropriate for the terrain and for the kind of mines expected. - The contractor is responsible for the mines removal in all of the worksite extension including that which is denominated not as high risk area.	
Materials	Not Applicable	
Measurement	- Code 121 - Lump sum. - Based on the properly supplied information about the risk levels. - MAE. - Code 122 - M² - Measurements are taken from the road and borrow pit areas with <i>free high mine risk</i> after the necessary demining process executed according to the standards mentioned above. - MAE.	

Activity: Compliance with the Special Conditions of Contract to promote the employment of female workers.		Code: 140												
Reference Code														
Application	Under the poverty reduction program, the Government policy is to encourage female participation in the road sector. This code has the proposal to stimulate the Contractor to take steps to ensure that all the females who wish have access to employment opportunities in road maintenance and rehabilitation. When the women labour is more than 25% (minimum percentage for female participation), the work value increases proportionally to the percentage of female labour.													
Tasks	• Outreach and communication activities with authorities and local communities with the aim of encouraging female participation. • Use of recruitment methods approved through the Contract that facilitate the employment of females. • Necessary measures to prevent discrimination and harassment against female workers.													
Standards	• A communication campaign should be carried out to inform the population throughout the road about the existence of vacancies for both male and female and encouraging the participation of women. • The recruitment process must privilege the registration of women until it reaches the minimum percentage required. • Men and women must have the same access to the information on the job chances and must be dealt with equally in the recruitment process. • The contractor shall not make any form of wage discrimination based on gender. • A workers’ union with representatives of the workers of both the Contractor ’s and temporary <i>staff</i> must be created for the resolution of any disputes at work.													
Materials	Not Applicable													
Measurement	• Provisional sum (PS). • Based on properly supplied information about female participation levels according to the <i>salary sheet</i>. • The percentage of female labour is calculated based on female and male worker-days. (Number of female temporary workers during the month) x 100% divided by the (total number of workers during the month: male + female), Table 9-1. MAE. Table 9-1 Increase in value of work for employing women <table><tr><th>% age of women above % minimum of female participation monthly</th><th>% increase of the value works</th></tr><tr><td><25%</td><td>0%</td></tr><tr><td>25 - <35%</td><td>0.5%</td></tr><tr><td>35 - < 40%</td><td>1.0%</td></tr><tr><td>45 - < 45%</td><td>1.5%</td></tr><tr><td>>45 %</td><td>2.0%</td></tr></table>		% age of women above % minimum of female participation monthly	% increase of the value works	<25%	0%	25 - <35%	0.5%	35 - < 40%	1.0%	45 - < 45%	1.5%	>45 %	2.0%
% age of women above % minimum of female participation monthly	% increase of the value works													
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Activity: Conformity with Special Conditions of the Contract to promote the employment of local workers		Code: 141												
Reference Code														
Application	Under the poverty reduction program, the Government policy is to increase work opportunities for the local communities in the road sector: This code has the provision to encourage the Contractor to give work opportunities to the people living in areas neighbouring the work site and to use more labour intensive methods to increase the work opportunities and at the same time adopt efficient techniques of work. The contract value shall be increased proportionally to the amount shown in the payroll for the casual workers as a proportion of the works.													
Tasks	• Awareness activities and communication with the local authorities and communities to encourage the recruitment of local labour from the local community. • Use of recruitment methods approved by the Provincial Directorate of Public Works and Habitation which facilitates the recruitment of local workers; replacing the labour from one community to other according to the progress of the works. • The use of construction techniques based on the use of intensive labour promoting the creation of jobs.													
Standards	• Awareness campaign should be carried out to inform the population along the road about the existence of employment opportunities for the local communities • There shall be liaison with the local authorities and communities to agree on transparent procedures for the selection and recruitment of local workers privileging the poorest. • The process of recruitment and demobilisation of the occasional workers should be continuous as work progresses from one community to another to avoid the need for occasional worker to be accommodated at the camp. • The contractor should have a work team with experience in the use of labour intensive (or labour based) techniques.													
Materials	Not applicable													
Measurement	• Provisional sum (PS); Based on the properly supplied information about local labour participation levels given in the <i>salary sheet</i> of the effective worker days. The value of the works is increased by the percentages defined below, in relation to the ratio of the value of the casuals wage bill to the value of work for the month in question. $(\text{Value of bill of wages of month of the casual workers}) \times 100\% \div (\text{Value of works for the month, without the VAT})$ It is applied as a percentage when the requirements of the work norms have been fulfilled, Table 9-2. MAE. Table 9-2 Increase in the value of works for employing local labour <table><tr><th>% of the value of the works for month paid out as wages for local workers.</th><th>% of increase of the value works for the month</th></tr><tr><td><25%</td><td>0%</td></tr><tr><td>25 - <35%</td><td>0.5%</td></tr><tr><td>35 - < 40%</td><td>1.0%</td></tr><tr><td>45 - < 45%</td><td>1.5%</td></tr><tr><td>>45 %</td><td>2.0%</td></tr></table>		% of the value of the works for month paid out as wages for local workers.	% of increase of the value works for the month	<25%	0%	25 - <35%	0.5%	35 - < 40%	1.0%	45 - < 45%	1.5%	>45 %	2.0%
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Activity: Occupational Health and Safety - Conformity with the Special Conditions of the Contract Program Against HIV/AIDS		Code: 142
Reference Code		
Application	The main purpose of the program of HIV/AIDS for the tertiary roads is the reduction of transmission of the HIV in the zone of influence of the road through the implementation of activities of prevention directed at all the workers of the road sector, as well as the local communities, and to reduce the impact of the epidemic in the referred area.	
Tasks	- Contract a specialist institution or person for the establishment of services in the area of HIV/AIDS for the workers and the communities around the road. - Stigma, discrimination and confidentiality: The reduction of the stigma and discrimination related to HIV/AIDS should be present in all activities of the program at all levels. Therefore, it will have to be a constant topic in each of the activities. - Prevention and reduction of the impact: The programme must include activities of prevention as well as initiatives aimed at the reduction of the impact of the epidemic. - The activities of prevention must include: - Free distribution of masculine and feminine condoms (10 condoms monthly per worker, on average). The condoms must also be placed in places of bigger concentrations of the workers and easy access and distributed to each worker on the day of payment. - Interventions for the change of behaviour through : - Distribution of promotional materials. - Formation of <i>educators of peers</i> in the workplace and community activists. - Discussions in small groups. - Sessions of theater, video and lectures. - Promotional games such as football. - Posters containing messages on the HIV and AIDS to be placed throughout the road. - Diagnosis and Treatment of Sexually Transmitted Infections (STI). - Voluntary consultation and tests. - To facilitate the access of the infected and affected workers, and respective families to the services of psychiatrist support and health (including prevention and treatment of opportunist infections). - The contracted specialist should promote the accomplishment of the activities, at least two hours every month within the schedule of the work. - The contracted specialist should work in close coordination with the accomplishment of activities of the organisation and ensure they have support.	
Standards	- The contracted specialist should provide a work plan and follow the guidelines of the MoWT's Department of HIV and AIDS and it must be approved by the Engineer. This specialist must work in close collaboration with this Department of HIV/AIDS and district units responsible for combating to the AIDS. - The magnitude and type of activities which shall be implemented will depend on the duration of the works and the number of workers involved. - The contracted specialist should submit monthly reports of the activities. This report will have to be submitted together with contractor's monthly report to the Engineer. - The program against HIV/AIDS will be initiated one month after the beginning of works and will be extended until the conclusion of the work (provisional reception).	

	- The following results of the activity are expected: - All the workers to have gratuitous access to the condoms. - All the workers to have conscience of the importance of practice of safe sex and to know the methods of prevention of HIV/AIDS. - All infected and affected workers should be directed to support and advice. - All the workers should have knowledge of the <i>law of protection of workers living with HIV/AIDS</i> (5/2002, of 5 of February), see activities in Table 9-3. Table 93 Activities for combating HIV/AIDS <table border="1"> <thead> <tr> <th>Activity</th><th>Amount</th></tr> </thead> <tbody> <tr> <td>Masculine condoms for workers</td><td>12 for month</td></tr> <tr> <td>Female condoms for workers</td><td>6 for month</td></tr> <tr> <td>Sessions of theater</td><td>1 time for 6 months</td></tr> <tr> <td>Sessions of video</td><td>1 time for 6 months</td></tr> <tr> <td>Discussions in small groups</td><td>1 time for 6 months</td></tr> <tr> <td>Promotional events</td><td>1 time for 6 months</td></tr> <tr> <td>T-shirts for workers</td><td>2</td></tr> <tr> <td>Hats for workers</td><td>2</td></tr> <tr> <td>Counselling of couples</td><td>10</td></tr> <tr> <td>Posters for workers</td><td>2</td></tr> <tr> <td>Brochures for workers</td><td>2</td></tr> <tr> <td>Advertising panels</td><td>2</td></tr> </tbody> </table>	Activity	Amount	Masculine condoms for workers	12 for month	Female condoms for workers	6 for month	Sessions of theater	1 time for 6 months	Sessions of video	1 time for 6 months	Discussions in small groups	1 time for 6 months	Promotional events	1 time for 6 months	T-shirts for workers	2	Hats for workers	2	Counselling of couples	10	Posters for workers	2	Brochures for workers	2	Advertising panels	2
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Hats for workers	2																										
Counselling of couples	10																										
Posters for workers	2																										
Brochures for workers	2																										
Advertising panels	2																										
Materials	- All new Material that will be produced and published shall be approved by the Inspector. - Promotional materials such as t-shirts, hats, door keys, stickers, posters, brochures - Other printed material. - Theater and videos. - Condoms. - Advertising Panels.																										
Measurement	- Provisional sum(PS). - On the basis of the supply of information duly based on the level of activities executed (monthly reports and payment receipts). - The financial proposal must include costs related with the program of combating HIV/AIDS in compliance with the bill of quantities and budget that constitutes part of this section. - MAE.																										

Activity: Production, placement and removal of Project Bill Boards at the work place	Code: 150
Reference Code	
Application	These bill boards are to inform the road users and the local community on the work in progress including the type of work, stage, duration and total budget of project, the Contractor's name, name of Supervising Consultant and origin of the financing, Figure 91. This code is used for any work in the road sector: maintenance, rehabilitation, construction or spot improvements.
Tasks	• Placement of signs for safety. • Making sure existing traffic can pass. • Supply, production and painting of the bill board and post. • Marking and preparation of the place to hoist the bill board. • Excavating foundations for the placement of the bill board. • Placement and fixing the bill board and straightening it vertically. • Backfilling and compaction of soils in the foundations of the posts. • Removing vegetation that may obstruct the visibility of the panel. • Removing the temporary signs/bill board
Standards	• In a maximum period of 14 days after the commencement of the contract, the contractor should raise the bill board illustrated in Fig 9-1 on either side of the road length covered in the contract. The panel must be placed perpendicularly. • The panel must not be placed more than 10 meters from either end of the road section covered by the contract. • The panel must be legible on both sides. • The panel must be placed in an area approved by the Engineer and at least 1 meter distance from the backslope of drain. • The bill board shall be constructed with plain steel with 0.6 mm thickness, 2.2m width and 2.7 m height on a treated wooden structure. • The maximum tolerance accepted for the width and the height of the plate is ± 20 mm. • The maximum tolerance accepted for the size of the letters is ± 5 mm. • The wooden post shall be securely embedded in the ground with of minimum depth of 0.75 m. • The bill boards shall be removed within a maximum period of 14 days after definitive reception of the work by the Client. • The bill board must have the following legend, appropriate for the Contract:
	Standard Billboard design  1. The working of the project signboard and the location to be installed to be as directed by the Engineer 2. Materials to be used for fabrication of signboard shall be pressure impregnated treated softwood timber sizes as indicated in the drawing 3. Working boards to be nailed to the posts using nails. 4. Project board posts and struts to be embedded in concrete ration 1:2:4 Figure 9-1 Technical specifications for project billboard The expanded version is given in the Appendix.
Materials	• All material must be approved by the Engineer. • Smooth steel plates of minimum thickness 0.6 mm. • Enamel and undercoat paint. • Wooden pole with minimum diameter of 15 cm and treated with creosote or a product of equivalent quality;
Measurement	Unit (NO) MAE

Activity:	Clearing and Grubbing 40 m <AHD< 100 m 100 m <AHD< 200 m 200 = <AHD< 500 m AHD = > 500 m	Code: 160 161 162 163
Reference Code		
Application	The code of clearing and grubbing must be used for works involving removal of vegetation and other objectionable material in the reserved area for the road: ○ Trees with trunk of less than 30 cm in diameter. ○ Shrubs, grass and roots. ○ Topsoil with organic material until depth of 5 cm. ○ Pebbles. Use codes 180, 181 and 182 for the improper ground removal with depth more than 5 cm and codes 170 and 171 for the removal of trees with diameter greater than 30 cm.	
Tasks	• Placement of safety signs. • Marking of the work area (including placement of pegs to mark the alignment of the road). • Cutting shrubs and grass.	
	• Removing organic soil and roots until the depth of 5 cm. • Removing boulders possible to move using manual tools. • Removing material cut/excavated away from the carriageway to a place approved by the Engineer. • Removal of temporary signs.	
Standards	• To guarantee safety, appropriate signs shall be placed on the road. • Cleaning and Grubbing shall be carried out within the width extending 1.5 m from the outer slopes of side drains or more as directed in the contract. • All vegetative materials (shrubs, grass), boulders, roots and organic soils within the depth of 5 cm must be removed. • Deleterious soil layers which may affect the future performance of the road (grit, sands, clay) shall be removed. • The removed material must be placed in a convenient place at least 40 m away from the road and spread in a layer with thickness not more than 20 cm. • The removed material should not be burnt. • Once the activities are concluded, the temporary signs should be removed.	
Materials	• Not applicable	
Measurement	• M² • Measurements are taken only in the area where vegetative and other objectionable materials were removed. MAE.	

Activity:	Cutting and removal of trees Diameter more than 30 cm and less than 70 cm Diameter more than 70 cm	Code: 170 171
Reference Code		
Application	This activity is used for the cutting and removal of all trees which obstruct the road as indicated by the Engineer. The code choice depends on the diameter of the tree to be cut down. This code is also used for the removal of other trees out of the road area, depending on the Engineer's instruction based on the understanding that the trees shall interfere in the construction of the drainage system of the road or for being considered a hazard leaving them. All the activities of rehabilitation should be approved and follow the environmental instructions for the Roads Sector.	
Tasks	• Safety Signs. • Selection and Identifying species of trees that can be used or that must be protected. • Identification of the trees to cut. • Cutting and removing the trees away of the road. • Excavating and removing of the trunks and roots. • Backfilling of holes and compacting.	
Standards	• To guarantee road safety, appropriate signs shall be placed in the road. • Cut down the indicated trees in the area reserved for the road, with special attention to the security of workers, population and properties. • Removed material shall be placed in a convenient place at least 40 m away from the carriageway and cut to pieces. • In case the Engineer decides to leave some trees because of their value, even if they are in the reserved area for the road, his instruction must be respected (e.g. trees with special value). Backfill the holes created through removal of stumps and roots with soils of the same or higher quality than the soil in the referred area. • Backfill in layers of not more than 15 cm in thickness. • Water, mix and compact the layers. • Once the activities are concluded, the temporary signs shall be removed.	
Materials	Not applicable	
Measurement	Number (No.) MAE	

Activity: Improper Ground Removal and Excavation 40 m <AHD< 100 m 100 m <AHD< 200 m 200 <AHD< 500 m AHD> 500 m		Code: 180 181 182 183
Reference Code		
Application	This code is for the excavation and removal of the improper and/or unnecessary material from the road platform and from the side drains due its poor quality or to ensure construction of good profile. The 5 cm of excavations mentioned in code 160 are not included in this code.	
Tasks	• Placing safety signs. • Marking of the area for improper soils and excavation works. • Control of the levels and the width of the excavation. • Excavating of the soil. • Removing, transporting and spreading of excavated material. • Removing temporary signs once work is completed.	
Standards	• To ensure appropriate road safety, signs shall be placed on the road. • Only remove soil layers that may: 1. Affect the future performance of the road due to higher content of organic material, marshy ground and badly graded. 2. Exceed the needed quantity and result in undesired alignment of the road platform. • Non-deleterious soil layers (to the future road) should not be removed. • Soil layers in the defined area for the roadway and drains should be removed. • The removed material must be placed in a convenient place at least 40 m away from the road, and spread in a layer with thickness not more than 20 cm. • Once the activities are concluded, the temporary signs should be removed.	
Materials	• Not applicable	
Measurement	• M³ • Measurements are taken on the volume of the soils excavated. MAE.	

Activity:	Demolition and removal Removal and disposal of the cars and other abandoned garbage Demolition of existing structures	Code: 190 191
Reference Code		
Application	Code 190 is used for the removal and disposal of vehicles, dead animals and other abandoned garbage in the reserved area of the road. Code 191 is used for the demolition of simple structures, existing concrete and reinforced concrete structures, metallic or wooden structures. The codes 191 and 192 are used according to the Engineer's instructions.	
Tasks	Code 190 • Placing temporary safety signs. • Inspection of the reserved area of the road. • Notice about irregularities to the Proprietors. • Removal of vehicles and other abandoned trash including contaminated soils (oil, etc.). • Transportation and placement of the trash to a location previous approved by the Engineer. • Clearing and leveling of the land where the abandoned cars and/or other garbage have been removed. • Removal of temporary signs.	
	Code 191 • Placing of temporary safety signs. • Definition of the procedures to be considered for the demolition of the structure(s) basing in structural stability. • Demolition of the structure. • Transportation and disposal of the rubbish to a location previous approved by the Engineer. • Clearing and leveling of the land where structures were demolished and removed. • Removal of temporary signs.	
Standards	Code 190 • Find the owners of the vehicles, dead animals and other rubbish that should be removed from reserved area. • In case the owners are found a written notice through the local authorities must be given informing them about the time limit to remove their goods. • In case the proprietors do not exist or do not remove their goods in the stipulated time the works should be allowed to continue. • To guarantee road safety appropriate signs shall be placed on the road. • The staff must be equipped with protective clothes such as helmets, safety goggles for eye protection, boots, gloves and overalls. • Removal of cars and other abandoned trash including contaminated soils (oil, etc.) shall be carried out.	
	Code 191 • The demolition work should only be authorised after installation of new or provisional structures in substitution of the ones to be removed in order to be able to facilitate the management of the regular traffic without interruption. • The staff should be equipped with protective clothing such as helmets, safety goggles for eye protection, boots, gloves and overalls. • Only the items indicated in the Engineer's instructions must be removed. • The removed material must be placed in a convenient place at least 40 m away from the road, and spread in a layer with thickness not more than 20 cm. • Once the activities are concluded, the temporary signs should be removed.	
Materials	• Not applicable	
Measurement	• No (code 190 removal and disposal of abandoned vehicles, dead animals and other garbage to a place indicated by the supervisor). • Removed a vehicle is considered a unit of measurement. • A quantity of 5 M³ of rubbish is considered a unit of measurement. • MAE. • M³ (code 191 Demolition of existing structures); • Measurements are taken from the volume of the rubbish transported and deposited in the specific place indicated by the Engineer. • MAE.	

9.5 Series 200 Construction of Drainage Structures

9.5.1 Introduction

- i. This Series covers standards and guidelines in the construction of drainage structures. The main categories of such structures include:
- ii. Drains (side, mitre and catchwater);
- iii. Lining of drains with concrete, masonry, etc;
- iv. Water crossing structures such as culverts, causeways, drifts, etc;
- v. Protection works using stone pitching, gabions, energy dissipaters, vegetation, etc; and

Sub-surface drainage systems are also covered in detail.

The construction of low volume roads and their sustainability hinges on the provision of good drainage, 'water is the enemy'.

Activity: Side Drains, Mitre Drains, and Catchwater Drains.		Code:
Opening and smoothening of Side Drains		210
Reference Code		
Application	This code is used for the excavation of side drains on the road during road construction and rehabilitation. The excavated soil is then used for the construction of the formation and camber of the road. The side drain directs the water that drains off the carriageway of the road to a safe place away from the road. In places where the soil has a natural inclination and where the quality allows, the construction of side drains is not always necessary. To reduce the effect of erosion in the side drain scour checks (cascades) can be constructed (codes 290 the 293). If the inclination of side drain is greater than 5%, then the side drain might be lined with concrete or wet stone pitching (codes 230 and 231). The choice of these options depends on the side drain inclination and type of soil.	
Tasks	• Placing safety signs • Making sure that existing traffic can pass. • Identification of side drains with the Engineer. • Setting out dimensions and inclinations of the side drain to be constructed. • Marking excavation tasks. • Excavation of side drains and smoothing the bottom of the drain and slopes. • Depositing the material obtained from the excavation of the side drain on the carriageway to be used in the construction of the formation and camber. • Excavation of excess materials to meet levels. • Controlling the shape, depth and inclination of the excavated side drain. • Constructing mitre drains. A 5 m bank should be left between one section of the side drain and the following section of the side drain to ensure the drainage of the water from the side drain to the mitre drain. • Removing temporary signs.	
Standards	• To guarantee road safety, appropriate signs shall be placed on the road. • The side drain excavation should be constructed as per the design drawings and should not have irregularities. • The side drain size should be bigger where the anticipated run-off volume is higher or where it is necessary to increase the material collected from the side drain excavation for the formation. • Rocks, stones and roots in the side drain profile should be removed. • At the start, the side drain should have a minimum inclination of 1 % and a maximum inclination of 3%. However, in undulating and mountainous areas an inclination greater than 3% can be considered but scour checks shall be placed as designed (see code 290 and 293). In mountainous areas with erosive soil, the side drain should be lined according to the Engineer's instruction (code 230 and 231). • The profile of the side drain should be controlled as per standard specifications or as designed and so must its inclination. • The inclination of the side drain should permit the draining of the water into the mitre drains, culverts or water courses. • If the soil quality is adequate the material obtained from excavating the side drains should be deposited on the road formation. If the soil is inappropriate it must be deposited out of the side drain area in a layer not thicker than 15 cm or transported away from the road area completely.	
Materials	Not applicable	
Measurement	• M³ • Measurements of the volume of the excavated drains are taken	

Activity:	Side Drains, Mitre Drains and Catchwater Drains	Code:												
	Opening and smoothening of Mitre Drains	211												
Reference Code														
Application	This code is used for the excavation of mitre drains. The mitre drains have the function of directing the water that drains off the side drain away from the road. To reduce erosion in the mitre drain, scour checks (cascades) shall be constructed where necessary (codes 290 the 293). If the gradient is greater than 5%, lining of the mitre drain with concrete or stone pitching may be necessary (codes 230 and 231). The choice of these options depends on the mitre drain gradient and type of soil.													
Tasks	• Placing safety signs. • Making sure that existing traffic can pass • Identifying the appropriate position of the mitre drain with the Engineer and owner of the property. • Demarcating dimensions and inclinations of the mitre drain to be constructed. • Marking of excavation tasks. • Excavating mitre drain and smoothing of the bottom of the drain and slopes. • Removing reject material. • Controlling the shape, depth and inclination of the excavated mitre drain. • Using the material obtained from the excavation of the mitre drain for the construction of the bank to direct the water away from the side drain to the mitre drain. • Removing temporary signs.													
Standards	• To guarantee road safety, appropriate signs should be placed on the road. • The mitre drains shall be constructed at regular intervals to permit the discharge of the water from the side drains. The mitre drains are placed along the side drain with spacing shown in Table 9-4. Table 94 Spacing between mitre drains <table><tr><th>Inclination/slope of the side drain</th><th>The spacing between mitre drains should not exceed</th></tr><tr><td>Gradient < 4 %</td><td>200 m</td></tr><tr><td>Between 4 and 6 %</td><td>160 m</td></tr><tr><td>Between 6 and 8 %</td><td>120 m</td></tr><tr><td>Between 8 and 10 %</td><td>80 m</td></tr><tr><td>Between 10 and 12 %</td><td>40 m</td></tr></table> • The excavation of the mitre drain shall be carried out as per the drawings and shall not have irregularities. • The size of the mitre drain shall be bigger where the anticipated run-off volume is higher. The transverse profile of the mitre drain should also be bigger than the transverse profile of the side drain. • The transverse profile of the mitre drain shall be bigger at the outlet extremity to allow water to spread out and away easily. • The angle of entry from the side drain to the mitre drain shall be 30 to 60 de grees or 90 degrees in flat areas. See Figure 92. • The mitre drain shall have a minimum inclination of 2 % and the bottom of the mitre drain should be on the same level as the side drain. The inclination shall be uniform. • The profile of the mitre drain should be regulated with a template of the section and the inclination must be regulated with metric tape and spirit level or an automatic level. • The mitre drain outlets should be made in accordance with 3 discharge ways: Superficial discharge in the existing land. 1) Discharge into a water course. 2) Discharge in an infiltration pit or sump. • Where possible discharge into a water course is preferred. • If the mitre drain conducts the water onto natural terrain, the invert level of the mitre at discharge point and the level of the terrain should be the same at the link.		Inclination/slope of the side drain	The spacing between mitre drains should not exceed	Gradient < 4 %	200 m	Between 4 and 6 %	160 m	Between 6 and 8 %	120 m	Between 8 and 10 %	80 m	Between 10 and 12 %	40 m
Inclination/slope of the side drain	The spacing between mitre drains should not exceed													
Gradient < 4 %	200 m													
Between 4 and 6 %	160 m													
Between 6 and 8 %	120 m													
Between 8 and 10 %	80 m													
Between 10 and 12 %	40 m													

	• Discharge of water in erosive soils should be avoided. In the extreme case, where this is unavoidable, the discharge area should be protected according to the protection code 250. • If the mitre discharges into a cultivated area, the discharged point should be agreed with the owner of the property. • Rocks, stones and roots in the drain profile should be removed. • The temporary signs shall be removed after completion of the works.
Materials	Not applicable
Measurement	• M³ • Measurements are taken of the volume of the excavation. • MAE



Figure 9-2 Mitre Drain discharging water from road structure

Activity: Side Drains, Mitre Drains, and Catch-water Drains		Code:
Opening and smoothening of Catch-water Drains		212
Reference Code		
Application	This code is used for the excavation of catch-water drains along the up-slope side of the road to prevent erosion of the backslope (in cut) and reduce the inflow of run-off into the side drains. To reduce the effect of erosion in the drains, scour checks (cascades) shall be constructed (codes 290 the 293). If the inclination of drain is greater than 5%, the drain should be lined with concrete or wet stone pitching (codes 230 and 231). The choice of these options depends on the gradient and type of soil.	
Tasks	• Placing safety signs. • Making sure that existing traffic can pass. • Identifying the positions of the outlets of the catch water drain with the Engineer and the owner of the property. • Demarcating dimensions and gradients of the drain to be excavated. • Marking excavation tasks. • Excavating catch water drain. • Controlling the shape, depth and gradient of the excavated drain. • Depositing the material obtained from the excavation of the drain away from the drain to avoid the same material falling or washing back into the drain. • Removing temporary signs.	
Standards	• To guarantee road safety, appropriate signs should be placed in the road. • The excavation of the drain should be carried out following the drawings and should have no irregularities. • Rocks, stones and roots in the drain profile should be removed. • The material obtained in the drain excavation should be deposited on the road formation or the area between the catch water drain and the backslope. • The surface of the drain should have a 1% gradient. • The profile of the drain shall be controlled with a template and its gradient shall be controlled with an automatic level or line level and a metric tape. • The gradient of the drain should allow free flow of water from drain into: 1. Simple discharge onto the land. 2. Water course. 3. Culvert inlet. • Preferably, the catchwater drain should discharge into the land upslope of the road or into a water course. • If the drain discharges the water onto natural terrain, the level of the drain's discharge and the level of the terrain should be the same at the link. • Discharge of water onto erodible soils should be avoided however, if this is unavoidable, the discharge area should be protected using vegetation, protection code 250, or with stone paving or stone pitching. • If the drain discharges the water onto cultivated terrain, the location should be agreed with the owner of the property. • The temporary signs shall be removed after completion of the works.	
Materials	Not applicable	
Measurement	• M³ • Measurements are taken of the volume of the excavated drain. • MAE	

Activity: Side Drains, Mitre Drains and Catch-water Drains		Code:
Opening and smoothening of the outlet drain of culverts		213
Reference Code		
Application	This code is used for the excavation and smoothening of unpaved outlet drains of culverts. This drain has the function of discharging water from the culvert away from the road. If the gradient of the culvert outlet drain is greater than 5%, then it must be paved with concrete or stone pitching (codes 230 and 231). If the gradient is greater than 10% special attention is required to prevent erosion. This may include the construction of energy dissipaters.	
Tasks	• Placing safety signs. • Making sure that existing traffic can pass. • Identifying culvert outlet drains with the Engineer and owner of the property. • Demarking dimensions and inclinations of the culvert outlet drain to be excavated. • Marking excavation tasks. • Excavating drain and smoothening its surface and slope. • Controlling the shape, depth and gradient of the excavated culvert outlet drain. • Removing temporary signs.	
Standards	• To guarantee road safety, appropriate signs should be placed in the road. • The excavations of the drain should be made according to the water flow in the area, the general characteristics of the area and environmental considerations. Determining the water flow is vital for design of the drain. • It is important that the dimension in the outlet of the culvert should be greater than the capacity of the culvert to avoid damming of the water. • The profile of the culvert outlet drain should be made following the drawings and should have no irregularities. • The mouth of the drain might be enlarged according to the Engineer's instructions if the anticipated water flow justifies outlets that are bigger. • The culvert outlet drain should have 2% minimum gradient and should be at the same level with the culvert outlet. The gradient should be uniform.	
	• The culvert outlet drain should be made to cater for 2 discharge ways: 1) Superficial discharge onto the existing land. 2) Discharge into a watercourse. • Preferably the drain should discharge into a water course. • If the culvert outlet drains the water into the natural terrain, the invert level of the drain at the discharge end should be the same as the natural ground. • The mouth of the culvert outlet drain should be bigger to disperse the water in order to avoid erosion and damming of the water. • Discharge of water in erosive soils should be avoided. If this cannot be avoided the discharge area should be protected with vegetation according to the protection code (Code 250), with stones paving and stone pitching. • The profile of the drain shall be regulated with a template and its gradient shall be regulated with an automatic level or line level and a metric tape. • The material obtained in the excavation of the drain should be transported and deposited to a predefined place indicated by the Engineer to ensure that the material cannot be washed back with the water. • The temporary signs shall be removed after completion of the works.	
Materials	Not applicable	
Measurement	• M³ • Measurement of the volume of the excavated culvert outlet drain. • MAE	

Activity: Installation of concrete culvert pipes (clevis) including excavation, foundation and backfill Diameter 60 cm Diameter 75 cm Diameter 90 cm Diameter 120 cm	Code: 220 221 222 223
Reference Code	
Application	This code is for the construction of culverts with one or more outlets using pre-fabricated concrete pipes. These codes are used only for quality control and for payment for excavating the trench, constructing the foundations, placing the culvert pipes and backfilling. The construction of walls, boxes, drains for outlets have their own codes, 213, 230, 240, 241 and 242.
Tasks	• Placing safety signs. • Making sure that existing traffic can pass. • Supplying all necessary material for the construction of slabs for the foundation, slab and placement of pipes. • Any necessary activities to divert water from the excavation and keep the excavations free of the water. • Excavating the trench for the construction of the slabs and for placing the pipes. • Compacting the bottom of the trench. • Controlling the gradient of the drain for the slab and for the pipes. • Constructing the bottom slab using concrete as indicated in the drawings. • Placing culvert pipes, Figure 9-3. • Placing and compacting concrete for the culvert surround. • Backfilling over the culvert with selected soil or gravel and compacting to the required density (minimum 93% Mod. AASHTO). • Constructing protection slabs as indicated in the drawings. • Clearing the area around the culvert • Removing the temporary signs.
Standards	• To guarantee road safety, appropriate signs should be placed in the road • The Contractor should take the necessary actions to ensure safe passage of normal traffic on the road during the placement of the pipes. • If the culverts are used to connect the side drains, partial placement of the pipes on one side of the road while the other side is open to traffic followed by construction of the other side once the completed side is ready to be trafficked is recommended in order to avoid interruption of traffic. • The culvert should be constructed according to the drawings. • The gradient and dimensions of the elements should be controlled with automatic or line level and a metric tape. • The transverse gradient should be 1.5 – 3%. • The tolerance for the maximum thickness of the pipes should be $\pm 5\text{mm}$, $\pm 10\text{mm}$ for the slab thickness and $\pm 0.5\%$ for the gradient. • If the culvert is on a <i>watercourse</i> it should be aligned to follow the natural orientation and level of the <i>watercourse</i>. • If the culvert discharges into natural ground, its outlet level should be the same as the ground. • The excavated trench should have the width and depth indicated in the drawings. • The bottom of the trench should be compacted to a minimum density of 93% Mod. AASHTO. • The slab where the pipes shall be placed shall be constructed using concrete according to codes 640 to 643. • The concrete pipes should be placed carefully to avoid damage, work accidents and to sit properly on top of the concrete slab. • The joints between the culverts should be smooth and closed with concrete to allow the transport of the water without leakage through the joints. • The backfill around the pipes can be carried out using the material excavated from the trench or gravel approved by the Engineer

	• The backfill material should be compacted in 10 cm layers to a density of 95% Mod. AASHTO and to avoid differential levels on the both sides on the pipes. • Compacting the backfill above the culvert should be carried out without vibration to avoid damaging the pipes. • The protection slab should be constructed according to the design. • The remaining material for the backfill should be placed in 15 cm layers at optimum moisture content and compacted to a minimum of 95% Mod. AASHTO. • If there is an existing gravel layer, it should be put back. • All inappropriate material shall be removed from the road reserve. • The temporary signs shall be removed after completion of the works.
Materials	All materials should be approved by the Engineer. • The pre-fabricated concrete pipes could be fabricated by the Contractor or by a supplier/provider: 1) Pipes produced by the Contractor should be fabricated using 25 MPa concrete and approved by the Engineer. 2) Pipes provided by any other supplier should follow the standard SABS 667 or equivalent and approved by the Engineer. 3) If there is need to reinforce the pipe culverts, it should be done using steel mesh (6 mm) all around the pipe. 4) The concrete pipes should have no defects that may cause damage in-service. The concrete used for the slab for paving and protection should be prepared according to the codes 640 and 643.
Measurement	• M • Measurements are taken of pipeline length. • MAE.



Placement of prefabricated concrete pipes including placement of reinforcement for wing and head walls

Figure 9-3 Construction of Precast Pipe Culverts

Activity: Diameter 80 cm Diameter 100 cm Diameter 120 cm	Installation of metallic culvert pipes (ARMCOs) including excavation, foundation and backfilling	Code: 225 226 227
Reference Code		
Application	This code is for the construction of metallic pipes with one or more inlets/outlets. These codes are used only for quality control and for payment for opening the trench, excavating the foundations, placing the pipes (metallic pipes) and backfilling. The construction of walls, boxes, drains for outlets have their own codes, 213, 230, 240, 241 and 242.	
Tasks	• Placing safety signs. • Making sure that existing traffic can pass. • Supplying all necessary material for the construction of the slab and placing of pipes. • Any necessary activities to divert the existing flow and keep the excavations free of the water. • Excavating the trench for the construction of the slab and placement of the pipes. • Compacting the bottom of the trench.	
	• Controlling the gradient of excavation for the slab and for the pipes. • Constructing the foundations with concrete as indicated in the drawings. • Mounting the metallic pieces in conformity with the instructions of the supplier. • Placing the culvert pipes on the slab. • Backfilling and compacting the area above the culvert using gravel or other approved soil. • Constructing protection slab as indicated in the drawings. • Placing, compacting and curing the concrete. • Placing gravel layer (s) on top of the slab. • Clearing the area around the culvert. • Removing the temporary signs.	
Standards	• To ensure road safety, appropriate signs should be placed in the road • The Contractor should take the necessary actions to ensure the safe passage of normal of traffic on the road during the placement of the pipes. • If the culverts are used to connect the side drains, partial placement of the pipes is recommended where part of the culvert length is constructed on one side of the road followed by the remaining length on the other side of the road when the completed side is open to traffic. • The culvert should be constructed according to the drawings. • The gradient and dimensions of the excavation should be regulated with automatic or line level and a metric tape. The transverse gradient should be 1.5 – 3%. • The tolerance should be ± 10 mm for the slab thickness and $\pm 0.5\%$ for the gradient. • If the culvert is on a <i>watercourse</i>, it should be placed to follow the natural direction of flow and level of the <i>watercourse</i>. • If the culvert discharges onto natural land, the invert level of the outlet should be the same as the existing ground. • The excavated trench should have the width and depth indicated in the drawings. • The bottom of the trench should be compacted to a minimum of 93% Mod. AASHTO. • The metallic culvert should be placed by the Contractor following the instructions of the supplier or be placed by the provider/supplier. • The concrete pipes should have no defects that might cause damage in the long term and should be placed carefully to avoid damages and work accidents. • The joints of the pipes should be smooth and closed to allow the flow of water without leakage through the joints. • The landfill around the pipes might be done using excavated material from the trench, or gravel, and should be approved by the Engineer. • The backfill material should be compacted in 10 cm layers to a minimum of 95% Mod. AASHTO and to avoid different layer levels on the both sides of the pipes.	

	• Compacting the fill above the culvert should be carried out without vibration to avoid damaging the pipes. • The protection slab should be constructed according to the drawings. • The remaining material from the backfill should be placed in 15 cm layers at optimum moisture content and compacted to a minimum of 95% Mod. AASHTO. • If there is an existing gravel layer, it should be put back. • The backfill material and height should be according to the design specifications and the backfill thickness must correspond to $\frac{1}{2}$ of the culvert diameter ($h=0.5 \times D$) • At the inlet and outlet, protection walls should be placed to avoid erosion and water returning to the culvert. • The temporary signs shall be removed after completion of the works.
Materials	All material should be approved by the Engineer. • The metallic pipes (ARMCO Superlit and MP 100 or equivalent with thickness specified in accordance to the design) in conformity with SABS 1461 and SATCC 2203 with galvanized screws and screw nut. The concrete used for the slab should be prepared according to the codes 640 and 643.
Measurement	• M • Measurements are taken of the pipeline length. • MAE

Activity: Construction of paved side drains		Code:
Concrete paved side drains (C 20)		230
Reference Code		
Application	This code is used for construction procedures for concrete paved side drains, Figure 94. Excavations of side drains and placing of concrete are included. The concrete should be prepared according to the codes 640 and 643.	
Tasks	• Placing safety signs. • Making sure that existing traffic can pass. • Excavation and leveling the bottom of the side drain. • Providing all necessary materials. • Controlling the level and shape of the side drain. • Placing formwork and mesh wire. • Concreting and curing. • Removing temporary signs.	
Standards	• works shall be executed in accordance with characteristics given in the design (dimensions, placement, production and finishing). • The maximum tolerance for the side drain measurements should be ± 20 mm thickness, ± 5 mm for the slab and ± 0.5 % for the gradient. • The drain surface should be floated and smooth and should not have irregularities of more than 2 cm. • The concrete side drain might be fabricated on the site or preferably moulded in-situ using moulding dishes. • Manual smoothening of the surface must be done through cutting, backfilling, etc. • All unsuitable material should be removed and replaced by appropriate material for the surface layer. A minimum thickness of 75 mm of the side drain's surface should be compacted. • All the excess material from the excavations should be removed and transported away from running water to a place indicated by the Engineer. • A light concrete mix or blinding should be used for the regularisation of the surface. • Wooden formwork shall be used during placement of the concrete lining of the side drain. The transversal section of the profiles should be the same as in the side drain and the spacing should be 2 m. • In joints between the profiles, polyethylene (0.15mm thick) should be placed in conformity with the design.	
	• After every 4 slabs, one expansion joint with 10 mm of polystyrene lining should be placed. Bitumen or other equivalent material approved by the Engineer should be used to seal the joints. • Where reinforcement is required, it shall be placed and fixed at a depth given in the design. • The strength of the concrete should be according to the design. • The placement and compaction of the concrete must be done using manual tools and a profile board to facilitate the shaping of the side drain (triangular or trapezoid). • The surface must be finished using a wooden straight edge in order to be uniform and be without irregularities. • If Portland cement is used, 24 hours after the batching and placement process the formers may be removed. The curing process should take 7 days. During this period all the exposed surfaces shall be kept moist. All exposed concrete surfaces shall be watered frequently after placement of sand or moistened sacks or plastic. • An expansion joint should be made for each segment with spacing of 2m, and it should be filled with worm asphaltic cement. • Protection works should be done in the terminal points of the side drain in accordance with codes 250, 251, 252, 253, 254 or 260. Stones masonry should be placed at the end to dissipate the energy of the water. • The geometrics on the side drain construction should be set out and controlled by a surveyor.	

	• All The gradient of the side drain should be more than 1 % to avoid siltation. • The side drain should allow for efficient discharge of the water without compromising the safety of the other adjacent devices and properties. • The concrete to be use should have compression strength of C20 (20 MPA after 28 days) In case of pre-fabricated slab, the following procedure must be used: • Use of wooden and plastics formwork. • The quality of concrete must be controlled through tests that should be in accordance with the requirements of the contract. • The transportation of the slabs should be carefully carried out and the Engineer must refuse the slabs with cracks. • In the joints of the side drain and in difficult terrain catchwater or energy dissipaters should be constructed (code 211)
Materials	All material should be approved by the Supervising Engineer. • The concrete used for the slab for paving should be prepared according to the codes 640 and 643. • Polystyrene 10 mm thickness. • Bitumen 80/100. • Polyethylene with 0.15 mm thickness.
Measurement	• M • MAE



Figure 9-4 Side drains lined with concrete

Activity: Construction of paved side drains: Lining of side drains using stone pitching.		Code: 231
Reference Code		
Application	This code is used for the procedure for the construction of side drains lined using stone pitching.	
Tasks	• Placing safety signs. • Making sure that existing traffic can pass. • Excavating and regularising the drain. • Cleaning the rocks before placing. • Preparing the mortar. • Bonding the stones with mortar. • Placing the stones in the gutter surface. • Smoothing the surface with impermeable material. • Removing the temporary signs.	
Standards	• All works shall be executed according to the drawings (dimensions, positioning, construction and finishing). • Regularisation of the surface shall be carried out through filling, cutting and adjusting. • All unsuitable material should be removed and replaced by appropriate material for the drain surface layer. • A minimum thickness of 75 mm of the drain's surface should be compacted. • All material spread on the surface of the drain should be compacted to a minimum of 95% of Modified AASHTO density. • The excavated material may be used for the regularisation of the drain surface. • The stone should be cleaned and free of any improper material such as clay and fixed with cement mortar mixed in proportions of 1:4 cement/sand. • First a layer of mortar of 50 mm minimum thickness should be placed on the surface of the drain. • Before the mortar gets dry the stones should be placed in the sloping side of the side drain in such way that their longitudinal axes are in the appropriate inclination. • The mortar must be placed in all verified holes and spaces on the surface. • The stone pitching should be cured by covering the surface with a humid substance such as moistened sacks for a period not less than 4 days and should be free of weight until it reaches the adequate strength. • The final layer surface of the side drain should be regular, smooth and impermeable. • The thickness of the stone pitching measured in 90° angles starting from the surface should not be less than 200 mm. • All the excess material from the excavations should be removed and transported away from the drains and watercourse to a place indicated by the Engineer. • Protection works should be done at the end points of the drain in accordance with codes 250, 251, 252, 253, 254 or 260. Stone masonry should be placed at the discharge end of the drain to cut the energy of the water. • The geometrics on the side drains construction shall be done by a surveyor. • The side drain must discharge the water in such a way that it does not compromise the safety of the other adjacent devices and properties. • The maximum tolerance for the side drain measurements should be ± 20 mm, ± 5 mm for the slab and ± 0.5 % for the gradient. • The drain surface should be floated and smooth and should not have irregularities in excess of 4 cm measured with 2 m straight edge. • The stones that are used should have dimensions less than 200 mm. • The shape of the stones should permit the stability of the layers in the required thickness.	
Materials	• All material shall be approved by the Engineer. • The cement should be Portland type and follow the SABS Standard 471 or equivalent. It should be stored in proper storage, protected from water and damp conditions and without contact with floor or walls. Cement stored for more than 8 weeks should not be used in the works. The sand and stones should be collected from the sources approved by the provincial laboratory and conform to SABS norm 1083 or equivalent and should be clean and free of clay and organic materials. • The water should be clean and free of acid, salt, sugar or any other organic material including chemicals which have the capability to weaken the quality of the concrete.	

	• The stone used should have a dimension not less than 20 cm and should have 5 to 20 kg weight, and should also be clean and tough. • Wooden (formwork) approved by the supervisor. The concrete should be mixed according to proper methods of quality control of the components, in relation to 1 unit of cement →4 units of sand. The quantity of water should only be enough to achieve adequate consistence. Water shall not be added to the mortar after 20 minutes from the time water is first added to the mix during the mixing process.
Measurement	• M • MAE

Activity: Construction of wings walls, catch pits (culvert boxes) and energy dissipaters for prefabricated culvert pipes using plain concrete (C 20).	Code: 240
Reference Code	
Application	This code is used for construction of wing walls (Figure 95), catch pits (Figure 95) and energy dissipaters in protection works of culvert inlets and for backfill. The concrete should be prepared according to the codes 640-643.
Enclosed tasks	• Placing safety signs. • Constructing a detour to guarantee the continuously passage of the traffic. • Supplying all necessary materials for the construction of wing walls, catch pits and energy dissipaters. • Setting out the foundation area. • Excavating the trench for the foundation of the wing walls, catch pits and the outlets in accordance with the drawings. • Compacting the wing wall foundation. • Applying blinding on the foundations of the wing walls. • Controlling the alignment of the walls. • Constructing the wing walls and catch pits with concrete and masonry in accordance with the design. • Compacting and curing of the concrete. • Testing the quality of concrete and mortar or, in the case of purchased items, evaluating the accompanying documentation to show compliance with the standards. • Backfilling and compacting the soils around the culvert and behind the wing walls. • Removing the detour. • Clearing the area around the catch pits (culvert boxes). • Removing the temporary signs.
Standards	• To guarantee road safety, appropriate signs shall be placed on the road. • The contractor should consider the necessary measures to guarantee the efficient passage of traffic without interruption on the road during the construction of the culverts. It might involve using a detour. • The construction should follow the form indicated in the designs. • The dimension and shape of the outlet and inlet boxes should be adjusted to allow the creation of a proper link with the drain and water channel. • The maxim tolerance accepted for the dimension of the walls and boxes is ± 10 mm, and for the thickness it is ± 20 mm. • The concrete should be mixed according to the method specifications. • The quantity of water should only be enough for adequate consistency. No more water should be added beyond 20 minutes after the first addition of water to the mix. • The concrete should be applied in less than 60 minutes after water is added. • The foundations for the walls must be excavated according to the drawings. • The foundation surface must be compacted to a minimum of 93% Mod. AASHTO. • Installing of the wooden formwork shall be according to the dimensions and drawings. • working area clean and firm. • Batching and mixing of the concrete (in a concrete mixer) shall be in accordance with the recommendations of the Engineer. • Slump tests shall be used to control the mix proportions. • Additional water shall be applied to make the mix more workable where necessary. • Only the quantity required for the work shall be mixed. • Concrete shall be placed in the formwork box and compact until all air bubbles have been removed. • A blinding of up to 5 cm of fine concrete 1:4:8 shall be applied to leave the The alignment of the wall shall be controlled using a plump and level. • The wall shall be constructed to the correct height. • If there is a need to drain water, drainage pipes of 6 to 8 mm diameter shall be placed through the concrete.

	- The concrete shall be covered completely and kept moist for adequate curing and to avoid shrinkage cracking. - Temporary signs shall be removed after completion of the works.
Materials	All material should be approved by the Engineer. The concrete for the boxes and walls shall be prepared in accordance with codes 640-643.
Measurement	- M³ of concrete - MAE



Wing walls should be constructed at wider angles to prevent water vortex which cause scour behind the walls (See specifications in Appendix)



Catch pits are prevent scour particularly when there is drop from the drain invert to the culvert invert

Figure 9-5 Wing walls and catch pit to collect water into the culvert

Activity: Construction of wing walls, catch pits and energy dissipaters for precast culvert pipes using reinforced stone masonry		Code: 241
Reference Code		
Application	This code is used for the construction of wing walls (Figure 96), catch pits and energy dissipaters in protection works for culvert inlets/outlets and for backfill.	
Tasks	• Placing safety signs. • Constructing a detour to guarantee the continuous passage of the traffic. • Supplying all necessary materials for the construction of walls, culvert boxes and energy dissipaters. • Setting out the foundation area. • Excavating the trench for the foundation of the wing walls, catch pits and the outlets according to the designs. • Compacting the wing wall foundation. • Applying blinding concrete for the wing walls. • Controlling the alignment of the walls. • Constructing the wing wall, inlet and outlet boxes using stone masonry in accordance with the drawings. • Backfilling and compacting soils above the culvert and behind the walls. • Closing the detour and reinstating the land. • Clearing the area around the culverts. • Removing the temporary signs.	
Standards	• To guarantee road safety, appropriate signs shall be placed on the road. • The contractor should consider the necessary measures to guarantee the efficient passage of traffic without interruption on the road during the construction of the culverts. It might involve using a detour. • The construction should follow the form indicated in the designs. • Place the bigger stones deeper in the blinding (or blinding concrete). • The smaller spaces should be filled using smaller stones and mortar. • Each stone should be fixed firmly. However, gaps shall be left between the stones and filled in with mortar. • During construction it is recommended to start with the edges of the walls followed by the middle part. • When constructing long walls build in sections of 7 to 10 m to facilitate construction. If necessary construct a column between the sections. • The stones must be placed horizontally to ensure stability. • The stones should be placed closely in order to minimize the empty spaces between them. The places may be fulfilled with small stones and concrete. • The mortar should be placed under supervision and must have a smooth finishing. • The mortar and concrete must be kept moist for 4 days after construction. • The dimension should be controlled using metric - tape, automatic or line level and string. • The maximum tolerance accepted for the dimension of the walls and catch pits is ± 10 mm for the thickness and ± 20 mm for length and height. • The surface of the walls and boxes should be smooth and free of irregularities bigger than 2 cm. • Removal of all unsuitable materials from the reserved area for the road to a place indicated by the Engineer. • The maximum stone size for the concrete should be 38 mm. • Actions should be taken to protect the inlets and the outlets of the culverts. • Outlets should be at least 1 m away from the road and water should never be discharged onto unprotected area or unstable slopes. • Protection and wing walls must be constructed to protect the area from the erosion and collapse of the outlets and inlets. • The outlets should be protected with stones. • Temporary signs shall be removed after completion of the works.	

Materials	• The cement should be Portland type and follow the SABS norm 471 or equivalent; it should be stored in proper storage, protected from water and damp conditions and avoid contact with floor or walls. Cement stored for more than 8 weeks should not be used in the works. The sand and stones should be collected from the sources approved by the provincial laboratory and should meet SABS norm 1083 or equivalent and should be clean and free of clay and organic materials. • The water should be clean and free of acid, salt, sugar or any other organic material/chemical which may affect the quality of the concrete. • The stones should have dimensions not less than 20 cm and should have 5 to 20 kg weight, and should also be clean and tough. The concrete should be mixed according to proper methods of quality control of the components, in relation to 1 unit of cement →4 units of sand. The quantity of water should only be enough to achieve adequate consistence. No additional water should be added to the mortar or concrete beyond 20 minutes from the time water is first applied to the mix.
Measurement	• M³ of concrete • MAE



Figure 9-6 Stone masonry wing walls and catch pits

Activity: Construction of wing walls, catch pits, including energy dissipaters for the prefabricated culverts using reinforced concrete including formwork (C20)		Code: 242
Reference Code		
Application	This code is used for the construction of wing walls, catch pits and energy dissipaters for the protection works for inlets/outlets and backfill for culverts built using reinforced concrete. The concrete should be prepared according to codes 640 – 643.	
Tasks	• Placing safety signs. • Construction of detour to guarantee the continuous and safe passage of normal traffic. • Supply of all necessary materials for the construction of walls, catch pits and energy dissipaters. • Setting out the alignment of the foundation area. • Excavating the trench for the foundation of the wing walls, catch pits and the outlets according to the drawings. • Compacting the foundations. • Applying blinding for the protection walls. • Controlling the alignment of the walls. • Constructing the wing walls, inlet and outlet concrete catch pits using concrete and stone masonry in accordance with to the designs. • Placing, compaction and curing of the concrete and stone masonry. • Testing the quality of concrete and mortar, assuming that the purchased units are delivered with the documentation to show their compliance to the standards. • Backfilling and compacting the soils around the culvert and behind the wing walls. • Closing the detour and reinstating the land. • Clearing of the area around the culverts. • Placing adequate signs to guarantee the safety of the road. • Removing the temporary signs.	
Standards	• The contractor should consider the necessary measures to guarantee the efficient passage of traffic without interruption on the road during the construction of the culverts. It might involve using a detour. • The construction should follow the form indicated in the designs. • The dimension should be controlled using a metric tape and an automatic or line level. • The dimensions and shape of the outlet and inlet boxes should be adjusted to permit the creation of a smooth link with drain and water channel. • The maximum tolerance accepted for the dimensions of the walls and boxes is ± 10 mm for the thickness and ± 20 mm for the length and height. • The concrete should be mixed according to approved methods for quality control. • The quantity of water should only be enough to ensure adequate consistence. No water should be added after 20mins from the time water is first added into the mix. • The concrete should be applied in less than 60 min after the water is added. • Formwork should be placed according to the dimensions and drawings. • Spread up to 5 cm of finishing concrete 1:4:8 to leave the batching area clean and firm. • Use batching boxes for the control of the mixing proportions. • Add water to make the mix workable. • Mix only the concrete required for the work. • Add the concrete mix in the box and compact to remove cavities and air bubbles.	
	• Control the alignment of the wall using a lead wire or string and level. • Complete the walls to the required height. • In case there is a need to drain water, place drainage pipes of 6 to 8 mm diameter. • Completely cover the concrete and keep it moist in order to achieve adequate curing and to avoid shrinkage cracking. • Remove all unsuitable materials from the reserved area of the road to a place indicated by the Supervising Engineer. • Place adequate signs to guarantee the safety of the road. • Remove all temporary signs	
Materials	• All material should be approved by the Supervising Engineer. • The concrete for the walls and culvert should be prepared according to the codes 640 and 643; • The steel formwork should be prepared according to codes 630 and 632.	
Measurement	• M³ of reinforced concrete • MAE	

Activity: Protection works		Code:
Protection using vegetation (grass, sort grass or vetiver)		250
Reference Code		
Application	This code is used for the protection works with vegetation of the areas susceptible to erosion; it can be with local material or specially cultivated plants (e.g. vetiver grass). This work should be executed in the critical zones of the slopes, side drains, mitre drains and road reserve areas.	
Tasks	• Placing safety signs. • Developing local sources of production of plants to promote and conserve vegetation for the control of erosion. • Analyzing the local climate and micro-climate, soils and fertilization necessities. • Defining the kind of vegetation to plant. • Setting out of the work area. • Clearing and regularising the area where the grass should be planted. • Defining cultivation methods and transplantation. • Preparing the holes for planting. • Planting. • Irrigating the plants daily for one month. • Removing temporary signs.	
Standards	• Adequate signs should be placed to guarantee road safety. • Actions of erosion control should be taken (planting of vegetation) before the rainy season and should be carried out just after the construction, periodic maintenance or rehabilitation of the road. • Mark the work area in the critical zone of erosion in the slopes and embankments, drains, gullies and the reserved area for the road. • Protection alternatives should be selected (planting with gram-grass, vetiver grass). • The vegetation should be selected according to the growing properties, resistance, cover capacity, type, and depth of roots (for stabilization of the soil). • Native vegetation with low growth rate may be used. It should be appropriate for the local weather and soil. • If there is no source of vegetation for transplantation, nurseries should be prepared under the supervision of an agronomist to avoid improper vegetation being established or planted. Vetiver grass is usually recommended. • The plants should be transplanted from the farms without affecting of the roots.	
	• The material should be prepared in plastic bags for safe transportation and planted in the areas susceptible to erosion, previously selected. • If the soils are poor, lacking in nutrients use fertilizer to promote the efficient development of the vegetation. Add water and irrigate as necessary. • The grass must be free of any noxious elements or disease. • The area should be cleared of any other grass. • The area where the grass is to be planted must be moist to at least 15 cm depth before the grass is planted. • The planting must be manual and the spacing between the plants must be 10 to 15 cm; • Initially the prepared holes for planting must be 10 cm deep and in lines. • The plants must also be planted to 10 cm depth • Each line should have the minimum length of 10 m and spacing between 50cm and 100 cm between the lines. The lines should follow natural contours.	
	• Planting of the vegetation should be followed immediately by the irrigation mentioned above. • The biotechnical methods such as vegetation cover are affordable and do not require much additional support in the long term. • Conserve the vegetation cover with organic material. • Remove temporary signs.	
Materials	• All materials should be approved by the Supervising Engineer • Vegetation material or vetiver grass preferentially.	
Measurement	• M² • Measurements are taken in the area where vegetation material has been planted. • MAE	

Activity: Protection of erosion prone slopes		Code:
Protection works using dry stone pitching		251
Reference Code		
Application	This code is used for the protection of the slopes using stone pitching without mortar. It is to protect localized areas that are prone to erosion such as cut slopes, embankments and inlets and outlets of culverts and drains.	
Tasks	• Placing safety signs • Marking out the working area. • Removing vegetation from the work area if necessary. • Cutting or filling the slopes to regularize the surfaces. • Regularisation and compacting the surface. • Supplying and transporting the stones. • Placing and leveling the stones. • Filling with soils and compacting the space between the stones. • Removing all improper materials from the road area. • Removing the temporary signs.	
Standards	• Adequate signs should be placed to guarantee road safety. • The area for the protection works shall be prepared properly including leveling and compacting with hand rammers or other appropriate means. • The dimensions should be controlled using metric tape and spirit level. • The maximum tolerance acceptable for the dimensions of the protection is ± 20 mm for the thickness and ± 50mm for the length.	
	• The created settlement surface should be smooth and plain and with irregularities less than 5 cm. • The layer of stone should be 20 cm minimum thickness. • The stones should be placed in such way that the space between them is the minimum possible and the irregular side should be face down. • The spaces between the stones should be filled with soil. If possible, organic soil should be used for filling the top 10 cm. Small stones might be used to fill the spaces between the bigger stones. • After the soil has been watered and compacted any depressions and empty spaces that still exist must be filled. • On the inlets and outlets of culverts place the stones between 15 and 30 cm in size and staggered to avoid the formation of single files or straight lines and protect the area around the stone pitching with vegetation. Flat stones shall be used for this purpose. • It is recommended that the stones are placed up to a minimum of 1.5 m from the outlet. • In the extreme end of drains where the water is discharged, walls of 45 cm depth and 30 cm width should be constructed for protection against scour. • Remove of all excess or unsuitable materials from the road reserve area of the road. • Remove temporary signs.	
Materials	• All material should be approved by the Engineer. • The handpacked stone should be 15 cm minimum sized, with 10 to 20 kg weight and should be clean and resistant.	
Measurement	• M² • Measurements are taken on stone pitched area. • MAE	

Activity: Protection works		Code:
Protection using wet stone pitching (mortar stones)		252
Reference Code		
Application	This code is used for protection works using wet stone pitching (Figure 97) and masonry for localized areas prone to erosion. Use this code for energy dissipaters, lined drains, swampy passages, protection structures for culvert outlets and protection of embankments, protection structures for the discharge of water in culverts and for protection of slopes.	
Tasks	• Placing safety signs • Setting out the work area. • Removing vegetation from the working area if necessary. • Cutting and filling the surfaces to level. • Leveling and compacting of surfaces that are prone to settlement. • Supplying and transporting the stones. • Placing and leveling the stones bonded by cement mortar. • Curing the mortar. • Testing the quality of the materials and mortar to ensure compliance with the standards. • Removing the temporary signs.	
Standards	• Prepare the area for the construction of lining and compact to a minimum density of 93% Mod. AAHSTO where possible. • The mortar should be applied within 60 min after the water is first added during the mixing process. Mix only the quantity of mortar that can be applied within the stipulated time (within 60min from the time water is first added). • Place the stones while controlling the leveling. • To allow adequate curing, the masonry should be kept moist for 3 days after construction.	
	• The layer of stone should be 20 cm minimum thickness. • The dimensions should be controlled using a metric tape and level. • The maximum tolerance acceptable for the dimensions of the protection is ± 10 mm for the thickness and ± 20mm for the length. • The surface of the stone pitching and masonry should be smooth and plain and with irregularities of less than 4 cm measured with a 2 m straight edge. • The stones must be placed in a layer of mortar. The spaces around the stones must be filled using mortar. Small stones might be used to fill the spaced between the bigger stones. • The stones must be overlapped horizontally and vertically on masonry walls to avoid placing stone in defined lines or single files to prevent creating lines of weakness. • After applying the mortar the joints must be pointed. Pointing is the construction a ridge of cement mortar on the surface of the walls along the joints between adjacent stones. • At the end of structures where water is discharged, protection walls of 45 cm depth and 30 cm width should be constructed. • If there is a need of repairing existing walls, work should be executed on structures that are structurally sound and structures in degraded condition should not be considered for filling cavities with mortar. • The deformed joints should be cleaned using appropriate tools. • In places where the joint must be renewed, the stone must be removed completely and replaced with a new mortared layer. • Mortar should be applied to moist surfaces. It is recommended that mortar is prepared in proportions of 1:4 ratio of cement to sand.	
	• Apply fresh mortar in the joints and fill completely. • Finishing should be done using appropriate tools on areas with tight curves. • The final surface of the applied mortar should be lower than the masonry surface, for aesthetic reasons. • During the dry season the mortar might dry quickly. It is necessary to keep the masonry moist until the mortar is fully cured by covering the area with wet bags or any other material. • For aesthetic reasons, clean the stone surfaces to remove the mortar. • Clean all the adjacent area of the work and remove the rubbish from the area reserved for the road (empty cement bags should be placed in a proper place). • Remove temporary signs.	

Materials	• All material should be approved by the Engineer. • The cement should be Portland type and follow the SABS standard 471 or equivalent. It should be stored in proper storage protected from water and damp conditions and without contact with floor or walls. Cement stored for more than 8 weeks should not be used. The sand and stones should be collected from the sources approved by the provincial laboratory and should meet with SABS norm 1083 or equivalent and should be clean and free of clay and organic materials. • The water should be clean and free of acid, salt, sugar or any other organic material/chemicals which could affect the quality of the concrete. • The stone should have dimensions not less than 20 cm and should have 5 to 20 kg weight and should also be clean and tough. • The mortar should be mixed in accordance with approved methods of quality control of the components, in relation 1 unit of cement →4 units of sand. The quantity of water should only be enough to achieve adequate consistence. No water should be added to the mortar after 20 min from the time water is first added to the mix.
Measurement	• M² • The volume of the stone pitching or masonry taken after construction. • MAE.



Figure 9-7 Embankment and slope protection using wet stone pitching (with cement mortar)

Activity: Protection works		Code:
Protection using sandbags		253
Reference Code		
Application	This standard defines procedures for the protection of the slopes, outlets and drains subject to erosion, using sandbags.	
Tasks	• Placing safety signs. • Setting out work area. • Removing vegetation from the work area if necessary. • Cutting and filling earthworks where necessary. • Leveling and compacting surfaces which are prone to settlement. • Selecting the required type of sand, stockpiling and washing. • Filling the sandbags. • Placing the sandbags. • Fixing the sandbags together using two strands of wire, 3 mm. • Cleaning the work area. • Removing all unsuitable materials from the road area. • Removing the temporary signs.	
Standards	• Adequate signs should be placed to guarantee road safety. • The work area should be properly prepared including opening, leveling and compacting the foundation where the sandbags will be placed. • The excavation of the foundation has to reach natural soils with adequate support capacity. If this is not possible suitable soil should be imported and placed to form a firm foundation for the sand bags. • The sand used for the filling of the bags should be clean. • The bags must be filled completely, closed and tied using two strands of wire, 3 mm diameter. • The bags should be organized to accomplish the desired alignment and should be placed as bricks with one bag closing the joints in the lower layer. • All the corners should be tied using two strands of wire, 3 mm diameter. • The alignment of the bags should be controlled using metric tape and level. • Clean all the adjacent area of the work and remove the rubbish from the area reserved for the road (empty cement bags should be placed in proper place). • It is recommended that this work be executed in dry season.	
Materials	• All material should be approved by the Engineer. • The sand has to be obtained from a source approved by the provincial laboratory and should be clean and not contain organic or clay materials. The grading of the sand should permit the infiltration of the water. • The bags must be synthetic, clean, impermeable and free of any noxious elements • The tying wire should be 3 mm diameter.	
Measurement	• M² • Measures are taken from the number of placed sandbags. • MAE.	

Activity: Protection works		Code:
Planting trees in the areas susceptible to erosion		254
Reference Code		
Application	This code describes procedures to be taken for the process of planting of trees in the area susceptible to erosion.	
Tasks	• Placing safety signs. • Clearing the area where the trees are to be planted. • Transporting the plants and obtaining approval from the Supervising Engineer before planting. • Making the holes for planting the trees. • Watering the holes before and after planting the trees. • Adding fertilizer, if necessary. • Protecting the trees with stakes. • Removing the temporary signs.	
Standards	• Adequate signs should be placed to guarantee road safety. • The kind of plants to be used must be in compliance with the Engineer's instructions. • The contractor should ensure that the plants are free of any disease and are in good condition. • Each plant should be carefully transported to the planting place. The vehicle used for transport should be equipped to protect the plants from wind and sunlight. • The area where the trees are to be planted should be cleared. • The trees should be placed in the area indicated in the project designs. • Holes should be opened, square or circular and a bit bigger than the diameter of the tree (normally 40 to 50 cm diameter and 50 to 60 cm depth). The spacing between the trees should be of 5 to 10 m. • The holes must be filled using selected soils and, depending on the soils, application of recommended fertilizers might be necessary. • Before removing a tree from the transport vessel, it should be well watered. • The holes must be watered before planting the trees. If the soil is not permeable, the hole must be filled to a depth of 15 cm using gravel before adding adequate soil to fill up the hole. • After the trees are planted in the correct places, they should be constantly watered to ensure the settlement of the soil. • After the settlement of the soil, additional soil should be added to a depth of 15 cm below ground level. • All trees must be tied to an adequate wooden stake fixed in the ground and protected with protecting stakes all round.	
	• The stake should be minimum 3.5 cm diameter and 30 cm longer than the tree and the maximum length should be 1.5 m from the soil surface. • After the planting process, the surface must be covered with grass to minimize evaporation. • The area should be irrigated daily for a period of 60 days.	
Materials	• All material should be approved by the Supervising Engineer. • The trees must be 2 to 3 years age (up to 1.5 m tall). • The roots must have no indication of deterioration. • The contractor should ensure that the plants are free of any disease and are in good condition. • Each plant should be carefully transported to the planting place. • The stakes must be minimum 3.5 cm diameter and should be 30 cm longer than the tree and the maximum length has to be 1.5 m from the soil surface.	
Measurement	• UN • MAE	

Activity: stones	Supplying, preparation and placement of gabions containing	Code: 260																						
Reference Code																								
Application	This code is used for the construction of gabion walls and aprons used in retaining walls, lining channels, revetments, energy dissipaters, lined drains, wet drifts and other anti-erosion structures.																							
Tasks	• Placing safety signs • Setting out work area. • Removing vegetation from the work area as necessary. • Selecting usable material. • Transporting the gabions. • Cutting and filling earthworks to level. • Leveling the foundations for the placement of the gabions. • Placing the gabion mesh. • Filling the gabions with stones. • Verifying the dimensions, stability and quality of the gabions. • Removing improper and excessive material away from the road reserve. • Removing the temporary signs.																							
Standards	• Adequate signs should be placed to guarantee road safety. • Plan the activity to be executed considering all factors that might influence and change the stability of the soil. Verify the drainage characteristics; control the usage of the water and definition of the kind of structure to be constructed. • It is recommended that this work be executed in the dry season. • Initially the foundation should be prepared. The foundations shall be excavated and aligned to the required level. If necessary good soil, stones or gravel shall be placed and compacted to a minimum of 93% modified AASHTO. • The gabions are fabricated using galvanized steel mesh filled with small rocks and stones in pre-defined sizes. • The gabion boxes are supplied and transported in pieces so that the load has the least volume possible. The standard dimensions of gabions are shown in Table 9-5. Table 9-5 Standard dimensions of gabions<table><tr><th>Gabion box</th><th>Standard dimensions</th></tr><tr><td>Length</td><td>1, 2, 3 or 4 m</td></tr><tr><td>Width</td><td>1.0 m</td></tr><tr><td>Height</td><td>0.3, 0.5 or 1.0 m</td></tr><tr><td colspan="2">Diaphragm (Cells Separation) = 1.0 m</td></tr><tr><td></td><td></td></tr><tr><td>Gabion mattress</td><td></td></tr><tr><td>Length</td><td>6 m</td></tr><tr><td>Width</td><td>2.0 m</td></tr><tr><td>Height</td><td>0.2, 0.3 to 0.5 m</td></tr><tr><td colspan="2">Diaphragm (Cells Separation) = 0.6 to 1.0 m</td></tr></table> • The maximum tolerance acceptable for the dimensions of the gabions is ± 10% for the length, ± 5 % for the width and ± 0.5 % for the height. • Place the gabions in the desired positions and verify the height and alignment of the gabions before filling with stones. • Filling up should be by hand and the stones should be placed carefully. • Only clean and strong stones must be used. In this way the gabions will be stable, with little or no internal sliding effects. • All the ends of the gabion meshes must be fixed together. • The dimensions should be controlled by using metric tape and spirit level. • The boxes must be placed together using 2.2 mm double intertwined wire along the corners at every 15 cm. • The tying wire should be stretched using proper tools (pliers) and tied with double intertwining strands.		Gabion box	Standard dimensions	Length	1, 2, 3 or 4 m	Width	1.0 m	Height	0.3, 0.5 or 1.0 m	Diaphragm (Cells Separation) = 1.0 m				Gabion mattress		Length	6 m	Width	2.0 m	Height	0.2, 0.3 to 0.5 m	Diaphragm (Cells Separation) = 0.6 to 1.0 m	
Gabion box	Standard dimensions																							
Length	1, 2, 3 or 4 m																							
Width	1.0 m																							
Height	0.3, 0.5 or 1.0 m																							
Diaphragm (Cells Separation) = 1.0 m																								
Gabion mattress																								
Length	6 m																							
Width	2.0 m																							
Height	0.2, 0.3 to 0.5 m																							
Diaphragm (Cells Separation) = 0.6 to 1.0 m																								

	• The gabions in the centre shall be filled first so that they support the others. • The boxes shall be moored and tied with wire and clamps to the desired shape. • Boxes with 1 m height must be filled 1/3 height. • The wire for the horizontal reinforcement must be fixed and tied with reel to maintain the vertical side plump and for it to be free of slack. • The stones must be arranged carefully leaving approximately 3 to 5 cm space from the walls of the box to allow for space for the fixing process. • The stones must be arranged by hand placing the bigger stones in the lower part. • Smaller material might be used to fill the spaces in the top surface. • The covers of gabion mesh are then placed (over the stones). • The corners should be fixed first to ensure that the mesh covers fit well to cover the top of the box. • The cover is then placed and interlaced safely on the top of the walls. • Clean all the adjacent area and remove the rubbish from the area reserved for the road. • Remove temporary signs.																	
Materials	• All material should be approved by the Supervising Engineer. • The gabions are constructed with galvanized steel mesh in conformity with SABS 675 or equivalent and with a minimum thickness of 3 mm and 2.2 mm for ties. • The stone should be clean and tough and not degraded, and must be supplied by a source approved by the Provincial Laboratory according to SABC 1083 or equivalent. • The Table 96below gives the size of the stones in relation to the box dimensions. Table 9-6 Stone sizes for gabions <table><tr><th rowspan="2">Depth of the Box (m)</th><th colspan="2">Dimension of the stone</th></tr><tr><th>Medium Size mm</th><th>Maximum Size (mm)</th></tr><tr><td>0.2</td><td>125</td><td>150</td></tr><tr><td>0.3</td><td>125</td><td>200</td></tr><tr><td>0.5</td><td>125</td><td>250</td></tr><tr><td>1.0</td><td>125</td><td>250</td></tr></table> Stones greater than the maximum sizes indicated in the sheet should not be used. At least 85% of the stones must be greater that the medium size indicated in the sheet.	Depth of the Box (m)	Dimension of the stone		Medium Size mm	Maximum Size (mm)	0.2	125	150	0.3	125	200	0.5	125	250	1.0	125	250
Depth of the Box (m)	Dimension of the stone																	
	Medium Size mm	Maximum Size (mm)																
0.2	125	150																
0.3	125	200																
0.5	125	250																
1.0	125	250																
Measurement	• M³ • Measurements are taken of the volume of the gabions. • MAE																	

Activity:	Construction of box culvert made of Reinforced Concrete Type A - 0.8 x 0.6 (0.48 M²) Type B - 1.2 x 0.8 (0.96 M²) Type C - 1.5 x 1.2 (1.80 M²)	Code: 270 271 272
Reference Code		
Application	This code is used for the construction of box culverts made of reinforced concrete including excavation, foundations, walls, roadway, formwork and protection works according to the design. All codes are used for the respective elements of box culvert.	
Tasks	• Placing safety signs • Constructing detour to guarantee the continuous and safe passage of normal traffic. • Supplying and transporting all necessary materials for the construction of the box culvert, Figure 98. • Confirming the alignment with the Engineer. • Excavating the trench for the construction of the culvert, foundation of the wing walls, inlet and outlet boxes, and outlet channel. • Controlling the gradient of the pipes and other elements of the box culvert. • Placing the formwork. • Placing and fixing the reinforcement in conformity with design. • Constructing the box culvert, wing walls, inlet and outlet boxes (catch pits), and outlet channel in reinforced concrete as indicated in the design. • Placing, compacting and curing of the concrete. • Backfilling and compacting the soils around the culvert boxes and behind the wing walls. • Closing the detour followed by rehabilitation of the land where the detour was built. • Clearing the area around the box culvert. • Removing temporary signs.	
Standards	Preparation • Adequate signs should be placed to guarantee road safety. The contractor should consider the necessary measures to guarantee the efficient passage of traffic without interruption on the road during the construction of the culvert. The traffic might use a detour or any other means. • In case of relief culverts (relieving flow in the side drains), it is recommended that partial construction of the box culvert is carried out on one side of the roadway to begin with and the other side constructed when the first side is complete to ensure that the traffic is not interrupted. • The box culvert shall be constructed according to the design. Alignment and Dimensions • The alignment, characteristics and dimensions of the box culvert shall be set out in conformity with the design. • The maximum tolerance acceptable for the dimensions of the waterway is ± 20 mm (walls and boxes), ± 10 mm (slab) and ± 0.5 % (gradient). • If the culvert is along a watercourse it must be aligned in such way that it follows as much as possible the direction of flow of the water. • The transverse gradient should be between 2 % and 5 %. • The culvert outlet shall be built to discharge water onto a convenient place. It might be into a watercourse or surface discharge onto an existing terrain. • Preferably the culvert should discharge into a watercourse. Foundations • After the foundation box is excavated, the base should be regularized and compacted using manual compactors (rammers) or a vibrating plate compactor. • The gradient of the foundation must be between 2 and 5 %; • In saturated conditions water should be pumped out to ensure dry conditions. • If the soil is not stable a riprap should be constructed according to the drawings and the Engineer's instructions. Formwork • The formwork might be constructed using metallic plates or wooden boards. • The formwork must be clean and with regular surfaces. • The formwork will be constructed in different steps, foundation, walls and roadway.	

	• Before the placement of the reinforcement the formwork structure should be approved by the Engineer. • See codes 640, 641, 642 and 643 regarding the formwork standards. Reinforcement • The reinforcement and reinforcement schedule shall be according to the drawings. • All elements of the culvert, base, support wall, front walls, wing walls and the roadway should be connected through the links (reinforcement bars linking different component of the structure). • The reinforcement must be tied using wire to create a framework strong enough to support its own weight and the weight of the workers during the placement of concrete.
	• Concrete spacers should be used to ensure adequate concrete cover over the reinforcement. • The Engineer should approve steel-fixing before pouring concrete. • See codes 630, 631, 632, 633 for reinforcement standards for provision, preparation, setting and steel fixing. Concrete • All elements of the culvert must be constructed using reinforced concrete. • See codes 640, 641, 642 and 643 for the concrete standards.
	Backfill • The box culvert must be protected with a gravel layer with minimum thickness of 15 cm (see code 340). • The backfill shall be compacted to a minimum of 93% modified AASHTO at OMC. Protection • Protection works against erosion must be built around the box culvert in conformity with design. • If the culvert discharges onto natural land the outlet should be at the same level the natural ground. • The outlet ditch must be large where it links with the natural terrain to allow the water to spread thus reducing the water energy. • If the culvert discharges into cultivated area, the outlet position should be decided with the owner of the property. Finishing • Unsuitable material shall be removed away from the road reserve. • Remove temporary signs.
Materials	• All material should be approved by the Supervising Engineer. • The concrete for the slab should be prepared in conformity with codes 640 to 643. • The reinforcement should be prepared in conformity with codes 630 to 632. • The steel bars must be prepared in accordance with the SABS 920 standards or equivalent. For each lot of steel bars delivered to the site, the contractor shall present the supplier's certificate with all details and technical specifications. The reinforcement bars and steel quality should be in conformity with drawings and SABS 82 (using symbol R or Y) • The cement should be Portland type and follow the SABS standard 471 or equivalent. It should be stored in proper storage, protected from water and damp conditions and without contact with floors or walls. Cement stored for more than 8 weeks should not be used for the works. The sand and stones should be collected from the sources approved by the provincial laboratory and follow SABS standard 1083 or equivalent and should also be clean and free of clay and organic materials. • The water should be clean and free of acid, salt, sugar or any other organic material/chemical which could affect the quality of the concrete. • The concrete should be mixed according to proper methods of quality control of the components, in proportions 1:3:4 for cement/sand/concrete stone unless specified. The quantity of water should only be enough to ensure adequate consistency. • The minimum thickness of wire should be 1.25 mm.
Measurement	• M • Measurements are taken of the total length of the box culvert and included in the code is concrete, formwork, reinforcement, walls and protection works. • MAE



Walls built using stone masonry and slab built of reinforced concrete



Walls and slabs built using reinforced concrete

Figure 9-8 Box Culvert built using Concrete and Stone Masonry

Activity: Construction of drift (climate resilient structure)		Code: 274
Reference Code		
Application	This code is used for construction of drifts made of reinforced concrete or stone masonry including excavation, foundations, key walls, approach slabs, roadway, formwork and protection according to the design.	
Tasks	• Placing safety signs • Constructing detour to guarantee the continuously and safe passage of normal traffic. • Supplying and transporting all necessary materials for the construction of the causeway. • Confirming the alignment with the Supervising Engineer. • Excavating the trench for the construction of the foundation, the side walls, inlet apron and key wall and outlet apron and approach slabs. • Controlling the gradient of the drift and approach slabs. • Placing the formwork. • Placing and fixing the reinforcement in conformity with design. • Constructing the foundations, slab, inlet apron and key wall, outlet apron, approach slabs using concrete and stone masonry and stone pitching as indicated in the design. • Placing, compacting and curing the concrete. • Constructing stone masonry and stone pitching elements as indicated in the design. • Backfilling and compacting the soils under the approach slabs. • Closing the detour followed by rehabilitation of the land where the detour was built. • Clearing the area around the causeway. • Removing temporary signs.	
Standards	Preparation • Adequate signs should be placed to guarantee road safety. • The contractor should consider the necessary measures to guarantee the efficient passage of traffic without interruption on the road during the construction of the culvert. The traffic might use a detour or any other means. • The drift shall be constructed according to the design. Alignment and Dimensions • The alignment and dimensions of the drift should be set out in conformity with design. • The maximum tolerance acceptable for the dimensions of the drift is ± 20 mm (walls and boxes), ± 10 mm (slab) and ± 0.5 % (gradient). • The drift must be aligned in such way that it is perpendicular to the direction of flow of the water. • The transverse gradient should be between 2 % and 5 %. Foundations • After the foundation box is excavated, the base should be regularized and compacted using manual compactors or a vibrating plate. • The gradient of the foundation must be between 2 and 5 %; • In saturated conditions water should be pumped out to ensure dry conditions. • If the soil is not stable a riprap upstream and downstream of the drift should be constructed according to the drawings and Supervising Engineer's instructions. Formwork • The formwork might be constructed using metallic plates or wooden boards. • The formwork must be clean and with regular surfaces. • The formwork will be constructed in different steps, foundation, walls and roadway. • Before the placement of the reinforcement the formwork structure should be approved by the Supervising Engineer. • See codes 640, 641, 642 and 643 about the formwork standards.	

	Reinforcement • The reinforcement and reinforcement schedule shall be according to the drawings. • Wire mesh of 6 mm bars welded at 200mm spacing shall be used for the reinforcement of the slabs. • If the foundation of the drift is on a flat rock bed, holes should be drilled and dowels (20 mm to 30 mm diameter x 0.5 m length) fixed in the rock with grout (cement paste) to hold the structure in place preventing wash away during flooding. The dowels should have a length of 0.3 m embedded in the rock. • Concrete spacers should be used to ensure adequate concrete cover over the reinforcement. • The Supervising Engineer should approve steel-fixing before pouring concrete. • See codes 630, 631, 632, 633 for reinforcement standards for provisioning, preparation, setting, application of wire mesh. Concrete • All load carrying elements of the causeway must be constructed using concrete reinforced with wire mesh. The concrete shall be C20. • See codes 640, 641, 642 and 643 about the concrete standards. Backfill • The gravel layer under the approach slabs shall be 15 cm (see code 340). • The gravel layer shall be compacted to a minimum of 93% modified AASHTO at OMC Protection • Drifts are designed to overflow and the downstream side of the drift is prone to scouring due to the vortex that is created. • Protection against erosion must be built on the drift in conformity with design. • The outlet should be at the same level as the natural ground or level of the watercourse. • The outlet ditch must be large where it links with the natural terrain to allow the water to spread thus reducing the water energy. Finishing • All unsuitable material shall be removed away from the reserved area for the road. • Remove temporary signs.
Materials	• All material should be approved by the Supervising Engineer. • The concrete for the slab should be prepared in conformity with codes 640 to 643. • The reinforcement should be prepared in conformity with codes 630 to 632. • The cement should be Portland type and follow the SABS standard 471 or equivalent; it should be stored in proper storage, protected from water and damp conditions and without contact with floors or walls. Cement stored for more than 8 weeks should not be used for the work. The sand and stones should be collected from the sources approved by the provincial laboratory and follow with SABS standard 1083 or equivalent and should also be clean and free of clay and organic materials. • The water should be clean and free of acid, salt, sugar or any other organic material/chemical which could affect the quality of the concrete. • The stone for masonry should have dimension less than 20 cm and should have 5 to 20 kg weight, and should be clean and tough. • The concrete should be mixed according to proper methods of quality control of the components, in relation to 1:3:4 for cement/sand/concrete stone unless specified otherwise. The quantity of water should only be enough to ensure adequate consistency.
Measurement	• M • Measurements are taken from the total length of the drift including concrete, formwork, reinforcement, walls and protection works. • MAE.

Activity:	Construction of causeway also called a vented drift or ford (climate resilient structures)	Code: 275
Reference Code		
Application	This code is used for construction of causeways (Figure 99) made of reinforced concrete or stone masonry including excavation, foundations, walls, roadway, formwork and protection works according to the design.	
Tasks	• Placing safety signs • Constructing a detour to guarantee the continuous and safe passage of normal traffic. • Supplying and transporting all necessary materials for the construction of the causeway. • Confirming the alignment by the Supervising Engineer. • Excavating the trench for the construction of the culvert, foundation of the side walls, inlet apron and key wall, outlet apron and outlet channel. • Controlling the gradient of the pipes and other elements of the causeway. • Placing the formwork. • Placing and fixing the reinforcement in conformity with design. • Constructing the foundations, bottom slab, pipes (precast or cast in-situ), side walls, inlet apron and key wall, outlet apron, surround concrete, top slab, and approach slabs using concrete and stone masonry and stone pitching as indicated in the design. • Backfilling and compacting the soils under the approach slabs before the construction of the approach slabs. • Placing, compacting and curing the concrete. • Constructing stone masonry and stone pitching elements as indicated in the design. • Closing the detour followed by rehabilitation of the land where the detour was built. • Clearing the area around the causeway. • Removing temporary signs.	
Standards	Preparation • Adequate signs should be placed to guarantee road safety. • The contractor should consider the necessary measures to guarantee the efficient passage of traffic without interruption on the road during the construction works. The traffic might use a detour or any other means. • The causeway shall be constructed according to the design. Alignment and Dimensions • The alignment and dimensions of the causeway should be set out in conformity with design. • The maximum tolerance acceptable for the dimensions of the structure is ± 20 mm (walls and boxes), ± 10 mm (slab) and ± 0.5 % (gradient). • If the causeway is along a watercourse it must be aligned in such way that it is perpendicular to the direction of flow of the water. • The transversal gradient should be between 2 % and 5 %. • Preferably the causeway should discharge into a watercourse Foundations • After the foundation box is excavated, the base should be regularized and compacted using manual compactors or a vibrating plate. • The gradient of the foundation must be between 2 and 5 %; • In saturated conditions water should be pumped out to ensure dry conditions. • If the soil is not stable rip rap should be constructed on the upstream and downstream of the structure according to the drawings and Supervising Engineer's instructions. Formwork • The formwork might be constructed using metallic plates or wooden boards. • The formwork must be clean and with regular surfaces. • The formwork will be constructed in different steps, foundation, walls and roadway. • Before the placement of the reinforcement the formwork structure should be approved by the Supervising Engineer. • See codes 640, 641, 642 and 643 about the formwork standards. Reinforcement • The reinforcement and reinforcement schedule shall be according to the drawings. If the causeway consists of pipes and concrete surround wire mesh of 6 mm bars welded at 200mm spacing shall be used for the reinforcement of the bottom and top slabs.	

	• If the foundation of the causeway is on a flat rock bed, holes should be drilled and dowels (20 mm to 30 mm dia x 1m length) fixed in the rock with grout (cement paste) to hold the structure in place preventing wash away during flooding. The dowels should have a length of 0.5 m embedded in the rock. • All elements of the culvert, base, support wall, front walls, wing walls and the roadway should be linked through the tie-bars. • Where the openings consist of box culverts, the reinforcement must be tied using wire to create a framework strong enough to support its own weight and the weight of the workers during the placement of concrete. • Concrete spacers should be used to ensure adequate concrete cover over the reinforcement. • The Supervising Engineer should approve steel-fixing before pouring concrete. • See codes 630, 631, 632, 633 for reinforcement standards for provisioning, preparation, setting, application of steel. Concrete • All load carrying elements of the causeway must be constructed using reinforced concrete. The concrete surround around the pipes can be constructed with plums i.e. large stone or boulders (300 mm maximum size) to reduce the amount of concrete required to fill the spaces. The concrete shall be C20. If the main structure is a box culvert the top slab should be C30. • See codes 640, 641, 642 and 643 about the concrete standards. Backfill • The gravel layer under the approach slabs shall be 15 cm (see code 340). • The gravel layer shall be compacted to a minimum of 93% modified AASHTO at OMC. Protection • Causeways are designed to overflow during flood and the downstream side of the causeway is prone to scouring due to the artificial waterfall that is created. • Protection against erosion must be built on the causeway in conformity with design. • The outlet ditch must be large where it links with the natural terrain to allow the water to spread thus reducing the water energy. Finishing • Remove all unsuitable material away from the reserved area for the road. • Remove temporary signs.
Materials	• All material should be approved by the Supervising Engineer. • The concrete for the slabs should be prepared in conformity with codes 640 to 643. • The reinforcement should be prepared in conformity with codes 630 to 632. • The reinforcement bars must be prepared in accordance with the SABS 920 standards or equivalent. For each lot of reinforcement bars delivered to the site, the contractor must present the supplier's certificate with all details and technical specifications. The reinforcement bars and steel quality should be in conformity with drawings and SABS 82 (using symbol R or Y indicating the type of steel) • The cement should be Portland type and follow the SABS standard 471 or equivalent; it should be stored in proper storage, protected from water and damp conditions and without contact with floors or walls. Cement stored for more than 8 weeks should not be used for the works. The sand and stones should be collected from the sources approved by the provincial laboratory and follow with SABS standard 1083 or equivalent and should also be clean and free of clay and organic materials. • The water should be clean and free of acid, salt, sugar or any other organic material/chemical which could affect the quality of the concrete. • The stone should have dimension less than 20 cm and should have 5 to 20 kg weight, and should be clean and tough. • The concrete should be mixed according to proper methods of quality control of the components, in relation to 1:3:4 for cement/sand/concrete stone unless specified otherwise. The quantity of water should only be enough to ensure adequate consistency • The minimum thickness of wire should be 1.25 mm.
Measurement	• M • Measurements are taken from the total length of the causeway including concrete, formwork, reinforcement, walls and protection works. • MDE.



Figure 9-9 Construction of causeway (left) and causeway discharging water-base-flow (right)

Activity:	Construction of small bridges	Code: 276
Reference Code		
Application	This code is used for construction of small bridges made of reinforced concrete or stone masonry including excavation, foundations, walls, roadway, formwork and protection according to the design. See Figure 910.	
Tasks	• Placing safety signs • Constructing a detour to guarantee the continuous and safe passage of normal traffic. • Supplying and transporting all necessary materials for the construction of the bridge. • Confirming the alignment and setting out with the Supervising Engineer. • Excavating the trench for the construction of the bridge foundations, abutments and wing or side walls. • Excavating hard rock including blasting and disposal of rock debris. • Controlling levels and alignment of bridge members by a surveyor. • Placing and eventually removing the formwork. • Placing and fixing the reinforcement in conformity with design. • Constructing the foundations, bridge abutments, piers, bridge decks, walkways, parapets/guide posts and approach slabs using concrete and stone masonry, stone pitching, gabions and other protection works as indicated in the design. • Placing, compacting and curing of the concrete. • Placing bridge bearing plates. • Rubbing of concrete surfaces and filling cavities. • Backfilling and compacting the soils on the bridge approaches and foundations as indicated in the designs. • Placing bridge name plates and road signs. • Closing the detour followed by rehabilitation of the land where the detour was built. • Clearing the area around the bridge. • Removing temporary signs.	
Standards	Preparation • Adequate signs should be placed to guarantee road safety. • The contractor should consider the necessary measures to guarantee the efficient passage of traffic without interruption during the construction of the bridge. The traffic might use a detour or any other means. • The bridge shall be constructed according to the design. Alignment and Dimensions • The alignment and dimensions of the bridge should be set out in conformity with the design and approved by the Supervising Engineer. • The maximum tolerance acceptable for the dimensions of the bridge elements is ± 20 mm (walls and decks), ± 10 mm and ± 0.5 % (gradient). • If the bridge is on a watercourse it must be aligned in such way that it is perpendicular to the direction of flow of the water and it must be constructed centrally across the water course. Foundations • The foundations shall be excavated to level and, if there is no rock bed, the foundation shall be compacted manually or with light equipment (vibrating plate compactor, 1.7 ton vibratory steel roller). • Any water shall be pumped out to ensure dry conditions. • If the soil is not stable, formwork shall be placed on the vertical walls to prevent collapse. This shall be done according to the Supervising Engineer's instructions. Formwork • The formwork shall be constructed using metallic plates or wooden boards. • The formwork must be clean and with regular surfaces. • The formwork will be constructed in different steps for foundation, walls, substructure and superstructure (abutment, piers, walls, deck sides and soffits). • The design and placement of the formwork should be approved by the Supervising Engineer. • Deck soffits shall be kept in place for not less than 14 days after placement of the concrete. • Side shutters may be removed within 3 to 4 days after placement of the concrete. • See codes 640, 641, 642 and 643 for the formwork standards.	

	Reinforcement - The reinforcement and reinforcement schedule shall be according to the drawings. If the foundation of the causeway is on a flat rock bed holes should be drilled and dowels (20 mm to 30 mm dia x 1m length) fixed in the rock with grout (cement paste) to hold the structure in place preventing wash away during flooding. The dowels should have a length of 0.5 m embedded in the rock. - The reinforcement must be tied using wire to create a framework strong enough to support its own weight and the weight of the workers during the placement of concrete. - Concrete spacers should be used to ensure adequate concrete cover over the reinforcement (concrete cover should not be less than 30 mm). - The Supervising Engineer should approve steel-fixing before pouring concrete. - See codes 630, 631, 632, 633 for reinforcement standards for provisioning, preparation, setting, application of steel. Curing - All members shall be cured adequately by covering the surfaces with hessian cloth or other materials approved by the Supervising Engineer. Water should be applied to keep the surfaces moist at all times. - Decks and beams shall be cured for 14 days and other members of the bridge shall be cured for 7 days. Backfill - The bridge approaches including the area behind the abutments shall be backfilled with freely draining materials (gravel) and sand may be used to backfill the area behind the abutments - The material shall be compacted in layers not more than 150 mm thick (or 100mm if light equipment is used (vibrating plate compactors, vibrating pedestrian steel rollers). The layers shall be compacted to a minimum density of 93% Mod. AASHTO. If sand is used for backfilling, water should be poured onto it because this is the most effective way to compact sand. - Any part of the substructure which shall have the function of retaining soils shall have weep holes to drain the retained material.
	Protection - Bridges are designed to have an adequate free board (the difference in height between the maximum flood level and the bottom of deck or beams). However, this means that the greater part of the substructure is prone to strong water currents which may undercut or scour foundations and bridge embankment and backfill. Protection works shall be constructed to prevent damage in conformity with the designs. - Any alterations or additions to the protection works shall be in accordance with the Engineer's instructions. - If the downstream side of the bridge is likely to be eroded, riprap shall be constructed to spread and dissipate the energy in the water. Finishing - Remove all unsuitable material including debris away from the reserved area for the road. - Remove temporary signs.
Materials	- All material should be approved by the Supervising Engineer. - The concrete for the slab should be prepared in conformity with codes 640 to 643. - The reinforcement should be prepared in conformity with codes 630 to 632. - The reinforcement bars must be prepared in accordance with the SABS 920 standards or equivalent. For each lot of reinforcement bars delivered to the site, the contractor must present the supplier's certificate with all details and technical specifications. The reinforcement bars and steel quality should be in conformity with drawings and SABS 82 (using symbol R or Y). - The minimum thickness of tying wire should be 1.25 mm. - The cement should be Portland type and follow the SABS standard 471 or equivalent; it should be stored in proper storage, protected from water and damp conditions and without contact with floors or walls. Cement stored for more than 8 weeks should not be used for the works. The sand and stones should be collected from the sources approved by the provincial laboratory and follow with SABS standard 1083 or equivalent and should also be clean and free of clay and organic materials.

	• The water should be clean and free of acid, salt, sugar or any other organic material/chemical which could affect the quality of the concrete. • The stone masonry works should have thickness not less than 20 cm and should be to 20 kg weight and should be clean and tough. • The concrete should be mixed according to proper and approved methods ensuring that the batching of the constituents of concrete (cement/sand/concrete stone) confirm to the concrete mix design approved by the Supervising Engineer. The quantity of water should only be enough to ensure adequate consistency and workability and should be controlled using the slump test.
Measurement	• M • Measurements are taken from the total quantities for the bridge components including excavations, disposal of waste, concrete, formwork, reinforcement, walls, earth-works and protection works. • MAE.



Figure 9-10 General Structure of a Small Bridge made of Reinforced Concrete

Activity: Subsoil drainage system		Code:
Construction of subsurface drain		280
Reference Code		
Application	This code is used for the procedures for the construction of subsurface/subsoil drain below the pavement layers of the road, side drains and in sub-layers in slopes. Normally constructed in areas of high groundwater level (Figure 911) and in slopes and in cuts for the control and collection of the water. Filters of sand and stones are applicable according to the drawings and the Engineer's instructions.	
Enclosed tasks	• Placing safety signs • Setting out of the shape, extension, depth and gradient of the drain to be excavated. • Setting out excavation tasks. • Excavating the drain. • Leveling the layer and slopes of the drain. • Controlling the shape, gradient and depth of the drain. • Placing sand or stone filter layers or porous pipes with sand surround. • Covering with selected material. • Removing the excavated material away from the area. • Removing the temporary signs	
Standards	• To guarantee road safety appropriate signs should be placed in the road. • The shape of the drain must be excavated according to the drawings and free of irregularities. • Rocks, stones and roots in the drains profile should be removed. • The surface of the drainage should have 1% gradient. • The profile of the drain should be controlled with a drain template and the gradient should be controlled with automatic level and a metric tape. • The natural permeable material obtained might consist of sand or gravel according to the Engineer's instructions. • Permeable material should be spread over the layer in accordance to the specifications given for the project. The thickness of the layer should be as defined in the project. • The permeable material layers should not exceed 300 mm of thickness and must be compacted only slightly. • The thickness of each layer of natural permeable material must be carefully controlled. • For thinner layers adequate material should be used for effective cover. • When spreading successive layers of natural impermeable material, avoided contact with the lower layer. • The contamination of permeable natural material must be avoided (if geotextile is used it will be paid according to code 371). • The gradient of the filter should allow the discharge of water into an appropriate outlet. The outlet might be: 1. Surface discharge onto the existing terrain. 2. Side drain in the road. 3. Watercourse. 4. Culvert pipe. • A surface discharge or discharge into a watercourse is preferred. • Overlay the filtering material with selected and compacted material. • In case the ditch conducts the water to a natural terrain, the level of the ditch discharge and the level of the terrain should be the same at the outlet. • The outlet must be protected according to the drawings. • Discharge of water in erodible soils should be avoided, if this is unavoidable, the discharge area should be protected according to the protection code 251 and 252 with stones and stones with concrete • Remove the temporary signs.	
Materials	• All material must be approved by the Engineer. • The gravel and grit should be supplied from sources approved by the provincial laboratory and meet the standard SABS 1083 or equivalent. It must be clean, without clay and organic material and the size must be between 4.75 mm and 0.20 mm. the stones must be sized between 19.0 mm – 13.2 mm and must be clean, strong and not degraded.	
Measurement	• M³ • Measurements of the volume of the excavated drain. • MAE	



Figure 9-11 High Water Table – subsurface drainage required

Activity:	Construction of scour checks/bolsters	Code:												
	Using wooden piles and hand packed stones	290												
Reference Code														
Application	This code is used for the construction of scour checks along the side or mitre drains in steep areas to avoid erosion. Simple wooden and hand-packed stone structures may be constructed creating short barriers which, when silted up, form a stepped descent on steep sections with the purpose of dissipating the energy of the water. This helps to protect the drains and the areas where the water is discharged.													
Tasks	• Placing safety signs. • Setting out the position and shape of the scour checks. • Supplying and transporting the material to be used. • Excavating the foundations for the scour checks. • Placing wood/stones on the vertical side. • Placing stones on the horizontal side.													
	• To guarantee road safety appropriate signs should be placed on the road. • The scour checks must be constructed according to the design after the completion of the side drain. • The scour checks should not be too high, otherwise the water will be forced out of the side drain damaging the slopes, or edge of the curbing or pavement. • The cascades should not be constructed in the side drain which has less than 3% gradient because this would facilitate sedimentation of soil in the side drain which will eventually damage the carriageway of the road due to reduced capacity of the side drains. • The spacing of the scour checks should be chosen based on the gradient of the side drain according to the following Table 9-7. <table><tr><th colspan="2">Table 9-7 Spacing between cascades (wooden pile and packed stones)</th></tr><tr><th>Gradient of the drain</th><th>Spacing between the cascades</th></tr><tr><td>To 3 %</td><td>Unnecessary</td></tr><tr><td>Between 3 and 5 %</td><td>20 m</td></tr><tr><td>Between 5 and 10 %</td><td>10 m</td></tr><tr><td>Superior than 10 %</td><td>5 m</td></tr></table>		Table 9-7 Spacing between cascades (wooden pile and packed stones)		Gradient of the drain	Spacing between the cascades	To 3 %	Unnecessary	Between 3 and 5 %	20 m	Between 5 and 10 %	10 m	Superior than 10 %	5 m
Table 9-7 Spacing between cascades (wooden pile and packed stones)														
Gradient of the drain	Spacing between the cascades													
To 3 %	Unnecessary													
Between 3 and 5 %	20 m													
Between 5 and 10 %	10 m													
Superior than 10 %	5 m													
Standards	• The gradient of the side drain should be measured using a spirit level and metric tape. • Bamboo or equivalent material with 5 to 10 cm of diameter and minimum length of 60 cm should be used. The length above ground shall be approximately 15 cm. Alternatively; hand packed stone may be used. • The shape of the scour check must be controlled using a spirit level and metric tape. • The foundations must be constructed to maintain the correct profile. • A marker must be placed at the edge of the side drain and a template shall be placed to determine the depth of foundation by excavating to the required depth at the break points of the drain profile. • Construction of the scour check shall be in alignment with the shape of the drain cross-section and the piles shall be placed with a spacing of 2 cm maximum. • The piles should be tied using wire. • The stones or piles of the cascade must be well arranged to avoid seepage of water and fine sand through the scour check. • An apron should be constructed on the downstream side of the vertical wall of the scour check for protection using stones or grass layer. • Clearing of the reserved area for the road should be carried out to the satisfaction of the Engineer. • Remove the temporary signs.													
Materials	• All material must be approved by the Supervising Engineer. • Stones, bamboo or wooden piles:													
Measurement	• Number (No.) • Measurements of number of constructed scour checks. • MAE													

Activity: Construction of scour checks/ bolsters using wet stone masonry		Code: 292										
Reference Code												
Application	This code is used for the construction of scour checks along the side or mitre drains in steep areas to prevent erosion. Simple wet stone masonry structures may be constructed creating short barriers which when silted form a stepped descent on steep sections with the purpose of dissipating the energy of the water. This helps to protect the drains and the areas where the water is discharged.											
Tasks	• Placing safety signs. • Setting out the position and shape of the scour checks to be constructed. • Supplying and transporting the material to be used. • Excavating the foundations for the scour checks. • Constructing wet stone masonry in the vertical side. • Placing the stones or wet masonry apron on the horizontal side. • Removing temporary signs.											
Standards	• To guarantee road safety appropriate signs should be placed on the road. • The scour checks must be constructed according to the design after the construction of the side drain. • The scour checks should not be too high, otherwise the water will be forced out of the side drain damaging the slopes, or edge of the curbing or pavement. • The scour checks should not be constructed in side drain which has less than 3% gradient because it would facilitate sedimentation of the soil in the side drain which would eventually cause damage to the carriageway of the road due to reduced capacity of the side drains. • The spacing of the scour checks should be chosen based on the gradient of the side drain according to the following Table 9-8. Table 9-8 Spacing between cascades/scour checks (wet stone masonry) <table><tr><th>Gradient of the drain</th><th>Spacing between the cascades</th></tr><tr><td>Up to 3 %</td><td>Unnecessary</td></tr><tr><td>Between 3 and 5 %</td><td>20 m</td></tr><tr><td>Between 5 and 10 %</td><td>10 m</td></tr><tr><td>Greater than 10 %</td><td>5 m</td></tr></table>		Gradient of the drain	Spacing between the cascades	Up to 3 %	Unnecessary	Between 3 and 5 %	20 m	Between 5 and 10 %	10 m	Greater than 10 %	5 m
Gradient of the drain	Spacing between the cascades											
Up to 3 %	Unnecessary											
Between 3 and 5 %	20 m											
Between 5 and 10 %	10 m											
Greater than 10 %	5 m											
	• The gradient of the side drain should be measured using a spirit level and metric tape. • The foundations must be constructed carefully through excavation and compaction with depth between 10 and 15 cm maintaining the correct profiles. • The shape of the scour checks must be controlled using a spirit level and metric tape. • A marker must be placed at the edge of the side drain and a template should be placed to determine the depth of foundation by excavating holes to the required depth at the break points of the drain profile. • The scour checks should be constructed in alignment with the shape of the drain cross-section. • The stone masonry of the scour checks must be well arranged to avoid the seepage of water and fine sand. • The size of the stones must be between 20 and 30 cm. • The width of the scour check must be 20 cm minimum. • An apron should be constructed on the downstream side of the stone masonry scour checks for protection using stones or grass layer. • Clearing of the reserved area for the road should be carried out to the satisfaction of the Engineer. • Remove the temporary signs.											
Materials	• All material must be approved by the Supervising Engineer. • Stones: 20 to 30 cm in size weighing 5 to 20 kg.											
Measurement	• Number (No) • Measurements of number of constructed scour checks. • MAE											

Activity:	Construction of scour checks/bolsters using Concrete C 20 (No-fines concrete)	Code: 293												
Reference Code														
Application	This code is used for the construction of scour checks along the side or mitre drains in steep areas to prevent erosion. Simple concrete structures may be constructed creating short barriers which when silted form a stepped descent on steep sections with the purpose of dissipating the energy of the water. This helps to protect the drains and the areas where the water is discharged.													
Enclosed tasks	• Placing safety signs • Setting out the position and shape of the scour checks to be constructed. • Supplying and transporting the material to be used • Excavating the foundations for the scour checks. • Placing the formwork (wood might be used for the formwork). • Preparing class C20 concrete. • Placing and curing the concrete. • Removing temporary signs.													
Standards	• To guarantee road safety appropriate signs should be placed in the road. • The scour checks must be constructed according to the design after the construction of the side drain. • The scour checks should not be too high, otherwise the water will be forced out of the side drain damaging the slopes, or edge of the curbing or pavement. • The scour checks should not be constructed in side drains which has less than 3% gradient. This facilitates sedimentation of the soil in the side drain which, in turn, will eventually damage the pavement. • The spacing of the scour checks should be chosen based on the gradient of the drain according to the Table 9-9. <table><tr><th colspan="2">Table 9-9 Spacing between cascades (no-fines concrete)</th></tr><tr><th>Inclination of the side drain</th><th>Spacing between the cascades</th></tr><tr><td>Up to 3 %</td><td>Unnecessary</td></tr><tr><td>Between 3 and 5 %</td><td>20 m</td></tr><tr><td>Between 5 and 10 %</td><td>10 m</td></tr><tr><td>Greater than 10 %</td><td>5 m</td></tr></table>		Table 9-9 Spacing between cascades (no-fines concrete)		Inclination of the side drain	Spacing between the cascades	Up to 3 %	Unnecessary	Between 3 and 5 %	20 m	Between 5 and 10 %	10 m	Greater than 10 %	5 m
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Inclination of the side drain	Spacing between the cascades													
Up to 3 %	Unnecessary													
Between 3 and 5 %	20 m													
Between 5 and 10 %	10 m													
Greater than 10 %	5 m													
	• The gradient of the side drain should be measured using a spirit leveling and metric tape. • The shape of the cascade must be controlled using a profile board and metric tape. • The foundations must be constructed carefully through excavation to level and compacting to maintain the correct profiles. • A marker must be placed at the edge of the side drain and a template shall be placed to determine the depth of foundation by excavating to level at the break points of drain profile. • Formwork shall be placed with a spacing of 20 cm and concrete shall be cast in the formwork, compacted properly and shaped to follow the profile given in the design. • After the application of the concrete it must be kept moist for a minimum period of 5 days. • The formwork must be removed after 5 days. • Support or aprons must be constructed for basic protection using stones or grass cover. • Clearing of the reserved area for the road should be carried out to the satisfaction of the Engineer. • Remove the temporary signs.													
Materials	• All material must be approved by the Engineer. • The concrete should be prepared according to the codes 640 and 643.													
Measurement	• Number (No) • Measurements of the number of constructed scour checks. • MAE													

9.6 Series 300 Earthworks

9.6.1 Introduction

Earthworks include all activities involving excavations on the right of way, opening of borrow pits, extraction of materials from borrow areas, placement, formation and construction of subgrade, subbases and road base.

This section covers the construction operations, standards and specifications which should be followed and the management of works, materials specifications, approval process and measurement of completed works for payment.

Earthworks are a crucial stage in road construction and in low volume roads provision the pavements provides the bulk or all of the structural strength of the roads because the surfacings are usually semi or non-structural. The construction procedures and specifications including tolerances stipulated in the activity codes will ensure compliance with specific and general conditions of contract with regards to the execution of earthworks. The standards and specifications cover setting out, processing of materials, application of materials, shaping and compaction including disposal of waste material.

Activity:	Excavation to level, levelling and compaction of roadbed and fill	Code: 310
Reference Code		
Application	This code is used for road construction and rehabilitation projects after the execution of the codes 160 (cleaning and grubbing) and 170 (cutting and removal of trees) and 180 (cut to waste and excavation) and before the execution of the code 210 (opening of side drains). In situations where the natural ground does not have the desired profile it will be necessary to carry out earthworks (balancing cut and fill) to achieve the desired transverse profile (formation) and longitudinal profile with smooth alignments. In uneven grounds the work might involve more earthworks including cutting and filling. If it is not possible to carry out the earthworks using materials within a distance of 75 m use code 320 (Excavation, transport, spreading, watering and compacting of the soil and fill) for the transportation of the soils from the borrow pit or other convenient source along the road for the fill or embankment.	
Tasks	- Placing safety signs LABOUR BASED OPERATIONS - Surveying the road and determining the longitudinal and transversal profiles with the aim of minimizing earthworks in conformity with the technical standards of the particular class of road. - Opening of slots each 10 m to indicate the extent of cutting or filling. - Setting out the earthworks tasks – 3m³/worker day for common excavation, 5m³/per worker day for loose soils and 1-2m³/worker day for hard excavation (cutting and fill). - Cutting, transporting (using labour and wheelbarrows for labour based methods) and filling. - Spreading, watering and compacting the fill (in layers not more than 15 cm); MACHINE BASED OPERATIONS - Setting out the section, placing pegs at 20 – 50 m intervals. - Cutting and filling in accordance with working drawings and Specifications. - Controlling profiles with boning rods, levels and total stations (survey equipment). - Spreading, watering, mixing, compaction and levelling. - Quality control of the soils used. - Removing the temporary signs.	
Standards	- To guarantee road safety, appropriate signs should be placed on the road. - Set out working areas. - Appropriate topographic techniques must be used for the determination of the longitudinal and transversal profile of the road, e.g. automatic level, theodolite or total station, profile rods, boning rods, metric tape, line level. - The surface of the road should not have irregularities greater than 5 cm, measured using a 2 m straight edge. - Any coarse material with diameter greater than 7.5 cm shall be removed or broken in smaller pieces before it is used. - The Engineer shall approve soils used for the construction using test results of grading, plasticity and compaction and other which may be considered necessary. - The contractor must monitor the quality of the excavated soils to ensure that there are no variations with the samples approved by the Engineer. - Spread the material in layers not more than 15 cm thick and water if necessary to reach the required level of moisture (optimum moisture content - OMC) and mix homogeneously. - The layers should be compacted to a minimum of 91 % Modified AASHTO or till refusal ensuring that the material does not degrade excessively. The compaction level (inspection of density in the field) shall be verified through the use of the sand replacement or Troxler tests. - For the application of the fill the required crossfall must be constructed in the transverse orientation according to the specifications. - The surface of the road bed must have no irregularities and must drain freely. - The contractor must control the longitudinal profile to ensure that the maximum limit gradients are not exceeded. The Engineer shall check the setting out, surveys and resultant horizontal and longitudinal profiles including gradient to ensure that they conform to the design and shall give approvals in writing. - Remove the temporary signs	

Materials	- The Engineer shall check the quality of the material through tests including grading, plasticity and compaction and shall give approval in writing. - The materials from borrow pit within 75 m or side drains shall conform to the quality specifications for fill as specified in the design (minimum CBR of 10% if not specified in the design).
Measurement	- M³ - Measurements of the volume of the compacted layers on the road. - MAE

Activity:	Excavation, transportation, spreading and watering and compaction of fill. AHD < 500 m 500 m ≤ AHD < 1000 m AHD > 1000 m	Code: 320 321 322
Reference Code		
Application	These codes are used for importing fill materials for the construction of road embankments and fill including identification of the borrow pits if it is necessary to increase the height of the road to suit the longitudinal alignment or improve the drainage. See Figure 912. The choice of the code depends on the average haulage distance from the borrow pit to the embankment and the mechanism of transportation. If the haul distance is less than 75 m then use code 310 which is also appropriate for use of manual means of transport e.g. by wheelbarrows.	
Tasks	- Placing safety signs. - Ensuring the safe passage of traffic. - Controlling the width of the road. - Surveying the road and determining the longitudinal and transversal profiles with the aim of minimizing earthworks in conformity with the technical standards of the particular class of road. - Identifying borrow pits along the road with soils approved by the Supervising Engineer. - Excavating, stockpiling, testing and loading of the soils. - Hauling, dumping, spreading, watering and compacting the material. - Collecting samples and testing for grading, plasticity and compaction. - Removing temporary signs.	
Standards	- To guarantee road safety appropriate signs should be placed on the road. - To guarantee safe passage of traffic, ensure that there is no danger posed by lack of traction and no interruption of the movement of traffic for periods of more than 10 minutes. - The dimensions and edges of embankments shall be built in accordance with the specifications of the project or as instructed by the Engineer. - The surface of the embankment shall be free of undulations or irregularities more than 2 cm measured using a 2 m straight edge and must drain freely. - A cross fall shall be formed on the embankment in conformity with the design. - Appropriate topographic techniques must be used for the determination of the longitudinal and transverse profiles of the road, e.g. automatic level, theodolite or total station, profile rods, boning rods, metric tape, line level. - Only excavate soils which are outside the profile of the formation. - The Engineer shall approve soils used for the construction using test results of grading, plasticity and compaction and other tests which may be considered necessary. - The contractor must monitor the quality of the excavated soils to ensure that there are no variations from the samples approved by the Engineer. - The soils shall be transported without damaging other areas of the completed works or affect sections of road where interventions were not initially anticipated.	

	• Remove all material with diameter greater than 7.5 cm before carrying out compaction. • Place soils, water, mix and compact in layers no more than 15 cm thick and apply sufficient water to reach the optimum moisture content. • The layers should be compacted to a minimum of 91 % Modified AASHTO or till refusal ensuring that the material does not degrade excessively. The compaction level (inspection of density in the field) shall be verified through the use of the sand replacement or Troxler tests. • Remove temporary signs.
Materials	• The Engineer shall control the quality of the soils being used through tests on grading, plasticity and compaction and give approval in writing. • The soils used for the embankment should be of CBR 15% or more.
Measurement	• M³ • Measurement of the volume of the soil after compaction. • MAE



Figure 9-12 Roadbed Preparation

Activity:	Formation, levelling, watering and compaction of the subgrade using material excavated from the drains	Code: 330
Reference Code		
Application	This code is used for the movement and leveling of soil involving the construction of formation/subgrade of the road using soils excavated from the side drains.	
Tasks	• Placing safety signs. • Setting out and placing pegs on the edges of the road indicating the level of fill. LABOUR BASED OPERATIONS • Setting out the earthworkS tasks - 3m³/worker day for common excavation, 5m³/per worker day for loose soils and 1-2m³/worker day for hard excavation (cutting and fill). • Spreading the soils excavated from the drains for the formation and camber. • Controlling the camber. • Watering, mixing and compaction. MACHINE BASED OPERATIONS • Cutting side drains using a grader or dozer in case of hard materials • Controlling drain profiles, camber and levels of the subgrade. • Watering, mixing and compacting the subgrade. • Testing thickness of subgrade and field densities. • Removing temporary signs.	
	• The contractor shall place adequate signs to guarantee safety • Under normal circumstances materials excavated from the side drains shall be used for the construction of the subgrade and must be approved by the Engineer. If needed, additional material can be obtained from the same source as the material used for the embankment if approved by the Engineer for the construction of subgrade. • Care should be taken to ensure that the material coming from the drains is adequate for the construction of the subgrade to achieve the required thickness. • The material that is dumped on the roadbed for the formation shall be spread and any undesirable material such as boulders in excess of 7.5 cm and vegetation or organic materials shall be removed. The soil shall be spread and the layer thickness shall not exceed 200 mm compacted (usually 150 mm is the preferred thickness of layers for compaction). • The subgrade layer shall be placed to conform to the designed transverse profile of the road (width and camber).	
Standards	• The bulking factor of soil i.e. the change in volume from loose to compacted is usually between 1.2 and 1.3. For example, if the bulking factor is 1.2, to achieve a 150 mm thickness of compacted layer the thickness of the layer before compaction should be 180 mm (150 x 1.2 = 180 mm). The actual bulking factor shall be determined through construction of a trial section. • The optimum moisture content for compaction must be determined. • Using labour-based methods of construction, the soil shall be watered at the top and allowed to soak overnight. The amount of water shall be adequate to bring the moisture content of the soil to the optimum moisture content. • For machine-based methods (i.e. use of disc harrow for mixing) the material shall be watered, mixed and compacted at optimum moisture content. • For compaction control, the compaction moisture shall be determined through manual tests. The material is at optimum moisture content if a sample of the soil is squeezed into a ball and the ball does not disintegrate when it falls back into the palm of the hand after being tossed up to a height of 1 m or the ball cracks when pressed with the thumb. If the ball disintegrates completely it means the moisture is below optimum and if it deforms without cracking or if moisture seeps out it means there is too much moisture. [Other field methods are also possible. For example, the sand bath method in which a dish is filled hot sand and a weighed wet soil sample is dried on the hot sand and reweighed)]. • Compaction of the material shall be carried out using pneumatic and vibratory steel rollers, (which may be the towed type). Also, pedestrian vibratory steel rollers may be used but the layer thickness should not be more than 75 mm.	

	• Compaction shall be carried out until no further densification can be achieved through subsequent passes (compaction to refusal). This can be determined through construction of a trial section. The procedure involves placing pegs firmly into the ground or subgrade after the material to be compacted has been watered, mixed, leveled and is ready for compaction. The spacing between the pegs shall be adequate for the roller to pass through without disturbing the pegs. A string is pulled tightly over the pages and the height of string is measured from the soil layer on pre-marked positions (5 positions minimum). The roller is passed and the height is measured again. This is repeated up to 15 passes. The height or preferably the change in height is recorded and this is plotted against the number of pass. The graph levels out when there is no more change in height on subsequent passes and the corresponding number of passes read from the graph is minimum number of passes to achieve refusal compaction and refusal density for that particular roller. Modern rollers determine the refusal compaction automatically. Also check the degradation of the materials by testing the grading before and after compaction. • The number of passes required depends on the capacity of roller. A trial section shall be constructed to determine the exact number of passes required to ensure that the desired level of compaction is achieved (95% Mod. AASHTO). • The level of compaction shall be determined through either the Sand Replacement Method (ensure that the sand is calibrated) or by using a Troxler or nuclear density gauge (also ensure that it is calibrated). • Additional tests shall be carried out on the materials on the road after mixing to ensure that they conform to the quality standards. • After compaction, the width, crossfall and thickness shall be measured and checked against the specifications indicated in the designs. It is obligatory to use the camber-board to control the crossfall and camber and a system to check the thickness (the holes dug up during the sand replacement method can be used for this). • The crossfall shall conform to the design and should allow efficient drainage of the water from the surface of the road. Damage to the surface or haulage of material over the surface shall be avoided. • The camber shall be 5% for unpaved roads and 2.5% to 3.5% for sealed roads and a maximum of 8% for super-elevations. Higher gradients or crossfall will encourage erosion of unsealed shoulders of the road. • The tolerance limits acceptable are ± 20 mm for the width and $\pm 0.5\%$ for the camber. • Control of the camber should be done using automatic levels or camber board and spirit level for labour-based methods. • The surface shall be plane and free from irregularities in excess of 2 cm measured using a 2 metre straight edge. • Avoid leaving loose material in the drainage system of the road. • Remove temporary signs.
Materials	• The Engineer shall control the quality of the soils being used through tests on grading, plasticity and compaction and give approval in writing. • The soils used for the embankment should have a minimum CBR of 15%. The soils shall be compacted to a minimum of 91% Mod. AASHTO or until refusal ensuring that degradation does not occur. The level of compaction shall be carried out using the sand replacement method or Nuclear Density Gauge (Troxler).
Measurement	• M^3 • Measurement of the volume of the soil after compaction. • MAE

Activity:	Construction of mechanically stabilised bases (roadbase) AHD 1 km: 0 to 2 km AHD 3 km: 2 to 4 km AHD 5 km: 4 to 6 km AHD 7 km: 6 to 8 km AHD 9 km: 8 to 10 km AHD 12.5 km: 10 to 15 km AHD 17.5 km: 15 to 20 km For extra haul distance beyond 20 km.	Code: 340 341 342 343 344 345 346 347
Reference Code		
Application	These codes are used for importing of (sub)base materials and the construction of the (sub)base layer(s) after works related to code 330 (formation leveling, watering and compaction of the subgrade) have been completed and approved. Codes 340 to 346 include excavation, transportation and placement of the material, while code 347 is used for payment of the extra haulage distance beyond 20 km which is the maximum covered under code 346. The choice of the code depends on the average haulage distance (AHD) from the borrow pit to the dumping place on the road. The code shall not be used when the dumping place is changed or there is change in the borrow pit to another with similar materials. The Engineer shall decide where the material shall be dumped on the road. The same codes shall be used for the placement of soils for stabilization. For the opening of the borrow pit, code 351 shall be used and code 352 shall be used for the reinstatement of the borrow pit.	
Tasks	• Acquiring the exploration license. • Locating borrow pit identified at planning stage. • Clearing bushes and grass. • Removing overburden and stockpiling on the periphery of the borrow pit. • Stockpiling the base material. • Carrying out the necessary laboratory testing to (i) verify that the quality of the soil meets the standards and (ii) verify that the results of the compaction and strength tests meet the standards. • Placing safety signs on the section earmarked for dumping. • Placing pegs to demarcate the lanes and the also the intended thickness of the layer. • Marking of tasks where labour-based methods. • Where necessary preparing the surface of the road before dumping including leveling and/or repairing, scarifying and watering (note that compaction may be necessary). • Loading, transporting, spreading, leveling, watering and compacting the material. • Placing gravel on the side of the road for future maintenance. • Removing temporary signs	
Standards	• To guarantee road safety, appropriate signs should be placed on the road. • To guarantee safe passage of traffic, ensure that there is no danger posed by lack of traction to vehicle without 4 wheel drive and no interruption of the movement of traffic for periods of more than 10 minutes. • Authority to prospect for materials should be obtained in order to conform to national regulations for the extraction of natural materials. • The borrow pit previously identified during the planning stage shall be located including the concrete pegs demarcating the area covered by the borrow pit. • The area shall be cleared of trees and other vegetation including grass and roots to a depth of 150mm, paid under Code 348. • The overburden material shall be removed and stockpiled on the peripheries of the borrow pit. This overburden material may be used for blending or fill if approved by the Engineer, paid under Code 348. • The selected material shall be stockpiled in heaps and the top and bottom must be flat and rectangular for ease of measurement of quantities of the material, paid under Code 348. • Samples shall be collected 3 – 5 samples per stockpile for quality tests. The actual sampling and testing shall be paid under Code 610.	

	• If the formation carried out under code 330 requires some repairs before placement of the sub-base or base layer the contractor shall take full responsibility for the repairs. • The materials shall be loaded, transported and dumped at predetermined spacing necessary to achieve the required thickness of the (sub) base layers. • The soil shall be spread and the layer thickness shall not exceed 200 mm compacted (usually 150 mm is the preferred thickness of layers for compaction). • The bulking factor of soil i.e. the change in volume from loose to compacted is usually between 1.2 and 1.3. For example, if the bulking factor is 1.2, to achieve a 150 mm thickness of compacted layer the thickness of the layer before compaction should be 180 mm ($150 \times 1.2 = 180$ mm). The actual bulking factor shall be determined through construction of a trial section. • The optimum moisture content for compaction must be determined. • If other methods are not available, the compaction moisture shall be determined through manual tests. The material is at optimum moisture content if a sample of the soil is squeezed into a ball and the ball does not disintegrate when it falls back into the palm of the hand after being tossed up to a height of 1 m or the ball cracks when pressed with the thumb. If the ball disintegrates completely it means the moisture is below optimum and if it deforms without cracking or if moisture seeps out it means there is too much moisture. [Other field methods are also possible. For example, the sand bath method in which a dish is filled hot sand and a weighed wet soil sample is dried on the hot sand and reweighed]]. • Compaction of the material shall be carried out using pneumatic and vibratory steel rollers, (which may be the towed type). Also pedestrian vibratory steel rollers. • Compaction shall be carried out until no further densification can be achieved through subsequent passes (compaction to refusal). This can be determined through construction of a trial section. The procedure involves placing pegs firmly into the ground or subgrade after the material to be compacted has been watered, mixed, leveled and is ready for compaction. The spacing between the pegs shall be adequate for roller to pass through without disturbing the pegs. A string is pulled tightly over the pegs and the height of string is measured from the soil layer on pre-marked positions (5 positions minimum). The roller is passed and the height is measured again. This is repeated up to 15 passes. The height or preferably the change in height is recorded and this is plotted against the number of pass. The graph levels out when there is no more change in height on subsequent passes and the corresponding number of passes read from the graph is minimum number of passes to achieve refusal compaction. Modern rollers determine the refusal compaction automatically. Also check the degradation of the materials by testing the grading before and after compaction. • The number of passes required depends on the capacity of roller. A trial section shall be constructed to determine the exact number of passes required to ensure that the desired level of compaction is achieved (95% Mod. AASHTO). • The level of compaction shall be determined through either the Sand Replacement method (ensure that the sand is calibrated) or by using a Troxler or nuclear density gauge (also ensure that it is calibrated). • Additional tests shall be carried out on the materials on the road after mixing to ensure that they conform to the quality standards. • After compaction, the width, crossfall and thickness shall be measured and checked against the specifications indicated in the designs. It is obligatory to use the camber-board to control the crossfall and camber, and a system to check the thickness (the holes dug up during the sand replacement method can be used for this). • The maximum tolerance limits are ± 20 mm for the width, ± 5 mm for the thickness, and ± 0.5 % for the crossfall. • The crossfall shall be free from undulations and irregularities in excess of 2 cm measured using a 2 m straight edge.
	• For labour-based construction, a heap of 3 m³ of material (gravel) shall be placed at 200m intervals for future maintenance activities. • Remove temporary signs once all activities have been concluded.

Materials

- The soil (gravel) shall always be from sources approved by the Engineer.
- UNPAVED ROADS AND UNSEALED SHOULDERS**
- For unpaved road the characteristics shall conform not only to TRH 20 specifications but also to performance based specifications. TRH20 specifications are:
 - Maximum aggregate size of 37.5 mm.
 - Reject index shall not exceed 5 %.
 - Shrinkage product shall be within range 100 – 365
 - Grading coefficient within range 16 – 34
 - AIV within range 20 - 65
 - Plot the values of material properties in the SP and Gc chart below. It should plot in E1 and E2, Figure 9-14.
- Note that this specification, based on TRH 20 - Figure 9-14, is for riding quality for wearing courses and not for bases on paved roads. It also does not indicate quality in terms of performance in the form of rate of gravel loss and roughness progression.
- The following are the performance based specifications (Figure 9-13) for wearing course materials.
- Performance based specifications should take precedence over riding quality specifications.
- Such material shall be used for unsealed shoulders.

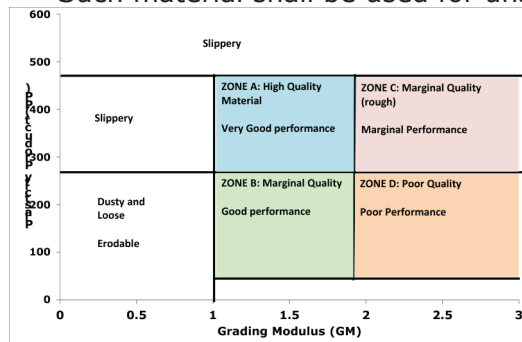


Figure 9-13 Performance based specifications for materials used for construction of unsealed shoulders

Where

- $PI_{0.075}$ = the Plasticity Index of the material passing the 0.075mm sieve
- Plasticity Product (PP) = $PI_{0.075} \times \text{\%age of material passing 0.075mm sieve}$
The preferred range is 280-480 (minimum PP = 50)
- Grading Modulus (GM) =

$$GM = 3 - \left(\frac{P_{2.36} + P_{0.425} + P_{0.075}}{100} \right)$$

Where: $P_{2.36}$ = percentage passing 2.36 mm sieve
 $P_{0.425}$ = percentage passing 0.425 mm sieve
 $P_{0.075}$ = percentage passing 0.075 mm sieve
 The preferred range is 1.0-1.9

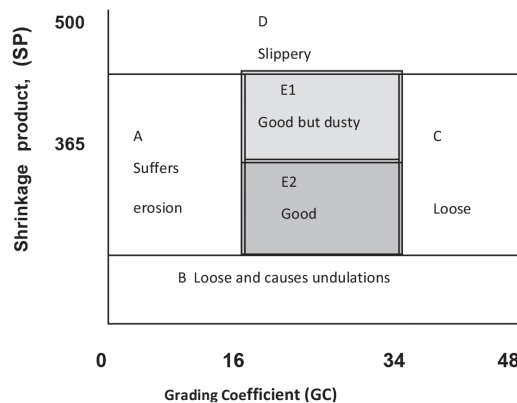


	Figure 9-14 Specifications for gravel wearing course for unsealed shoulders (based on riding quality) Where a) Shrinkage Product (SP) = Linear Shrinkage_{0.425} x %passing 0.425mm b) Grading Coefficient (GC) = (%passing 26.5mm - %passing 2.0mm) x %passing 4.75mm/100 - The soil shall be free of inappropriate materials and vegetation. - The layer shall be compacted to a minimum density of 95% Mod. AASHTO. - The CBR shall be minimum 15%. SUBBASE/BASE MATERIALS FOR PAVED ROADS For paved roads the material shall conform to the following specifications: - Minimum CBR of 40% - Maximum PI of 12 for non-pedogenic materials - Maximum PI of 20 for pedogenic materials. - The standards given above are appropriate for low volume paved roads.
Measurement	- M³ - Measurement of the volume of the soil after compaction. - MAE Code 347 M³KM If the distance from the borrow pit to the section where the material will be dumped is in excess of 20 km the work shall be reimbursed through code 347. This shall be worked out as the extra distance (km) multiplied by the quantity of material in M³. Example: If the haul distance is 30 km and the volume of the load is 2000 M³ the quantity for payment is calculated as follows: Extra haul distance in excess of 20 km = 30-20 km = 10km The quantity of soil = 2000 M³ The measurement for payment = 2000 x 10 = 20,000 M³KM - Materials applied in excess of the required thickness shall not be paid for.

Haul distances given in italics are not appropriate for labour based construction and low cost sealing and should only be considered when there is no other option.

Activity: Borrow pits Opening of a borrow pit Rehabilitation and/closure of borrow pit	Code: 348 349
Reference Code	
Application	Code 348 This code for all activities involving the opening and preparation of a borrow pit (Figure 915). Code 349 This code shall be used for all activities necessary for the rehabilitation and closure of borrow pit for the purpose of minimizing the environmental impact and the hazards it presents.
Tasks	Code 348 • Placing safety signs. • Obtaining the list and details of the borrow pits identified during the planning phase of the project. • Obtaining the test results of the materials tested during the prospecting stage. • Demarcating the excavation area (marking the tasks when using the labour based method). • Clearing the area, removing the vegetation and demolishing undesirable features within the work area and the access route to the borrow pit. • For labour based works - setting out the earthworks tasks - 3m³/worker day for common excavation, 5m³/per worker day for loose soils and 1-2m³/worker day for hard excavation • Excavating and removing the overburden. • Excavating trial pits to mark the depth of excavation for labour based works. • Excavating the material. • Stockpiling the material. • Constructing an access to the borrow pit. • Removing temporary signs. Code 349 • Placing safety signs • Reducing the inclination of the side slopes. • Spreading the soil which was removed during the opening of the borrow pit. • Planting lawn or grass cover. • Nurturing the planted lawn or grass by watering for at least 30 days. • Removing temporary signs.
Standards	Code 348 • To guarantee road safety, appropriate signs should be placed on the road. • The borrow pit shall be located at least 50 m from the road reserve. • The plan shall ensure minimum earthworks and removal of trees and vegetation when preparing the access to the borrow pit. • The overburden shall be removed outside the demarcation of the borrow pit and stockpiled. It may be used for blending with the gravel and will also be used for the rehabilitation of the borrow pit after extraction of the material (gravel) is completed. • Once the required material has been stockpiled, appropriate samples shall be collected and sent immediately to the laboratory for testing, paid for under Code 710. The tests shall include plasticity, grading, compaction, CBR and others as specified by the Engineer. • The quantities of the material in the borrow pit shall be verified. The quantity of the material available can be estimated by measuring the volumes of materials in the stockpiles. The bulking factors of materials given in Code 340 to 347 shall be used to determine whether the material is adequate for the intended section as given in the mass-haul diagram (a diagram showing which borrow pit covers each section on the road). • Remove temporary signs. Code 349 • To guarantee road safety, appropriate signs should be placed on the road. • The inclination of the side slopes of the borrow pit shall be reduced to a maximum of 45 degrees.

	- Based on the decision of the Engineer, the top soils which were removed during the opening of the borrow pit shall be spread in the borrow pit to encourage the growth of vegetation. - Lawn or grass cover shall be planted and watered for a month. - Remove temporary signs.
Materials	Not Applicable
Measurement	- Number (No) - Measurement is on the number of borrow pits opened and rehabilitated during the course of the contract. - MAE



Figure 9-15 Borrow Pits showing shallow and deep extraction

Activity	Construction of base/sub-base Stabilisation of in-situ soils with cement/lime – (thickness and percentage of cement/lime are predetermined). Stabilisation agent for code 350 (cement/lime)	Code: 350 351
Reference Code		
Application	This code is used for the stabilization of material used as base or sub-base of road pavements through mixing with a stabilising agent (Portland cement/Lime). The cement/lime stabilised soil is a mixture of soil, cement/lime and water in proportions given in the mix design. The cement/lime content is influenced by the grading of the soil and the required strength of the stabilised material. This code includes the activities of mixing, smoothing, leveling, watering and compaction of the base or sub-base, Figure 916. The quantity of the cement/lime is reimbursed under code 351. It is important to note that the soils for the (sub) base are reimbursed under code 340-347. It is important to carry out tests to ensure that the quality of the cement stabilised base or subbase conforms to specifications. This shall be reimbursed under Series 600.	
Tasks	• Placing safety signs. • Ensuring safe and continuous passage of traffic. • Supplying and storing stabilising materials. • Setting out the site including placement of pegs for control of levels and dimensions. • For LB operation - Setting out group tasks, 20 -30m3 for a gang of 5 workers. • Constructing trial section and identifying the materials and the cement content that complies with the specifications of the project together with the Engineer. • Marking the centerline and the edges of the lanes with pegs. • Cleaning and preparing the surface of the subbase. • Placing the base materials and spreading.	
	• Distributing cement over the (sub) base layer. • Mixing the (sub) base soil with cement. • Watering, leveling and compacting the (sub) base paid under Codes 340 – 347. • Protecting and curing the stabilised (sub) base. • Clearing the work area. • Removing temporary signs.	
	• In order to guarantee road safety, adequate signs shall be placed on the road. • The contractor shall put in place measures to ensure safe and uninterrupted passage of regular traffic. This may include construction of a detour as may be instructed by the Engineer. • The contractor shall stockpile the materials, avoiding contamination. • Quality control is essential. Samples shall be collected from different parts of each stockpile (at least 3 - 5 samples per stockpile). The tests shall include grading, plasticity and compaction and CBR/DCP-DN. • The cement shall be the Portland type conforming to standard SABS 471 or equivalent. The cement shall be stored appropriately; protected from water and damp conditions and contact with walls and floors or ground. The sand and stone shall be collected from sources that have been approved by an approved laboratory and in conformity with standard SABS 1083 or equivalent. The sand and stone shall be clean; without clay and organic material. • The equipment and tools to be used shall be approved by the Engineer. • Topographical surveys shall be used control the works, e.g. pegs string shall be placed to mark the levels for the (sub) base layer to guide the labourers and towed grader operators during final cutting. • The workers involved in the works shall be provided with protective equipment including work suits/overalls, gloves, protective eyeglasses, respiratory dust masks and boots. • It is important to ensure that weather conditions will not affect the execution of the works. No works shall be carried out during rain, during periods of strong winds or soon after the rains when there is an accumulation of moisture on the surface. • An experimental (trial) section shall be constructed to identify the materials, the mixtures, to calibrate equipment for mixing and to establish a method statement together with the Engineer. • It is necessary to carry out laboratory tests in order to ensure good quality works in conformity with the specifications. This shall be paid under Series 600. • The work area shall be demarcated and delimited.	

Standards	• Clean and prepare the surface of the (sub) base to remove inappropriate material. • The bags of cement/lime shall be distributed uniformly on the surface throughout the whole width of the lane of the road, following specifications for the cement/lime content (the spacing of the cement/lime bags shall be varied according to the cement/lime content). The position for placing the cement/lime bags should be measured out and marked. • Before spreading and mixing of the cement/lime with soil, the bags shall be counted and registered by the Engineer. • The distribution of cement/lime bags, mixing, leveling, watering, compaction of the soils, and final cutting and finishing shall be carried out in a time period of 8 hrs or less from the time the cement comes into contact with the soils. The process from mixing to completion from the time water is first added shall be less than 6 hrs. This is important for achievement of the strength given in the design. • Precautions shall be taken in handling cement and the correct approach shall be used which includes prevention of damage to the environment and potential health risks to the staff involved in the works and the road users. • Immediately after distribution of the cement, proceed to mix the cement and the soil using the disc harrows. The equipment shall be approved by the Engineer to cope with the quantity of soils being stabilised. • The maximum tolerance limits for the cement content after mixing shall be $\pm 10\%$ of the specified values and the minimum value shall not be less than 70 % of the value specified. • Care shall be taken to mix the full depth of the layer without contaminating it with the material from the layer below. • Care shall be taken to avoid lamination and poor mixing at construction joints.
	• Care shall be taken to control the moisture. The moisture content shall not exceed the optimum moisture content by more than 2 %. • Compaction shall follow immediately after the leveling is completed. • The type of rollers shall be chosen based on the type of soil being compacted.
	• Pneumatic and vibratory steel rollers for plastic materials. • Pneumatic and vibratory steel rollers for fine and granular materials including sandy soils. • Grid roller followed by pneumatic and vibratory steel roller for very coarse material. • Compaction shall be carried out from the shoulder to the centerline for normal camber or from the lower side to the upper side on a super-elevation. • The mix shall be leveled and final cutting shall be carried out by an experienced towed grader operator in order to achieve the required levels and the tolerance limits for the dimensions and final levels of the base given in the design. • The period from the start of compaction to completion shall not exceed 3 hrs. • The level of compaction shall be 100 % Mod. AASHTO of the dry density obtained in the laboratory. Where this is not achievable due to the characteristic of the materials, the actual achievable densities shall be determined through a trial/experimental section and changes to the lower limit specification for the density shall be given in writing by the Engineer. • Any completed section shall be prepared so that the curing process can begin immediately. The section shall be protected and cured for a period not less than 7 days. • The process of curing shall consist of a method that keeps the surface of the layer always moist. Water for curing shall be applied 24hrs after completion of the construction process. • Apply a sand cover over the layer and keep the sand always moist for the duration of the curing period. • <i>Prime shall not be considered as a means of curing because it has been shown that the cement treated base still dries out after prime has been applied.</i> • <i>Application of water directly on the cement-stabilised base i.e. without the sand cover or equivalent shall not be permitted at all.</i> • Regular traffic shall not be allowed on the cement-stabilised base until after it has been sealed. Limited use by construction traffic may be allowed but only after 7 days of curing.

	- Where a detour is not provided, the construction of a cement-treated base shall be carried out in one lane until the seal is applied and the lane has been opened to traffic. Only then should construction of the cement-treated base on the other lane be carried out. This avoids the trafficking of the cement treated based for long periods of time which would result in serious damages to the surface of the base. The defects that occur as a result of trafficking an unsealed base can lead to premature failures in the base and in the surfacing not long after the road is completed because wearing of the cement stabilised base is difficult to correct and this also affects the bond between the base and the surfacing. - Once the curing has been completed, lamination resulting mainly from compacting thin layers of material over a smooth surface of the cement-treated base shall be checked by dragging a chain over the surface. A reduction in the pitch of the sound from the chain is indicative of lamination in the base. The lamination shall be dug out completely and the excavation shall be filled with cold mix asphalt.
	- The works area shall be cleared and any materials removed shall be placed at a location approved by the Engineer not less than 10 m outside the road reserve and shall be spread in a layer not more than 20 cm thick. - Remove temporary sign on conclusion of the work activities.
Materials	- All materials shall be approved by the Engineer. - Cement shall be the Portland type conforming to standard SABS 471 or equivalent. - The water shall be clean and free from concentration of acids, salt, sugar, and other organic materials or chemicals which may weaken the strength of the concrete.
Measurement	Code 350 - M³ - Measurements are taken on the volume of the compacted stabilised soil. - MAE Code 351 - Tons - Measurements are taken on the quantity of cement used in Tons.



Sub base spreading



Compacted sub base

Figure 9-16 Construction of Sub-base and Base using Labour Based Technology

Activity:	Construction of emulsion treated bases (ETB) In-situ mixing Plant mixing Stabilising agent, Emulsion SS60	Code: 352 353 354
Reference Code		
Application	These codes are used for the preparation and placement of emulsion-treated bases in the construction of low volume roads, Figure 917. The construction of Emulsion Treated Bases (ETB) involves the selection of materials, mixing with emulsion in the proportions given in the mix design, application of the ETB in or on the base, mixing, compaction and curing.	
Tasks	- Placing safety signs - Setting out of the work site, including placement of pegs for control of levels and dimensions - Excavating materials from borrow pit and stockpiling. - Quality testing (grading, plasticity, compaction and CBR) Mixing in-situ (code 352) - Loading, dumping and spreading the material to achieve the thickness given in the pavement design. - Applying compaction water. The moisture should be below optimum moisture content by the percentage equal to the emulsion required as given in the mix design. - Mixing of the materials in-situ to achieve a homogeneous spread of moisture in the material. - Spraying the emulsion on the moist material and mixing the moist soil with the emulsion using a tractor attached disc harrow.	
	Plant mix (code 353) - Preparing the batching site including compacting the soil and constructing concrete platforms for stockpiling materials. - Loading and dumping materials at the batching site. - Placing and mixing the materials (soils) with water and emulsion in proportions given in the design. - Transporting the ETB from the plant to the site and dumping. Compaction - Spreading, final cutting and compacting the ETB. - Curing the ETB. - Removal of temporary signs	
Standards	- In order to guarantee road safety, adequate signs shall be placed on the road. - The contractor shall put in place measures to ensure the safe and uninterrupted passage of regular traffic. This may include construction of a detour as may be instructed by the Supervising Engineer. - The contractor shall stockpile the materials, avoiding contamination. - Quality control of the material is essential. Samples shall be collected from different parts of each stockpile (at least 3 – 5 samples per stockpile). The tests shall include grading, plasticity and compaction. In-situ mixing (code 352) - The material shall be hauled, dumped and spread to achieve the thickness given in the design when compacted. - Water shall be applied such that the resultant moisture content is below Optimum fluid content (OFC) by a percentage equal to the quantity of the emulsion required for the ETB. OFC = OMC = Moisture in the soil + added moisture + emulsion - It is possible to dilute the emulsion with water to delay breaking but care should be taken not to exceed the OFC. - The material shall be mixed thoroughly with water to form a homogeneous mix using disc harrow.	

	- Emulsion shall be sprayed on the moist material in such a way that the quantities required for stabilization are achieved with a tolerance of +/- 5%. - The material shall be mixed thoroughly as follows: Tractor towed disc harrow - run a disc harrow at high speed thorough the material immediately after applying the emulsion so that mixing is carried out before the emulsion breaks. Mixing should continue until the ETB is homogeneous. Plant mixing (code 353) - The material shall be added into the mixing plant followed by the addition of water. Emulsion is added last after a thorough mixing of the material and water in a concrete mixer. The added water shall be enough to increase the moisture content of the ETB mix to OFC. - The ETB shall be mixed until it is homogeneous. - Transportation to site shall be carried out immediately and the material shall be covered to prevent the emulsion from breaking too quickly. - Prior to mixing, tests should be carried out to determine the breaking time of the emulsion in the ETB to check how long the mix can last before the emulsion begins to break. - The material shall be dumped and spread immediately ready for compaction. Compaction - The ETB shall be spread to the full width and levels of the base given in the design. Final cutting shall be carried out after 2 to 3 passes of the roller. - The surface of the ETB shall be free of irregularities; a maximum tolerance of 20 mm shall be used for irregularities measured with a 2 m straight edge. - If the material is fine-grained e.g. sand, light compaction equipment shall be used. This may include a 1.7 tonne vibratory steel roller or a light pneumatic roller. This is because fine materials are not stable enough and the ETB layer tends to deform under the weight of heavy compaction plant. - Compact to refusal unless there are indications of degradation of the materials or excessive deformation of the layer. - However, heavy plant can be used on coarse materials which are more stable when stabilised with emulsion. - For quality control of compaction, the field densities shall be measured using a Troxler or nuclear density gauge. The Compaction judgment chart shall be used for assessment and approval of compaction. - The quality of the ETB applied shall be determined by carrying out bitumen content tests. The quantity of the bitumen in the emulsion shall conform to the designs.
Materials	- All materials shall be approved by the Supervising Engineer. - Coastal sand with 90 – 95% passing 2 mm or coarser can be used for the construction of ETB. - The emulsion shall be SS60 and certified by the manufacturer.
Measurement	- M³ - Measurements are taken on the volume of compacted ETB. - MAE



Mixing of emulsion and sand base using tractor and disc harrow

Application of ETB using a paver

Figure 9-17 Construction of Emulsion Treated Base

Activity:	Construction of blended bases	Code: 355
Reference Code		
Application	This code is used for the preparation and construction of blended bases in the construction of roads, Figure 918. Blending of bases is the mixing of two or more materials which are inferior in quality individually but produce a better quality material when mixed. The blended material should meet the minimum specifications for base course.	
Tasks	• Placing safety signs • Setting out of the work site, placing pegs for control of dimensions and levels. • Excavating materials from borrow pits and stockpiling. • Quality testing (grading, plasticity and compaction) • Loading, dumping and spreading the first material to achieve the thickness given in the blending design. • Loading, dumping and spreading the second material to the thickness specified in the blending design. • Blending the materials by mixing thoroughly with disc harrows. • Watering, mixing and compacting the blended materials. • Final cutting to remove irregularities. • Removing temporary signs.	
Standards	• In order to guarantee road safety, adequate signs shall be placed on the road. • The contractor shall put in place measures to ensure safe and uninterrupted passage of regular traffic. This may include construction of a detour as may be instructed by the Supervising Engineer. • The contractor shall stockpile the materials, avoiding contamination. • Quality control is essential. Samples shall be collected from different parts of each stockpile (at least 3 samples per stockpile). The tests shall include grading, plasticity and compaction. • A blending design should be carried using proportion of 70:30, 50:50, 30:70 or other proportions as directed by the materials laboratory. • The materials shall be loaded, transported, dumped and spread one at a time. The layers of the two materials shall, after spreading, be one upon the other. The proportions of the thicknesses shall be the same as the proportions given in the blending design. • The materials shall be mixed before watering. When the blend is homogeneous, water shall be applied and mixing shall be continued until the compaction moisture is evenly distributed and the mix is at optimum moisture content. • The blend shall be compacted to a minimum density of 95% Mod. AASHTO or until refusal using a roller of minimum weight of 1.7 - 8 tons. • For quality control of compaction, the field densities shall be measured using the Sand Replacement method or the Troxler or nuclear density gauge. The Compaction judgment chart shall be used for assessing and approving compaction.	
Materials	• All materials shall be approved by the Supervising Engineer. • The soils earmarked for blending shall meet the required minimum CBRs given in the blending design. The blend shall be tested and its properties should meet the minimum standards.	
Measurement	• M³ • Measurements are taken on the volume of compacted soil. • MAE	



Blending of sand and calcrete using excavator – blending can be carried out using machines in borrow pit while construction can be by labour based technology

Note: blending can also be carried out by applying the materials in 2 layers followed by mixing with disc harrow.



Completed blended base

Figure 9-18 Construction of Blended Bases

Activity:		Code:
Construction of Armoured Bases.		356
Aggregate for armouring		357
Reference Code		
Application	This code is used for the preparation and placement of armoured bases in the construction of roads, Fig 9-19. Armouring of bases involves the application of coarse aggregate on top of weak bases. This code provides the procedures for the application of the armouring. The purpose of armouring is to increase the bearing strength of the base and also strengthen the interface between the base and the surfacing.	
Tasks	• Placing safety signs • Setting out of the work site; placing pegs for control of levels and dimensions. • Excavating materials from borrow pits and stockpiling • Quality testing (grading, plasticity, compaction and CBR) • Loading, dumping and spreading the material to achieve the thickness given in the pavement design. • Watering and compacting the base to 91% of mdd compaction. • Final cutting to remove irregularities. • Watering of the surface of the base. • Applying aggregate and spreading to form a uniform cover. • Rolling-in of aggregate into the base. • Removing temporary signs.	
Standards	• In order to guarantee road safety, adequate signs shall be placed on the road. • The contractor shall put in place measures to ensure the safe and uninterrupted passage of regular traffic. This may include construction of a detour as may be instructed by the Supervising Engineer. • The contractor shall stockpile the materials, avoiding contamination. • Quality control is essential. Samples shall be collected from different parts of each stockpile (at least 3 samples per stockpile). The tests shall include grading, plasticity and compaction and CBR/DCP-DN. • The materials shall be loaded, transported, dumped, spread, watered and compacted to an approximate density of 91% Mod. AASHTO. • The aggregate 30-40 mm in size shall be placed and spread on the surface of the compacted base. The armouring shall be inspected and all gaps in the aggregate filled by hand. • The aggregate shall be rolled into the base using a heavy vibratory steel roller at high amplitude vibratory mode. Passes of the roller shall be applied uniformly until refusal. This means the aggregate cannot penetrate any further after successive applications. At this stage the weak base is fully armoured.	
	• For quality control of compaction, the field densities shall be measured using a Troxler gauge. The Compaction judgment chart shall be used for assessing and approving compaction.	
Materials	• All materials shall be approved by the Supervising Engineer. • The CBR of the base material shall be tested and shall not be lower than 20% and the PI shall not be more than 22. • The aggregate shall be 30-40 mm in size and a maximum size of 50 mm is allowable. The aggregate must be lean and without clay coating. It could be crushed stone, sieved coarse gravel (including calcrete or limestone, quartz, rhyolite, etc.)	
Measurement	• M³ • Measurements are taken on the volume of compacted soil. • MAE	



Spreading aggregate for armouring; sand base is compacted and the top 50 mm was scarified before applying aggregate (1)



Completed section of armoured base after rolling (3)



Figure 9-19 Construction of Blended Base

Activity:	Construction of strip road in conformity with the design Plain concrete Reinforced concrete Stone masonry	Code: 360 361 362
Reference Code		
Application	This code is for the construction of strips using plain concrete (code 380) or reinforced concrete (code 361) or stone masonry (code 362) on the road pavement. The strips are used as paving on the road to support the wheels of the vehicles. They are used on low volume roads particularly on slopes (with gradient in excess of 7% and slippery soils) with the objective of improving the transitability and road safety. The choice of material, the dimensions and positions shall be as indicated by the Engineer. Placement of the strips is commonly used for spot improvements.	
Tasks	• Placing temporary safety signs. • Ensuring the safety passage of traffic. • Supplying and transporting of the materials to the work site. • Identifying, marking and establish positions for the strips with the Engineer. • Excavating the foundations and compacting the base in conformity with the design. • Verifying the vertical and horizontal alignment using the spirit level metric tape. Code 360 and 361 • Placing formwork. • (In case of code 381; placing wire mesh reinforcement to reinforce the concrete for the strips). • Producing, placing, compacting and curing the concrete. • Removing formwork. • Filling and compacting the base on the edges of the strips. Code 362 • Placing and curing the stone masonry. • Filling and compacting the base on the edges of the strips. • Clean the work area. • Removing temporary signs once activities have been concluded.	
Standards	• In order to guarantee site safety, adequate signs shall be placed on the road. • The works shall be organized in such a way that the passage of traffic will not be interrupted by more than 10 min. There must always be a demarcated layout to facilitate movement of traffic and ensure safety without causing hazard for any vehicles. It is recommended to execute the work on one side of the road while the other side is open to traffic then on the other in order to avoid interrupting the traffic. • The excavation of the strips shall be carried out according to the dimensions of strips, excavating vertically with shovels and spades ensuring that the materials do not fall back into the excavations. Excavation shall be carried to firm ground free of marsh and organic material.	

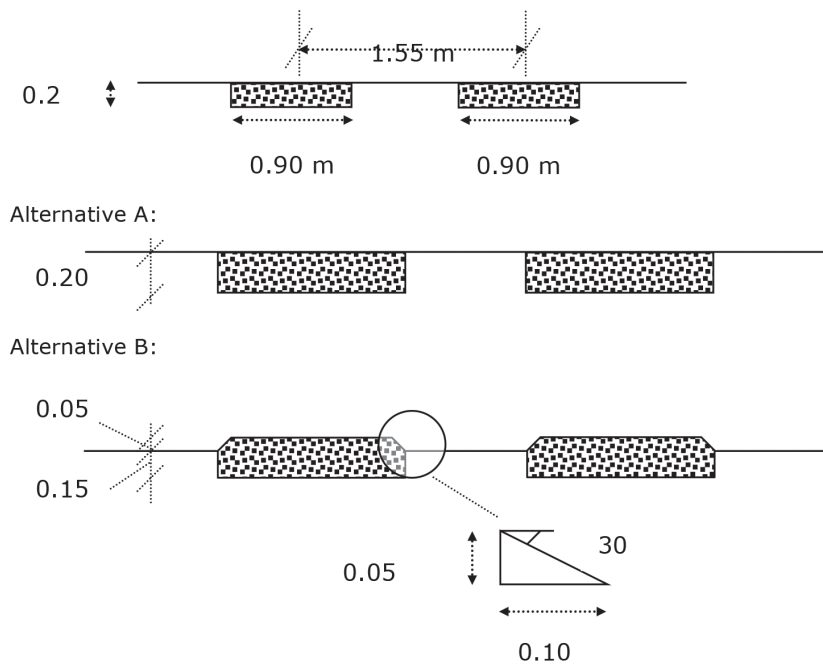
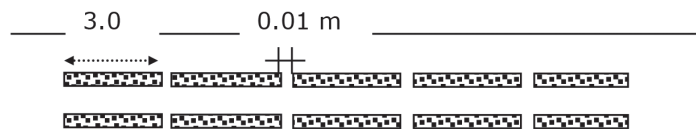
Transverse profile**Longitudinal Profile**

Figure 9-20 Drawings showing layout and dimensions of for the construction of strip roads

- The depth of the excavation should be verified. If the foundation material is weak (marshy, clayey, etc.) backfill with stronger material preferably granular applying a layer of 10 cm minimum compacted.
- The foundations should be leveled.

Concrete

- The formwork shall be carried out using wooden planks with a minimum thickness of 2 cm to form boxes and the formwork shall be fixed with adequate rigidity to prevent excess movement during the placement of concrete.
- The concrete shall be of class C20 with proportions of 1:2:4 and it shall be cast in the foundation to a thickness of 0.2 m and a width of 0.9 m and a length of 3 m, Figure 9-20, Figure 9-22.
- For the production of concrete class C20, **see code 642**.
- Where reinforcement is included a wire mesh (6 mm diameter @ 200 mm spacing) shall be placed as shown in Figure 9-21.

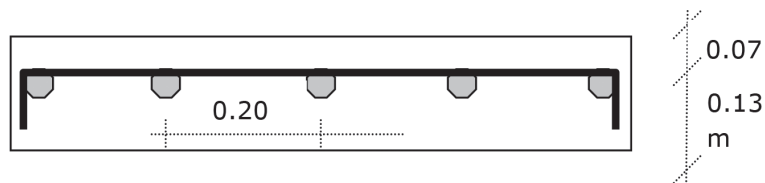


Figure 9-21 Wire mesh reinforcement for concrete strip on a strip road

- For the standard for the wire reinforcement **see code 633**.

Stone masonry

- See code 252 for standards for protection works using stone masonry.
- When applying stone masonry a layer of cement mortar (1:4, cement to sand) shall be applied on the foundation and the stones shall be placed to sit in the mortar. The upper layer of stones shall be placed in such a way that the most regular or plain surfaces of the stones are at the top of the strips.
- The spaces between the big stone shall be filled with smaller stone embedded in cement mortar.
- Traffic shall not be allowed on the strips during the curing period (7 days) for the concrete and stone masonry.
- Clean the work area.
- Remove the temporary signs once the activities have been completed.

Materials

- The soils of foundations must be compacted to a minimum density of 93% Mod. AASHTO.
- For the standards of concrete materials, stone masonry and wire mesh reinforcement see codes 642, 252 and 633.

Measurement

- M²
- Measurements are taken of the length of road where strips have been constructed.
- MAE



Figure 9-22 Strip Roads – showing strips constructed using concrete

9.7 Series 400 Construction of Surfacing

9.7.1 Introduction

This section covers activity codes for the construction of surfacings. The codes set out the standards and specifications which should be followed and fulfilled during the execution of surfacing works.

Key components of the codes include:

1. The description of the activity code and its application.
2. The list of tasks which should be carried out and where necessary these are described briefly for clarity. It is important that all the activities are listed in order for the works to be priced properly.
3. The standards and specifications for the execution of the works
4. The specifications and handling of the materials.

The construction of low cost seals is delicate and should be carried out properly and the specifications should be adhered to strictly in order for the expected performance to be realised.

Activity: Application of Prime Coat MC 30 (or Emulsion or Tar Prime TP7)		Code: 410
Reference Code		
Application	This code covers the application of bitumen prime before the application of a bituminous surfacing. Cut-back bitumen MC30 (mixing bitumen 85/100 at the rate of 80% of petroleum and 20% of bitumen) or emulsion diluted 50:50 with water or tar prime (TP7)	
Tasks	• Placing safety signs. • Ensuring the continuous safe passage of regular traffic during the works. • Testing the equipment and the application rates. • Setting out the working area. • Supplying and storing the prime. • Providing protection equipment for the staff. • Cleaning and brooming the surface of the road base of all loose material and debris. • Placing markers to show the edge of carriageway and also to guide the sprayer during application. • Sprinkling water to moisten the surface of the base slightly and suppress dust. • Calibrating and carrying out test runs of the equipment before application. • Applying the prime coat. • Protecting the primed area during the curing. • Cleaning the working area. • Removing the temporary signs.	
Standards	• To guarantee road safety appropriate signs should be placed on the road. • The work activity should be organized in such way that the traffic movement shall not be interrupted for more than 10 min. A passage should be available allowing vehicles to use safely without causing danger to any vehicle. • The contractor should notify the supervisor 1 day before applying the prime coat for the application rates to be tested and confirmed by the Supervising Engineer. • The storage of the bituminous material should be in well closed drums or tanks approved by the Supervising Engineer. • The equipment and tools to be used must be approved by the Supervising Engineer. • Tests must be made on the equipment to control the rates and uniformity of application. • The base layer should be placed, smoothened, compacted and tested before the prime is sprayed. • The prime should be applied no longer than 24 hrs after the base has been broomed and cleaned of all loose and deleterious material. • Care must be taken to ensure that the weather does not interfere with the execution of the work. The application of bitumen should not be done during or when it is threatening to rain, after sun set, during strong winds or after the rain when the base is too moist. • Before the application of the prime, water should be sprinkled on the base in order to break the surface tension. Care should be taken to avoid over application of the water which may delay the commencement of prime application. • The application should be done in such way that the prime distribution is uniform throughout the sprayed area. • The staff should be well equipped with helmets, eyeglasses for protection, boots, gloves and overalls. Contact of bitumen with skin must be avoided. • Mark the limits of the working area referenced from centreline of the road with predefined lane widths in conformity with the design and the Supervising Engineer's instructions. • The prime is normally applied on one lane at a time. • If the application is done in several lanes, overlapped should be 100 mm minimum. Place resistant paper strip at the beginning and end of each spray to ensure that the cross-sectional joints are well positioned. • The prime should not have a penetration of less than 3 mm but preferably it should be 5 mm. • The temperature for the application of the prime coat (cut-back bitumen MC-30) shall be as specified in the designs or according to the Supervising Engineer's instructions. For MC-30 prime, the recommended storage temperatures are 30-65°C and spraying temperatures should be between 45 - 60°C. • The temperature of the prime should be tested and verified. It is important to ensure that all the jets of the distribution bar/hand lance are working well; this test should be made out of the road area before spraying.	

- The height of the distributor bar must be set or adjusted before the application of the bituminous material so that each point of the road is sprayed by 2 nozzles and thus receives 2 flows separately to obtain the indicated application rate. This is illustrated in Figure 9-23.

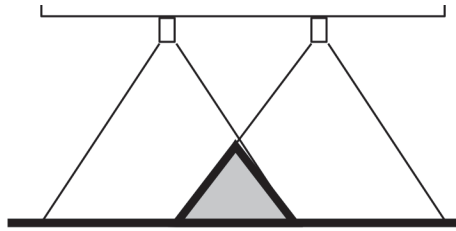


Figure 9-23 Illustration of application of prime

Setting up the height of binder spray bar for a single overlap

- Attach or weld a chain on the hand lance to guide the height of the nozzle in order to achieve the application configuration given in Figure 9-23.
- The edge limits of the primed area should be parallel lines with tolerance of not more than 25 mm. The edges should be marked with small stones or nails and string set out to give good straight lines and curvature conforming to the geometric design requirements. To avoid spraying outside the demarcated area shutter boards should be held in place on the edges of the area being sprayed and moved along in sequence with the movement of the hand lance during spraying.
- Vehicles shall not be allowed to pass on the freshly primed sections.
- The prime should be cured completely before allowing traffic on it only when it is absolutely necessary (normally 7 days should be allowed but 5 days may be sufficient depending on weather conditions). If this is not possible, sand, quarry sand or quarry dust should be applied over the prime but only after the prime has fully penetrated (24 to 48hrs after spraying) otherwise the sand will soak away the prime and prevent penetration.
- The working area must be cleaned appropriately.
- Any defects must be corrected so that all primed areas are covered adequately. Emulsion, stable 60 or SS 60 should be used for this.
- No material must be left in the carriage way, shoulders, side drains and mitre drains which may prevent efficient drainage.
- Bituminous waste material must be removed from the site to a place indicated by the Supervising Engineer. Other waste material should be removed and placed in a convenient place a minimum of 10 m away from the roadway, and spread in a layer no thicker than 20 cm.
- Remove the temporary signs.

Materials	- All material should be approved by the Supervising Engineer. - Prime is obtained as follows: - Dilution of asphaltic emulsion (SS60 – 60% of bitumen) with water in portion of 50/50. Clean water and free of acid, salt, sugar or metal concentration shall be used for dilution. - Cutting back 85/100 bitumen with 80% kerosene to get MC30. - For tendering purposes fog spray shall be applied at a rate of 0.6L/m². - Clean water free of acid, salt, sugar or metal concentration shall be sprinkled on the surface before the application of prime.
Measurement	- M² - Measurements are taken on the primed area. - MAE

Activity:		Code:
Application of fog spray for priming and rejuvenation		411
Reference Code		
Application	The fog spray is used to rejuvenate an asphaltic surfacing that shows loss of aggregate, for correction of defects, for sealing of very thin cracks (<3 mm) on existing pavements and/or priming of new bases stabilised with emulsion (ETB) or cement. The application is done with bitumen emulsion (SS60) diluted 50:50 with water.	
Tasks	• Placing safety signs • Ensuring the continuous and safe passage of traffic during the execution of works. • Setting out the work area. • Supplying and storing the material. • Providing protection equipment for the staff. • Cleaning and brooming the surface of the road base of all loose material and debris. • Testing the equipment and the application rates. • Applying the fog spray • Protecting the sprayed area during curing. • Cleaning the working area • Removing the temporary signs.	
Standards	• To guarantee road safety, appropriate signs should be placed in the road. • The work activity should be organized in such way that the passage of traffic shall not be interrupted for more than 10 min. A passage should be available allowing vehicles to use safely without causing danger to others. • The contractor should notify the supervisor 1 day before applying the fog spray (prime) for the application rates to be tested and confirmed by the Supervising Engineer. • The storage of the bituminous material should be in well closed drums or tanks approved by the Supervising Engineer. The emulsion shall be agitated once every 2 weeks to mix the material and prevent the development of lumps which occurs when the emulsion is kept still for long periods of time. • The equipment and tools must be approved by the Supervising Engineer. • Tests must be made on the equipment to control the rates and uniformity of application. • The base layer should be placed, leveled, compacted and tested before the fog spray is applied. • The fog spray should be applied no longer than 24 hrs after the base has been broomed and cleaned of all loose and deleterious material. • Care must be taken to ensure that the weather does not interfere with the execution of the work. The application of fog spray should not be done during or when it is threatening to rain, after sun set, during strong winds or after the rain when the base is too moist. • Before the application of the fog spray, water should be sprinkled on the base in order to break the surface tension. Care should be taken to avoid over application of the water which may delay the commencement of the application of the fog spray. • The application should be done in such way that the distribution of emulsion is uniform throughout the sprayed area. • The involved staff should be well equipped with eye glasses for protection, boots, gloves and overalls. Contact with skin must be avoided. • Mark the limits of the working area referenced from the centreline of the road with predefined lane widths to the edges in conformity with the design and the Supervising Engineer's instructions. • The fog spray is normally applied on one lane at a time. • If the application is done in several lines, overlapped should be 100 mm minimum. Place resistant paper strip at the beginning and end of each spray to ensure that the cross-sectional joints are well positioned. • When used on a new base, the fog spray should not have a penetration of less than 3 mm but preferentially it should be 5 mm. • The temperature for the application of the fog spray (SS60) shall be as specified in the designs or according to the Supervising Engineer's instructions. For SS60 spray, it is recommended that storage and spraying is at ambient temperatures. • It is important to ensure that all the jets of the hand lance are working well; this test should be carried out off the road area before spraying. • The height of the hand lance must be set or adjusted before the application of the bituminous material so that each point of the road is sprayed by 2 nozzles and thus receives 2 flows separately to obtain the indicated application rate. • Attach or weld a chain on the hand lance to guide the height of the nozzle.	

	• The edge limits of the primed area should be parallel lines with tolerance of not more than 25 mm. The edges should be marked with small stones or nails and string set out to give good straight lines and curvature conforming to the geometric design requirements. To avoid spraying outside the demarcated area shutter boards should be held in place on the edges of the area being sprayed and moved along in sequence with the movement of the hand lance during spraying. • Vehicle passage is not allowed in the freshly sprayed sections. • The fog spray should be cured completely before allowing traffic onto it only when it is absolutely necessary (normally 7 days should be allowed but 5 days may be sufficient depending on weather conditions). If this is not possible, sand, quarry sand or quarry dust should be applied over the sprayed area but only after the bitumen has penetrated adequately (24 to 48hrs after spraying) otherwise the sand or quarry sand and dust will soak away the emulsion and prevent penetration. • The working area must be cleaned appropriately. • Any defects must be corrected so that all sprayed areas are covered adequately. Emulsion stable 60 or SS 60 should be used for this. • No material must be left in the carriage way, shoulders, side drains and mitre drains which may prevent efficient drainage. • Bituminous waste must be removed from the site to a place indicated by the Supervising Engineer. Other waste material should be removed and placed in a convenient place a minimum 10 m away from the roadway and spread in a layer no thicker than 20 cm. • Remove the temporary signs.
Materials	• All material should be approved by the supervisor; • The fog spray is obtained through the dilution of asphaltic emulsion (SS60 – 60% of bitumen) with water in portion of 50/50. • For the tender effects the fog spray shall be applied in a proportion of 0.6 L/m². When used as a prime the application rate should be 1.0 – 1.2L/m². • Clean water and free of acid, salt, sugar or metal concentration.
Measurement	• M² • Measurements are taken of the sprayed area. • MAE

Activity:	Construction of Cold Mix Asphalt with defined thickness	Code: 420
Reference Code		
Application	This code is used for the construction of cold mix asphalt concrete pavement and may be used for sealing or repair of sections of damaged paved road or for the overlay of sections of compacted roads in semi-urban areas after the placement of sub-base. It must not be used to repair pavement with signs of deformation, structural defects, etc. This activity should follow the binder and prime coat application paid under codes 4-10 and 4-11.	
Tasks	• Placing safety signs • Guaranteeing the continuous and safe passage of traffic during the works. • Supplying and storing the materials. • Setting out the working area, marking segments on the cleaned primed base. • Cutting, removing and cleaning of all damaged material (paid under code 416). (If necessary, repairing the base or sub-base) • Applying binder coat or prime coat paid under the codes 4-10 and 4-11. • Preparing, laying and compacting the bituminous mix, Figure 9-24. • Controlling the levels of the surfacing using rails, or slides. • Preventing the diffusion or scattering of bituminous material out of the marked area. • Cleaning the working area. • Curing the cold mix asphalt. • Removing the temporary signs	
Standards	• To guarantee road safety appropriate signs should be placed on the road. • The work activity should be organized in such way that the passage of traffic shall not be interrupted for more than 10 min. A passage should be available allowing vehicles to use safely without causing danger to others. • The storage of the bituminous material should be in well closed drums or tanks approved by the Supervising Engineer. • The equipment and tools to be used must be approved by the Supervising Engineer. • Care must be taken to ensure that the weather does not interfere with the execution of the work. The application of asphalt concrete should not be done during or when it is threatening to rain, after sun set, during strong winds or after the rain when the base is too moist. • The staff involved in the works must be well equipped with helmets, eyeglasses for protection, boots, gloves and overalls. Contact with skin must be avoided. • Mark the limits of the working area referenced from centreline of the road with predefined lane widths to the edges in conformity with the design and the Engineer's instructions. • Application of binder and prime coat shall be according to codes 4-10 and 4-11. • The equipment and tools for the application cold mix asphalt concrete should be approved by the Supervising Engineer and should be in good condition. • Rails of predetermined height shall be placed and fixed on the edges with nails and these shall be used to guide the thickness of cold mix asphalt (loose) before compaction. Trial applications should be made to determine bulking factor of the cold mix asphalt (i.e. the difference between loose and compacted thickness) which is usually 1.2 -1.3. • The aggregate shall be placed in demarcated areas or marked out boxes in predetermined quantities which will achieve the thickness of cold mix stipulated in the designs. This process is called <i>spotting</i>. • Predetermined volumes of emulsion shall be poured on the heaps of aggregate followed immediately by mixing. • Generally, the cold mix asphalt shall be mixed in-situ by labourers equipped with appropriate hand tools but concrete mixers may also be used. • Cold mix asphalt shall be prepared at ambient temperatures. (However, when hot binders are used both the binder and aggregates and filler shall be heated and mixed and the temperature of the asphalt concrete shall be between 130 and 155 deg C and should not exceed 170 deg C.)	

	- The aggregate, filler and binder shall be mixed until a homogeneous mixture is obtained in which all particles are uniformly coated. Care shall be taken to avoid excessively long mixing times which can cause hardening of the binder (or overheating). - If the cold mix asphalt is prepared at a batching area the haulage of the asphalt concrete from the production area to the site shall be done using equipped canvas covered trucks or tractor trailers for the protection of the asphalt and to minimize loss moisture. - No vehicles should be allowed over freshly placed asphaltic concrete. - No asphalt concrete overlay shall be applied over a cracked surface or similar structure after rains. - Once mixed thoroughly the cold mix asphalt shall be spread and leveled with a straight edge guided by the side rails. - The supply of the material must be done in such a manner that all the operation is concluded without the necessity of working after sun set. - No pavement must be constructed during night time, with exceptions where there is artificial lighting and under the Supervising Engineer's authority. - No staff member is allowed to step on fresh pavement. - The mix shall be rolled using a pneumatic or vibratory steel roller in non-vibratory mode, applying a minimum of six passes or as directed by the Engineer.
	- Rolling shall start longitudinally at the sides and gradually progress towards the centre of the pavement, except on super-elevated curves, or where the area to be paved has a straight cross-fall where rolling shall begin on the low side and progress to the higher side, uniformly overlapping each preceding track, covering the entire surface. - During rolling, the rollers shall move at a slow but uniform speed (not to exceed 5 km/h). - The transverse and longitudinal joints should be placed with vertical edges and in straight lines and angles cut using an asphalt cutter. - When the operation is interrupted for insufficiency of asphaltic material, a joint should be made according to the previous descriptions.
	- The joint must coincide with horizontal markers or pegs on the road as much as possible in order for the daily work output to be easily determined. - The final layer shall be applied in such manner that it follows the profile of the existing pavement, must have no irregularities or defects. For any patching or correction drainage of water shall be enhanced by ensuring that the final level is slightly above the existing road level (+3 mm) to avoid accumulation of water (verify with a 2m straight edge). - The contractor must protect the working area and avoid any damage to the pavement. - The working area must be cleaned appropriately. - No material must be left in the carriage way, shoulders, side drains and mitre drains because it may impede drainage. - Asphaltic waste must be removed from the site to a place indicated by the Engineer. Other waste material should be removed and placed in convenient place a minimum 10 m away from the roadway and spread in layers no thicker than 20 cm. - Remove the temporary signs.
Materials	- All material should be approved by the Supervising Engineer. - The cold mix asphalt concrete should be made of aggregate, bitumen and fines. - The amount of voids must be carefully controlled and in general term should be 3 – 4%. Excessive voids make the concrete porous when there are little or no voids then the concrete becomes prone to shoving. Bitumen Emulsion K3-60 or K2-60 shall be used. - The emulsion should be stored in drums or binder tank and should be agitated once every 2 weeks to prevent coagulation of the bitumen which results in lumps. Once this happens the emulsion shall be considered spoiled unsuitable for use. - The emulsion shall be stored at ambient temperature and in a shed. Aggregate - The grading of aggregate must be according to Table 910 for low volume roads or Table 911 for higher traffic volumes (Table 4202/3 of the SATCC Standards). - The aggregate must be clean of dust, mud and other impurities with 2% maximum of inactive material (stone powder) or 1% of active material (cement).

Table 9-10 Grading of aggregate and cold mix specifications for low volume sealed roads

Sieve size (mm)	Coarse	Fine
20	100	
14	80 – 100	100
5	54 – 72	62 – 80
2.36	42 – 58	44 – 60
1.18	34 – 48	36 – 50
0.6	26 – 38	28 – 40
0.3	18 – 28	20 – 30
0.15	12 – 20	12 – 20
0.075	6 – 12	6 – 12
Bitumen Content (80/100) by weight of aggregate	5 – 7%	5.5 – 7.4%

Table 9-11 Grading specifications for aggregate for asphalt concrete

Sieve Size (mm)	Continuously Graded		
	Coarse	Medium	Fine
25	100	-	-
20	85 - 100	-	-
14	71 - 84	100	-
10	62 - 76	82 - 100	100
5	42 - 60	54 - 75	64 - 88
2.36	30 - 48	35 - 50	45 - 60
1.18	22 - 38	27 - 42	35 - 54
0.600	16 - 28	18 - 32	24 - 40
0.300	12 - 20	11 - 23	16 - 28
0.150	8 - 15	7 - 16	10 - 20
0.075	4 - 10	4 - 10	4 - 12
Aggregate	93.5%	93.5%	93.5%
Bitumen*	5.5%	5.5%	6.0%
Active Filler	1.0%	1.0%	1.0%

- The percentage of sand (nominal diameter 0.075 – 2 mm) in the aggregate must be 35% minimum.
- The hardness of the aggregate should be checked according to the TMH1 B2 method. The value of 10% FACT should be 80kN for low volume roads and 210kN for high traffic volumes and the minimum ratio between the wet and dry test results should be 75%.
- The absorption of water should be according to TMH1 B14 and C15 and not more than 1% (by weight) for the coarse grading and 1.5% (by weight) for the fine material.
- The shape of aggregate should be in accordance with TMH1 method B3 and the value should be in accordance with Table 9-12.

Table 9-12 Maximum flakiness for asphalt concrete aggregate

Nominal size of aggregate (mm)	Maximum flakiness index %
	Surfacing aggregate
	Grade 2/3
19.0	30
13.2	30
9.5	35
6.7	35

Supplementary fine material (nominal diameter <0.075 mm)

- If there is lack of fine material in the aggregate, quarry dust might be added (supplementary inactive material).
- The supplementary active material might be Portland cement and should meet SABS 471 or equivalent specifications.
- No more than 2% of the supplementary inactive material should be added.
- Clean water free of acid, salt, sugar or metal concentration shall be used.

Measurement

- M³ of the asphaltic concrete
- Measurements are taken of the volume of asphalt concrete from test holes for depth and surface dimensions of the asphalt layer.
- MAE



Mixing and laying cold mix asphalt (1)



Levelling cold mix asphalt (2)



Rolling cold mix asphalt (3)



Completed cold mix asphalt (4)

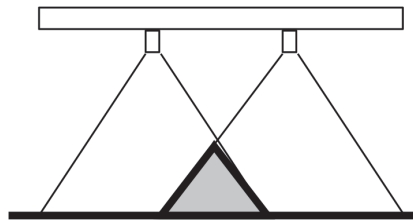
Figure 9-24 Construction of Cold Mix Asphalt using Emulsion

Activity:	Construction of single surface dressing using Bitumen Emulsion or 150/200 pen. or 85/100 pen or MC3000 and aggregate: 10 or 14 mm Variation of aggregate 10 or 14 mm Variation of bitumen emulsion or 150/200 or 85/100 pen or MC3000.	Code: 430 431 432
Reference Code		
Application	This code is for the construction of single surface dressing for sealing roads or as reseals. This application of single seal follows after works paid under code 410 (priming) or code 411 (application of fog spray) for resealing operations. This code involves the application of bitumen to hold the surfacing layer followed by the application of aggregate. Use reseal code (code 450) for repair of defects through aggregate loss or to seal defective surfacing or small surface defects on surfaced roads. The surfacing involves a single spray of bitumen and application of a single layer of aggregate (10 or 14 mm) Codes 431 and 432 are used for the reimbursement for variations in the application rates for the bituminous materials and aggregate according to the Supervising Engineer's instructions.	
Tasks	• Placing safety signs • Guarantee the continuous and safe passage of traffic during the execution of works. • Setting out the work area. • Supplying and storing the material. • Providing protection equipment for the staff. • Cleaning, brooming and removing all loose materials and debris. • Testing the equipment and the application rates. • (Applying prime or fog spray paid under codes 410 and 411) • Applying tack coat using emulsion or 150/200 pen or 85/100 or MC3000. • Spreading the aggregate. • Carrying out initial compaction of the surfacing. • Inspecting the surfacing and correcting defects or deficiencies and removing excess aggregate. • Rolling of surfacing. • Application of binder on the surfacing • Completing the rolling of surfacing. • Checking the final road levels. • Cleaning the work area. • Removing the temporary signs.	
Standards	• To guarantee road safety, appropriate signs should be placed on the road. • The work activity should be organized in such way that the passage of traffic shall not be interrupted for more than 10 min. A passage should be available allowing vehicles to use safely without causing danger to others. • The storage of the bituminous material should be in well closed drums or tanks approved by the Supervising Engineer. • The equipment and tools to be used must be approved by the Supervising Engineer. • Equipment shall be tested for the control of application rates and uniformity of spray. • The staff involved in the works must be well equipped with helmets, eyeglasses for protection, boots, gloves and overalls. Contact with skin must be avoided. • After using bitumen or similar products it is necessary to remove the bitumen from equipment and tools by heating and for those items that are not resistant to heat like glasses and boots the bitumen shall be removed using solvents such as kerosene and diesel. • Mark the edges of the area to be surfaced measuring from the centerline conforming to the width as given in the specifications or in accordance with the Supervising Engineer's instructions. • The surface of the base or existing surfacing should be free of irregularities or defects that might impede the free drainage of water. • Care must be taken to ensure that the weather does not interfere with the execution of the work. The application of bitumen should not be done during or when it is threatening to rain, after sun set, during strong winds or after the rain when the base is too moist. • For tender purposes, the application rates given in Table 913 shall be used.	

Table 9-13 Binder application rates for single surface dressing

Aggregate	Application rate for hot binder	Application rate for emulsions
10 mm	1.0L/m ²	1.6 L/m ²
14 mm	1.2L/m ²	2.0L/m ²

- A binder distributor shall be used for the application of the bitumen 150/200 or 85/100 or MC3000. The binder shall be applied at the temperature given in the materials section below or according to the Supervising Engineer's instructions and the temperature shall be measured and verified using a thermometer and recorded.
- The bitumen shall be applied uniformly without areas with less or excess binder.
- Under normal circumstances the binder shall be applied to cover one lane at a time.
- If the application is done on several lanes, overlaps should be 100 mm minimum. Place resistant paper strip at the beginning and end of each spray run to ensure that the cross-sectional joints are well positioned.
- It is important to ensure that all the jets of the distributor bar are working properly; this test should be made away from the road area before spraying.
- The height of the distributor bar must be set or adjusted before the application of the bituminous material so that each point of the road is sprayed by 2 nozzles and thus receives 2 flows separately to obtain the indicated application rate, Figure 95.

**Figure 9-25 Setting up the binder distributor spray bar for a single overlap**

- The height of the hand lance must be set or adjusted before the application of the bituminous material so that each point of the road is sprayed by 2 nozzles and thus receives 2 flows separately to obtain the indicated application rate.
- Attach or weld a chain on the hand lance to guide the height of the nozzle.
- The edge limits of the primed area should be parallel lines with tolerance of not more than 25 mm. The edges should be marked with small stones or nails and string set out to give good straight lines and curvature conforming to the geometric design requirements. To avoid spraying outside the demarcated area shutter boards should be held in place on the edges of the area being sprayed and moved along in sequence with the movement of the hand lance during spraying.
- Vehicle passage is not allowed on the freshly applied binder.
- 9.5 mm or 13.2 mm aggregate shall be used for a single surface dressing.
- The aggregate shall be spotted and labourers positioned on each heap ready to spread the aggregate.
- The aggregate shall be applied and spread immediately after the application of binder.
- In spreading the bitumen, care should be taken to avoid situations where there is no contact between aggregate or where there is excess aggregate forming an extra layer on top.
- The application rate for the aggregate shall conform to the values given in Table 9-14.

Table 9-14 Aggregate application rates for single surface dressing

Aggregate	Application rate for aggregate
10 mm	0.008m ³ /m ²
14 mm	0.012m ³ /m ²

	• The aggregate must dry up before application on the road. • Compaction shall be carried out properly, applying passes from the edge towards the centerline of the road immediately after the application of the aggregate using a pneumatic roller with minimum 1.7 tonnes in weight. • Once the bitumen has set, the surfacing shall be inspected. To produce a uniform distribution of the aggregate, care must be taken to add material where there is shortage (under application) and to remove material where there is excess that is forming an extra layer on top • Rolling shall be continued until a total of 8-12 passes of a pneumatic or steel drum roller in non-vibratory mode. • The second coat of binder shall be applied (and left to break in the case of emulsion). • Rolling shall continue and a minimum of 8 passes should be applied once the emulsion has broken. • If hot binders are used then rolling shall continue for 3 more days applying a minimum of 8 passes of the roller during the hottest part of the day. Bitumen should start showing in the interspaces between the aggregate. • After completing the construction of the surfacing, check the profile of the road and ensure the pavement is free of irregularities in order to allow good drainage. • No material must be left on the carriage way, shoulders, side drains and mitre drains which may prevent efficient drainage. • Traffic shall be allowed on the completed lane after completion of compaction. • Bituminous waste must be removed from the site to a place indicated by the Supervising Engineer. Other waste material shall be removed and placed in a convenient place a minimum 10 m away from the roadway, and spread in a layer no thicker than 20 cm.																			
Materials	All materials shall be approved by the Supervising Engineer. Bitumen • Emulsion is the most preferred binder and K2-60 and K3-60 should be used for this purpose. K1-60 may be used on steep slopes and on aggregate with low fines content in order to aid adequate penetration of the emulsion. • Bitumen 85/100 penetration grade shall be in accordance with SABS 307. • MC3000 shall be prepared from cutting back 85/100 pen with 8 to 10% kerosene or purchased directly. • For better performance it is recommended that 150/200 pen bitumen is prepared by cutting back 85/100 pen with 6-10% diesel. This binder is more durable and hardens slowly thus giving longer life to surfacings. 150/200 is good for Otta seals, surface dressing, etc. • Limit of temperatures for heating the binders and for application shall be according to the designs or the instructions of the Supervising Engineer. If not the specifications given in Table 9-15 shall be used. Table 9-15 Spray temperature for cutback binders <table><tr><th rowspan="2">Material</th><th colspan="3">Temperature limits for spraying (°C)</th></tr><tr><th>Minimum</th><th>Maximum</th><th>Recommended</th></tr><tr><td>85/100 pen</td><td>165</td><td>190</td><td>175</td></tr><tr><td>150/200 pen</td><td>135</td><td>155</td><td>145</td></tr><tr><td>MC3000</td><td>135</td><td>155</td><td>145</td></tr></table> • Aggregate • The grading of the aggregate shall be in accordanceTable 916. If the class has not been specified then use class 2 for calculating unit price. • Maximum flakiness is given in Table 9-17. • Crushed stone shall be single nominal size of 14 or 10 mm free of dust, soil coating, sand and any impurities with fines content less than 1.5% by weight of material passing 0.425 mm sieve. • If the grading of the aggregate fails to meet specifications given in Table 9 -16, the spray rates shall be adjusted to suit the available material, Table 9-16.	Material	Temperature limits for spraying (°C)			Minimum	Maximum	Recommended	85/100 pen	165	190	175	150/200 pen	135	155	145	MC3000	135	155	145
Material	Temperature limits for spraying (°C)																			
	Minimum	Maximum	Recommended																	
85/100 pen	165	190	175																	
150/200 pen	135	155	145																	
MC3000	135	155	145																	

- The strength of the aggregate shall be determined according to TMH 1 Section B2. A minimum value of 10% FACT is 80kN for low volume roads and 210kN for higher traffic and the minimum ratio of wet test to dry test should be 75%. The maximum values for the aggregate are given in Table 9-16.

Table 9-16 Grading specifications for single surface dressing

Sieve size (mm)	Class	Percentage passing sieves		
		14 mm	10 mm	7 mm
37.5	Grade 1 or 2			
25				
20		100		
14		85-100	100	
10		0-30*	85-100	100
7		0-5**	0-30*	85-100
4.75			0-5**	0-30*
2.36				0-5**
	Grade 3	This grade satisfies all prerequisites of grade 1 and 2 except *0-50 and **0-10		
% passing 4.75mm	Grade 1	0.5	0.5	0.5
	Grade 2	1.5	1.5	2.0
	Grade 3	2.0	2.0	3.0
% passing 0.075mm	Grade 1	N/A	N/A	N/A
	Grade 2	0.5	0.5	1.0
	Grade 3	1.5	1.5	1.5

Table 9-17 Aggregate specifications for single surface dressing – maximum flakiness values

Nominal size of aggregate (mm)	Maximum flakiness index %
	Surfacing aggregate
	Grade 2/3
20	30
14	30
10	35
7	35

Measurement

440 M²
441 M³
442 L

For code 440 measurements are taken on the surfaced area.

For code 441 and 442 measurements are taken on volume of aggregate and the volume of bitumen applied compared with the application rates given in the work standards.

MAE

Activity:	Construction of single surface dressing with sand capping using bitumen emulsion, 150/200 pen or 85/100 pen or MC3000 and aggregate: 9.5 or 13.2 mm. Variation of aggregate 9.5 or 13.2 mm. Variation of bitumen 150/200; 85/100 pen; MC3000 or Emulsion (K3-60). Variation of sand.	Code: 440 441 442 443
Reference Code		
Application	This code is for the construction of single surface dressing for sealing roads or as reseals followed by the application of a sand seal capping, Figure 926. This application of single surface dressing with sand capping follows after works paid under code 410 (priming) or code 411 (application of fog spray) for resealing operations. This code involves the application of bitumen to hold the surfacing layer, application of aggregate, rolling, and another application of binder followed by application of sand and rolling over the sand. Use reseal code (code 450) for repair of defects due to aggregate loss or defective surfacing or small surface defects on surfaced roads. The surfacing involves application bitumen and application of a single layer of aggregate (9.5 or 13.2 mm) Codes 441, 442 and 443 are used for the reimbursement for variations in the application rates for the bituminous materials, aggregate and sand according to the Supervising Engineer's instructions.	
Tasks	• Placing safety signs • Guaranteeing the continuous and safe passage of traffic during the execution of works. • Setting out the work area. • Supplying and storing the material. • Providing protection equipment for the staff. • Cleaning, brooming and removing all loose materials and debris. • Testing the equipment and the application rates. • (Applying prime or fog spray using codes 410 and 411) • Applying tack coat using 150/200 pen or 80/100 pen or MC3000 or Emulsion. • Spreading the aggregate. • Carrying out initial compaction of the surfacing. • Inspecting the surfacing and correcting defects of deficiencies and removing excess aggregate. • Applying a second coat of binder in the case of emulsions. • Applying sand capping. • Rolling over the sand capping. • Checking the final road levels. • Cleaning work area. • Removing the temporary signs.	
Standards	• To guarantee road safety, appropriate signs should be placed on the road. • The work activity should be organized in such way that the passage of traffic shall not be interrupted for more than 10 min. A passage should be available allowing vehicles to use safely without causing danger to others. • The storage of the bituminous material should be in well closed drums or tanks approved by the Supervising Engineer. • The equipment and tools to be used must be approved by the Supervising Engineer. • Equipment shall be tested for the control of application rates and uniformity of spray. • The staff involved in the works must be well equipped with helmets, eyeglasses for protection, boots, gloves and overalls. Contact of bitumen with skin must be avoided. • After using bitumen or similar products it is necessary to remove from equipment and tools the bitumen by heating and for those items that are not resistant to heat like glasses and boots the bitumen shall be removed using solvents such as kerosene and diesel. • Mark the edges of the area to be surfaced measuring from the centerline conforming to the width as given in the specifications or in accordance with the Supervising Engineer's instructions. • The surface of the base or existing surfacing should be free of irregularities or defects that might impede the free drainage of water.	

- Care must be taken to ensure that the weather does not interfere with the execution of the work. The application of bitumen should not be done during or when it is threatening to rain, after sun set, during strong winds or after the rain when the base is too moist.
- For tender purposes, the application rates in Table 918 shall be used.

Table 9-18 Binder application rates for single surface dressing with sand capping

Aggregate	Application rate for hot binder	Application rate for the emulsion
9.5 mm	1.6L/m ²	2.2L/m ²
13.2 mm	1.8L/m ²	2.4L/m ²

- A binder distributor fitted with a hand lance shall be used for the application of the bitumen emulsion or 150/200 pen or 80/100 pen or MC3000. The binder shall be applied at the temperature given in the materials section below or according to the Supervising Engineer's instructions and the temperature shall be measured and verified using a thermometer and recorded.
- The bitumen shall be applied uniformly with no areas deficient or with excess binder.
- Under normal circumstances the binder shall be applied to cover one lane at a time.
- If the application is done on several lanes, overlaps should be 100 mm minimum. Place resistant paper strip at the beginning and end of each spray run to ensure that the cross-sectional joints are well positioned.
- It is important to ensure that all the jets of the distributor bar are working properly; this test should be made away from the road area before spraying.
- The height of the distributor bar must be set or adjusted before the application of the bituminous material so that each point of the road is sprayed by 2 nozzles and thus receives 2 flows separately to obtain the indicated application rate.
- Attach or weld a chain on the hand lance to guide the height of the nozzle.
- The edge limits of the primed area should be parallel lines with tolerance of not more than 25 mm. The edges should be marked with small stones or nails and string set out to give good straight lines and curvature conforming to the geometric design requirements. To avoid spraying outside the demarcated area shutter boards should be held in place on the edges of the area being sprayed and moved along in sequence with the movement of the hand lance during spraying.
- Vehicle passage is not allowed on the freshly applied binder.
- 9.5 mm or 13.2 mm aggregate shall be used for a single surface dressing.
- Use a chip spreader (called the Chippy) or labourers with hand tools to apply aggregate immediately after the application of the binder.
- In spreading the bitumen, care should be taken to avoid situations where there is no contact between aggregate or where there is excess aggregate forming an extra layer on top.

Table 9-19 Aggregate application rate for single surface dressing with sand capping

Aggregate	Application rate for aggregate
10 mm	0.008m ³ /m ²
14 mm	0.012m ³ /m ²

- The aggregate shall be free of impurities and dust. For pre-coated aggregate MC30 (85/100 pen bitumen cut back with 80% kerosene) or dilute emulsion (diluted 50:50 with water) shall be mixed with the aggregate ensuring that the aggregate is coated completely and appears black.
- The aggregate must dry up before application on the road.
- Compaction shall be carried out properly, applying passes from the edge towards the centerline of the road immediately after the application of the aggregate using a pneumatic roller with minimum 1.7 tonnes in weight.
- Once the bitumen has set, the surfacing shall be inspected. To produce a uniform distribution of the aggregate, care must be taken to add material where there is shortage (under application) and to remove material where there is excess that is forming an extra layer on top.

	• The second coat of binder shall be applied in accordance with the design and immediately followed by the application of sand. • Rolling shall be carried out on the sand capping until a total of 8-12 passes. • Rolling shall be continue for 3 more days applying a minimum of 8 passes of the roller during the hottest part of the day, avoiding or minimizing aggregate crushing. • After completing the construction of the surfacing, check the profile of the road and ensure the pavement is free of irregularities in order to allow good drainage. • No material must be left on the carriage way, shoulders, side drains and mitre drains which may prevent efficient drainage. • Traffic shall be allowed on the completed lane after completion of compaction. • Bituminous waste must be removed from the site to a place indicated by the Supervising Engineer. Other waste material shall be removed and placed in a convenient place a minimum of 10 m away from the roadway, and spread in a layer no thicker than 20 cm. • The application rate for the aggregate shall conform to the values given in Table 9-19.																								
Materials	All materials shall be approved by the Supervising Engineer. Bitumen • Emulsion should be used for surfacing operations which are purely labour based. Emulsion K3-60 is recommended. Alternatively K2-60 may be used.K1-60is not recommended for this work because it sets too quickly and is difficult to use. • Alternatively, hot binders may be used i.e. 85/100 pen or 150/200 pen or MC3000. • Bitumen 85/100 penetration grade shall conform to the manufacturer’s specifications and should have penetration range of 85 to 100 dmm. • MC3000 shall be prepared from cutting back 85/100 pen with 8 to 10% kerosene. • For better performance it is recommended that 150/200 pen bitumen is used (prepared by cutting back 85/100 pen with 6-10% diesel). This binder is more durable and hardens slowly thus giving longer life to surfacings. 150/200 is good for Otta seals, surface dressing, etc. • Limit of temperatures for heating the binders and for application shall be according to the designs or the instructions of the Supervising Engineer. If not the specifications given in Table 9-20 shall be used. Table 9-20 Binder application temperatures for single surface dressing with sand capping <table><tr><th>Material</th><th colspan="3">Temperature limits for spraying (°C)</th></tr><tr><th></th><th>Minimum</th><th>Maximum</th><th>Recommended</th></tr><tr><td>85/100 pen</td><td>165</td><td>190</td><td>175</td></tr><tr><td>150/200 pen</td><td>135</td><td>155</td><td>145</td></tr><tr><td>MC3000</td><td>135</td><td>155</td><td>145</td></tr><tr><td>Emulsion</td><td></td><td></td><td>Ambient</td></tr></table> Aggregate • The grading of the aggregate shall be in accordanceTable 9-21 . If the class has not been specified then use class 2 for calculating unit price. • Crushed stone shall be single nominal size of 13.2 or 9.5 mm free of dust, soil coat- ing sand and any impurities with a fines content less than 1.5% by weight of materi- al passing 0.425 mm sieve. • If the grading of the aggregate fails to meet specifications given in Table 9-21 the spray rates shall be adjusted to suit the available material. Maximum flakiness values are given in Table 9-22.	Material	Temperature limits for spraying (°C)				Minimum	Maximum	Recommended	85/100 pen	165	190	175	150/200 pen	135	155	145	MC3000	135	155	145	Emulsion			Ambient
Material	Temperature limits for spraying (°C)																								
	Minimum	Maximum	Recommended																						
85/100 pen	165	190	175																						
150/200 pen	135	155	145																						
MC3000	135	155	145																						
Emulsion			Ambient																						

Table 9-21 Aggregate grading specifications for single surface dressing with sand capping

Sieve size (mm)	Class	Percentage passing sieves		
		14 mm	10 mm	7 mm
37.5	Grade 1 or 2			
25				
20		100		
14		85-100	100	
10		0-30*	85-100	100
7		0-5**	0-30*	85-100
4.75			0-5**	0-30*
2.36				0-5**
	Grade 3	This grade satisfies all prerequisites of grade 1 and 2 except *0-50 and **0-10		
% passing 4.75mm	Grade 1	0.5	0.5	0.5
	Grade 2	1.5	1.5	2.0
	Grade 3	2.0	2.0	3.0
% passing 0.075mm	Grade 1	N/A	N/A	N/A
	Grade 2	0.5	0.5	1.0
	Grade 3	1.5	1.5	1.5

The strength of the aggregate shall be determined according to TMH 1 Section B2. Minimum value of 10% FACT is 210kN and the ratio of wet test to dry test should be lower than 75%.

Table 9-22 Aggregate specification for single surface dressing with sand capping– maximum flakiness**Table 4302/10 SATCC**

Nominal size of aggregate (mm)	Maximum flakiness index %
	Surfacing aggregate
	Grade 2/3
200	30
14	30
10	35
7	35

Measurement

443 M²
 444 M³
 445 L
 446 M³

For code 443 measurements are taken on surfaced area.

For code 444, 445 and 446 measurements are taken on volume of aggregate and the volume of bitumen and volume of sand applied compared with the application rates given in the work norm.

MAE



Application of sand on SSD (3)

Completed SSD with sand seal capping (4)

Figure 9-26 Construction of single surface dressing with sand capping

Activity:	Construction of double surface dressing using bitumen Emulsion or 150/200 pen or 85/100 pen or MC3000 and aggregate: 10/20 mm Variation of aggregate 10/20 mm Variation of bitumen Emulsion or 150/200 or 85/100 pen or MC3000.	Code: 450 451 452
Reference Code		
Application	This code is for the construction of double surface dressing for sealing roads, Figure 927 Application of double surface dressing – curing process. This application of double surface dressing follows after works paid under code 410 (priming) or code 411 (application of fog spray) for resealing operations. This code involves the application of bitumen tack coat to hold the surfacing layer of 20mm aggregate and a bitumen seal coat followed by application of fine aggregate, 10 mm. Codes 451 and 452 are used for the reimbursement for variations in the application rates for the aggregate and bituminous materials according to the Supervising Engineer's instructions.	
Tasks	• Placing safety signs • Guaranteeing the continuous and safe passage of traffic during the execution of works. • Setting out the work area. • Supplying and storing the material. • Providing protection equipment for the staff. • Cleaning, brooming and removing all loose materials and debris. • Testing of the equipment and the application rates. • (Applying prime or fog spray using codes 410 and 411) • Applying tack coat using emulsion or 150/200 pen or 80/100 pen or MC3000. • Spreading the aggregate. • Carrying out initial compaction of the surfacing. • Inspecting the surfacing and correcting defects of deficiencies and removing excess aggregate. • Completing rolling of the surfacing. • Repeating the process for the second layer (seal coat). • Checking the final road levels. • Cleaning the work area. • Removing the temporary signs.	
Standards	• To guarantee road safety, appropriate signs should be placed on the road. • The work activity should be organized in such way that the passage of traffic shall not be interrupted for more than 10 min. A passage should be available allowing vehicles to use safely without causing danger to others. • The storage of the bituminous material should be in well closed drums or tanks approved by the Supervising Engineer. • The equipment and tools to be used must be approved by the Supervising Engineer. • Equipment shall be tested for the control of application rates and uniformity of spray. • The staff that is involved in the works must be well equipped with helmets, eyeglasses for protection, boots, gloves and overalls. Contact with skin must be avoided. • After using bitumen or similar products it is necessary to remove the bitumen from clothes and tools by heating or, for those items that are not resistant to heat like glasses and boots, the bitumen shall be removed using solvents such as kerosene and diesel. • Mark the edges of the area to be surfaced using measuring from the centerline conforming to the width as given in the specifications or in accordance with the Supervising Engineer's instructions. • The surface of the base or existing surfacing should be free of irregularities or defects to ensure free drainage of water. • Care must be taken to ensure that the weather does not interfere with the execution of the work. The application of the bitumen should not be done during or when it is threatening to rain, after sun set, during strong winds or after the rain when the base is too moist. • For tender purposes, the application rates in Table 9-23 shall be used.	

Table 9-23 Binder application rates for tack coats (Double Surface Dressing)

Aggregate	Application rate for hot binder	Application rate for emulsion
10 mm	1.0L/m ²	1.6L/m ²
14 mm	1.2L/m ²	2.0L/m ²
20 mm	1.4L/m ²	2.4L/m ²

- A binder distributor fitted with a hand lance shall be used for the application of the bitumen 150/200, 85/100 or MC3000. The binder shall be applied at the temperature given in the materials section below or according to the Supervising Engineer's instructions and the temperature shall be measured and verified using a thermometer and recorded.
- The bitumen shall be applied uniformly with no areas deficient or with excess binder.
- Under normal circumstances the binder shall be applied to cover one lane at a time.
- If the application is done on several lanes, overlap should be 100 mm minimum. Place resistant paper strip at the beginning and end of each spray to ensure that the cross-sectional joints are well positioned.
- It is important to ensure that all the nozzles of the hand lance are working well; this test should be made away from the road area before spraying begins.
- The height of the hand lance must be set or adjusted before the application of the bituminous material so that each point of the road is sprayed by 2 nozzles and thus receives 2 flows separately to obtain the indicated application rate.
- Attach or weld a chain on the hand lance to guide the height of the nozzle.
- The edge limits of the primed area should be parallel lines with tolerance of not more than 25 mm. The edges should be marked with small stones or nails and string set out to give good straight lines and curvature conforming to the geometric design requirements. To avoid spraying outside the demarcated area shutter boards should be held in place on the edges of the area being sprayed and moved along in sequence with the movement of the hand lance during spraying.
- Vehicle passage is not allowed on the freshly applied binder.
- 10 mm and 14 mm aggregate shall be used for the surface dressings.
- Use a chip spreader (the Chippy) or labourers to apply aggregate immediately after the application of the binder.
- In spreading the aggregate care should be taken to avoid situations where there is no contact between aggregate or where there is excess aggregate forming an extra layer on top.
- The application rate for the aggregate shall conform to the values given in Table 9-24.

Table 9-24 Aggregate application rates for double surface dressing

Aggregate	Application rate for aggregate
10 mm	0.008m ³ /m ²
14 mm	0.012m ³ /m ²
20 mm	0.014m ³ /m ²

- For small areas (< 50m²) aggregate shall be applied by hand.
- The aggregate shall be free of impurities and dust. For pre-coated aggregate MC30 (85/100 pen bitumen cut back with 80% kerosene) shall be mixed with the aggregate ensuring that the aggregate is coated completely and appears black.
- The aggregate must be dry before application on the road.
- Compaction shall be carried out properly by applying passes from the edge towards the centerline of the road immediately after the application of the aggregate using a pneumatic roller with minimum weight of 8 tonnes.
- Once the bitumen has set the surfacing shall be inspected. To produce a uniform distribution of the aggregate, care must be taken to add material where there is shortage (under application) and to remove material where there is excess (i.e. forming an extra layer on top)

	- Rolling shall be continued until a total of 12-15 passes on the first day. - Rolling shall continue for 3 more days applying 8 passes of the roller during the hottest part of the day. Bitumen should start showing in the interspaces between the aggregate. - After completing the construction of the surfacing check the profile of the road and ensure the pavement is free of irregularities that might inhibit good drainage. - No material must be left in the carriage way, shoulders, side drains and mitre drains which may prevent efficient drainage. - Traffic shall be allowed on the completed lane after completion of compaction. - Repeat the process for the second layer of the surface dressing (sometimes called the 'seal coat'. The first layer (sometimes but erroneously called a tack coat) should be opened to traffic for 2 weeks or more before the second layer is applied. - Use 10 mm aggregate (1/2 the dimension of the aggregate used for the first layer,20 mm). - Use a lower application rate for the second layer (see Table of binder application rates above). - Asphaltic waste must be removed from the site to a place indicated by the Supervising Engineer. Other waste material should be removed and placed in a convenient place minimum 10 m away from the roadway, and spread in a layer no thicker than 20 cm.																			
Materials	All materials shall be approved by the Supervising Engineer. Bitumen - Emulsion K3-60 and K2-60 are recommended for use. K1-60 is difficult to use because it breaks too quickly and is only recommended for construction of steep slopes and when using aggregate with very low fines content. - The emulsion shall be kept under shed and should be agitated every 2 weeks to prevent coagulation the binder coming out of suspension. - Bitumen 85/100 penetration grade shall be in accordance with SABS 307. - MC3000 shall be prepared from cutting back 85/100 pen with 8 to 10% kerosene. - For better performance 150/200 pen bitumen is recommended prepared by cutting back 85/100 pen with 6-10% diesel. This binder is more durable and hardens slowly thus giving longer life to surfacings. 150/200 is good for Otta seals, surface dressing, etc. - Limit of temperatures for heating the binders and for application shall be according to the designs or the instructions of the Supervising Engineer. If not the specifications given in the Table 9-25 shall be used. Table 9-25 Binder spray temperature for double surface dressing <table><tr><th rowspan="2">Material</th><th colspan="3">Temperature limits for spraying (°C)</th></tr><tr><th>Minimum</th><th>Maximum</th><th>Recommended</th></tr><tr><td>85/100 pen</td><td>165</td><td>190</td><td>175</td></tr><tr><td>150/200 pen</td><td>135</td><td>155</td><td>145</td></tr><tr><td>MC3000</td><td>135</td><td>155</td><td>145</td></tr></table> Aggregate - The grading of the aggregate shall be in accordance Table 9-26 . If the Grade has not been specified then use Grade 2 for calculating the unit price. - Crushed stone shall be single nominal size of 14 or 10 mm free of dust, soil coating sand and any impurities with a fines content less than 1.5% by weight of material passing 0.425 mm sieve. - If the grading of the aggregate fails to meet specifications given in Table 9-27 the spray rates shall be adjusted to suit the available material. - The strength of the aggregate shall be determined according to TMH 1 Section B2. The minimum value of 10% FACT is 210kN and the ratio of wet test to dry test should be greater than 75%.	Material	Temperature limits for spraying (°C)			Minimum	Maximum	Recommended	85/100 pen	165	190	175	150/200 pen	135	155	145	MC3000	135	155	145
Material	Temperature limits for spraying (°C)																			
	Minimum	Maximum	Recommended																	
85/100 pen	165	190	175																	
150/200 pen	135	155	145																	
MC3000	135	155	145																	

Table 9-26 Aggregate grading specifications for double surface dressing

Sieve size (mm)	Class	Percentage passing sieves		
		14 mm	10 mm	7 mm
37.5	Grade 1 or 2			
25				
20		100		
14		85-100	100	
10		0-30*	85-100	100
7		0-5**	0-30*	85-100
5			0-5**	0-30*
2.36				0-5**
	Grade 3	This grade satisfies all prerequisites of grade 1 and 2 except *0-50 and **0-10		
% passing 4.75mm	Grade 1	0.5	0.5	0.5
	Grade 2	1.5	1.5	2.0
	Grade 3	2.0	2.0	3.0
% passing 0.075mm	Grade 1	N/A	N/A	N/A
	Grade 2	0.5	0.5	1.0
	Grade 3	1.5	1.5	1.5

Table 9-27 Flakiness specification for aggregate for double surface dressing

Nominal size of aggregate (mm)	Maximum flakiness index %
	Surfacing aggregate
	Grade 2/3
20	30
14	30
10	35
7	35

Measurement

440 M²
 451 M³
 452 L

For code 450 measurements are taken on surfaced area.

For code 451 and 452 measurements are taken on volume of aggregate and the volume of bitumen applied compared with the application rates given in the work norm.

MAE

**Figure 9-27 Application of double surface dressing – curing process**

Activity: Construction of slurry seal		Code: 460
Reference Code		
Application	This code is used for the construction of slurry seal, Figure 928. Slurry seal is a mixture of fine aggregate (sand), cement, water and cationic emulsion. Slurry seal is usually built to a thickness of 5-10 mm. Where traffic is likely to be heavy 15 mm thickness can be specified. Slurry seal can be used as a maintenance intervention to provide a seal over an existing old surfacing that requires rejuvenation or to seal small cracks and defects or to repair pavements which may be disintegrating and ravelling. This code involves the concept of sealing large areas of surfacing with equipment. For repairing small defects, refer to codes 863 and 893.	
Tasks	• Placing safety signs. • Guaranteeing the continuous and safe passage of traffic during the execution of works. • Setting out the work area. • Supplying and storing the material. • Providing protection equipment for the staff. • Constructing a trial section to test equipment and application rates. • Verifying that the preceding activities involving the preparation of the bases have been completed. (Applying prime , code 410 and applying fog spray, code 411 if there is an existing surfacing) • Cleaning, brooming and removing all loose materials and debris. • Batching of the slurry seal. • Transporting the slurry to work area. • Applying and compacting the surfacing. • Protecting and curing the slurry seal. • Controlling the corrections on any areas with deficiencies. • Completing rolling of surfacing. • Checking the final road levels. • Cleaning of work area. • Removing the temporary signs.	
Standards	• To guarantee road safety appropriate signs should be placed on the road. • The work activity should be organized in such way that the passage of traffic shall not be interrupted for more than 10 min. A passage should be available allowing vehicles to use safely without causing danger to others. • The storage of the bituminous material should be in well closed drums or tanks approved by the Supervising Engineer. • The equipment and tools to be used must be approved by the Supervising Engineer. • The cement shall be stored in such a way that it is not in direct contact with the floors, walls or in places of high humidity. • The emulsion shall be used within 6 months and that which has exceeded this period should not be used. • All equipment and tools shall be approved by the Supervising Engineer. • A trial mix shall be carried out together with the Supervising Engineer to determine the rates of application and proportion of the mix, to calibrate the equipment and establish the work method. • The staff involved in the works must be well equipped with helmets, eyeglasses for protection, boots, gloves and overalls. Contact with skin must be avoided. • After using bitumen or similar products it is necessary to remove the bitumen from tools and equipment by heating. For those items that are not resistant to heat like glasses and boots the bitumen shall be removed using solvents such as kerosene and diesel. • The surface shall be inspected. It must be clean and any dust and deleterious substances shall be removed. • Mark the edges of the area to be surfaced measuring from the centerline and conforming to the width as given in the specifications or in accordance with the Supervising Engineer's instructions. • The surface of the base or existing surfacing should be free of irregularities or defects that might prevent free drainage of water.	

	• Care must be taken to ensure that the weather does not interfere with the execution of the work. The application of the slurry should not be done during or when it is threatening to rain, after sun set, during strong winds or after the rain when the base is too moist. • After completing the trial section, define the proportions for the slurry mix. • After determining the proportions, the mixing time shall also be defined to ensure adequate mixing and thorough coating of the aggregate and also the time it takes to open to traffic can be determined. • If the slurry is to be applied to an unbound roadbase, apply prime or fog spray paid under code 410 and 411. • Rails of predetermined thickness which is equal to the required thickness of the slurry before compaction shall be placed on the edges and fixed firmly of the base with nails. • Slurry shall be mixed in concrete mixers. Alternatively a mobile mixing plant (truck) may be used and the slurry should be applied within 12hrs of mixing. • The slurry shall be applied at ambient temperature. • The slurry shall be transported in wheel barrows on to specially prepared areas with side rails and laid in the middle of the lane. • When a mechanical distributor is used it shall be driven at a constant slow speed. • The consistency of the slurry shall be monitored to ensure that it is homogeneous. • Ensure that the batching process is consistent and accurate in order for the quality of the slurry to be consistent. • All areas that are missed or are seen to have inadequate slurry and any irregularities shall be corrected immediately by moving the slurry on the surface with a straight edge such as a straight wooden plank or rubber squeegees.
	• After the corrections the surface shall be smoothed with passes of hessian with addition of fluid slurry or emulsion. • The slurry shall be cured according to the type of slurry used. With slow setting slurry (made from K3-60) traffic can be allowed to pass after 4 hrs. • For medium setting slurry traffic shall be allowed to pass after 1.5 hrs. • After completing the application of the slurry and in accordance with the instructions of the Supervising Engineer, the lane shall be opened to traffic. • The traffic shall be controlled for a period of 24 hrs. • If necessary, and in conformity with the instructions of the Supervising Engineer, compaction of the slurry shall be carried out with steel roller with a minimum weight of 1.7 tons for labour based works. • The edges of the slurry shall conform to the design width as given in the designs or as specified by the Supervising Engineer with a tolerance of not more than 25mm and should not have irregularities.
	• After completing the construction of the surfacing check the profile of the road and ensure the pavement is free of irregularities that might prevent good drainage. • If not specified in the contract, the application rates below shall be used. • If the final levels of the surfacing are not satisfactory more slurry shall be applied to correct the levels. • No material must be left in the carriage way, shoulders, side drains and mitre drains which may prevent efficient drainage. • Asphaltic waste must be removed from the site to a place indicated by the Supervising Engineer. Other waste material should be removed and placed in a convenient place minimum 10 m away from the roadway, and spread in a layer no thicker than 20 cm. • Remove temporary signs

Materials	All materials shall be approved by the Supervising Engineer. Emulsion - The emulsion that is used for the slurry shall be anionic with 60% bitumen (K3-60) and conform to SABS 309. - The emulsion shall be supplied with a certificate indicating clearly the type, composition and quantities. Sand - The sand used for coarse slurry shall have a minimum of 75% sand size aggregate with strength that is consistent with clean natural sand not that of weathered sand. The second requirement is that it should conform to the grading specifications given in Table 9-28 Table 9-28 Aggregate grading specifications for slurry seal <table border="1" data-bbox="408 651 1112 1187"> <thead> <tr> <th>Sieve sizes (mm)</th><th>Percentage passing (by weight) (%)</th></tr> </thead> <tbody> <tr><td>14</td><td></td></tr> <tr><td>10</td><td></td></tr> <tr><td>5</td><td>100</td></tr> <tr><td>5</td><td>70-90</td></tr> <tr><td>2.36</td><td>45-70</td></tr> <tr><td>1.18</td><td>28-50</td></tr> <tr><td>0.6</td><td>19-34</td></tr> <tr><td>0.3</td><td>12-25</td></tr> <tr><td>0.15</td><td>7-18</td></tr> <tr><td>0.075</td><td>2-8</td></tr> </tbody> </table> Cement: - Ordinary Portland cement shall be used conforming to AASHTO M85, SABS 471 or equivalent or Portland blast furnace cement (PBFC) in compliance with AASHTO M240, SABS 626 or equivalent. Water: - Water shall be free of concentrations of acids, salt, sugar or organic mineral. Slurry: - The slurry shall be produced by adding sand, emulsion, cement and water in a concrete mixer or mobile mixing plant (truck). For tender purposes the following shall be considered on in the production of the slurry seal. - Inert (sand) 1m³ - Emulsion 260 litres - Cement 0.01m³ - Water 235 litres (adjust to get good workability)	Sieve sizes (mm)	Percentage passing (by weight) (%)	14		10		5	100	5	70-90	2.36	45-70	1.18	28-50	0.6	19-34	0.3	12-25	0.15	7-18	0.075	2-8
Sieve sizes (mm)	Percentage passing (by weight) (%)																						
14																							
10																							
5	100																						
5	70-90																						
2.36	45-70																						
1.18	28-50																						
0.6	19-34																						
0.3	12-25																						
0.15	7-18																						
0.075	2-8																						

- Alternatively the proportions 1 m³ slurry with 1.5% of cement and 145 litres of water can be used.
- For the purpose of tender the following application in Table 929 and Table 930 shall be used.

Table 9-29 Application rates by volume of slurry for Cape Seal (slurry on single surface dressing)

Aggregate sizes of the surface dressing(mm)	Application rates for slurry (m ³)
14 mm	0.006 m ³ /m ²
20 mm	0.008m ³ /m ²

Table 9-30 Application rates by mass of slurry for Cape Seal (slurry on single surface dressing)

Aggregate sizes (mm)	Application rates for slurry (m ³)
14 mm	7.5kg/m ²
20 mm	9.5kg/m ²

- Slurry shall be mixed until it is homogeneous
- The time required for the mixing process shall be adequate for all the aggregate to be properly coated and the spread of the emulsion in the mix to be homogeneous.
- There shall be no delay in applying the slurry after it is mixed.

Measurement

- M²
- Measurements are taken on the surfaced area



Mechanised construction of slurry seal



Labour based construction of slurry seal

Figure 9-28 Construction of slurry seal

Activity:	Construction of double sand seal using bitumen emulsion or 150/200 pen or MC3000. Variation of sand. Variation of bitumen Emulsion or 150/200 or MC3000.	Code: 461 462 463
Reference Code		
Application	This code is for the construction of sand seal surfacing, Figure 9-29. This application of sand seal follows after works paid under code 410 (priming) or code 411 (application of fog spray) for resealing operations. This code involves the application of sand seal on a new base on low volume roads and sand seal as a reseal intervention on existing surfacing. For emulsion treated bases see code 350 and for slurry seal see code 460. The concept of this code is for application on large areas requiring surfacing by labour based methods. For small areas use code 892. Codes 462 and 463 are used for the reimbursement for variations in the application rates for the sand and bituminous materials according to the Supervising Engineer's instructions.	
Tasks	• Placing safety signs • Guaranteeing the continuous and safe passage of traffic during the execution of works. • Setting out the work area. • Supplying and storing the material. • Providing protection equipment for the staff. • Cleaning, brooming and removing all loose materials and debris. • Testing the equipment and the application rates. • (Applying prime or fog spray using codes 410 and 411) • Applying tack coat using emulsion or 150/200 pen or SS60 or MC3000. • Spreading the sand. • Carrying out initial compaction of the surfacing. • Inspecting the surfacing and correcting defects, deficiencies or removing excess sand. • Completing rolling of surfacing. • Curing sand seal while open to traffic. • Repeating procedure for the second seal after 2 weeks of trafficking. • Checking the final road levels. • Cleaning of work area. • Removing the temporary signs.	
Standards	• To guarantee road safety, appropriate signs should be placed on the road. • The work activity should be organized in such way that the passage of traffic shall not be interrupted for more than 10 min. A passage should be available allowing vehicles to use safely without causing danger to others. • The storage of the bituminous material should be in well closed drums or tanks approved by the Supervising Engineer. • The equipment and tools to be used must be approved by the Supervising Engineer. • Equipment shall be tested for the control of application rates and uniformity of spray. • A trial section shall be constructed to determine the application rates and develop a work method. • The staff that is involved in the works must be well equipped with helmets, eyeglasses for protection, boots, gloves and overalls. Contact with skin must be avoided. • After using bitumen or similar products it is necessary to remove the bitumen from tools and equipment by heating. For those items that are not resistant to heat, like glasses and boots, the bitumen shall be removed using solvents such as kerosene and diesel. • Mark the edges of the area to be surfaced by measuring from the centerline conforming to the width as given in the specifications or in accordance with the Supervising Engineer's instructions. • The surface of the base or existing surfacing should be free of irregularities or defects which could prevent free drainage of water. • Care must be taken to ensure that the weather does not interfere with the execution of the work. The application of the sand seal should not be done during or when it is threatening to rain, after sun set, during strong winds or after the rain when the base is too moist. • A binder distributor fitted with a hand lance shall be used for the application of the bitumen emulsion or 150/200 or 85/100 or MC3000. The binder shall be applied at the temperature given in the materials section or according to the Supervising Engineer's instructions and the temperature shall be verified using a thermometer and recorded, Table 931.	

Table 9-31 Spray temperatures for binders (double sand seal)

Material	Temperature limits for spraying (°C)		
	Minimum	Maximum	Recommended
Emulsion SS60			Ambient
150/200 pen	135	155	145
MC3000	135	155	145

- If not specified in the contract, the application rates in Table 932 shall be used.

Table 9-32 Binder application rates for sand seal

Binder	Application rate for the binder
Emulsion	1.6L/m ²
150/200	1.0L/m ²
MC3000	1.0L/m ²

- The application rate of sand is given in Table 9-33.

Table 9-33 Application rate for sand seal

Aggregate	Application rate for the binder
Sand	16kg/m ² (7.5 L/m ²)

- Remove temporary signs
- The bitumen shall be applied uniformly with no areas deficient or with excess binder.
- Under normal circumstances the binder shall be applied to cover one lane at a time.
- If the application is done on several lanes, overlap should be 100 mm minimum. Place resistant paper strip at the beginning and end of each spray to ensure that the cross-sectional joints are well positioned.
- It is important to ensure that all the jets of the distributor bar are working well; this test should be made away from the road area before spraying.
- The height of the hand lance must be set or adjusted before the application of the bituminous material so that each point of the road is sprayed by 2 nozzles and thus receives 2 flows separately to obtain the indicated application rate.
- Attach or weld a chain on the hand lance to guide the height of the nozzle.
- The edge limits of the primed area should be parallel lines with tolerance of not more than 25 mm. The edges should be marked with small stones or nails and string set out to give good straight lines and curvature conforming to the geometric design requirements. To avoid spraying outside the demarcated area shutter boards should be held in place on the edges of the area being sprayed and moved along in sequence with the movement of the hand lance during spraying.
- Vehicle passage is not allowed in the freshly applied binder.
- Use chip spreader (the Chippy) or labour to spread the sand immediately after the application of the binder. It is important to note that application of fine sand with a chip spreader is difficult and sometimes impossible in which case labour should be used.
- The sand shall be free of impurities and dust.
- The sand shall be moistened before application on the road.
- Immediately after the application of the aggregate, compaction shall be carried out properly using a pneumatic roller with minimum weight of 8 tonnes or vibratory steel roller in non-vibratory mode applying passes from the edge towards the centerline of the road.

	- Rolling shall be continued until a total of 8-12 passes on the first day. - Rolling shall continue for 3 more days applying 8 passes of the roller during the hot test part of the day. Bitumen should start showing within a week after application. Applying thick sand seal using emulsion: - When applying sand seal using emulsion the application of the emulsion and the sand shall be split both for the first layer and second layer. The process is illustrated in Figure 929. - The first spray onto the primed base shall be 0.6L/m² followed by the application of sand at a rate of 5.0L/m². - Before rolling a second spray of binder on the sand is applied at a rate of 1.0L/m² and the remainder of the sand is applied emulsion is applied in a single spray and emulsion tends to flow and spread thinly over the																											
	- primed base leaving a very thin film of binder to hold the sand. Consequently, very little of the sand applied is bound and the resultant sand seal layer is very thin. - This innovative process will make it possible to used fine sands that are locally available in many areas. - The process is repeated for the second seal of the double sand seal. - Traffic shall be allowed after 3 days of rolling. - During curing of the sand seal, the sand that is pushed off the carriageway shall be broomed back. - After completing the construction of the surfacing, check the profile of the road and ensure the pavement is free of irregularities that might inhibit good drainage. - No material must be left in the carriage way, shoulders, side drains and mitre drains which may prevent efficient drainage. - Asphaltic waste must be removed from the site to a place indicated by the Supervising Engineer. Other waste material should be removed and placed in a convenient place minimum 10 m away from the roadway, and spread in a layer no thicker than 20 cm.																											
Materials	All materials shall be approved by the Supervising Engineer. Bitumen - Emulsion shall be cationic with 60% bitumen. The emulsion shall be supplied with a certificate clearly indicating the type, composition, quantity and the supplier. - MC3000 shall be prepared from cutting back 85/100 pen with 8 to 10% kerosene. The emulsion shall conform to SABS 309. - For better performance 150/200 pen bitumen is recommended prepared by cutting back 85/100 pen with 6-10% diesel. This binder is more durable and hardens slowly thus giving longer life to surfacings. 150/200 is good for Otta seals, surface dressing, sand seals, etc. - Limit of temperatures for heating the binders and for application shall be according to the designs or the instructions of the Supervising Engineer. If not the specifications given in Table 934 shall be used. Table 9-34 Spray temperatures for binders (double surface dressing) <table><tr><th rowspan="2">Material</th><th colspan="3">Temperature limits for spraying (°C)</th></tr><tr><th>Minimum</th><th>Maximum</th><th>Recommended</th></tr><tr><td>85/100 pen</td><td>165</td><td>190</td><td>175</td></tr><tr><td>150/200 pen</td><td>135</td><td>155</td><td>145</td></tr><tr><td>MC3000</td><td>135</td><td>155</td><td>145</td></tr></table> Aggregate - The grading of the sand shall have maximum size of 6.7 mm and shall be in accordance Table 4302/1. Table 9-35 Grading specification for sand <table><tr><th>Sieve size (mm)</th><th>Percentage Passing</th></tr><tr><td>6.7</td><td>100</td></tr><tr><td>0.3</td><td>0 – 15</td></tr><tr><td>0.15</td><td>0 - 2</td></tr></table>	Material	Temperature limits for spraying (°C)			Minimum	Maximum	Recommended	85/100 pen	165	190	175	150/200 pen	135	155	145	MC3000	135	155	145	Sieve size (mm)	Percentage Passing	6.7	100	0.3	0 – 15	0.15	0 - 2
Material	Temperature limits for spraying (°C)																											
	Minimum	Maximum	Recommended																									
85/100 pen	165	190	175																									
150/200 pen	135	155	145																									
MC3000	135	155	145																									
Sieve size (mm)	Percentage Passing																											
6.7	100																											
0.3	0 – 15																											
0.15	0 - 2																											

Measurement	461	M ²
	462	M ³
	463	L
	For code 461 measurements are taken on surfaced area. For code 462 and 463 measurements are taken on volume of sand and the volume of bitumen applied compared with the application rates given in the work norm.	



Applying emulsion on primed base at 0.5L/m² (1)



Applying sand is applied at 5L/m² (2)



Applying emulsion on sand at 1.0L/m² and applying sand at 2.5L/m² (3)



Rolling of sand layer producing a thick layer of sand seal even with very fine sand (4)

Figure 9-29 Construction of Thick Sand Seal (Cold Sand Asphalt) – showing split application of emulsion

Activity:	Construction of Otta seal surfacing using 150/200 pen or MC3000 or Emulsion (recommended K3-60) Variation on aggregate. Variation on Bitumen Emulsion or 150/200 or MC3000	Code: 464 465 466
Reference Code		
Application	This code is for the construction of Otta seal surfacing. The application of Otta seal follows after works paid under code 410 (priming) or code 411 (application of fog spray) for resealing operations or sealing stabilised bases. This code involves the application of binder followed by the application of naturally occurring gravel; quartzitic or laterite or calcrete aggregate sieved through 19 mm and 5 mm sieves to produce graded aggregate with a small proportion of fines (< 0.075 mm) less than 13%. Construction of Otta seals is commonly used on low volume roads and has the advantage of using local materials. The concept of this code involves the use of MC3000 and 150/200 or emulsion as the binder. Otta seal can be constructed using both equipment and labour based methods. Codes 465 and 466 are used for the reimbursement for variations in the application rates for the sand and bituminous materials according to the Supervising Engineer's instructions. It is recommended to apply a second layer of Otta seal or sand seal to prolong the service life of the Otta seal surfacing.	
Tasks	• Placing safety signs • Guaranteeing the continuous and safe passage of traffic during the execution of works. • Setting out the work area. • Supplying and storing the material. • Providing protection equipment for the staff. • Constructing a trial section to test the equipment and the application rates. • Verifying the preparation of the base and applying priming or fog spray. (Applying prime or fog spray using codes 410 and 411) • Cleaning, brooming and removing all loose materials and debris. • Preparing, heating and applying tack coat using 150/200 pen, MC3000 or emulsion (note that the K3-60, K2-60 and K1-60 are applied at ambient temperature). K1-60 is recommended for use on steep gradients. • Transporting, applying and spreading the aggregate. • Carrying out initial compaction of the surfacing. • Inspecting the surfacing and correcting defects or deficiencies or excess aggregate or emulsion. • Removing excess aggregate. • Completing the rolling of surfacing. • Curing the Otta seal through trafficking. • For double Otta seal: Repeating the procedure for the second seal after 6-12 weeks of curing through trafficking. • For single Otta seal with sand capping: Applying a sand capping on top of the single Otta seal after 2 weeks of curing through trafficking. • For emulsion based Otta seal/graded aggregate seal: Applying emulsion in split application (0.6L/m² tack and 1.8L/m² on top) • Checking the final road levels. • Cleaning of work area. • Removing the temporary signs. • Maintaining the Otta seal during the maintenance or defects liability period (1 year).	
Standards	• To guarantee road safety, appropriate signs should be placed on the road. • The work activity should be organized in such way that the passage of traffic shall not be interrupted for more than 10 min. A passage should be available allowing vehicles to use safely without causing danger to others. • The storage of the bituminous material should be in well closed drums or tanks approved by the Supervising Engineer. • The equipment and tools to be used must be approved by the Supervising Engineer. • A trial section shall be constructed to calibrate the equipment, control the application rates and uniformity of spray and develop a work method.	

- The staff involved in the works must be well equipped with eyeglasses for protection, boots, gloves and overalls. Contact with skin must be avoided.
- If hot bitumen or similar products are used it is important to provide heat resistant gloves and boots and respiratory equipment to prevent inhalation of fumes.
- The base must be broomed or cleaned using compressor before application of the binder.
- Mark the edges of the area to be surfaced measuring from the centerline and conforming to the width as given in the specifications or in accordance with the Supervising Engineer's instructions.
- The surface of the base or existing surfacing should be free of any irregularities or defects which could inhibit free drainage of water.
- Care must be taken to ensure that the weather does not interfere with the execution of the work. The application of the Otta seal should not be done during or when it is threatening to rain, after sun set, during strong winds or after the rain when the base is too moist.
- If necessary prime shall be applied and paid under code 410.
- For tender purposes the application rate below shall be used for an Otta seal surfacing.
- Apply a tack coat of fluid binder 150/200 pen, MC3000 or MC800 at the temperature shown in Table 9-36 and according to the Engineer's instructions.
- A binder distributor fitted with a hand lance shall be used for the application of the bitumen 150/200 or SS60 or MC3000. The binder shall be applied at the temperature given in Table 936 or according to the Supervising Engineer's instructions and the temperature shall be verified or measured using a thermometer.
- The height of the hand lance must be set or adjusted before the application of the bituminous material so that each point of the road is sprayed by 2 nozzles and thus receives 2 flows separately to obtain the indicated application rate.

Table 9-36 Spaying Temperatures for Binders (Otta Seal Surfacing)

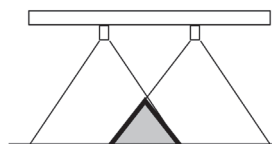
Material	Temperature limits for spraying (°C)		
	Minimum	Maximum	Recommended
Emulsion SS60			Ambient
150/200 pen	135	155	145
MC3000	135	155	145

- For tender purposes the following application rates for binder and aggregate shall be used if not specified otherwise, Table 9-37 and Table 9-38.

Table 9-37 Application Rates for Binders for Otta Seal

Binder	Application rate of binder for Coarse Aggregate: 19mm (max)	Application rate of binder for Fine Aggregate: 13mm (max)
Emulsion	3.3 L/m ²	2.6 L/m ²
150/200	2.0L/m ²	1.6 - 1.8L/m ²
MC3000	2.0L/m ²	1.6 - 1.8L/m ²

- The bitumen shall be applied uniformly without areas deficient in bitumen or areas with excess.
- Under normal circumstances the binder shall be applied to cover one lane at a time.
- If the application is done on several lanes, overlap should be 100 mm minimum. Place resistant paper strip at the beginning and end of each spray to ensure that the cross-sectional joints are well positioned.
- The temperature of the binder should be verified and recorded.
- It is important to ensure that all the jets of the hand lance are working well; this test should be carried out away from the road area before spraying.



- Attach or weld a chain on the hand lance to guide the height of the nozzle.
- The edge limits of the primed area should be parallel lines with tolerance of not more than 25 mm. The edges should be marked with small stones or nails and string set out to give good straight lines and curvature conforming to the geometric design requirements. To avoid spraying outside the demarcated area shutter boards should be held in place on the edges of the area being sprayed and moved along in sequence with the movement of the hand lance during spraying.
- Vehicle passage is not allowed on the freshly applied binder.
- Use chip spreader (the Chippy) or labour to spread the aggregate immediately after the application of the binder. When using the labour based method, aggregate shall be spotted along the shoulders in predetermined quantities measured using half drums (100L of aggregate). The aggregate shall be applied in predetermined areas which give the required application rates.
- Aggregate shall be applied uniformly avoiding over or under-application.
- For the tender purposes use the application rate of aggregate given in Table 9-38.

Table 9-38 Application Rate for Aggregate for Otta Seal

Application rate for the aggregate
0.016m ³ /m ²

- The aggregate shall be applied moist.
- The aggregate shall be free of impurities and dust.
- After application of the aggregate, compaction shall be carried out immediately, applying passes from the edge towards the centreline of the road using a pneumatic roller with minimum weight of 8 tonnes.
- Rolling shall be continued until a total of 12-15 passes on the first day.
- Rolling shall continue for 5 more days applying 12 passes of the roller during the hottest part of the day. Bitumen should start showing within 3 weeks after application.
- Traffic shall be allowed after 3 days of rolling but the speed shall be controlled to be below 50km/hr.
- During curing of the Otta seal, aggregate that has been pushed off the carriageway by traffic shall be broomed back.
- Where there are signs of bleeding, sand or quarry dust shall be applied according the Supervising Engineer's instructions.
- It is important to ensure that the full lane width cures properly and that the aggregate is embedded in the binder to give a good texture.
- **Construction of Otta seal using Emulsion SS60, MS60 and RS60:**
 - Graded aggregate seal/Otta seal can be constructed using emulsion, Figure 930.
 - The primed surface shall be cleaned properly with brooms and/or air compressor.
 - The section is marked to demarcate the lanes and segments as described above.
 - The application of the emulsion shall be split into two sprays. The first spray shall be applied at approximately 30% of the full rate, in this case 0.8 L/m².
 - Aggregate shall be applied at a rate of 16 L/m². Fine grained aggregate (nominal maximum size of 13.2 mm) is more suitable but the conventional size (nominal maximum size of 19 mm) may also be used.
 - The second application of binder shall be applied at a rate of 1.8 L/m² if not specified (approx. 70% of the full rate)

- For natural aggregate or aggregate with a high content of dust rolling shall be carried out after the second application of binder and when the emulsion has broken and set. This may take an hour or two.
- Apply a minimum of 8 passes of 1.7 ton steel drum roller or 12 passes of pneumatic a roller on the first day. 8 passes shall be applied during the hot part of the second day. If adequate compaction can be applied on the first day it may not be necessary to roll again. The Engineer shall inspect the Otta seal surfacing for the purpose of approval.
- After-care such as brooming back and controlling traffic using cones is not necessary unless traffic is very low (<20 vehicles per day).
- The road may be opened to traffic on the second or third day.
- For a double Otta seal, the second layer may be applied after one to two weeks depending on the level of traffic.

	○ This approach is for both machine and labour based application and has a unique advantage that it is safer for labour based methods because emulsion is applied at room temperature. ○ Also the construction process is less laborious and less expensive than the Otta seals based on hot binders. There is also no aggregate loss or the common problems of poor curing of the Otta seal surfacing even at low temperatures and low traffic. • After completing the construction of the surfacing, check the profile of the road and ensure the pavement is free of irregularities that might inhibit good drainage. • No material must be left in the carriage way, shoulders, side drains and mitre drains which may prevent efficient drainage. • Bituminous waste must be removed from the site to a place indicated by the Supervising Engineer. Other waste material must be removed and placed in a convenient place minimum 10 m away from the roadway, and spread in a layer no thicker than 20 cm. • Remove temporary signs																																																																			
Materials	All materials shall be approved by the Supervising Engineer. Bitumen • Emulsion shall be cationic with 60% bitumen for acidic aggregate (quartz) and anionic for basic aggregate (calcrete). The emulsion shall be supplied with a certificate clearly indicating the type, composition, quantity and the supplier. • MC3000 shall be prepared from cutting back 85/100 pen with 8 to 10% kerosene. The emulsion shall conform to SABS 309. • For better performance, 150/200 pen bitumen prepared by cutting back 85/100 pen with 6-10% diesel is recommended. This binder is more durable and hardens slowly thus giving longer life to surfacings. 150/200 is good for Otta seals, surface dressing, sand seals, etc. • Limit of temperatures for heating the binders and for application shall be according to the designs or the instructions of the Supervising Engineer. If not, the specifications given in Table 9-39 shall be used. Table 9-39 Spraying Temperature of Binders for Otta Seal <table><tr><th>Material</th><th colspan="3">Temperature limits for spraying (°C)</th></tr><tr><th></th><th>Minimum</th><th>Maximum</th><th>Recommended</th></tr><tr><td>85/100 pen</td><td>165</td><td>190</td><td>175</td></tr><tr><td>150/200 pen</td><td>135</td><td>155</td><td>145</td></tr><tr><td>MC3000</td><td>135</td><td>155</td><td>145</td></tr></table> Aggregate • Aggregate for Otta seal comprises natural aggregate obtained from sieving natural gravels such as calcretes, laterites and quartzite, etc. See Table 9-40. • The aggregate can also be crushed graded stone. Table 9-40 Grading Specifications for Otta Seal (Coarse Grained) <table><tr><th rowspan="2">Sieve sizes (mm)</th><th colspan="3">Percentage passing</th></tr><tr><th>Dense (500<ADT)</th><th>Medium (500<ADT<100)</th><th>Open (ADT<100)</th></tr><tr><td>20</td><td>100</td><td>100</td><td>100</td></tr><tr><td>16</td><td>92-100</td><td>85-100</td><td>80-100</td></tr><tr><td>14</td><td>83-100</td><td>70-83</td><td>50-80</td></tr><tr><td>10</td><td>68-98</td><td>45-68</td><td>35-60</td></tr><tr><td>7</td><td>50-80</td><td>30-55</td><td>20-40</td></tr><tr><td>5</td><td>42-68</td><td>20-42</td><td>10-30</td></tr><tr><td>2.36</td><td>20-48</td><td>4-20</td><td>0-8</td></tr><tr><td>1.18</td><td>15-33</td><td>2-15</td><td>0-6</td></tr><tr><td>0.425</td><td>7-25</td><td>0-7</td><td>0-3</td></tr><tr><td>0.075</td><td>2-10</td><td>0-2</td><td>0-1</td></tr></table>	Material	Temperature limits for spraying (°C)				Minimum	Maximum	Recommended	85/100 pen	165	190	175	150/200 pen	135	155	145	MC3000	135	155	145	Sieve sizes (mm)	Percentage passing			Dense (500<ADT)	Medium (500<ADT<100)	Open (ADT<100)	20	100	100	100	16	92-100	85-100	80-100	14	83-100	70-83	50-80	10	68-98	45-68	35-60	7	50-80	30-55	20-40	5	42-68	20-42	10-30	2.36	20-48	4-20	0-8	1.18	15-33	2-15	0-6	0.425	7-25	0-7	0-3	0.075	2-10	0-2	0-1
Material	Temperature limits for spraying (°C)																																																																			
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16	92-100	85-100	80-100																																																																	
14	83-100	70-83	50-80																																																																	
10	68-98	45-68	35-60																																																																	
7	50-80	30-55	20-40																																																																	
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2.36	20-48	4-20	0-8																																																																	
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0.425	7-25	0-7	0-3																																																																	
0.075	2-10	0-2	0-1																																																																	

- Grading for a fine-grained Otta seal surfacing is given in Table 9-41. This Otta seal has the advantage of curing quickly and adequately even at low traffic.
- If the material does not satisfy the grading requirements it shall be sieved through 20mm and 5mm sieves for coarse-grained Otta seals and through 14mm and 5 mm sieves for fine-grained Otta seals to remove reject fractions of coarser aggregate and fines.
- The aggregate shall be free of organic materials, mud and other impurities.
- The fines content (<0.075 mm) must not exceed 10% for coarse grained Otta seals and 13% for fine grained Otta seals.

Table 9-41 Grading Specifications for Fine Grained Otta Seal

Sieve sizes (mm)	Fine grained Otta seal for LVRs.
20	
16	100
14	55-100
10	33-70
7	20-52
5	12-40
2.36	5-25
1.18	2-18
0.425	1-15
0.075	0-13

- The strength of the aggregate shall be determined according to TMH 1 section B2. The value of 10% FACT (dry) shall be minimum 90kN (ADT<100) and 110kN (ADT>100). The ratio of wet to dry strength is given in Table 9-42.

Table 9-42 Aggregate Strength Specifications for Otta Seal

Strength requirements for Otta seal aggregate	Traffic at time of construction	
	ADT<100	ADT>100
Minimum strength 10% FACT (dry)	90kN	110kN
Strength ratio (wet/dry)	0.60	0.75
ACV	30	26

Measurement464 M²465 M³

466 L

For code 464 measurements are taken on surfaced area.

For code 465 and 466 measurements are taken on volume of sand and the volume of bitumen applied compared with the application rates given in the work norm.

MAE



Applying emulsion at 1.0L/m² after rolling (3)



Spreading Otta seal aggregate (2)



Application of Otta seal using emulsion– application of emulsion (0.8L/m²) and aggregate (16L/m²) (1)



Left: Otta seal with emulsion
Right: Cold mix with emulsion (4)

Figure 9-30 Construction of Emulsion based Otta Seal/Graded Aggregate Seals

Activity:	Construction of penetration macadam surfacing using Emulsion or 150/200 pen or MC3000 Variation of aggregate Variation of bitumen Emulsion or 150/200 or MC3000.	Code: 467 468 469
Reference Code		
Applica- tion	This code is for the construction of penetration macadam surfacing on unsealed bases, Figure 9-31. This application of penetration macadam seal follows after works paid under code 410 (priming) or code 411 (application of fog spray) for sealing stabilised bases. This code involves the application of aggregate followed by the application of binder; the binder penetrates from the top. This code involves the use of emulsion K3-60 but other binders such as 150/200 or 85/100 pen or MC3000 can be used too. Penetration macadam is suitable for low volume roads and in severe conditions such as mountainous areas. Penetration macadam involves the use of intensive labour or labour based methods. Codes 468 and 469 are used for the reimbursement for variations in the application rates for the bituminous materials and aggregate according to the Supervising Engineer's instructions.	
Tasks	• Placing safety signs • Guaranteeing the continuous and safe passage of traffic during the execution of works. • Setting out the work area. • Supplying and storing the material. • Providing protection equipment for the staff. • Constructing a trial section to test the equipment and the application rates. • Verifying the preparation of the base and application of priming or application of fog spray. (Applying prime or fog spray is paid under codes 410 and 411) • Cleaning, brooming and removing all loose materials and debris. • Preparing, heating and light application of first spray using 150/200 pen, or MC3000 or emulsion, K3-60 or K2-60 (note that emulsion is applied at ambient temperature). • Transporting, applying and spreading the aggregate. • Inspecting the surfacing and correcting defects or deficiencies or excess aggregate • Carrying out initial compaction of the surfacing. • Applying aggregate for second layer. • Removing excess aggregate. • Spraying second coat of binder. • Completing the rolling of the surfacing. • Checking the final road levels. • Cleaning the work area. • Removing the temporary signs.	
Standards	• To guarantee road safety, appropriate signs should be placed on the road. • The work activity should be organized in such way that the passage of traffic shall not be interrupted for more than 10 min. A passage should be available allowing vehicles to use safely without causing danger to others. • The storage of the bituminous material should be in well closed drums or tanks approved by the Supervising Engineer. • The equipment and tools to be used must be approved by the Supervising Engineer. • A trial section shall be constructed to calibrate the equipment, control the application rates and uniformity of spray and develop a work method. • The staff involved in the works must be well equipped with helmets, eyeglasses for protection, boots, gloves and overalls. Contact with skin must be avoided. • If hot bitumen or similar products are used it is important to provide heat resistant gloves and boots and respiratory equipment to prevent inhalation of fumes. • The base must be broomed or cleaned using compressor before application of the binder. • Mark the edges of the area to be surfaced measuring from the centerline conforming to the width as given in the specifications or in accordance with the Supervising Engineer's instructions. • The surface of the base or existing surfacing should be free of irregularities or defects which might inhibit free drainage of water. • Care must be taken to ensure that the weather does not interfere with the execution of the work. The application of bitumen should not be done during or when it is threatening to rain, after sun set, during strong winds or after the rain when the base is too moist.	

- The binder shall be applied lightly (0.8L/m^2) before the application of aggregate. This helps in providing a waterproofing seal over the base. It also helps in bonding the aggregate to the base.
- Aggregate shall be spread using a chip spreader (the Chippy) or labour immediately after the light application of the binder. When using labour based methods aggregate shall be spotted along the shoulders in predetermined quantities measured using half drums (100L of aggregate) which give the required application rates.
- For tender purposes, for a 3-layer penetration macadam surfacing, the application rate below shall be used, Table 9.43.

Table 9.43 Aggregate Application Rates for a 3-Layer Penetration Macadam Surfacing

Aggregate size (mm)	Percentage (%)	Application rate
40	60%	$0.036\text{m}^3/\text{m}^2$ (36L/m ²)
28	30%	$0.018\text{m}^3/\text{m}^2$ (18L/m ²)
14-20	10%	$0.006\text{m}^3/\text{m}^2$ (6L/m ²)

- A 2 layer penetration macadam surfacing is usually adequate for low volume roads. For tender purpose the application rates given in Table 944 shall be used.

Table 944 Aggregate Application Rates for a 2-Layer Penetration Macadam Surfacing

Aggregate size (mm)	Percentage (%)	Application rate
20-40	55%	$0.027\text{m}^3/\text{m}^2$ (27L/m ²)
5-14	45%	$0.018\text{m}^3/\text{m}^2$ (18L/m ²)

- Care shall be taken to prevent segregation of the aggregate during application.
- Aggregate shall be applied uniformly avoiding over or under application. After inspection, where aggregate is deficient, more shall be added and where there is excess, aggregate shall be removed.
- After surfacing, the profile of the road shall conform to the designs or instructions of the Supervising Engineer. The recommended crossfall shall be 2.5 - 3% and shall be free of irregularities or defects that might inhibit good drainage.
- The aggregate shall be rolled using a cylinder of minimum weight of 8 tonnes. 8 passes of the cylinder shall be applied. Compaction shall be carried out from the edges towards the centerline.
- The aggregate shall be slightly moistened before application of the binder to reduce dust and surface tension.
- Binder shall then be applied using the rates given in Table 9.45.

Table 9-45 Application rate for Binder Emulsion for Penetration Macadam

Penetration Macadam structure	Binder	Application rate
3 Layer	K3-60/K2-60	6 L/m ²
2 Layer	K3-60/K2-60	2.5-3.0 L/m ²

- A towed binder tank with hand lance shall be used for the application of the bitumen (150/200 or K2-60/K3-60 or MC3000). The binder shall be applied at the temperature given in Table 946 or according to the Supervising Engineer's instructions. The temperature shall be verified using a thermometer and recorded.

Table 9-46 Spraying Temperature of Binders for Penetration Macadam

Material	Temperature limits for spraying (°C)		
	Minimum	Maximum	Recommended
Emulsion SS60/ RS60			Ambient
150/200 pen	135	155	145
MC3000	135	155	145
80/100	165	190	175

- The bitumen shall be applied uniformly without areas deficient in bitumen or areas with excess.
- The application of binder shall be split into three applications of i.e. 0.8L/m² before application of aggregate for the first layer each followed by the application of aggregate for the first layer.
- The first layer is then rolled with 8-12 passes of a steel drum roller in non-vibratory mode.
- The aggregate for the second layer is then applied and spread uniformly to fill the interspaces between the large aggregate.
- The second coat of binder sprayed and it penetrates into both layers.
- Rolling is continued when the emulsion has broken. At least 12 passes of a steel drum roller with a minimum weight of 1.7 tonnes is used. Heavier rollers may be used but care should be taken to prevent excessive crushing of aggregate. The rollers must be wetted to prevent pick up of the surfacing. Alternatively, a thin layer of sand (2-4 mm) may be applied to prevent pick up of the surfacing.
- The application of binder shall be uniform and care shall be taken to avoid over or under application of the binder.
- Under normal circumstances the binder shall be applied to cover one lane at a time.
- If the application is done on several lanes, overlap should be 100 mm minimum. Place resistant paper strip at the beginning and end of each spray to ensure that the cross-sectional joints are well positioned.
- The temperature of the binder should be verified and recorded.
- It is important to ensure that all the jets of the distributor bar are working well; this test should be carried out away from the road area before spraying.
- The height of the distributor bar must be set or adjusted before the application of the bituminous material so that each point of the road is sprayed by 2 nozzles and thus receives 2 flows separately to obtain the indicated application rate.
- Attach or weld a chain on the hand lance to guide the height of the nozzle.
- The edge limits of the primed area should be parallel lines with tolerance of not more than 25 mm. The edges should be marked with small stones or nails and string set out to give good straight lines and curvature conforming to the geometric design requirements. To avoid spraying outside the demarcated area shutter boards should be held in place on the edges of the area being sprayed and moved along in sequence with the movement of the hand lance during spraying.
- Vehicle passage is not allowed on the freshly applied binder.
- After completing the construction of the surfacing, check the profile of the road and ensure the pavement is free of irregularities that might inhibit good drainage.

Materials

All materials shall be approved by the Supervising Engineer.

Bitumen

- Emulsion shall be cationic with 60% bitumen for acidic aggregate (e.g. quartz) and anionic for basic aggregate (e.g. calccrete). The emulsion shall be supplied with a certificate clearly indicating the type, composition, quantity and the supplier.
- MC3000 shall be prepared from cutting back 85/100 pen with 8 to 10% kerosene. The emulsion shall conform to SABS 309.
- For better performance 150/200 pen bitumen prepared by cutting back 85/100 pen with 6-10% diesel is recommended. This binder is more durable and hardens more slowly thus giving longer life to surfacings. 150/200 is good for Otta seals, surface dressing, sand seals, etc.
- Limit of temperatures for heating the binders and for application shall be according to the designs or the instructions of the Supervising Engineer. If not, the specifications given in the Table 9-46 shall be used.

	Aggregate: - The aggregate shall be free of organic materials, mud, dust and other impurities. Aggregate with impurities shall be washed. - The aggregate shall be crushed stone of sizes 40, 28, 10-14 mm free of dust mud and other impurities with less than 1.5% of by weight of material passing 0.425 mm sieve. - The strength of the aggregate shall be determined according to TMH 1 section B2. The value of 10% FACT (dry) shall be minimum 210kN. The ratio of wet to dry strength shall be greater than 75%. - The shape shall be determined according THM 1 section B3. Maximum flakiness shall be 30. Water: - Water must be clean and free of concentrations of acids, salts, sugar, or other organic material.
Measurement	467 M² 468 M³ 469 L For code 467 measurements are taken on surfaced area. For code 468 and 469 measurements are taken on volume of aggregate and the volume of bitumen applied compared with the application rates given in the work norm.MAE



Applying binder and coarse aggregate (first layer)- (1)



First layer rolled and completed (2)



Applying second layer followed by emulsion at a rate of 2.5L/m² (3)



Rolling penetration macadam using steel drum roller (wetted to prevent pick up of binder) – (4)

Figure 9-31 Construction of a 2-layer Penetration Macadam

9.8 Series 500 Auxiliary Works

Auxiliary works involve the activities related to providing the road user with information that is necessary to use the road safely and appropriately. The main aspects include:

1. Road signs (the vertical signs)
2. Road markings (informative, regulatory and warning signs and lines painted on the road to guide the user.
3. Traffic calming measures (humps, rumble strips, etc.).

The geometric design for low volume roads involves following the existing road profile as much as possible and this compromises safety to a certain extent. Placement of signs is therefore of paramount importance.

Activity:	Production and placement of kilometre markers	Code: 510
Reference Code		
Application	This code is used for the production and placement of kilometre markers made of reinforced concrete with dimensions conforming to the design type. The position shall be as instructed by the Supervision Engineer.	
Tasks	• Producing kilometer markers using reinforced concrete moulded into prefabricated units with dimensions indicated in the designs. • Transporting to the work place. • Identifying the sites where the kilometer markers will be placed together with the Supervising Engineer. • Placing temporary signs in the work area. • Excavating the foundations in accordance with the designs. • Placing kilometer markers and controlling the horizontal and vertical alignment using spirit level and metric tape. • Backfilling the foundations using granular material, or stabilised non-granular material. • Cleaning the work area. • Painting the markers in dimensions and colours indicated in the design indicating the class of road and the kilometres. • Removing temporary signs.	
Standards	• Place temporary signs and the work area. • The production and placement of the kilometer markers shall conform to Uganda specifications stipulated in the manual for 'Road Traffic Signs'. • The prefabricated markers shall be produced using class C20 concrete with a smooth finish. • The dimensions, colours, letters and numbers shall conform to the design. • The tolerance shall be ± 20 mm for the width and height and ± 10 mm for the thickness. • Identification of the positions to place the markers shall be carried out together with Supervising Engineer. • The foundations shall be excavated according to the specified dimensions (300 mm x 500 mm) excavated vertically avoiding sliding back of soils into the hole. The foundations shall be excavated to firm soil free from organic material. • If necessary place a blinding layer of aggregate 10 cm thick. • Verify that the excavation has been carried out properly to the required level of the foundation. • The marker is then placed in the hole and aggregate is placed to achieve the required levels; horizontal and vertical. • The foundation shall be backfilled with granular materials in layers not thicker than 15 cm. If necessary material from the borrow pits shall be used for backfilling. • Reflective paint shall be applied at the top and oil paint shall be applied on other parts of the markers. • Remove temporary signs.	
Materials	• All materials must be approved by the Supervising Engineer. • The reinforcements shall be prepared in conformity with the reinforcement and bending schedule. • Reflective paint and oil paint.	
Measurement	• Unit • The price includes all works related to production of markers, transportation to site, application of the markers and painting. • MAE	

Activity:	Production and placement of protection barrier	Code: 520
Reference Code		
Application	This code involves supply and placement of metallic protection barriers (Armco type) supported by posts of wood or metal conforming to the designs in positions indicated by the Supervising Engineer. Protection barriers are used to guard walls, bridges and curves, Figure 932.	
Tasks	- Identifying the sites, alignment and marking the locations where the protection barriers will be placed together with the Supervising Engineer. - Supplying and transporting the materials to the work place. - Opening foundations for support posts. - Placing posts while controlling the vertical and horizontal alignment using spirit level and metric tape. - Placing spacer blocks. - Fixing and aligning the protection barrier on the posts. - Placing reflective tags on the barriers. - Finishing at the end of the barriers. - Clearing the work area. - Removing temporary signs.	
Standards	Safety - Temporary signs shall be placed in the work area to guarantee safety. Alignment and marking - All locations and alignment of the protection barriers shall be as indicated by the Supervising Engineer. Excavating foundations - Excavating the foundations shall be according to the design of the foundation and geometry, and excavated vertically with a shovel or spade avoiding materials falling back into the excavation. Excavate till firm ground free of water and organic materials. - The bottom of the foundation should be level. - If no firm ground is reached concrete blinding shall be placed with thickness of 5 cm (class C15, codes 640, 641, 642, 643, and 644 for standards for the production of concrete) Wooden posts - The wooden posts (pine or eucalyptus) shall have diameter 150 and 230 mm and treated with creosote in conformity with SABS 538 or other material for treatment of equivalent quality. - Use wooden post with diameter similar to the section of the barrier to facilitate the fixing of the spacer blocks and the metallic rails. - The wooden rails shall be 1.75 m long. - Approximately 1.0 m of the post should be below ground level and the post shall be placed such that the top of the posts is at uniform levels; horizontal or sloping. - The foundations shall be backfilled with granular or fine stabilised material and compacted manually with rammers in layer not more than 100 mm thick. - The spacing between posts shall be 3.81 m. - In the case where the barriers are constructed on culverts galvanized metal posts of 168 mm diameter shall be used (25.3 kg/m). - The metal posts shall have a metal plate welded at the base with dimension 300 mm x 300 mm x 12 mm. - The base plate shall be fixed to the concrete using bolts or dowels 300 mm long and 16 mm diameter. - The metallic rails shall be of the Armco type, galvanized, with width of 300 mm, 12 mm thickness and 4128 mm long in accordance with SABS 763. - The tolerance limits for the dimensions of the protection barriers shall be $\pm 1\%$. - The top of the barrier shall be plane and free of undulations or irregularities more than 1 cm measured using a 2 m straight edge. - The rail shall be about 0.53 m above the ground level. - The ends of the rails shall be tapered into the ground and outward from the road at 45 degrees into concrete posts which shall be as per the Supervising Engineer's instructions.	

	Fixing accessories - The bolts which are used for fixing the rails to the posts shall be galvanized (class A40), 320 mm long and 16 mm diameter. - The washers shall be 3mm thick. - The nuts shall be galvanized and 16 mm in diameter (class A40). - The spacer blocks shall be pine or eucalyptus with dimensions 350 x 150 x 100 mm. - Fix the rails to the posts with a reinforcement plate of dimensions 75 x 45 x 3 mm. - The rails shall be connected using 8 bolts and nuts and where they are attached to the posts. - The connectors connecting the 2 rails shall both have their surfaces in full contact with the rails. Reflective tags - Reflective tags shall be galvanized supplied and fixed on the barrier in a V-shape with a thickness of 1.5 mm. - The reflective tags shall be fixed on the rail in alternating colours. - The reflective tags shall be fixed to the rails using galvanized bolts and nuts. - The reflective tags shall have retro reflective paint (or coloured using auto-coloured plastic) white and red in conformity with the design and SATCC standard – SATCC 5200. Finishing - Clear the work area. - Remove the temporary signs.
Materials	- All materials must be approved by the Supervising Engineer. - The galvanized metallic rails shall conform to standards SABS 763. - The wooden posts (pine or eucalyptus) shall be treated with creosote (conform to SABS 538). - The metallic post shall be galvanized class A40 (25.3 kg/m). - Bolts, nuts and washers and accessories shall be galvanized class A40. Reflective tags shall conform to standard SATCC 5200. Concrete for foundation blinding shall conform to codes 640 to 643.
Measurement	M MDE



Barrier rails on steep slopes to prevent vehicles from falling over



Barrier on a rail/road level crossing to stop vehicles when train is passing

Figure 9-32 Protection Barriers – showing measures to protect users against severe accidents

Activity: Placement of Vertical Road Traffic Signs		Code: 530
Reference Code		
Application	Road signs provide information on routes, distances and services, information on hazards and limits and prohibitions in using the road, to guarantee road safety, Figure 933. The work consist of supplying and placement of the permanent vertical traffic signs including warning, information and regulatory signs conforming to the design and instructions of the Supervising Engineer.	
Tasks	• Producing and supplying the signs. • Identifying and marking locations for placement of the signs together with the Supervising Engineer. • Placing temporary signs in work area. • Clearing the area around the sign to guarantee visibility of the message on the sign. • Excavating foundation. • Placing the signs. • Backfilling the foundation with granular or stabilised fine soils. • Cleaning the area and removing vegetation to make the sign visible. • Removing temporary signs.	
Standards	• The dimensions and colours shall conform to the Road Signs Standards of SADC. • Maximum tolerance for the dimensions of the signs shall be $\pm 1\%$. • All colours on the signs except black and grey shall be reflective. • The selection and placement of the signs shall meet the following basic requisites. ◦ Must be necessary. ◦ Attract the attention of the road users. ◦ Transmit a clear message. ◦ Allow adequate time for reaction. • All signs shall be placed on the left hand side of the road and the front of the sign shall be at 85 to 90 degrees to the centerline of the roads to allow for maximum visibility to the road users. • The distance between the inner side of the sign to the outer edge of the drain shall be between 1.50 m and 3.60 m. • The height of the bottom of the sign (plate) to the ground level shall be 2.0m. • Temporary signs shall be placed in work area. • Demarcation of the locations of the works and identification of the sites for placing the signs shall be carried out together with the Supervising Engineer. • Excavating the foundations shall be according to the dimensions of the foundation excavated vertically with a shovel or spade avoiding materials falling back into the excavation. Excavate till firm ground free of water and organic materials. If this is not possible then a layer of aggregate 10 cm thick shall be placed at the bottom of the foundation. • The excavation shall be cylindrical and square with cross-sectional dimensions less than 90 cm and a depth of 80 cm. • Level out the foundation. • Align the post vertically with the spirit level using aggregate and a wooden support before backfilling. • Weld a metal plate at the bottom of the sign post to prevent rotation of the sign once fixed. • Backfill with soil stabilised with 8% (1:12) cement in layers not more than 10 cm thick. • Remove temporary signs.	

Materials	• All materials must be approved by the Supervising Engineer. • Concrete for the blinding shall conform to codes 640 to 643. • The metal plates for the signs shall be galvanized and 1.4 mm thick covering the whole perimeter of the boards. The metal plates shall be galvanized and finished or covered with coloured plastic (such material shall be applied cold over the plate which shall be clean and free from oil and dust. • The signs shall be fixed to the posts using methods approved by the Supervising Engineer in conformity with the design. • The posts shall be tubular with a diameter of 2" (50 cm) with 2 mm thick metal or wooden (eucalyptus) 100 mm in diameter and treated with creosote or equivalent materials.
	• The metallic pipes shall have a lid at the top to prevent water entering inside the pipe. • Metal used for fixing such as braces, bolts and nuts shall be galvanised.
Measurement	M² The vertical traffic signs shall be measured as the effective area covered by sign in m². MDE



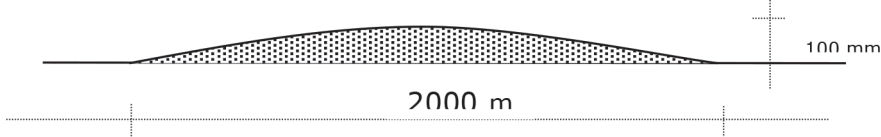
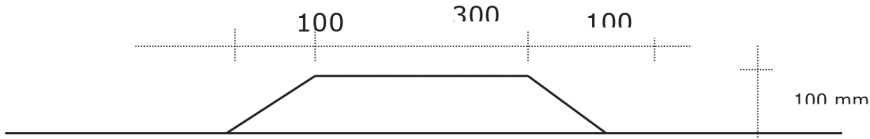
Figure 9-33 Vertical Signs – Warning, regulatory and Informative

Activity: Marking of Horizontal Road Traffic Signs		Code: 540
Reference Code		
Application	Road signs provide information on routes, distances and services, information on hazards and limits and prohibitions in using the road, to guarantee road safety, Figure 934. The horizontal signs together with markings, symbols, and writings on the surface of sealed roads should meet the project designs to achieve safety requirements and comfort for the road users. The supply of the horizontal warning, informative and regulatory signs shall be as per the Supervising Engineer.	
Tasks	- Identifying the sites and pre-marking (alignments and points) where horizontal signs should be placed together with the Supervising Engineer and indicating the extent, width and dimensions of the lanes. - Placing temporary signs in the work area. - Cleaning the surface of the pavement ensuring that it is thoroughly clean, dry and free from debris, oils, chemicals and acids. - Carrying out a trial application of the paint on a small section for approval by the Supervising Engineer. - Applying the paint by hand or appropriate equipment. - Removing temporary signs.	
Standards	- The dimensions and colours of the signs shall conform to the standards of SADC/ SATCC Road Signs given in the Road Traffic Signs Manual. - The maximum tolerance of the dimensions of the horizontal signs is $\pm 5\%$. - The alignment of the horizontal signs shall be smooth and free of irregularities more than 0.5 cm measured using a 2 m straight edge. - The horizontal signs shall meet the following basic requirements: - They should serve a real purpose. - Should attract the attention of the road users. - Should send a clear and simple message. - Should allow adequate time for reaction. - The horizontal signs shall only be applied on bituminous surfacing or concrete pavement. - Temporary signs shall be placed in the work area. - Such horizontal signs shall be pre-marked with points using the same colour as that to the horizontal signs. - The spacing of the pre-marking shall be in the same positions as those of the actual horizontal signs and the spacing shall not be less than 1.5 m. - Approval of the pre-marking by the Supervising Engineer shall be controlled using the straight lines and tangents and the associated angles of the curves. - The area set for application of the horizontal signs shall be cleaned using hand brooms or one pass of the mechanical broom or an air compressor if necessary. - If the application is by mechanical means the paint shall be applied in one coat. - If application is by hand then dual application of the coating shall be used. - The application rate shall be 0.42 L/m² and the paint shall be applied without dilution. - In the case of reflective paint the micro-spheres shall consist of high quality glass of soda-cal type and shall be applied immediately after the application of the paint at a rate of 0.8kg/Lt of paint. - Thermoplastic paint shall be sprayed to a thickness of 1.5 mm and if applied by extrusion the thickness shall be 3 mm. - The thermoplastic shall be mixed with 20% (by weight) glass micro-spheres. - The horizontal signs shall not be applied on a humid surface, or when temperature below 10 degrees Celsius or during strong winds. - Tolerance limits: - Width should not be less than the specifications and not more than 10 mm extra. - The position shall not deviate by more than 10 mm longitudinally and 20 mm transversal. - The alignment of the lanes shall not deviate by more than 10 mm in every 15 m. - The length of gaps in the lines demarcating lanes should not deviate by more than 150 mm.	

	- Equipment shall meet all conditions necessary for proper application and include storage for the paint and glass micro spheres, spray guns capable of applying the paint strips and gaps simultaneously, air compressor, mixer, automatic direction control mechanism, guidance lance for control of end points, adjustments, control system for spacing and width of strips, rare lights, rotating light, indicators, and pressure regulators. - For the application of thermoplastic, the equipment shall have reservoir with heaters of the type of boilers that regulate temperature. - Remove temporary signs.
Materials	- The type of paint used for the horizontal signs shall conform to the following: - The paint shall conform with technical specifications of standards SABS 731. - The paint shall be reflective. - The thermoplastic paint shall conform to BS 3662 and BS 6088. - The choice of material shall be based on the design of the project, traffic volume and the expected life of markings such as thermoplastic paint which is more resistant. - In the case where the glass micro-spheres are added and mixed with the paint the viscosity shall be reduced by adding a solvent up to a maximum of 5% by volume. - The paint shall be white, yellow and red and shall conform to the SATCC Road Traffic Signs Manual.
Measurement	M² The code on horizontal signs shall be measured as the effective area applied in m². MDE



Figure 9-34 Horizontal Road Signs – Road Markings

Activity:	Construction of humps on the surface of the road	Code: 550
Reference Code		
Application	This code is used for the construction of humps on the road surface or ground indicated by the Supervising Engineer. Humps are transverse undulations constructed on top of the pavement with the intention of reducing traffic speed passing through this location, Figure 935. The humps serve to increase safety in populated areas.	
Tasks	• Identifying sites and marking positions where the humps will be constructed together with the Supervising Engineer and local authorities. • Placing temporary signs in the work area. • Marking longitudinal and transverse alignments of the hump using pegs, metric tape and level together with the Supervising Engineer. • Watering and scarifying the place set out for the construction of the hump. • Transporting the materials from the borrow pit close to the site if it is necessary to import the material. • Compacting loose material at optimum moisture content. • Controlling the shape of the hump. • Clearing the work area. • Removing temporary signs.	
Standards	• Place temporary sign in the work area. • The transverse profile of the hump shall conform to the designs type with low height, Figure 9-35. • The maximum tolerance for the height of the hump shall be ± 10 mm and length $\pm 5\%$. • The surface of the hump in transverse section shall be plane and free of undulations or irregularities in excess of 2 cm measured using a 2 m straight edge. Alternative A- Circular  Alternative B- Plateau  Figure 9-35 Specifications for humps – traffic calming measures	
	• The width of the hump depends on the transverse profile of the road. The hump shall be constructed to the shoulder of the road such that the vehicles cannot ride on and damage the shoulders and drains while avoiding the humps. • The minimum distance between humps shall be 100 m. • A hump shall be constructed and completed within one day. • Fill the area for the hump with soil and compact at optimum moisture content. • The imported soil shall be of the same quality as a mechanically stabilised base, code 340. • Remove temporary signs.	
Materials	Not applicable	
Measurement	M Measured as the length of the hump in M. MDE	

9.9 Series 700 Quality Control

Quality control is important for the delivery of works that provide good value for money for the client. Quality control is part of quality assurance where the quality of works is assured through procedures involving:

- i. Appropriate and standardised materials testing procedures;
- ii. Method specification: Stipulating the proper procedures for the execution of works which, when followed, should produce works of the desired quality fit for the intended purpose;
- iii. Self-assessment by the contractor: This means that the contractor should control the works and ensure that the quality conforms to the standards set in the contract;
- iv. Supervision: This includes assessment of the works, identification of defects in the works, and delivery of site instructions for corrective measures to be taken;
- v. Field and laboratory quality tests for dimensions, compaction, consistency, irregularities and other quality aspects checked against the specifications and tolerance limits;
- vi. Carrying out remedial measures to ensure that specifications are met; and
- vii. Approval of works given in writing and tied to payment certificates to ensure good value for money for the client.

The sustainable provision of low volume roads and low cost seals is largely dependent on the quality control and the quality assurance systems including enforcement.

Activity:	Preparation and implementation of the Quality Control Plan	Code: 710
Reference Code		
Application	This code is used for the preparation and execution of the Quality Control Plan by the Contractors in conformity with the Contract Management Procedures. Contractually the Contractor is responsible for the quality of the works. This is an integral part of the contractual obligations through the Quality Control Plan (QCP). The importance of the Quality Control Plan is that it establishes the tests and inspections that should be carried out for the purpose of Quality Assurance of the works. In the same manner it informs the Engineer on the standards and specifications, including the tests and evaluation of the results, for the approval of works. The approval system helps to standardise the quality of the works outputs for the different activities. It is the responsibility of the Engineer to ensure that the required tests are carried out and to check that they have been carried out properly by an approved laboratory and qualified personnel. The Engineer's assessment of the results and the approval and rejection of the works shall be documented and given to the contractor and the client as part of the quality assurance system. To this effect the Engineer shall complete the relevant approval forms, which shall be attached to the Interim Payment Certificates. This way good value for money shall be assured for the client (MoWT/District Local Government, Urban Councils, Road Authorities, etc.). All tests and retests (where there is doubt about the accuracy of the results) or after remedial or corrective works by the contractor shall be reimbursable under the codes 720 to 740.	
Tasks	• Preparing the Quality Control Plan to establish the mechanism for controlling the quality of the work. • Outlining the Contractor's and the Engineer's responsibilities for quality control. • Nominating the laboratory or laboratories that will carry out the quality tests that are stipulated in the contract and the Quality Control Plan. • Establishing a plan and procedures for inspections and laboratory testing. • Inspecting the works by the Engineer to verify the dimensions, quality and standards. • Sampling and carrying out the laboratory tests as indicated in the Quality Control Plan by the nominated laboratories under the supervision of the Engineer. • Submitting the test results, completed approval forms and instructions by the Engineer to the Contractor and the client.	
Standards	• The Engineer shall prepare the Quality Control Plan and discuss it with the contractor and the client. The Quality Control plan shall be finalized within 14 days of commencement of the works. The model of the Quality Control Plan shall conform to the Tender Documents and the Contract Management Procedures. If necessary, the Quality Control Plan shall be revised during the course of the execution of contract. • The Contractor and the Engineer shall inspect the work on the basis of the Quality Control Plan, verifying the physical dimensions, quality and standards in conformity with the contract conditions and specifications. The inspections and quality control activities, such as inspections and quality testing, shall continue as long as the contract is in force. These activities shall be reimbursed under code 610. If the quality control budget is provided under the contract then the contractor shall pay for the tests and any costs related the testing as directed by the Engineer. • The Engineer shall instruct the appointed laboratory to carry out the tests and, whenever necessary, shall take responsibility for the quality of the testing and accuracy of the results. The Engineer shall supervise the sampling and also randomly check the testing of the materials in the laboratory. • The prospecting of materials and location of borrow areas should have been carried out as part of the Project Planning covered in PART A. • The tests on materials that shall be carried out under this activity shall be on samples collected after stockpiling of the materials in the borrow pits to verify the quality of the material which will be hauled and used on site. • The list of materials quality tests is given in Table 947.	

Table 9-47 List of Materials and Works Quality Tests

Description	Frequency of the sampling and testing requirements
Test on soils	
Collecting, preparing, packaging and transporting of the samples to the laboratory for tests.	Samples shall be collected from each stockpile. Collect 5 samples from different parts of each stockpile. Each sample shall be 50kg or more. Collect 3 samples per stockpile if the material is sand.
Grading tests	One sieve analysis for each sample
Atterberg limits	2 tests for each sample
Determination of mode: laboratory dry density and optimum moisture content.	Mix the material for each stockpile and carry out 2 similar tests on the mixed material to check for accuracy of the test results.
Determination of CBR (soaked, OMC and 0.75 OMC (optional))	Carry out 2 similar tests on the mixed samples for each stockpile.
Tests for concrete	
Slump test on fresh concrete	One test for every 3 batch mixes
Cube strength tests	Six cubes for every pour (every lift or continuous pour)
Test on aggregate in the laboratory	
Grading	Samples shall be collected from 3 or more different positions in the truck or stockpile and the samples shall be tested separately.
ACV	Minimum of 3 tests and plot a grading envelop for each delivery on site.
10% FACT	Minimum of 3 tests, record and take an average of the values
Water absorption	Minimum of 3 tests, record and take an average of the values
Bitumen tests	
Penetration (suitable for straight run and cutback bitumen, 60/70 pen, 80/100 pen, 150/200 pen, MC3000, MC800)	Collect samples of each truck load delivery and test separately. Minimum of 3 tests, record and take an average of the values

	Softening point (suitable for straight run and cutback bitumen, 60/70 pen, 80/100 pen, 150/200 pen, MC3000, MC800)	Minimum of 3 tests, record and take an average of the values
	Viscosity tests (suitable for straight run and cutback bitumen, 60/70 pen, 80/100 pen, 150/200 pen, MC3000, MC800)	Minimum of 3 tests, record and take an average of the values. District laboratories that do not have capacity shall send samples to MoWT laboratory in Kireka.
	Bitumen content (suitable emulsions)	Minimum of 3 tests, record and take an average of the values
	Compaction tests	
	Testing of field compaction using sand replacement or Troxler.	9 tests per 200m (standard) 9 tests per 400m for remote projects. 9 tests per km and DCP tests every 50m with some of the test carried out within 0.5 m of the sand replacement or Troxler tests for very remote projects The average values of density and the standard deviation shall be plotted on the compaction judgment chart given above, Figure.....
Materials	Not Applicable	
Measurement	- Measurement contract. - The contractor shall pay for the cost of tests in accordance with the Engineer's instructions (in the case where budget is under the works contract) - MAE	

Activity: Sampling from stockpiles in borrow pits, recording and transporting samples to the laboratory for testing		Code: 720
Reference Code		
Application	This code is for the sampling of soils from the stockpiles in the borrow pits, method of collection, packaging to preserve the material and transporting to the laboratory.	
Tasks	• Identifying the position for sampling on the stockpiles • Excavating and sampling of materials from the test hole excavated on top of the stockpile and packaging the material in 50 kg samples bags • Recording information regarding the project, borrow pit and sample. • Marking the samples • Transporting to the laboratory • Testing	
Standards	• The sampling process follows after the completion of stockpiling of material in the borrow pit using dozers or excavators. On completion of heaping the material in the stockpile, the top of the stockpile should be leveled. • The positions where the samples will be collected should be identified such that the samples are representative of the material in the stockpile. Usually 5 positions are sufficient. If the surface of the stockpile is rectangular then samples shall be collected from each of the 4 corners and the middle. • The material shall be sampled from a depth of 1 to 1.5 m from the surface of the stockpile. • A 50 kg sample of material shall be collected and placed in a sample bag. Metal or hard card labels shall be placed in the sample before tying the bags with plastic ties. The information that should be written on the tags shall include; Project Name, Pit Number, Stockpile Number, Sample Number and Materials Description. • The information about the sample and the material shall be written in the sampling forms and shall include; Project Name, Pit Number, Sample Number and Materials Description (in more detail). • The process shall be repeated for the other stockpiles. • The samples shall be transported to the laboratories for testing and care shall be taken to prevent alterations due to rain, impact or other adverse conditions including contamination.	
Materials	• Not applicable	
Measurement	• Quantities • The contractor shall pay for the costs of labour, transport and per diems based on the Engineer's instruction. • MAE	

Activity:	Sieve Analysis Testing the grading of materials	Code: 730
Reference Code		
Application	This code is used in the testing of the materials for their particle size distribution and comparison with the specifications. The sampling of the materials, registering, packaging and transportation to the laboratories is paid under code 710.	
Tasks	• Registering the samples. • Preparing the samples. • Defining the quantity of the soil required • Carrying out the sieve analysis. • Completing the test form. • Presenting the results in the results forms.	
Standards	• All samples shall be registered by the laboratory including; the project name, source of the sample, date and time received. • Once the samples are registered and prepared, the laboratory shall fill in form for Testing Instructions, where the specifications for the tests shall be obtained. Qualified and experienced laboratory technicians shall be tasked with carrying out the tests and, once completed, they shall complete forms with details of the test data and results. • For determination of the grading of coarse soils, use sieves. For fine materials hydrometer tests shall be used. It is recommended that dispersion agents are used in order to improve in the accuracy of the grading tests. The dispersion agent helps to detach very fine particles clinging on the larger particles. • Preparation of samples: ○ The material shall be air dried and, after drying, the material is broken up and worked by hand or with the help of a spatula. ○ Through successive riffing to get a representative sample, obtain a sample size of 2000 g which is predominantly coarse and 1500 g which is predominantly fine. ○ After weighing, collect a sample of the materials passing No. 10 (2.0 mm sieve size). ○ Break up the material retained and sieve it through sieve No. 10 and this should be repeated until the material has been sufficiently broken and sieved. ○ The material retained shall then be sufficiently wet sieved on sieve No. 10. ○ The material retained on sieve No. 10 after wet sieving shall be collected and oven dried. ○ For the material passing 2.0 mm sieve, separate a portion for sedimentation test and (2.0 mm >particle size fractions>0.075 mm): a) Collect 200 g of material which is predominantly coarse. b) Collect 70 g of materials which are predominantly fine. ○ The remaining material shall be sieved through sieve size No.40 (0.425 mm sieve size). ○ Material retained on 0.425 mm sieve shall be broken up by hand and sieved through the same sieve. ○ 100 to 120 g of the material passing the 0.425 mm sieve shall be used in carrying out the Liquid Limit (LL) test and 50 g for Plastic Limit (PL). Plasticity Index, PI = LL - PL • SIEVING COARSE MATERIAL 1) The material produced in the preparation of the sample in step (6), which was place in the oven to dry shall be removed from oven after drying and placed in a desiccator. 2) After cooling the material is sieved through a series of standards sieves. 3) After sieving adequately the material retained in each sieve shall be weighed.	

	• SEDIMENTATION 1) The material resulting from the preparation of the sample in step (7) shall be placed in the beaker and a deflocculent shall also be added and left for 24 hrs. 2) The mixture of deflocculent/soil is passed to a dispersion glass. 3) The dispersion shall be carried out in a) Minimum of 10 min for material that is predominantly coarse. b) Minimum of 15 min for material which is predominantly fine. 4) Transfer the mixture to a graduated beaker with a volume of 1000cm³. 5) Agitate the mix for 1 min and start the test of determining or reading off the densities at predetermined times. 6) When the test is completed repeat the steps three times. 7) Pass all the material through sieve No. 200 (0.075 mm sieve size) and wash the material retained. 8) Once sufficiently washed, the material shall be placed in the oven to dry. The percentage of the material retained in each sieve and percentage passing by weight shall be calculated and reported.
	9) Agitate the mix for 1 min and start the test of determining or reading off the densities at predetermined times. 10) When the test is completed repeat the steps three times. 11) Pass all the material through sieve No. 200 (0.075 mm sieve size) and wash the material retained. 12) Once sufficiently washed, the material shall be placed in the oven to dry. The percentage of the material retained in each sieve and percentage passing by weight shall be calculated and reported. • Complete the forms of tests register in conformity with laboratory specifications, recording all data and results obtained from the respective tests. • All results shall be presented in standard laboratory forms, and the Engineer shall coordinate with the laboratory for request for retest or approval of the results. • The Engineer shall use the results to approve or reject the materials and to give a written communication to the contractor and the client. • SIEVING FINES (2.0 mm > particle sizes > 0.075 mm) 1) The material resulting from sedimentation step (8) shall be removed from the oven and taken to the desiccator. 2) After cooling the material is sieved through a series of sieves indicated (from 2 mm to 0.075 mm). 3) After sieving adequately the material retained in each sieve shall be weighed.
Materials	• Not applicable
Measurement	• UN • MAE

Activity:	DCP Test	Code: 740
Reference Code		
Application	This test is used for the field determination of the mechanical properties of natural and slightly stabilised soils. It is a measure of the in-situ strength of the material in their natural state or in an existing pavement.	
Tasks	• Placing of signs for safety. • Establishing the places to carry out the tests. • Preparing the work area. • Carrying out the test. • Filling the test forms. • Presenting the results in the result forms and graphs.	
Standards	• Adequate signs shall be placed to guarantee the safety of the workers where necessary. • The locations for the tests shall be identified and recorded. • Procedure: 1) This DCP penetration is an in-situ test which helps in the identification of the geotechnical profile and properties of the soil. 2) The DCP consists of a cone for penetration (30 or 60 degrees at the tip but the 60 degrees cone is the most commonly used). The cone is fastened to a 1 m steel rod which is also connected to an anvil. Above the anvil is a rod with an 8 kg hammer and a handle on the other end (see diagram below). 3) A metre rule is attached to the anvil extending all the way to a plate at the bottom of the instrument. The plate has a hole through which the cone and rod can penetrate into the ground. 4) The assembled DCP equipment as shown in Figure 936.	

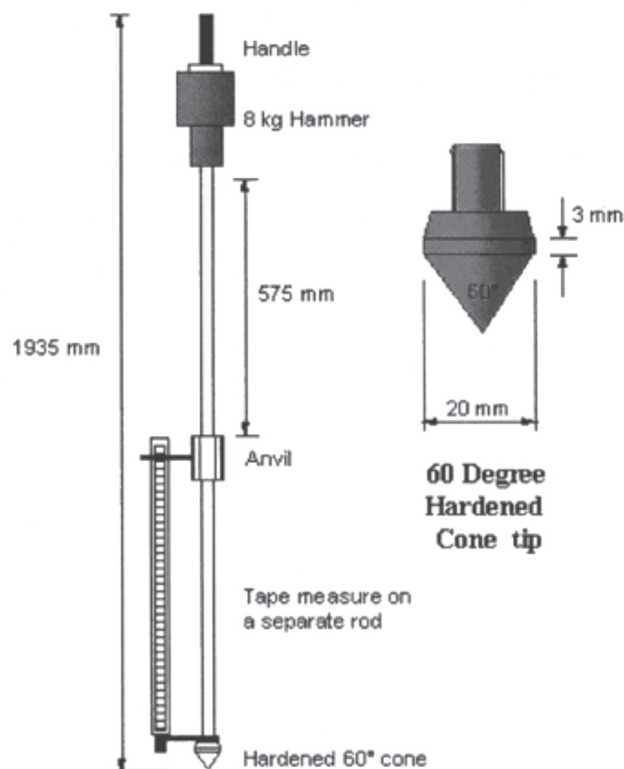


Figure 9-36 DCP Equipment

	- Place the DCP on the test point. - Fill in the form for the test including the project, the chainage for the test point, the off-sets from the centreline or shoulder, the lane number and the zero error. Note that the test may be carried out on virgin ground. - Apply the number of blows by dropping the hammer freely from the top most position to the anvil. Record the readings on the metre rule corresponding to the number of blows applied. The choice of the number of blows that shall be applied before taking a reading shall be guided by the penetration which should be approx. 10 mm. The number of blow can be reduced if the penetration is much more than 10 mm. - Stop the test once adequate penetration has been reached, normally 800mm is sufficient. However, where deeper penetrations are required extension rods can be added to achieve greater depths of penetration. - On completion of the test apply one blow upwards and try to turn the DCP clockwise or anticlockwise. If it turns without difficulty the penetration data is acceptable for analysis. If it feels significantly stiff then the test shall be discarded because it shows that there is high friction on the road. This is typical in situations where the rod is caught up between large stones during the test and there is additional resistance from side friction on the rod. - Once the test has been accepted the form for the test that was filled in during the test shall be ticked and filed. - Repeat the test procedure for other test positions. Data entry: - Check the data on the field forms for obvious errors. - Enter the information for the tests including the project name, test position (chainage, offset, zero error, etc.) into the software UKDCP3.1 or equivalent - Select the options for whether the test position has surfacing or not and whether the surfacing was removed or not. - Enter the blows/penetration data. - Select the moisture condition for automatic adjustment of the data for different moisture levels. Analysis: - Click on analysis and select the number of layers based on the plot of the blows/penetration data. - Click Structural Number and select structural class for each layer (base, sub-base, subgrade). The software will calculate the CBRs and the structural numbers for each layer and for the whole pavement. - Compare the results with the structural requirements of the design.
	Alternatively: - The results of the DCP blow/penetration data is expressed as the DN value which is the penetration/blow. The DN values are calculated for each layer and these are compared with the DCP design catalogue in order to verify the adequacy of the strength of the layers and the pavement as constructed. - The DN values shall be adjusted for in-situ moisture. - Presentation of the results: All results shall be presented in conformity with DCP standard forms and the DCP software outputs. - The results shall be used by the Engineer for assessment of the in-situ strengths and written communication shall be given to the contractor and the client.
Materials	Not applicable
Measurement	- UN - MAE

Activity: Field Density Tests		Code: 750
Reference Code		
Application	This test is used for the field determination of relative density of pavement layers constructed using either labour based or machine based methods. It is important to ensure that the compaction of the materials in the field meets specifications. This test shall be considered after the accomplishment of construction and compaction of fill or wearing course or subgrade or subbase or base. Covered in this Activity Code are two main tests • Sand replacement method • Nuclear density test method This code also covers the assessment and the approval process for compaction.	
Tasks	• Placing of signs for safety. • Establishing the places to carry out the tests. • Preparing the work area. • Carrying out the test. • Filling the test forms. • Presenting the results in the result forms and graphs.	
Standards	• Adequate signs shall be placed to guarantee the safety of the workers where necessary. • The tests shall be carried out by a qualified and experienced laboratory technician or engineer assisted by general hands/trained labourers. • The locations for the tests shall be identified and recorded. ○ Under normal circumstances the completed section shall be segmented into 200 m sections and each section shall be assessed for weak spots such as sponges, undesirable irregularities and inappropriate profiles. The section should be approved for quality testing by the Engineer. ○ Where projects are remote and laboratories are a long way away from site the length of the individual test sections may be increased to 400m or 500 m. The number of tests shall remain the same (9 tests/section) the test positions shall be spaced equally along the full length of section. ○ Tests points shall be marked out in a left-right-centre in a zig-zag manner ○ The test points which are close to the shoulders shall be located such they are not less than 50 cm from the edge of the compacted pavement layer. PROCEDURE: Sand Replacement Method 1) The apparatus (Figure 938) including sealed jars or thick plastic bags for collecting samples, the base plate, nails for fixing the base plate, the metallic cylinder with a cone and valve, a chisel and spatula shall be checked and taken to the site. 2) The test involves placing and fixing the base plate and excavating a cylindrical hole into the layer through the hole in the base plate, collecting the material (soil) sample quickly and under shed when its hot, placing and sealing the samples in jars or plastic bags to prevent excessive loss of moisture. The metallic cylinder with a cone shape at the bottom is placed over the hole and the cylinder is filled with specially prepared fine sand. The valve between the cylinder and the cone at the bottom is opened to allow the sand into the hole in the layer and the cone. 3) The volume of sand that goes to fill up the hole in the layer is calculated and this gives the volume of the hole excavated into the layer. This is equal to the volume of the material excavated out of the compacted layer. 4) At the laboratory, the samples are weighed to determine their mass with moisture. 5) The small and representative samples, a minimum of 3 in number are collected and weighed separately to get the wet mass. 6) After weighing the samples are placed in an oven at 105 °C to dry and weight regularly until the measured mass is constant. This is taken as the dry mass. Average values are calculated for each sample (minimum of 3 values is required). The difference between the wet mass and dry mass is the mass of moisture in the samples.	

	7) The moisture content (w) of the material is the mass of water as a fraction of the dry mass of soil. 8) The dry mass of the material from the hole (Md) is calculated from the wet mass (Mw) using the following equation $Md = \frac{Mw}{(1+w)}$ 9) The dry density (Yd) of the materials from the hole excavated from the compacted layer is the dry mass of the materials divided by the volume of the hole (V) $\gamma_d = \frac{Md}{V}$ 10) The dry densities for each of the 9 test positions are determined in the same manner. 11) A standard procedure for determining laboratory maximum dry density (mdd) for the same material is carried out. The field dry density is then expressed as a percentage of the maximum dry density and this is referred to as relative density or level of compaction achieved in the field. 12) There are compaction lower limit specifications e.g. 91 % mdd for subgrade, 93% of mdd for subbase and 95% of mdd for base course. Analysis and evaluation 13) Of the 9 relative density results 3 shall be taken as outliers and taken from the data set. 14) From the remaining 6 relative density values, the mean value and the standard deviation are calculated. The values are plotted on the Compaction Judgment Form given in 15) Figure 937. Depending on which zone it plots in the section is scientifically and statistically evaluated on whether it has passed or failed or can be conditionally accepted. For conditional acceptance the reasons should be stated, such as material has high content of mica and cannot be densified any future. 16) The uniformity of densities is as important as the average density. Nuclear density test method The nuclear density gauge (Figure 939) is an instrument which uses radioactive probes to determine the field density of pavement layers. The instrument has a probe that releases radioactive particles in very minute quantities and a sensor which aggregates hits by such particles. Highly density material absorb more of the particles before they reach the sensor and this way the hit counts are calibrated for different densities. The sand replacement method is used for the calibration of the Nuclear Density Gauge. Once the calibration factors are entered the gauge automatically calculates the densities and moisture contents. This is a rapid method but it is not as accurate as the sand replacement methods. It's completely harmless if used properly.
Materials	Not applicable
Measurement	- UN - MAE

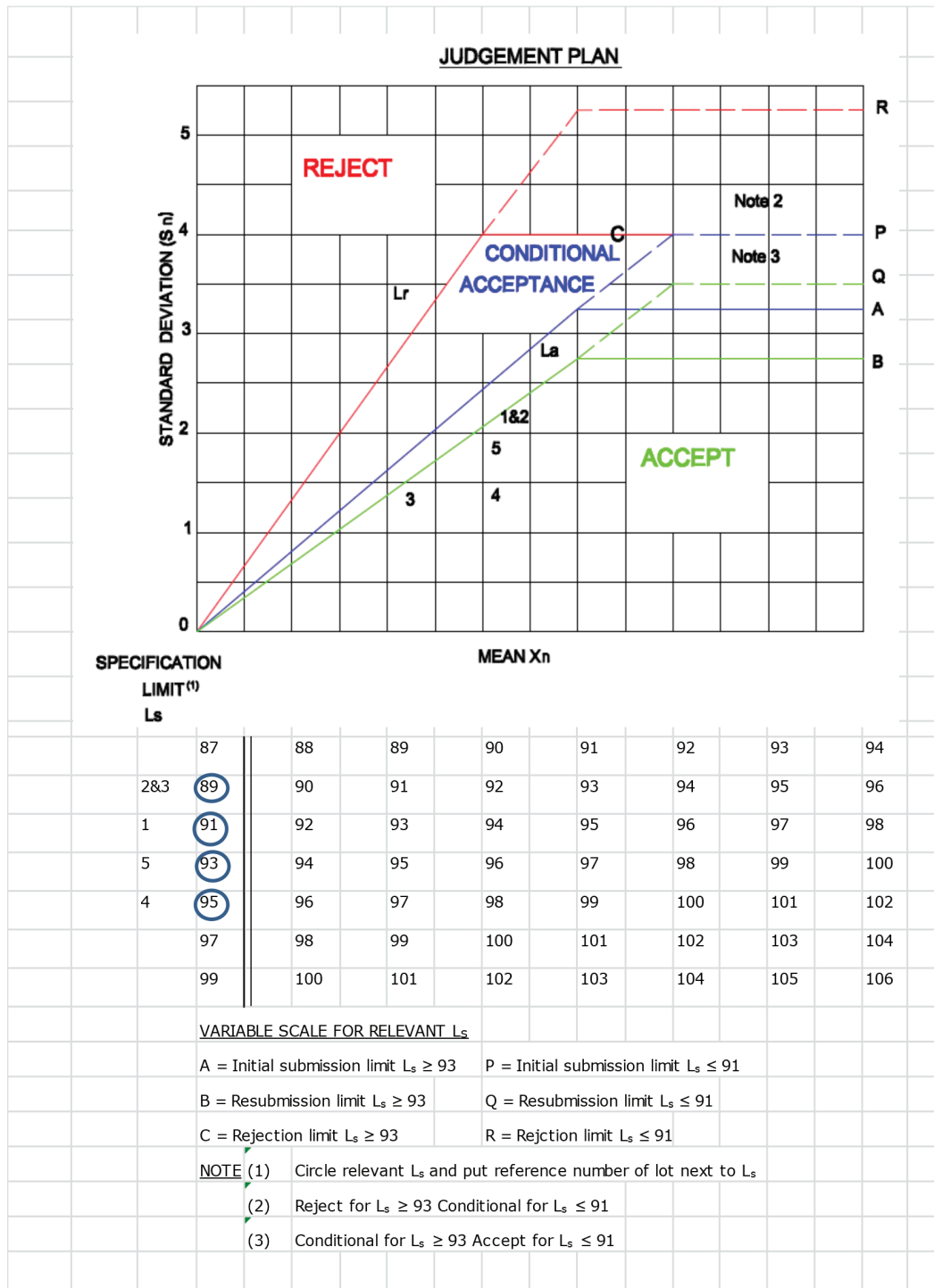


Figure 9-37 Compaction Judgment Form



Figure 9-38 Apparatus for Sand replacement Method



Figure 9-39 Nuclear Density Gauge

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GLOSSARY OF TERMS

PAVEMENT

Base (or base course)	The layer of the pavement structure resting upon and through which the load is transmitted to the subbase, subgrade and foundation.
Formation:	The surface of the improved subgrade, in its final shape, upon which the pavement structure, consisting typically of the base course and surfacing is constructed.
Subgrade:	All the material below the pavement and may include the improved subgrade and the in situ subgrade.
Improved subgrade:	The layer immediately below the pavement which would typically be the re-shaped and compacted wearing course of the unpaved road.
Subbase:	The layer of the pavement structure that lies above the subgrade and below the base course.
Base course	This is the layer below the surfacing and on low volume roads it is the main bearing layer. The base course is generally made of natural gravel or modified material like Emulsion Treated Bases (ETB) and lime stabilised bases.
Surfacing:	The uppermost pavement layer which provides the running surface for traffic. It can consist of a bituminous or non-bituminous type of surfacing.
GENERAL MATERIALS Borrow area:	A site from which natural material, other than solid stone, is removed for construction of the road works.
Naturally occurring gravel/soil:	Material from natural sources used in its original state for the construction of pavement layers.
Quarry:	An open surface from which natural material, other than solid stone, is removed by drilling and blasting, for construction works
BITUMINOUS MATERIALS Asphalt concrete:	An admixture of bitumen-bound materials normally consisting of a mixture of coarse aggregate, fine aggregate and filler bound with straight-run bitumen. The proportions and grading of the coarse aggregate may be varied to produce different types of mix with differing properties.
Bituminous binders:	Petroleum derived adhesives used to stick chippings on to a road surface, in surface dressing or to bind together a layer of surfacing or base material.
Bitumen emulsion:	A binder in which petroleum bitumen, in finely divided droplets, dispersed in water by means of an emulsifying agent to form a stable mixture.
Bituminous seal:	A general term for thin bituminous wearing courses made of surface treatments or slurry seals or a combination of both.
Cut-back bitumen	Bitumen whose viscosity has been reduced by the addition of a volatile diluent.
Fog spray:	A light application of bitumen emulsion or cut-back, on top of surface dressing to improve the waterproofness of the surfacing and to assist in holding the chippings.
Prime coat:	An application of low viscosity bituminous binder to an absorbent surface, usually the top of the base with the aim of waterproofing the surface being sprayed and to help bind it to the overlying bituminous courses.
Straight-run bitumen:	Bitumen whose viscosity or composition has not been adjusted by blending with solvents or other substance
Surface dressing:	A surface treatment made of single sized aggregates of crushed material. Can be constructed in single or multiple layers.
Surface treatment:	A general term for thin bituminous wearing courses made by rolling aggregate into a sprayed thin film of bitumen. Aggregates can be made of crushed or natural gravel with a grading depending on the desired type of surface treatment to be produced. Can be constructed in single or multiple layers.
Slurry seal:	A surfacing material which consists of fine aggregate, mineral filler and bitumen emulsion, used in one or two layers, or on top of a single surface dressing (Cape seal).
Sand bitumen:	A base material consisting of a cold mix-in-place combination of sand and either bitumen emulsion or cutback.
Tack coat:	A light application of bituminous binder to a bituminous surface to provide a bond between this surface and the overlying bituminous course.

Wearing course (bituminous):	The uppermost surfacing layer. Can consist of a bituminous mix or a bituminous seal.
SOILS AND SOIL TESTS	Collective name for Liquid Limit and Plastic Limit tests. Originally proposed by A. Atterberg for determining soil plasticity.
Atterberg Limits:	
California Bearing Ratio (CBR):	The value given to an ad-hoc penetration test where the value 100% applies to a standard sample of good quality crushed material.
Clay:	The finest soil fraction. Comprises colloiddally fine, complex silicates formed by the natural decomposition of igneous rocks.
Clay fraction:	The fraction of soil composed of particles smaller in size than 0.002 mm. This is the fraction that generally imparts plasticity to a soil and in appropriate quantities helps bind the material together. Too much clay can lead to unacceptable swelling and shrinkage in response to moisture.
Cohesive soil:	Soil containing sufficient clay or silt particles to impart significant plasticity and cohesion.
Compaction:	The process whereby soil particles are densified, by rolling or other means, to pack more closely together, thus increasing the dry density of the soil.
Dry density (DD):	The mass of the dry material after drying to constant mass at 105 degrees Celsius, contained in a unit volume of moist material.
Dry density/moisture content relationship:	The relationship between the dry density and moisture content of a soil when a given amount of compaction is applied.
Equilibrium moisture content (EMC):	The moisture content at any point in a soil after moisture movements have stabilised in a constructed pavement.
Lateritic:	Soils and rocks containing iron oxides as a major constituent in concentrated form, remaining behind when more soluble products have been leached away.
Linear Shrinkage (LS):	The decrease in one dimension, expressed as a percentage of the original dimension of the soil mass, when the moisture content is reduced from the liquid limit to the oven-fry state.
Liquid Limit (LL):.	The moisture content, expressed as a percentage, at which the wet soil fines passes from a plastic to a liquid condition
Heavy compaction:	Laboratory compaction of a soil sample using a 4.5 kg hammer falling through a height of 115.5 mm to compact a soil in a 150 mm diameter mould
Moisture content:	The loss in mass, expressed as a percentage of the dry material, when a soil is dried to a constant mass of 105 degrees Celsius.
Optimum moisture content (OMC):	That moisture content at which a specified amount of compaction will produce the maximum dry density.
Particle size distribution (grading):	The spread of soil sizes (fraction) in a soil sample from gravel to sand to silt/clay as measured by passing the sample through a nest of standard sieves.
Plastic Limit (PL):	The moisture content at which the damp soil fines pass from a plastic to a solid condition, i.e. when the soil ceases to behave as a plastic material.
Plasticity Index (PI):	LL – PL, an indication of the quality of clay soils; the larger the PI, the plastic the clay content.
Plasticity Modulus (PM):	A measure of PI x % passing the 0.425 mm sieve.
Plasticity Product (PP):	A measure of the quantity of clay: PI x % passing the 0.075 mm sieve.
Residual soils:	These are the weathered remains of rocks that have undergone no transportation. They are normally sandy or gravelly, with high concentrations of oxides resulting from a leaching process
Rock:	Hard rigid coherent deposit forming part of the earth's crust, which may be of igneous (e.g. granite, basalt), sedimentary (e.g. limestone, sandstone) or metamorphic (e.g. slate, hornfels) origin and which normally requires blasting. Soft, more easily excavatable materials such as clays, shales and sands which geologically are rocks will be termed soil in an engineering classification.
Sand fraction:	That portion in a soil mass that will pass between the sieve sizes of 2.0 mm and 0.075 mm. In geological terms a sand is between 2.0 and 0.06 mm
Silt:	Mineral particles deposited as sediment in water and of such size that all will pass the 0.075 mm sieve. Individual particles are indistinguishable to the naked eye.

Silt fraction:	That portion of the soil mass that lies between the sizes 0.06 mm and 0.002 mm.
Soil:	A mixture of inorganic mineral particles together with some water and air, which may be further described as sands, gravels, silts and clays.
Soil moisture suction:	The suction forces which generate the cohesion in fine grained soils below the Liquid Limit.
TRAFFIC Design period:	Is the period during which the proposed pavement must carry the estimated cumulative number of standard axles without the need for major reconstruction work, except for re-sealing. At the end of this period the pavement should still be in a sufficiently good condition that strengthening will result in a further period of satisfactory traffic-carrying
Equivalence Factor:	of an axle or vehicle is the number of passages of an Equivalent Standard Axle which would cause the same damage to a road as one passage of the axle or vehicle in question.
Equivalent Standard Axle (ESA):	A design concept to enable the damaging effect of a range and number of different axle loads, to be considered in the structural design of a pavement. The equivalent standard axle imposes a load of 8,200 kg.

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